NOT MEASUREMENT SENSITIVE

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01 February 2017

DEPARTMENT OF DEFENSE INTERFACE STANDARD

NATIONAL IMAGERY TRANSMISSION FORMAT VERSION 2.1

FOR THE
NATIONAL IMAGERY TRANSMISSION FORMAT STANDARD

AMSC N/A **AREA GINT**

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FOREWORD

1. This standard is approved for use by all departments and agencies of the Department of Defense (DOD).

2. The National Imagery Transmission Format (NITF) Standard (NITFS) is the suite of standards documents for formatting digital imagery and imagery-related products and exchanging them among the Department of Defense (DOD), Intelligence Community (IC), and other United States (US) Government departments and agencies. Resulting from a collaborative US Government and industry effort, it is the common standard used to exchange and store files comprising images, symbols and graphics, text, and associated data.

3. The NITFS Technical Board (NTB) develops, coordinates, reviews, and plans for the NITFS, including this standard. It is a consensus-based government/industry forum that responds to the Geospatial Intelligence Standards Working Group (GWG). The GWG manages geospatial and imagery standards for the DOD and IC encompassed by the National System for Geospatial Intelligence (NSG).

4. The DOD and the IC are committed to the interoperability of systems used for formatting, transmitting, receiving, and processing imagery and imagery-related information. This standard describes the core of the National Imagery Transmission Format (NITF) and establishes its application within the NITFS.

5. Comments, suggestions or questions should be addressed to the National Geospatial-Intelligence Agency (NGA) National Center for Geospatial Intelligence Standards (NCGIS), Mail Stop N32, 7500 GEOINT Drive, Springfield, VA 22150, or emailed to ncgis-mail@nga.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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Change Log

Date	Pages Affected	Mechanism
01 May 2006 MIL-STD2500c	All	Incorporated the following approved RFCs: RFC-NTB-038 RFC-NTB-046 RFC-NTB-050 RFC-NTB-065 RFC-NTB-066
01 Aug 2016	All	Updated version from 2500c to 2500c w/CHANGE 1
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1. SCOPE

1.1 Scope. This standard establishes the requirements for the file format component of the National Imagery Transmission Format (NITF) Standard (NITFS). The file format described in this document is called the NITF. The NITFS is a collection of standard specifications that form the foundation for interoperability in the dissemination of still and motion imagery and imagery associated data among different computer systems. The use of NITF is independent of the domain (air, land, sea, space, etc.) in which the collecting platform operates.

1.2 Purpose. This document provides a detailed description of the standard file format structure for NITF version 2.1 (NITF2.1). It specifies the data content and format for all fields defined within the NITF file. Several NITF implementation issues are addressed in the appendices. An example of NITF as the basis for file formation in tactical communications is provided in Appendix E. Certifiable implementation of the NITF for support of interoperability is subject to constraints not specified in this standard. Pertinent compliance requirements are defined in DODI 8330.01.

1.3 Applicability. This standard is applicable to Department of Defense (DOD) and Intelligence Community (IC) members. It is mandatory for all Secondary Imagery Dissemination Systems (SIDS) in accordance with the Assistant Secretary of Defense for Command, Control, Communications, and Intelligence (ASD/C3I)) memorandum, Subject: National Imagery Transmission Format Standard (NITFS), 12 August 1991, and is applicable also to all types of primary imagery systems such as Unmanned Aerial Systems (UAS), archives, and libraries where imagery will be made available to the NSG. New equipment and systems, those undergoing major modification, or those capable of rehabilitation, will conform to this standard.

2. APPLICABLE DOCUMENTS

See Appendix C for References.

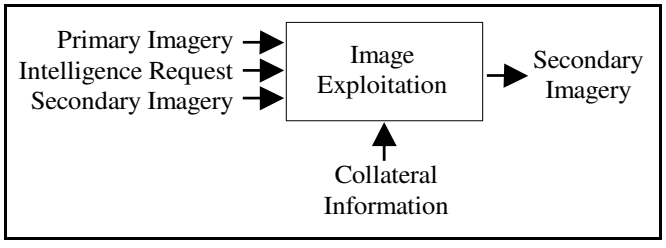
3. DEFINITIONS

See Appendix A for Acronyms and Abbreviations; See Appendix B for Terms and Definitions.

4. GENERAL REQUIREMENTS

4.1 Background. The DOD and the IC use many types of systems for the reception, transmission, storage, and processing of images, graphics, text, and other associated data. Without special efforts, the file format used in systems of one service or agency are likely to be incompatible with the format of another system. Since each system may use a unique, internal data representation, a common format for exchange of information across systems is needed for interoperability of systems within and among DOD and IC organizations. As the need for imagery-related systems grows, their diversity is anticipated to increase. The need to exchange data is also anticipated to increase, even though systems of each organization must retain their own individual characteristics and capabilities. This document defines the NITF, the standard file format for still and motion imagery and imagery-related products to be used by the DOD and IC. The NITF provides a common basis for storage and interchange of images and associated data among existing and future systems. The NITF can be used to support interoperability by providing a data format for shared access applications, while also serving as a standard file format for dissemination of images, graphics, text, and associated data.

4.2 NITF Operations Concept. The NITF shall be used as an interoperability format for transmission and storage of electronic imagery within and among DOD and IC organizations. The NITF has direct application to the dissemination of imagery to requesters of imagery derived intelligence. Multimedia intelligence reports will be composed and packaged into a single file that answers the Essential Elements of Information (EEI) of a particular requester. Intelligence reports may be composed of textual reports along with images, annotated images, graphics, and maps. Intelligence reports are generated after an interpreter exploits primary images or further exploits secondary images pulled out of an archive. The NITF is suitable for archiving imagery required to support the collection process in the reconnaissance cycle. FIGURE 1 illustrates the elements used in the exploitation process of the reconnaissance cycle.



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FIGURE 1. NITF Operational Concept.

In the NITF concept, imagery data interchange between systems is organized in files and is enabled by a potential cross-translation process. When systems use other formats than NITF, such as an internal imagery format, each system will have to translate between the system's internal representation for files, and the NITF file format. A system from which imagery data is to be transferred is envisioned to have a translation module that accepts information, structured according to the system's internal representation for images, graphics, text, and other associated data, and assembles this information into one file in the standard NITF file format. Then the file will be exchanged with one or more recipients. The receiving systems will reformat the file, converting it into one or more files structured as required by the internal representation of the receiving station. The functional architecture of this cross-translation process is shown on FIGURE 2. In the diagram, the terms "Native$_1$ File Format" and "Native$_2$ File Format" refer to files represented in a way potentially unique to the sending or receiving system. Using the NITF, each system must be compliant with only one external file format that will be used for interchange with all other participating systems. The standard format allows a system to send data to several other systems since each receiving system converts the file into its own native file format. Each receiving system can translate selectively and permanently store only those portions of data in the received file that are of interest. This allows a system to transmit all of its data in one file, even though some of the receiving systems may be unable to process certain elements of the data usefully. NITF can also serve as the internal native file format so any translation would be eliminated.

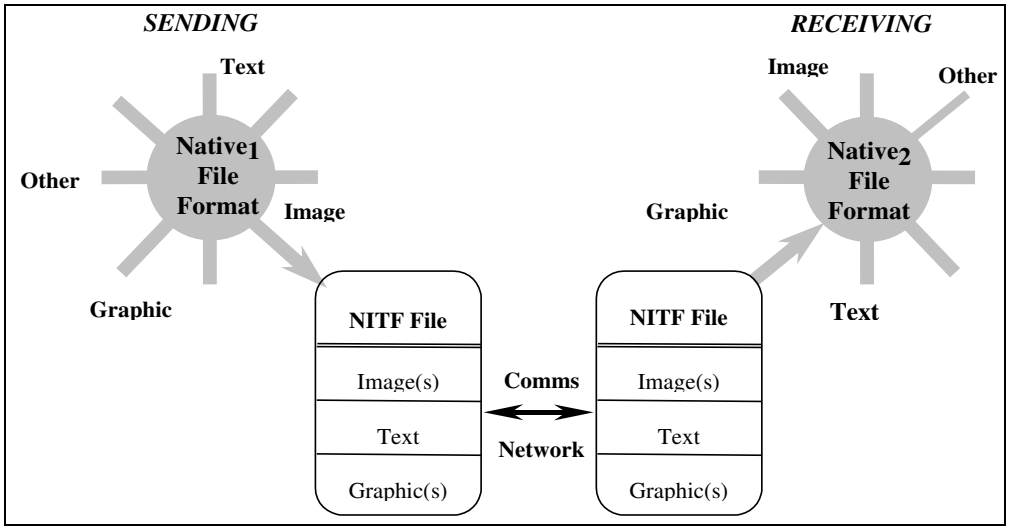


FIGURE 2. NITF Functional Architecture.

4.3 NITF Design Objectives. The design objectives of the NITF are as follows:

a. To provide a means whereby diverse systems can share imagery and associated data.

b. To allow a system to send comprehensive information within one file to users with diverse needs or capabilities, allowing each user to select only those portions of data that correspond to their needs and capabilities.

c. To minimize the cost and schedule required to achieve such capability.

4.4 NITF General Requirements. The NITF is specified to satisfy several general requirements in response to the role it plays in the NITF functional architecture. These requirements are:

a. To be comprehensive in the kinds of data permitted in the file within the image-related objectives of the format, including geo-located imagery or image related products.

b. To be implementable across a wide range of computer systems without reduction of available features.

c. To provide extensibility to accommodate data types and functional requirements not foreseen.

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d. To provide useful capability with limited formatting overhead.

4.5 NITF Characteristics. To serve a varied group of users exchanging multiple types of imagery and associated data who are using differing hardware and software systems, the NITF strives to possess the following characteristics:

a. Completeness - allows exchange of all needed imagery and associated data.

b. Simplicity - requires minimal preprocessing and post processing of transmitted data.

c. Minimal overhead - minimized formatting overhead, particularly for those users transmitting only a small amount of data and for bandwidth-limited users.

d. Universality - provides universal features and functions without requiring commonality of hardware or software.

4.6 NITF File Structure. The NITF file consists of the NITF file header and one or more segment(s). A segment consists of a subheader and data fields as shown on FIGURE 3. An example is provided in Appendix E, TABLE E-11.

<table>
<tr><td colspan="7">NITF File</td></tr>
<tr><td rowspan="2">NITF
File Header</td><td colspan="2">Segment 1</td><td colspan="2">…</td><td colspan="2">Segment n</td></tr>
<tr><td>Sub-
Header</td><td>Data
Field</td><td>…</td><td>…</td><td>Sub-
Header</td><td>Data
Field</td></tr>
</table>

FIGURE 3. NITF File Structure.

4.7 Common Coordinate System (CCS). The Common Coordinate System (CCS) is the virtual two dimensional Cartesian-like coordinate space that shall be common for determining the placement and orientation of displayable data within a specific NITF file and among correlated NITF files which comprise an integrated product.

4.7.1 CCS Structure. The virtual CCS structure can be conceived of as a two dimensional drawing space with a coordinate system similar in structure to the lower right quadrant of the Cartesian coordinate system. The CCS has two perpendicular coordinate axes, the horizontal column axis and the vertical row axis as depicted on FIGURE 4. The positive directions of the axes are based on the predominate scan (column) and line (row) directions used by the digital imagery community. The intersection of the axes is designated as the origin point with the coordinates (0, 0). Given the orientation of the axes on FIGURE 4, the positive direction for the column axis is from (0, 0) and to the right; the positive direction for the row axis is from (0, 0) downward. The quadrant represented by the positive column and positive row axes is the only coordinate space in which NITF displayable data may be located.

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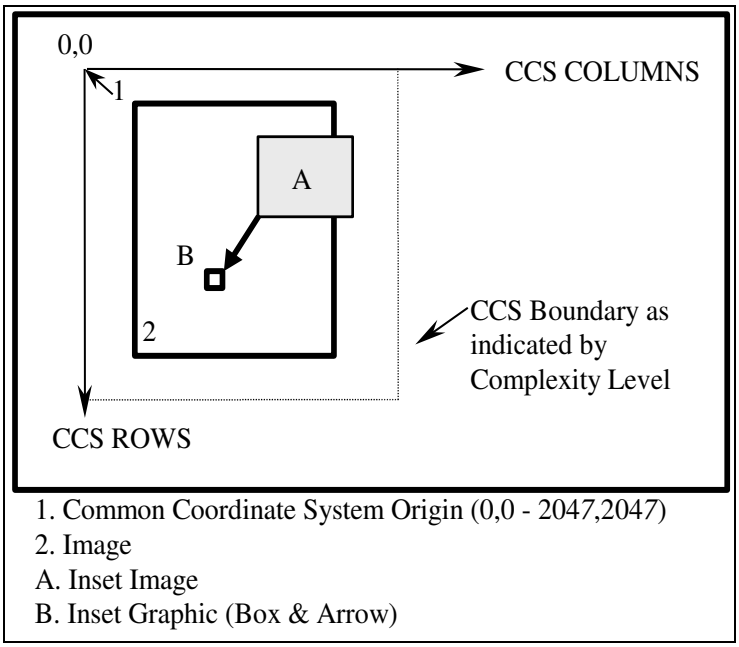


FIGURE 4. Common Coordinate System (CCS) Example.

4.7.2 Row and Column Coordinates. Displayable data shall be placed in the CCS according to the row and column coordinates placed in subheader location fields (for example, Image Location (ILOC), Graphic Location (SLOC)). The location coordinates of a specific data item (as shown on FIGURE 4.) represent row and column offsets from either the CCS origin point (when 'unattached'), or the location point in the CCS of the data item to which the image or graphic is attached. Other means used to locate displayable data shall be directly correlated to row and column coordinates. (For example, displayable tagged extension data might have geo-location data correlated with row and column indices.) When location coordinates are relative to the CCS origin coordinates, they shall always have a positive value. When location coordinates are relative to the location coordinates of an "item" to which they are attached, both positive and negative offset values are possible. In all cases, the location coordinates selected for any "data item" shall ensure that none of the displayable item extends outside of the quadrant defined by the axes of the CCS.

4.7.3 Common Coordinate System and Motion Imagery. The Motion Imagery Extension for NITF 2.1, NGA.STND.0044 (MIE4NITF) provides an extension to MIL-STD-2500C that allows image segments and blocks within image segments to contain multiple, temporally related, images called frames. MIE4NITF extends some of the NITFS concepts regarding the common coordinate system (CCS) and overlay of imagery. During viewing of a motion imagery sequence, frames occurring at a later point in time are meant to replace earlier frames. Frames with the same CCS coordinates, display levels and attachment levels, but a later temporal index are to be displayed on top of frames with an earlier temporal index as the motion imagery sequence is viewed.

Furthermore, the MIE4NITF allows coordination of the CCS, display levels and attachment levels across NITF files. This is allowed for cases where very large frames, which may not readily fit within a single NITF file, may be spatially segmented into multiple files through the use of virtual cameras and camera sets. Some systems may use this feature to create dashboard views of the motion imagery (MI) collection as well. See NGA.STND.0044 for further details and examples.

4.7.4 Complexity Level (CLEVEL) Constraints. The upper and left boundaries of the CCS are explicitly constrained in the specification. When CLEVEL constraints are specified, one of the key attributes for specification shall be to identify the lower and right boundary drawing space constraints for a given CLEVEL. CLEVEL constraints are defined in TABLE 9, TABLE 9(A).

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5. DETAILED REQUIREMENTS

5.1 Format Description.

5.1.1 Header, Segments, And Fields. The format contains a file header and segments. A segment contains a subheader, and data fields. The NITF file header and subheader fields are byte aligned. A file header carries information about the identification, classification, structure, content, size of the file as a whole, and the number and size of the major segments within the file. For each type of data segment, as shown on FIGURE 5, supported by the format, there is an associated subheader and data fields. A subheader contains information that describes characteristics of data fields that contain the actual data.

5.1.2 Extension Segments, Conditional Fields. Flexibility to add support for the data and data characteristics not explicitly defined in this standard is provided within the format. This is accomplished by providing for conditional fields in the file header and in each subheader indicating the presence of Tagged Record Extensions (TRE) and providing for a group of Data Extension Segments (DES). A TRE in a header or subheader may contain additional characteristics about the corresponding data segment, while the extension segments are intended primarily to provide a vehicle for adding support for new kinds of data. The identifier (tag name) for the TRE, and extension segment identifiers, will be coordinated centrally to avoid conflicting use. All Support Data Extensions (SDEs that is, TREs or DESs) shall be registered with the NGA NITF Technical Board (NTB). Any SDE not registered with the NTB is not, and shall not be, valid. A current listing of the TREs and DESs that have been registered with NGA is provided in the NITFS TRE Registry and NITFS DES Registry maintained by the NTB and listed on the Joint Interoperability Test Command (JITC) web site or in the NSG Standards Registry. All NITF implementations should handle the receipt of unknown extensions by at least recognizing that they are unknown extension types and ignoring them. This can be accomplished using the byte count, extension identifier, and data length field. Using these length offsets, the unknown extension can be ignored and the user can be informed that extension data has been skipped.

5.1.3 Supported Types of NITF Segments. A single NITF file may comprise different types of segments. A segment containing information of a standard data type (see sub-paragraphs 5.1.3.1 to 5.1.3.5) is called a standard data segment. The organization of the different types of segments is described below and on FIGURE 5.

5.1.3.1 Image Segments (IS). An Image Segment (IS) supports the standard image type of data.

5.1.3.2 Graphic Segments (GS). A Graphic Segment (GS) supports the standard graphic type of data.

5.1.3.3 Reserved Segments (RS). The Reserved Segments (RS) are place holders to support a future standard type of data that has yet to be defined.

5.1.3.4 Text Segments (TS). A Text Segment (TS) supports the standard text type of data.

5.1.3.5 Data Extension Segments (DES). A DES allows for the addition of different data types with each type encapsulated in its own DES.

5.1.3.6 Reserved Extension Segments (RES). A Reserved Extension Segment (RES) is a non-standard data segment that is user-defined. An NITF file can support different user-defined types of segments called RES.

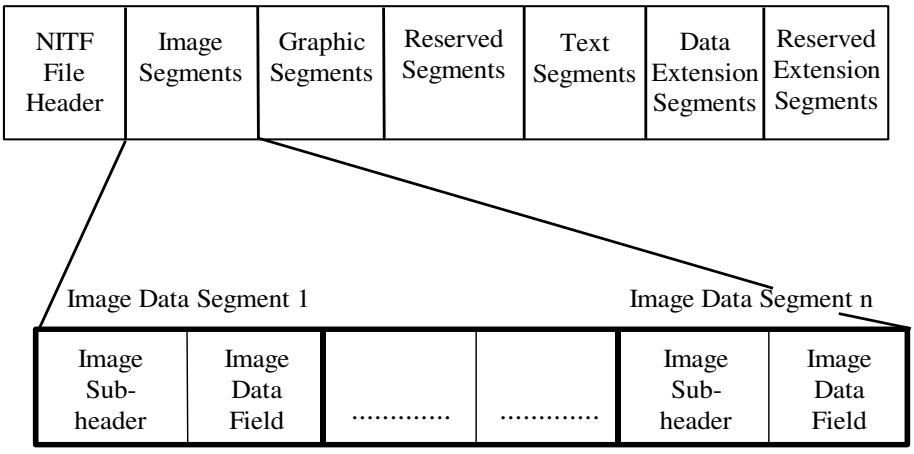


FIGURE 5. NITF Segment Order and Structure

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5.1.4 Application Guidance. The NITF file supports inclusion of standard data segments of information in a single file: image, graphic, and text. It is possible to include zero, one, or multiples of each standard data segment in a single file (for example: several images, but no graphics). Standard data segments shall be placed in the file in the following order: all ISs, followed by all GSs, followed by all TSs.

5.1.5 Standard Data Segment Subheaders. Each individual, standard data segment included in an NITF file, such as an IS or a GS, shall be preceded by a subheader corresponding to that data segment. This subheader shall contain details of that particular data segment. If no segments of a given type are included in the file, a subheader for that information type shall not be included in the file. All segments and associated subheaders of a single type shall precede the first segment for the next data type. The ordering of multiple segments of one type is arbitrary. A diagram of the overall NITF file structure is shown on FIGURE 5.

5.1.6 Header/Subheader Field Specification. The specification of the fields in the various headers/subheaders found within an NITF file is provided in a series of tables in the body of this document (an example is provided in Appendix E, TABLE E-11). Each table includes a mnemonic identifier for each FIELD within a header/subheader, the field's NAME, a description of the valid contents of the field, and constraints on the field's use, the field SIZE in bytes, the VALUE RANGE it may contain, and an indication of its TYPE (paragraph 5.1.8). The NITF file header fields are specified in TABLE 1. The standard data type segment subheader fields are specified in Tables 3, 4, and 5. The TRE subheaders (paragraph 5.8.1) are defined in TABLE 6. Finally, the DES subheader fields (paragraph 5.8.2) and RES subheader fields (paragraph 5.8.4) are defined in Tables 7 and 8. Submitting new values to header, subheader, and TRE fields can be done through the Custodian or at http://jitc.fhu.disa.mil/projects/nitf/registers/nitfcte/trereg.aspx.

5.1.7 Field Structure and Default Values. The NITF uses byte counts to delimit header and subheader fields, as opposed to special end-of-field characters or codes or direct addressing. These counts are provided in the tables, found in appendix A, that detail the NITF header and subheader field specifications.

a. Character Set. To provide simple communication among NITF stations, data within NITF are mostly represented using characters. Numbers represented by characters eliminate problems caused by word length and machine internal representation differences. Humans can easily read header and subheader fields populated with characters. The character sets used in NITF are:

(1) Universal Multiple Octet Coded Character Set (UCS) Transformation Format 8 (UTF-8) Subset (U8S). The NITF U8S is a subset of the UCS character set limited to 1-byte and 2-byte UTF-8 encoded characters (Basic Latin and Latin Supplement 1). The 1-byte encoded characters of the UTF-8 Subset (U8S) are the BCS characters. Their most significant bit is necessarily set to 0. The 2-byte encoded characters of U8S are described in TABLE D-2. The most significant bit of their first byte is set to 1, indicating that a second byte follows.

(2) Extended Character Set (ECS). The NITF ECS is a set of 1-byte encoded characters. Valid ECS character codes range from 0x20 to 0x7E, and 0xA0 to 0xFF, as well as Line Feed (0x0A), Form Feed (0x0C), and Carriage Return (0x0D). The ECS characters are described in TABLE D-1.

(3) Extended Character Set – Alphanumeric (ECS-A). The NITF ECS-A is a subset of the ECS. Valid ECS-A character codes range from 0x20 to 0x7E, and 0xA0 to 0xFF. Line Feed (0x0A), Form Feed (0x0C), and Carriage Return (0x0D) are not valid ECS-A characters.

(4) Basic Character Set (BCS). The NITF BCS is a subset of the ECS. The most significant bit of the BCS characters is set to 0. Valid BCS characters code shall range from 0x20 to 0x7E, plus Line Feed (0x0A), Form Feed (0x0C), and Carriage Return (0x0D).

(5) Basic Character Set – Alphanumeric (BCS-A). The NITF BCS-A is a subset of the BCS. Valid BCS-A character codes range from 0x20 to 0x7E.

(6) Basic Character Set-Numeric (BCS-N). The NITF BCS-N is a subset of the BCS that consists of the digits ‘0’ to ‘9’ (codes 0x30 to 0x39), plus sign (0x2B), minus sign (0x2D), decimal point (0x2E), and slash (0x2F).

(7) Basic Character Set-Numeric Integer (BCS-N integer). A subset of the BCS-N that consists of the digits ‘0’ to ‘9’ (codes 0x30 to 0x39), plus sign (0x2B), and minus sign (0x2D).

(8) Basic Character Set-Numeric Positive Integer (BCS-N positive integer). A subset of the BCS-N that consists of the digits ‘0’ to ‘9’ (codes 0x30 to 0x39).

b. Use of NITF Character Sets. All data in ECS-A or BCS-A populated NITF Header and Subheader Fields shall be left justified and padded to the right boundary with BCS Spaces (0x20). BCS-N positive integer fields and BCS-N integer Fields may contain one or more integer values. Each of these NITF encoded values has a fixed length and

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position within the field. Each NITF encoded integer value is right justified and padded to the left boundary with leading BCS Zeros (0x30)). However, where a BCS-N field allows a plus sign (0x2B) or minus sign (0x2D), it is the left most character of the integer value.

c. Standard Default Value. The standard default value shall be BCS Spaces (0BCS Spaces (0x20) for alphanumeric fields and BCS Zero (0x30) for numeric fields. For a few fields, a specific default may be indicated in the field description. In this case, the field description shall take precedence. All NITF Header and Subheader Fields contained in an NITF file shall contain either valid data (that is, data in accordance with the restrictions specified for the contents of the field in this document) or the specified default value.

d. Date and Time Field Values. Date and time fields within an NITF file often represent currency and sequence of information that is critical to the exploitation and interpretation of data. For those occasions when portions of the date and/or time entry are not obtainable or complete, the following convention will apply. Populate the unknown date/time two character subfield with two hyphen-minus characters (hexadecimal code "2D") indicating the portion of the date or time that is unknown. For example, populating a date and time field when the Century (CC), Year (YY), Month (MM), and Day (DD) are known, but the hour (hh), minute (mm), and second (ss) values are unknown, appears as: 20020425------. In another example such as a birthday of 14 Feb 47, where the CC is unknown or not expressed by the source of the information, the value would appear as: --470214.

e. Country Codes. Country codes found within the NITF File fall into the following categories:

(1) Bare Country Codes — a country code without explicitly specified or linked codespace information in the NITF file itself. For example, the IS.TGTID field contains a bare country code because the codespace specifying how that code should be interpreted is not contained in the NITF file. There are two types of bare codes:

(a) An a priori code is by definition a country code that is defined by some external unspecified country code standard, and the data consumer must already know which country code standard is in use to correctly identify the country associated with that code. For example, the 2-character code “US”, with no other codespace or version information right within the data file is an a priori code. All a priori codes contain only letters (A-Z).

- The 2-character a priori code “XN” is reserved for use in the File and Segment Security Classification System fields to indicate that classified data within the file or segment was generated by a component using NATO security marking guidance.

- The 2-character code “XF” is reserved for use in the File and Segment Security Classification System fields to indicate the de facto classification system represented by the security marking indicator NATO/ISAF”.

- The 2-character code “YY” is reserved for use when a country code is required but no other valid country code accurately reflects the geopolitical or geographic entity described.

- The 2-character code “UU” is reserved for use when a country code is required but the geopolitical entity to be described cannot be indicated or is an undesignated sovereignty.

- Other a priori 2-character codes are defined by the Geopolitical Entities and Codes (GEC) Registry, including all updates. The short URN-based codespace identifier for 2-character a priori codes as defined by GEC Update 18 is “ge:GEC:2:U18”. The GEC 2-character codespace is an explicit superset of the legacy FIPS PUB 10-4 standard (documented in the GENC registry using the “ge:F104:2:CN14” short URN-based codespace identifier.

- 3-character a priori codes are defined in the GENC geopolitical entity 3-character codespace. If a data consumer needs to translate an a priori 3-character code to the name of that country using a GENC service or local GENC database, it is up to the data consumer to generate the URI describing that country in whatever form is necessary for indexing into the database or service. For example, to determine the formal full name of country associated with the 3-character code “MMR”, using the most recent update to the GENC country data, the data consumer could combine this code with the GENC short URN-based codespace identifier “ge:GENC:3:now”, resulting in the short URN-based individual item identifier “ge:GENC:3:now:MMR”.

- Tetragraphs for the US information security marking system are defined by Information Security Marking Codes and Tetragraphs (ISMCAT).

(b) Ad hoc codes are defined by an individual data provider, potentially on a file-by-file basis. All ad-hoc codes contain at least one letter (A-Z or a-z) and one number (0-9). Data consumers must not assume that these identifiers have a constant interpretation across data files from different data producers, or even across data files from the same data producer. Interpretation of ad hoc codes is outside the scope of MIL-STD-2500C; however, a Tagged Record

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Extension (TRE) or Data Extension Segment (DES) can be defined to provide the translation of ad hoc codes, for example by providing a translation of those codes to GENC Short URN-based individual item identifiers.

(2) Fully-specified codes include a full specification of the codespace within the NITF file itself, potentially on a country code by country code basis. All new and updated country code fields are recommended to be formatted as a fully specified code using the short URN-based individual item identifier, as defined in section 5.4.4.4 of the GENC standard (e.g., "ge:GENC:3:3-5:USA"). This will allow the data provider to independently refer to the definition of a country and its name as appropriate for each individual field; relying on a shared codespace (e.g., storing "ge:GENC:3:3-5" in one field that applies to several other fields, which would contain just the bare code itself (e.g., "USA", "AUS") might force a data consumer to use a version of a country and country name definition that is inappropriate for that individual field.

5.1.8 Field Types. The NITF file header and various subheaders have two types of fields: required and conditional. A required field shall be present and shall contain valid data or the specified default value. A conditional field may or may not be present depending on the value of one or more preceding (required) fields. If a conditional field is present, it shall contain valid data. When a field is conditional, its description identifies what conditions and which preceding field or fields are used to determine whether or not to include it in the file. For example, in the NITF header, if the Number of Image Segments (NUMI) field contains the value of 2, the Length of the 1st Image Subheader (LISH1), Length of the 1st Image Segment (LI1), Length of the 2nd Image Subheader (LISH2), and Length of the 2nd Image Segment (LI2) fields will be present and must be filled with valid data. However, if the NUMI field contains a BCS zero (0x30), the subheader length and image length fields are omitted.

5.1.9 Logical Recording Formats.

5.1.9.1 Bit and Byte Order.

a. The method of recording numeric data on interchange media shall adhere to the "big endian" convention. In big endian format, the most significant byte in each numeric field shall be recorded and read first, and successive bytes recorded and read in order of decreasing significance. That is, if an n-byte field F is stored in memory beginning at address A, then the most significant byte of F shall be stored at A, the next at A+1, and so on. The least significant byte shall be stored at address A+n-1.

b. BCS character strings shall be recorded in the order in which the data is generated.

c. The most significant bit in each byte of every field, regardless of data type, shall be recorded and read first, and successive bits shall be recorded and read in order of decreasing significance.

d. Pixel arrays shall be recorded in the order specified in the Image Mode (IMODE) field and as discussed in paragraph 5.4.3.3. Pixel arrays shall be recorded from left to right starting at the top, and non-interlaced raster scanning downward. The top left pixel shall be recorded first, and the bottom right pixel shall be recorded last.

5.1.9.2 Row Column Relationship. NITF imagery is displayed by mapping each image pixel to a specific row "r" and column "c" within the bottom right quadrant shown on FIGURE 6. Rows are represented on the vertical (y-axis) and columns are represented on the horizontal (x-axis). Moving from location 0,0 down and to the right is considered moving in a positive direction. If the first pixel of an image is placed at r0,c0, it would be followed by pixels r0,c1; r0,c2 and so on until the end of the row. The first pixel of the second row of image pixels would be located at r1,c0.

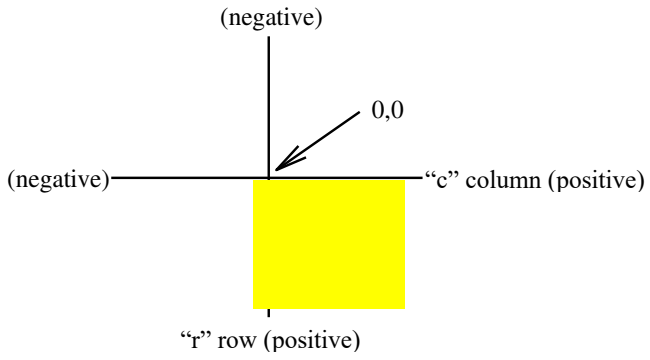


FIGURE 6. Row Column Relationship.

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5.2 The NITF File Header. Each NITF file shall begin with the file header, whose fields contain identification and origination information, file-level security information, and the number and size of segments of each type, such as IS(s), GS(s), and TS(s), contained in the file. FIGURE 7 depicts the NITF file header. It depicts the types of information contained in the header and shows the header's organization as a sequence of groups of related fields. The expansion of the "Image Group" illustrates how the header's overall length and content may expand or contract depending on the number of data segments of each type included in the file. The NITF header is detailed in TABLE 1. An example is provided in Appendix E, TABLE E-11.

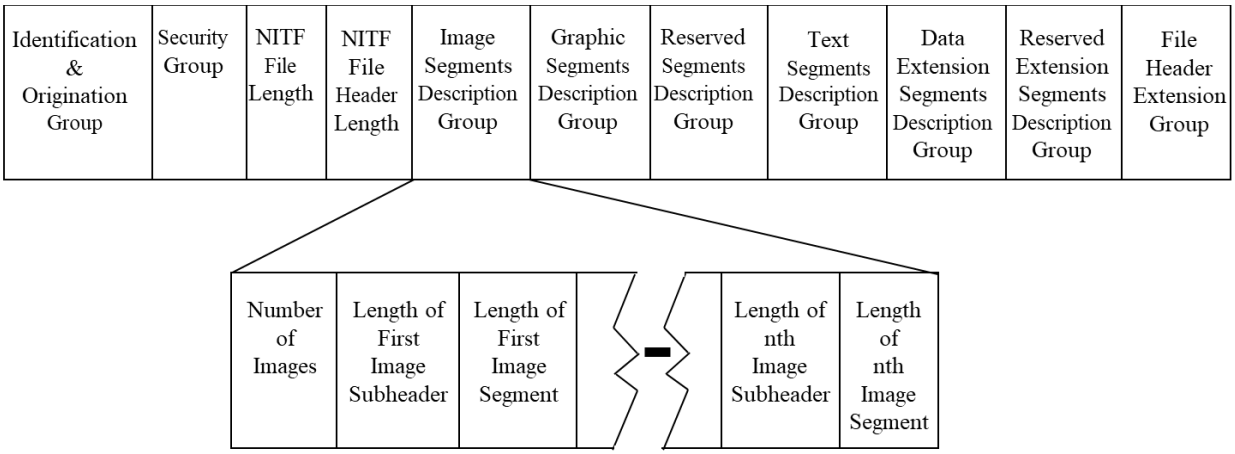


FIGURE 7. NITF File Header Structure and Content

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TABLE 1. NITF File Header.

(TYPE "R" = Required, "C" = Conditional, "< >" = BCS spaces allowed for entire field)
("†" annotations are explained at the end of the table)

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
FHDR	File Profile Name. This field shall contain the BCS-A character string uniquely denoting that the file is formatted using NITF. The valid value for this field is NITF.	4	BCS-A NITF	R
FVER	File Version. This field shall contain a BCS-A character string uniquely denoting the version. The valid value for this field is 02.10.	5	BCS-A 02.10	R
CLEVEL	Complexity Level. This field shall contain the complexity level required to interpret fully all components of the file. Valid entries are assigned in accordance with complexity levels established in TABLE 9. and 9(A)	2	BCS-N positive integer 01 to 99	R
STYPE	Standard Type. Standard type or capability. A BCS-A character string BF01 which indicates that this file is formatted using ISO/IEC IS 12087-5. NITF02.10 is intended to be registered as a profile of ISO/IEC IS 12087-5.	4	BCS-A BF01	R
OSTAID	Originating Station ID. This field shall contain the identification code or name of the originating organization, system, station, or product. It shall not be filled with BCS spaces (0x20).	10	BCS-A	R
FDT	File Date and Time. This field shall contain the time (UTC) (Zulu) of the file's origination in the format CCYYMMDDhhmmss, where CC is the century (00 to 99), YY is the last two digits of the year (00 to 99), MM is the month (01 to 12), DD is the day (01 to 31), hh is the hour (00 to 23), mm is the minute (00 to 59), and ss is the second (00 to 59). UTC is assumed to be the time zone designator to express the time of day.	14	BCS-N integer CCYYMMDDhhmmss	R
FTITLE	File Title. This field shall contain the title of the file or shall be filled with ECS spaces (0x20).	80	ECS-A (Default is ECS spaces (0x20))	<R>
FSCLAS	File Security Classification. This field shall contain a valid value representing the classification level of the entire file. Valid values are T (=Top Secret), S (=Secret), C (=Confidential), R (=Restricted), U (=Unclassified).	1	ECS-A T, S, C, R, or U	R
NOTE: If FSCLAS is T, S, C, or R, then FSCLSY must be populated with a valid code for the security classification system used.				

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TABLE 1. NITF File Header – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
FSCLSY	File Security Classification System. This field shall contain valid values indicating the national or multinational security system using a geopolitical (country) code digraph to classify the file. For US producers a geopolitical code (Country Code) as specified by Paragraph 5.1.7e shall be used to indicate their national security systems. The designator "XN" is for classified data generated by a component using NATO security system classification policy marking guidance. The designator "XF" indicates the de facto classification system policy represented by the security marking indicator NATO/ISAF". These codes were selected to not duplicate codes within the GEC codespace. If this field is populated with all ECS spaces (0x20), it shall imply that no security classification system applies to the file.	2	ECS-A (Default is ECS spaces (0x20)) The most up to date list of registered digraph geopolitical code values can be found on the NITF Field Value Registry, specified in section C.2.1 Online Resources.	<R>
NOTE: If any of the following fields are populated with anything other than spaces, then FSCLSY must be populated with a valid code for the security classification system used: FSCODE, FSCTLH, FSREL, FSDCTP, FSDCDT, FSDCXM, FSDG, FSDGDT, FSCLTX, FSCATP, FSCAUT, FSCRSN, FSSRDT, and FSCTLN.				
FSCODE	File Codewords. This field shall contain a word(s) that has been assigned a classification and a classified meaning to safeguard intentions and information regarding a specific classified plan or operation. This codeword may be a cryptonym used to identify sensitive intelligence data. Multiple entries shall be separated by a single ECS space (0x20): The selection of a relevant set of codewords is application specific.). If this field is populated with all ECS spaces (0x20), it shall imply that no codewords apply to the file. Past codeword examples are located in the NITF registry, specified in section C.2.1 Online Resources.	11	ECS-A (Default is BCS spaces (0x20))	<R>
FSCTLH	File Control and Handling. This field shall contain valid additional security control and/or handling instructions (caveats) associated with the file. Control and Handling is an indicator governing the distribution and use of information or material. Such indicators are the subject of intelligence community agreement and are specifically defined in appropriate regulations. Example values include digraphs found in the NITF Field Value Registry, specified in section C.2.1 Online Resources. The digraph may indicate single or multiple caveats. The selection of a relevant caveat(s) is application specific. If this field is populated with all ECS spaces (0x20), it shall imply that no additional control and handling instructions apply to the file.	2	ECS-A (Default is ECS spaces (0x20))	<R>
FSREL	File Releasing Instructions. This field shall contain a valid list of bare country and/or multilateral entity codes to which countries and/or multilateral entities the file is authorized for release. Valid itemsentities in the list are one or more bare country codes as found in FIPS PUB 10-4specified by Paragraph 5.1.7e, separated by a single ECS space (0x20). If this field is populated with all	20	ECS-A (Default is ECS spaces (0x20))	<R>

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TABLE 1. NITF File Header – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
	ECS spaces (0x20), it shall imply that no file release instructions apply.			
FSDCTP	File Declassification Type. This field shall contain a valid indicator of the type of security declassification or downgrading instructions which apply to the file. Valid values are DD (=declassify on a specific date), DE (=declassify upon occurrence of an event), GD (=downgrade to a specified level on a specific date), GE (=downgrade to a specified level upon occurrence of an event), O (=OADR), and X (= exempt from automatic declassification). If this field is populated with all ECS spaces (0x20), it shall imply that no file security declassification or downgrading instructions apply.	2	ECS-A DD, DE, GD, GE, O, X (Default is ECS spaces (0x20))	<R>
FSDCDT	File Declassification Date. This field shall indicate the date on which a file is to be declassified if the value in the File Declassification Type field is populated with DD. If this field is populated with all ECS spaces (0x20), it shall imply that no file declassification date applies.	8	ECS-A CCYYMMDD (Default is ECS spaces (0x20))	<R>
FSDCXM	File Declassification Exemption. This field shall indicate the reason the file is exempt from automatic declassification if the value in File Declassification Type is X. Valid values are X1 to X8 and 25X1 to 25X9. Values X1 to X8 correspond to the declassification exemptions found in DOD 5200.01-V1, paragraphs 4-202b(1) to (8) for material exempt from the 10-year rule. Values 25X1 to 25X9 correspond to the declassification exemptions found in DOD 5200.01-V1, paragraphs 4-301a(1) to (9) for permanently valuable material exempt from the 25-year declassification system. If this field is populated with all ESC spaces (0x20), it shall imply that a file declassification exemption does not apply. Value DNIO is Exempt from automatic declassification and can be declassified only by the Director of National Intelligence (DNI).	4	ECS-A X1 to X8, 25X1 to 25X9, DNIO, (Default is ECS spaces (0x20))	<R>
FSDG	File Downgrade. This field shall indicate the classification level to which a file is to be downgraded if the values in File Declassification Type are GD or GE. Valid values are S (=Secret), C (=Confidential), R (= Restricted). If this field contains an ECS space (0x20), it shall imply that file security downgrading does not apply.	1	ECS-A S, C, R (Default is ECS space (0x20))	<R>
FSDGDT	File Downgrade Date. This field shall indicate the date on which a file is to be downgraded if the value in File Declassification Type is GD. If this field is populated with all ECS spaces (0x20), it shall imply that a file security downgrading date does not apply.	8	ECS-A CCYYMMDD (Default is ECS spaces (0x20))	<R>
FSCLTX	File Classification Text. This field shall be used to provide additional information about file classification to include identification of a declassification or downgrading event if the values in File Declassification Type are DE or GE. It may also be used to identify	43	ECS-A (Default is ECS spaces (0x20))	<R>

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TABLE 1. NITF File Header – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
	multiple classification sources and/or any other special handling rules. Values are user defined free text. If this field is populated with all ECS spaces (0x20), it shall imply that additional information about file classification does not apply. This field may also contain overflow values from the FSCTLH or FSREL fields. Consult with the NITF Field Value Registry for overflow population guidance, specified in section C.2.1 Online Resources.			
FSCATP	File Classification Authority Type. This field shall indicate the type of authority used to classify the file. Valid values are O (= original classification authority), D (= derivative from a single source), and M (= derivative from multiple sources). If this field contains an ECS space (0x20), it shall imply that file classification authority type does not apply.	1	ECS-A O, D, M (Default is ECS space (0x20))	<R>
FSCAUT	File Classification Authority. This field shall identify the classification authority for the file dependent upon the value in File Classification Authority Type. Values are user defined free text which should contain the following information: original classification authority name and position or personal identifier if the value in File Classification Authority Type is O; title of the document or security classification guide used to classify the file if the value in File Classification Authority Type is D; and Derive-Multiple if the file classification was derived from multiple sources and the value of the FSCATP field is populated with M. In the latter case, the file originator will maintain a record of the sources used in accordance with existing security directives. One of the multiple sources may also be identified in File Classification Text if desired. If this field is populated with all ECS spaces (0x20), it shall imply that no file classification authority applies.	40	ECS-A (Default is ECS spaces (0x20))	<R>
FSCRSN	File Classification Reason. This field shall contain values indicating the reason for classifying the file. Valid values are A to H. These correspond to the reasons for original classification per E.O. 13526, Section 1.5.(a) to (h). If this field contains an ECS space (0x20), it shall imply that no file classification reason applies.	1	ECS-A A to H (Default is ECS space (0x20)) The most up to date list of registered values can be found on the NITF Field Value Registry, specified in section C.2.1 Online Resources.	<R>
FSSRDT	File Security Source Date. This field shall indicate the date of the source used to derive the classification of the file. In the case of multiple sources, the date of the most recent source shall be used. If this field is populated with all ECS spaces (0x20), it shall imply that a file security source date does not apply.	8	ECS-A CCYYMMDD (Default is ECS spaces (0x20))	<R>

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TABLE 1. NITF File Header – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
FSCTLN	File Security Control Number. This field shall contain a valid security control number associated with the file. The format of the security control number shall be in accordance with the regulations governing the appropriate security channel(s). If this field is populated with all ECS spaces (0x20), it shall imply that no file security control number applies.	15	ECS-A (Default is ECS spaces (0x20))	<R>
FSCOP	File Copy Number. This field shall contain the copy number of the file. If this field is populated with all BCS zeros (0x30), it shall imply that there is no tracking of numbered file copies.	5	BCS-N positive integer 00000 to 99999 (Default is BCS zeros (0x30))	R
FSCPYS	File Number of Copies. This field shall contain the total number of copies of the file. If this field is populated with all BCS zeros (0x30), it shall imply that there is no tracking of numbered file copies.	5	BCS-N positive integer 00000 to 99999 (Default is BCS zeros (0x30))	R
ENCRYP	Encryption. This field shall contain the value BCS zero (0x30) until such time as this specification is updated to define the use of other values.	1	BCS zero (0x30) = Not Encrypted	R
FBKGC	File Background Color. This field shall contain the three color components of the file background in the order Red, Green, Blue. Where (0x00, 0x00, 0x00) is black and (0xFF, 0xFF, 0xFF) is white.	3	Unsigned binary integer (0x00 to 0xFF, 0x00 to 0xFF, 0x00 to 0xFF)	R
ONAME	Originator's Name. This field shall contain a valid name for the operator who originated the file. If the field is populated with all ECS spaces (0x20), it shall represent that no operator is assigned responsibility for origination.	24	ECS-A (Default is ECS spaces (0x20))	<R>
OPHONE	Originator's Phone Number. This field shall contain a valid phone number for the operator who originated the file. If the field is populated with all ECS spaces (0x20), it shall represent that no phone number is available for the operator assigned responsibility for origination.	18	ECS-A (Default is ECS spaces (0x20))	<R>
FL	File Length. This field shall contain the length in bytes of the entire file including all headers, subheaders, and data. Note: The largest file is limited to 999999999998 (10^{12} -2) bytes. A value of 999999999999 in this field indicates that the actual file length was not available when the header was created; this concept called STREAMING_FILE_HEADER is no longer supported by 2500C.	12	BCS-N positive integer 000000000388 to 999999999998	R
HL	NITF File Header Length. This field shall contain a valid length in bytes of the NITF file header.	6	BCS-N positive integer 000388 to 999999	R
NUMI	Number of Image Segments. This field shall contain the number of separate image segments included in the file. This field shall be BCS zeros (0x30) if no image segments are included in the file.	3	BCS-N positive integer (Default is BCS zeros (0x30)) 000 to 999	R
. . . . . Start for each IS LISHn, LIn.				
NOTE: LISHn and LIn fields repeat in pairs such that LISH001, LI01; LISH002, LI002; .....LISHn,LIn.				

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TABLE 1. NITF File Header – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
LISHn	Length of nth Image Subheader. This field shall contain a valid length in bytes for the nth image subheader, where n is the number of the IS counting from the first IS (n=001) in order of the image segments' appearance in the file. Possible values for n are 001 to 999. This field shall occur as many times as specified in the NUMI field. This field is conditional and shall be omitted if the NUMI field contains BCS zeros (0x30). Note: The largest image subheader is limited to 999998 (10^6 -2) bytes. A value of 999999 in this field indicates that the actual subheader length was not available when the header was created; this concept called STREAMING_FILE_HEADER is no longer supported by 2500C.	6	BCS-N positive integer 000439 to 999998, 999999	C
LIn	Length of nth Image Segment. This field shall contain a valid length in bytes of the nth IS, where n is the number of the IS counting from the first IS(n=001) in order of the IS appearance in the file. Possible values for n are 001 to 999. If the IS is compressed, the length after compression shall be used. This field shall occur as many times as specified in the NUMI field. This field is conditional and shall be omitted if the NUMI field contains BCS zeros (0x30). Note: The largest image is limited to 9999999998 (10^{10} -2) bytes. A value of 9999999999 in this field indicates that the actual image length was not available when the header was created; this concept called STREAMING_FILE_HEADER is no longer supported by 2500C.	10	BCS-N positive integer 0000000001 to 9999999998, 9999999999	C
. . . . End for each IS LISHn, LIn; the number of loop repetitions is the value specified in the NUMI field.				
NUMS	Number of Graphic Segments. This field shall contain the number of separate graphic segments included in the file. This field shall be BCS zeros (0x30) if no graphic segments are included in the file.	3	BCS-N positive integer 000 to 999	R
. . . . . Start for each GS LSSHn, LSn.				
NOTE: LSSHn and LSn fields repeat in pairs such that LSSH001, LS00; LSSH001, LS002; .....LSSHn,LSn.				
LSSHn	Length of nth Graphic Subheader. This field shall contain a valid length in bytes for the nth graphic subheader, where n is the number of the graphic segment counting from the first GS (n=001) in the order of the graphic segments' appearance in the file. Possible values for n are 001 to 999. This field shall occur as many times as specified in the NUMS field. This field is conditional and shall be omitted if the NUMS contains BCS zeros (0x30). Note: The largest subheader is limited to 9998 (10^4 -2) bytes. A value of 9999 in this field indicates that the actual subheader length was not available when the header was created; this concept called STREAMING_FILE_HEADER is no longer supported by 2500C.	4	BCS-N positive integer 0258 to 9998, 9999	C
LSn	Length of nth Graphic Segment. This field shall contain a valid length in bytes of the nth GS, where n is the number of the GS, counting from the first GS (n=001) in the order of the graphic segments' appearance in the file.	6	BCS-N positive integer 000001 to 999998, 999999	C

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TABLE 1. NITF File Header – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
	Possible values for n are 001 to 999. This field shall occur as many times as specified in the NUMS field. This field is conditional and shall be omitted if NUMS field contains BCS zeros (0x30). Note: The largest graphic is limited to 999998 (106 -2) bytes. A value of 999999 in this field indicates that the actual graphic length was not available when the header was created; this concept called STREAMING_FILE_HEADER is no longer supported by 2500C.			
. . . . End for each GS LSSHn, LSn; the number of loop repetitions is the value specified in the NUMS field.				
NUMX	Reserved for Future Use. This field is reserved for future use and shall be populated with BCS zeros (0x30).	3	BCS zero (0x30)	R
NUMT	Number of Text Segments. This field shall contain the number of separate text segment(s) included in the file. This field shall be BCS zeros (0x30) if no text segments are included in the file.	3	BCS-N positive integer (Default is BCS zeros (0x30)) 000 to 999	R
. . . . . Start for each TS LTSHn, LTn.				
NOTE: LTSHn and LTn fields repeat in pairs such that LTSH001, LT00; LTSH001, LT002; .....LTSHn,LTn.				
LTSHn	Length of nth text subheader. This field shall contain a valid length in bytes for the nth text subheader, where n is the number of the text segment, counting from the first text segment (n=001) in the order of the text segments' appearance in the file. Possible values for n are 001 to 999. This field shall occur as many times as specified in the NUMT field. This field is conditional and shall be omitted if the NUMT field contains BCS zeros (0x30). Note: The largest subheader is limited to 9998 (10^4 -2) bytes. A value of 9999 in this field indicates that the actual subheader length was not available when the header was created; this concept called STREAMING_FILE_HEADER is no longer supported by 2500C.	4	BCS-N positive integer 0282 to 9998, 9999	C
LTn	Length of nth Text Segment. This field shall contain a valid length in bytes of the nth text segment, where n is the number of the text segment, counting from the first text segment (n=001) in the order of the text segments' appearance in the file. Possible values for n are 001 to 999. This field shall occur as many times as specified in the NUMT field. This field is conditional and shall be omitted if the NUMT field contains BCS zeros (0x30). Note: The largest text file is limited to 99998 (10^5 -2) bytes. A value of 99999 in this field indicates that the actual text file length was not available when the header was created; this concept called STREAMING_FILE_HEADER is no longer supported by 2500C.	5	BCS-N positive integer 00001 to 99998, 99999	C
. . . . End for each TS LTSHn, LTn; the number of loop repetitions is the value specified in the NUMT field.				
NUMDES	Number of Data Extension Segments. This field shall contain the number of separate DES included in the file. This field shall be BCS zeros (0x30) if no DES are included in the file.	3	BCS-N positive integer (Default is BCS zeros (0x30)) 000 to 999	R

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TABLE 1. NITF File Header – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
. . . . . Start for each DES LDSHn, LDn.				
NOTE: LDSHn and LDn fields repeat in pairs such that LDSH001, LD00; LDSH001, LD002; .....LDSHn,LDn.				
LDSHn	Length of nth Data Extension Segment Subheader. This field shall contain a valid length in bytes for the nth DES subheader, where n is the number of the DES counting from the first DES (n = 001) in order of the DES's appearance in the file. Possible values for n are 001 to 999. This field shall occur as many times as are specified in the NUMDES field. This field is conditional and shall be omitted if the NUMDES field contains BCS zeros (0x30). Note: The largest subheader is limited to 9998 (10^4 -2) bytes. A value of 9999 in this field indicates that the actual subheader length was not available when the header was created; this concept called STREAMING_FILE_HEADER is no longer supported by 2500C.	4	BCS-N positive integer 0200 to 9998, 9999	C
LDn	Length of nth Data Extension Segment. This field shall contain a valid length in bytes of the data in the nth DES, where n is the number of the DES counting from the first DES (n=001) in order of the DES' appearance in the file. This field shall occur as many times as are specified in the NUMDES field. This field is conditional and shall be omitted if the NUMDES fields contains BCS zeros (0x30). Note: The largest DES is limited to 999999998 (10^9 -2) bytes. A value of 999999999 in this field indicates that the actual DES length was not available when the header was created; this concept called STREAMING_FILE_HEADER is no longer supported by 2500C.	9	BCS-N positive integer 000000001 to 999999998, 999999999	C
. . . . End for each DES LDSHn, LDn; the number of loop repetitions is the value specified in the NUMDES field.				
NUMRES	Number of Reserved Extension Segments. This field shall contain the number of separate RES included in the file. This field shall be BCS zeros (0x30) if no RES are included in the file.	3	BCS-N positive integer (Default is BCS zeros (0x30)) 000 to 999	R
. . . . . Start for each RES LRESHn, LREn.				
NOTE: LRESHn and LREn fields repeat in pairs such that LRESH001, LRE001; LRESH001, LRE002; .....LRESHn,LREn.				
LRESHn	Length of nth Reserved Extension Segment Subheader. This field shall contain a valid length in bytes for the nth RES subheader, where n is the number of the RES counting from the first RES (n = 001) in order for RES' appearance in the file. This field shall occur as many times as are specified in the NUMRES field. This field is conditional and shall be omitted if the NUMRES field contains BCS zeros (0x30).	4	BCS-N positive integer 0200 to 9999	C
LREn	Length of nth Reserved Extension Segment. This field shall contain a valid length in bytes for the nth RES subheader, where n is the number of the RES counting from the first RES (n=001) in order of the RES appearance in the file. This field shall occur as many times as are specified in the NUMRES field. This field	7	BCS-N positive integer 0000001 to 9999999	C

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TABLE 1. NITF File Header – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
	is conditional and shall be omitted if the NUMRES field contains BCS zeros (0x30).			
. . . . End for each RES LRESHn, LREn; the number of loop repetitions is the value specified in the NUMRES field.				
UDHDL	User Defined Header Data Length. This field shall be populated with all BCS zeros (0x30) if no TRE are included in the UDHD. If a TRE exists, the field shall contain the sum of the length of all the TRE (paragraph 5.8.1) appearing in the UDHD field plus 3 bytes (length of UDHOFL field). If a TRE is too long to fit in the UDHD field, it shall be put in the TRE overflow DES with DESID set to the value TRE_OVERFLOW (paragraph 5.8.3.1).	5	BCS-N positive integer (Default is BCS zeros (0x30)) 00000, 00003 to 99999	R
UDHOFL	User Defined Header Overflow. This field shall contain BCS zeros (0x30) if the TRE in UDHD do not overflow into a DES, or shall contain the sequence number of the DES into which they do overflow. This field shall be omitted if the field UDHDL contains BCS zeros (0x30).	3	BCS-N positive integer (Default is BCS zeros (0x30)) 000 to 999	C
UDHD	User-Defined Header Data. If present, this field shall contain user-defined TRE data (paragraph 5.8.1). The length of this field shall be the value contained by the UDHDL field minus 3 bytes. Tagged record extensions shall appear one after the other with no intervening bytes. The first byte of this field shall be the first byte of the first tagged record extension appearing in the field. The last byte of this field shall be the last byte of the last tagged record extension to appear in the field. This field shall be omitted if the UDHDL field contains BCS zeros (0x30).	†[1]	User-defined	C
XHDL	Extended Header Data Length. This field shall be populated with all BCS zeros (0x30) if no TRE are included in the XHD. If a TRE exists, the field shall contain the sum of the length of all the TRE (paragraph 5.8.1) appearing in the XHD field plus 3 bytes (length of XHDLOFL field). If a TRE is too long to fit in the XHD field or the UDHD field it shall be put in the TRE overflow DES with DESID set to the value TRE_OVERFLOW (paragraph 5.8.3.1).	5	BCS-N positive integer (Default is BCS zeros (0x30)) 00000, 00003 to 99999	R
XHDLOFL	Extended Header Data Overflow. This field shall contain BCS zeros (0x30) if the TRE in XHD do not overflow into a DES, or shall contain the sequence number of the DES into which they do overflow. This field shall be omitted if the XHDL field contains BCS zeros (0x30).	3	BCS-N positive integer (Default is BCS zeros (0x30)) 000 to 999	C
XHD	Extended Header Data. If present, this field shall contain TRE (paragraph 5.8.1) approved and under configuration management of the NTB, a focus group of the Geospatial-intelligence Standards Working Group (GWG). The length of this field shall be the length specified by the field XHDL minus 3 bytes. TREs shall appear one after the other with no intervening bytes. The first byte of this field shall be the first byte of the first TRE appearing in the field. The last byte of this field shall be the last TRE to appear in the field. This field	††[1]	TRE	C

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TABLE 1. NITF File Header – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
	shall be omitted if the XHDL field contains BCS zeros (0x30).			

†[1] A value as specified in the UDHDL field minus 3 (in bytes)
††[1] A value as specified in the XHDL field minus 3 (in bytes)

5.3 NITF Product and Overlay Concept. The following subsections describe the non-destructive nature of the NITF structure, the relationships anticipated to exist among the segments in an NITF file, and the representations of these relationships in the file. An image product may conceivably consist of the following.

a. A correlated set of multiple NITF files.

b. A single NITF file with multiple images, each with their own overlays and associated data.

c. An NITF file with no image.

d. A single NITF file with a single image and its overlays and associated data.

To facilitate description of the NITF overlay concept, only the latter case will be addressed in the context of this subsection.

5.3.1 Image Overlay Relationships. Each single file image product is composed of one or more NITF standard data segments plus associated data. The association and portrayal of displayable segments is accomplished through the use of location indices, Display Levels (DLVLs), and Attachment Levels (ALVLs). The placement of displayable data segments in the CCS (paragraph 4.7) is recorded in the location field of the segment's subheader. The relative visibility, when displayed, of the various displayable segments in the file is recorded in the file by use of the display level (the "DLVL" field in the standard information type subheaders, specifically IDLVL for images and SDLVL for graphics). Groupings of related segments may be formed by use of the attachment level (the "ALVL" field in the standard information type subheaders, specifically IALVL for images and SALVL for graphics). For example, when a base IS is present, it may form the basis for using the other data contained in the product. Images other than the base image may be associated with the base image via the use of the ILOC, IDLVL, and IALVL fields of their image subheaders. All images and graphics associated with the base image define overlays to the base image in the sense that, when displayed, they will overwrite the underlying portion (if any) of the base image. Images and graphics associated (attached to) with the base image may be positioned such that they are completely on the base image, are partially on the base image, or completely off (adjacent to) the base image.

5.3.2 Overlays and Display Level (DLVL). The order in which images and graphics are "stacked" visually when displayed is determined by their display level (the DLVL field in the standard information type subheaders, specifically IDLVL for images and SDLVL for graphics), not by their relative position within the NITF file. The display level is a positive integer less than 1000. Every IS and GS in an NITF file shall have a unique display level. That is, no two segments may have the same display level. This requirement allows display appearance to be independent of data processing or file sequence order.

5.3.3 Display Level (DLVL) Interpretation. The display level determines the display precedence of images and graphics within an NITF file when they are output to a display device. That is, at any pixel location shared by more than one image or graphic, the value displayed there is that determined from the segment with the highest numbered display level. FIGURE 8 illustrates a sample "output presentation" from an NITF file that illustrates the effects of display level assignment. The DLVL of each segment shown on FIGURE 8 is indicated in the list of items on FIGURE 8. In the case shown, the segment with DLVL one is not an image but rather an opaque CGM rectangle (graphic data, not image data). Because the CGM rectangle is larger than the base image (which, in this case, serves as the first overlay because its display level is two), it provides a border to the image. Following increasing DLVL value, the border is overlaid by Text Graphic 1 which is, in turn, overlaid by arrow one, etc. The ALVL values on FIGURE 8 refer to "Attachment Levels.

5.3.4 Attachment Level (ALVL). ALVL provides a way to associate displayable segments (images and graphics) with one another so they may be treated together for certain operations such as moving them, rotating them, or displaying them as a group. The attachment level of a displayable segment shall be equal to the display level of the segment to which it is "attached." This value is stored in the "ALVL" field (specifically IALVL for images, SALVL for graphics) of the segment's subheader. An attachment level of zero shall be interpreted as "unattached." The segment having minimum display level shall have attachment level zero. Any other segment may also have ALVL zero, i.e., unattached.

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An overlay's display level shall always be numerically greater than its attachment level (that is, an overlay must be attached to something previously displayed or it is unattached). FIGURE 9 shows the attachment relationships of the overlays on FIGURE 8. A segment with DLVL 1 (DLVL 001) (the minimum DLVL in this example), must have an ALVL of zero. When an overlay or base is edited (moved, deleted, rotated), all overlays attached to it, directly or indirectly, may be affected by the same operation. For example, on FIGURE 9, if the image (DLVL 002, ALVL 001) were moved one centimeter to the left, the Text Graphic 1 (DLVL 003, ALVL 002) with its associated Arrow 1 (DLVL 004, ALVL 003), and the image inset (DLVL 005, ALVL 002) with its associated Arrow 3, (DLVL 008, ALVL 005), and the Text Graphic 2 (DLVL 006, ALVL 005) with its associated Arrow 2 (DLVL 007, ALVL 006) all would also be moved one centimeter to the left. If the Image Inset were deleted, so would be its associated Arrow 3 and Text Graphic 2 with Arrow 2. Although the ALVL provides the means to group or associate display items, the provision of user operations (such as moving, rotating, etc.) to act on or use ALVL information is an implementer's choice.

5.4 Image data.

5.4.1 General. For the NITFS, image data encompasses still and motion imagery; single band as well as multi-band imagery; imagery-derived products; images intended to be displayed as monochrome (shades of grey), color-mapped (pseudo color), or true color; grid or matrix data intended to provide additional geographic or geo-referencing information; bad pixel grids; and per-pixel metadata.

5.4.1.1 Image Representation (IREP). The Image Representation (IREP) field contains a valid indicator for the general kind of image represented by the data. It is an indication of the processing required in order to display an image. Valid representation indicators are MONO for monochrome; RGB for red, green, or blue true color, RGB/LUT for mapped color; MULTI for multiband imagery, NODISPLY for an image not intended for display, NVECTOR and POLAR for vectors with Cartesian and polar coordinates respectively, and VPH for Synthetic Aperture Radar (SAR) Video Phase History. In addition, compressed imagery can have this field set to YCbCr601 when represented in the ITU-R BT.6/270 color space prior to using JPEG compression (if the value of the Image Compression (IC) field is equal to C3, C8, M3, or M8). An image may include multiple data bands and color Look-Up Tables (LUTs), the latter within its header fields. True color images (three band) may be specified to be interpreted using either the Red, Green, Blue (RGB) or the YCbCr601 (Y = Brightness of signal, Cb = Chrominance (blue), Cr = Chrominance (red)) color system. YCbCr represents color as brightness and two color difference signals. The Y is the brightness (luma), Cb is blue minus luma (B-Y) and Cr is red minus luma (R-Y). Grids or matrix data may include one, two, or several bands of attribute values intended to provide additional geographic or geo-referenced information. VPH requires SAR processing to produce a displayable image. Vectors with Cartesian coordinates (NVECTOR) and vectors with polar coordinates (POLAR) require appropriate vector calculations to produce a displayable image. The processing required to display each band of the image is indicated in the nth Band Representation (IREPBANDn) field. TABLE 2 shows representative IREP examples and some of its associated fields.

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TABLE 2. Display Dependent Parameters. (Normative)

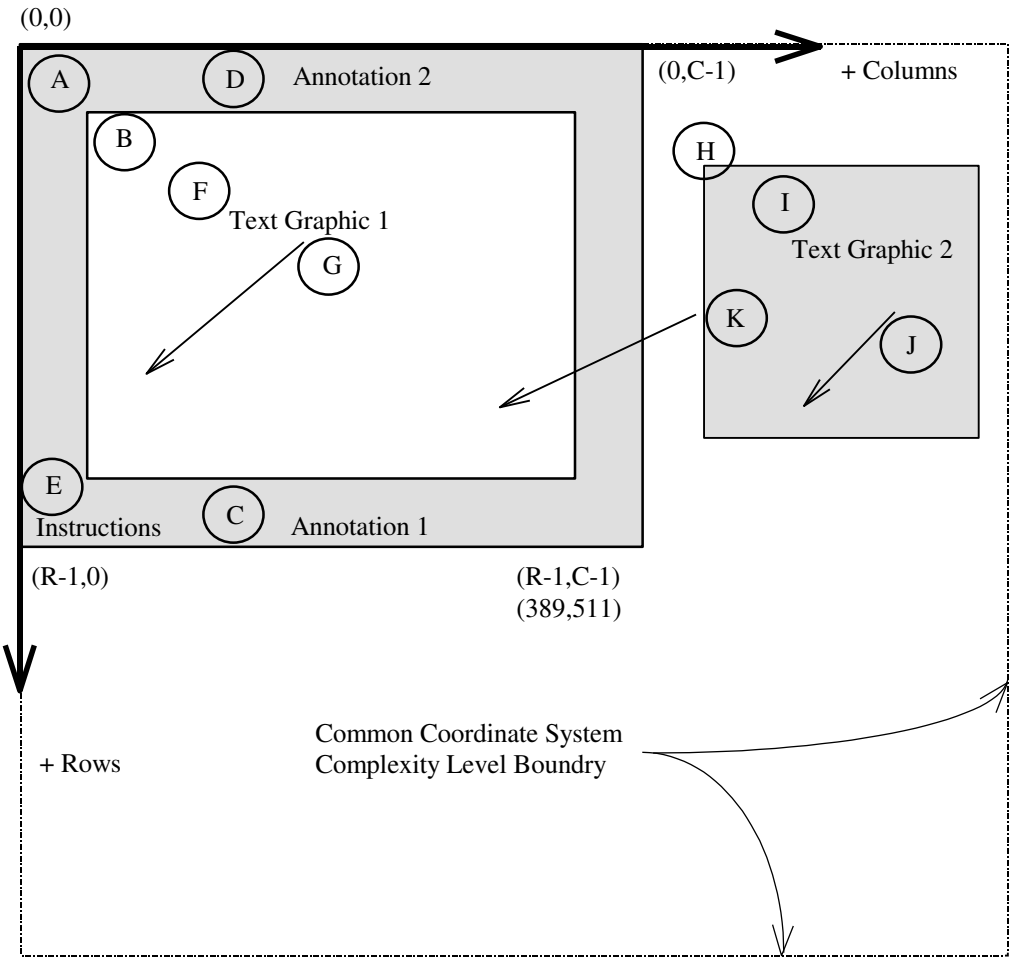
This is a table of common usages. It is not meant to limit the field values. Additional values and associated values may be registered through the NTB*.

IREP*	**IREPBANDn***	**NBANDS**	**PVTYPE**	**NLUTSn**	**Associated ICAT(s)***
NODISPLY	BCS spaces (0x20)	1 to 9, 0†2	INT, R, C, B, SI	0	BARO, CAVIS, CLOUD, DED, DEPTH, DTEM, ISAR, LOCG, MATR, PIXQUAL, SAR, SARIQ
MONO	LU, M, BCS spaces (0x20)	1	INT, R, B, SI	0, 1, 2	BP, CAT, CCD, DTV, EO, EOVIS, FL, FP, HR, HS, IR, ISAR, LEG, LWIR, MAP, MRI, MS, MWIR, NIR, OP, PAN, RD, SAR, SL, SWIR, TI, UV, VD, VIS, VNIR, XRAY
RGB	R, G, B	3	INT, R	0	CP, EOVIS, DTV, LEG, MAP, OP, PAT, VIS
RGB/LUT	LU	1	INT, B	3	CCD, CP, DTV, EOVIS, LEG, MAP, OP, PAT, VIS
YCbCr601	Y, Cb, Cr	3	INT	0	CP, DTV, EOVIS, LEG, MAP, OP, PAT, VIS
NVECTOR	BCS spaces (0x20)	1 to 9, 0†2	INT, R, C	0	None currently identified
POLAR	BCS spaces (0x20), M	2	INT, R, C	0	CCD , CURRENT, WIND
VPH	BCS spaces (0x20)	2	INT, R, C	0	None currently identified
MULTI	BCS spaces (0x20), M, R, G, B, LU	1 to 9, 0†2	INT, R, C, B, SI	0, 1, 2, 3	EOVIS, HS, LWIR, MS, MWIR, NIR, SWIR, UV, VNIR

†2 If NBANDS field contains 0 then XBANDS field is required where XBANDS>9
*Other variations of IREP, IREPBANDn, and ICAT can be registered online with approval of NTB

ICAT values and their definitions are found in the NITF Field Value Registry for the ICAT field, which is one of the IMAGE SUBHEADER metadata fields. The ICAT, ISUBCATn, PVTYPE, NBPP, ABPP, and number of bands in the image segment are all related. To see the allowed values for particular ICAT values, see TABLE 2(A) included with the ICAT field values in the NITF Field Value Registry;http://jitc.fhu.disa.mil/projects/nitf/registers/nitfcte/trereg.aspx

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		Location Offsets			
		DLVL	ALVL	Row	Column
A.	Border (Opaque Box)	001	000	00000	00000
B.	Exploited Image	002	001	00025	00025
C.	Annotation 1	999	000	00375	00156
D.	Annotation 2	998	000	00030	00156
E.	Instructions	997	000	00375	00008
F.	Text Graphic 1	003	002	00080	00100
G.	Arrow 1	004	003	00010	00065
H.	Image Inset	005	002	00080	00400
I.	Text Graphic 2	006	005	00025	00032
J.	Arrow 2	007	006	00065	00070
K	Arrow 3	008	005	00070	-0005

FIGURE 8. NITF Display Level (DLVL) Illustration.†

† This example uses a CGM rectangle as a border for the NITF composition. This method may be incompatible with some printers. These printers do not allow for transparent pixels in imagery. If an NITF composition uses CGM elements under images with NITF image padding (transparent pixels) the CGM will not be visible in any areas under the pad pixels. (For work around see paragraph D.2.11.)

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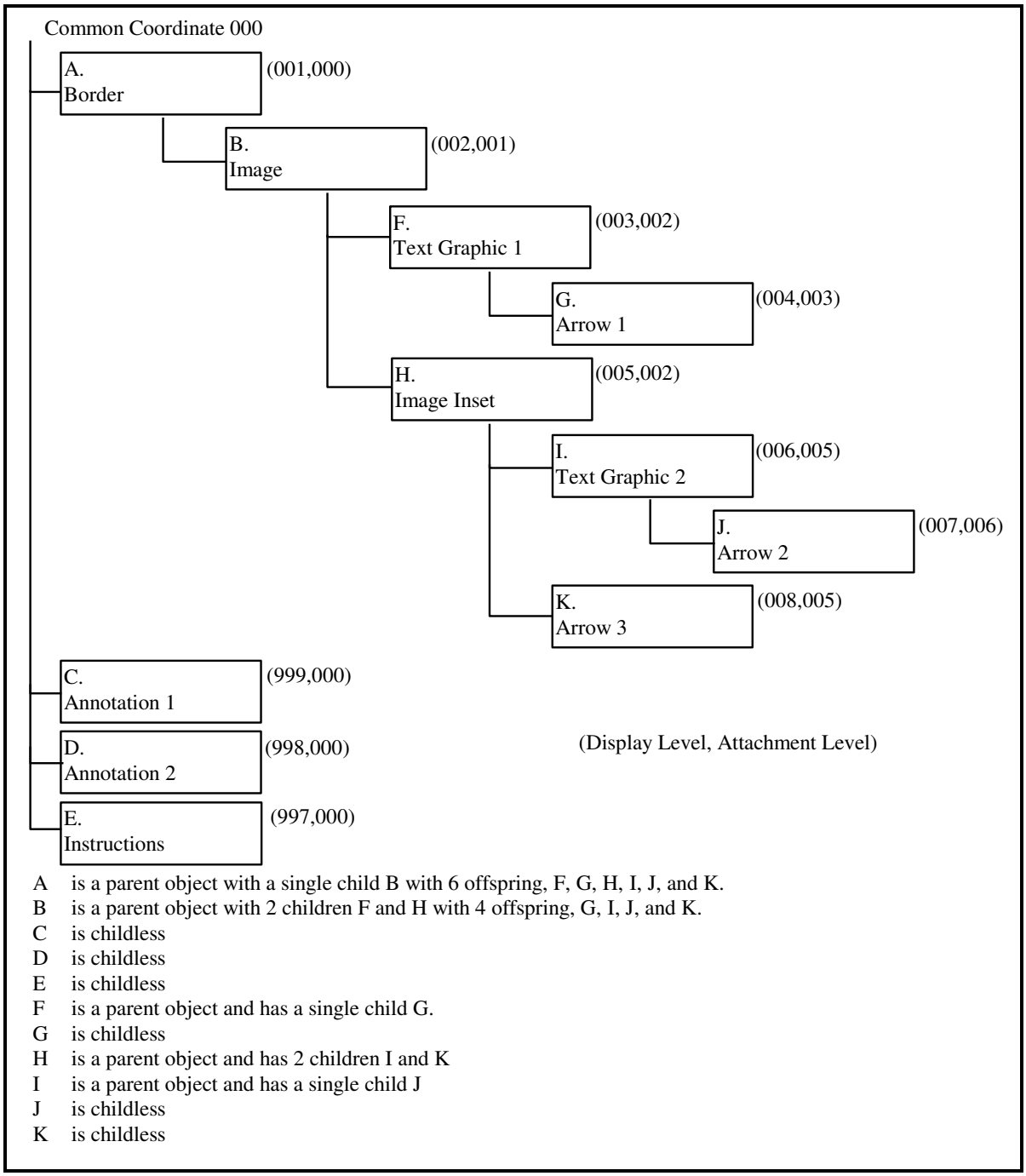


FIGURE 9. Attachment Level (ALVL) Relationships.

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5.4.1.2 Image Category (ICAT). The specific category of an IS reveals its intended use or the nature of its collector. The default is "VIS"; for additional values refer to the NITF Field Value Registry, specified in Appendix C.

5.4.2 Image Model. For the NITF, a still image is a two-dimensional rectangular array of pixels indexed by row and column. A pixel is represented by an n-vector of sample values; where n corresponds to the number of bands comprising the image. The ith entry of the pixel (vector) is the pixel value for the ith band sample of the image. Therefore, the ith band of the image is the rectangular array of ith sample values from the pixel vectors. For an image I with R rows and C columns, the coordinates of the image pixel located in the cth column of the rth row shall be denoted by an ordered pair (r,c), $0 \le r < R, 0 \le c < C$, where the first number, r, indicates the row and the second number, c, indicates the column in the image array. This notation is standard for addressing arrays and matrices. The pixel located at (r,c) is denoted by I(r,c). For example, a typical 24-bit RGB image is an array of R rows and C columns, where each set of indices (r,c), $0 \le r < R, 0 \le c < C$, identifies a pixel I(r,c) consisting of three single byte values (a three-vector) corresponding to the red, green, and blue samples. The image has three bands, each consisting of a R-by-C array of single byte sample values. One band comprises the red, one band comprises the green, and the third band comprises the blue pixel sample values. Specifically, the value at position r,c in the green band, for example, contains the green byte from the pixel I(r,c) three-vector at position r,c in the image.

5.4.2.1 Display of NITF Images. When an image with R rows and C columns is displayed, a mapping is accomplished from the stored image pixel value array I to a rectangular array S of physical picture elements, for example a Cathode Ray Tube (CRT) display. This mapping will be called the display mapping. Usually, the resulting display has an identified top, bottom, and left and right side. In a particular application, the display mapping may be defined explicitly. However, lacking this, an image stored in an NITF file shall be interpreted so that pixel I(0,0) is at the upper left corner, and pixel I(R-1,C-1) is at the lower right corner. The rth row of the image array I shall form the rth row of the display, counting from the top, $0 \le r < R$. Within the rth row, the pixels shall appear beginning on the left with I(r,0) and proceeding from left to right with I(r,1), I(r,2), and so on, ending with I(r, C-1). FIGURE 10 illustrates the display mapping just described. This mapping of pixel values to physical picture elements is typical of non-interleaved raster pattern of picture elements. The relationship of the pixels I(r,c) in the array to up, down, left, and right implicit in this diagram is used freely in later descriptions to simplify exposition.

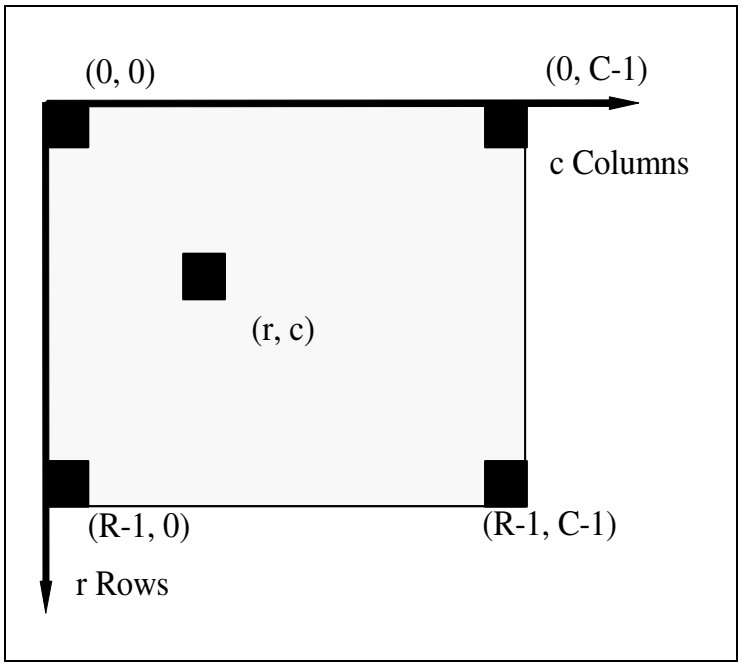


FIGURE 10. Image Coordinate Representation.

5.4.2.2 Blocked Images. The concept of blocked images extends the image model for NITF presented above to support the representation of an image in terms of an orderly set of subimages (or subarrays) called blocks. For large images (e.g., those having more horizontal and vertical pixel values than typical display devices), the performance of an

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imagery implementation can be potentially improved by “blocking” the image; that is, ordering the pixel values in the NITF file as a series of concatenated pixel arrays. For multi-blocked imagery, each block must start on a byte boundary.

a. The idea behind a blocked image is analogous to a rectangular tiled floor. Regard the overall floor cover as the image and each individual tile as a block. To make this more precise, let I be an image of R rows and C columns, and let the Number of Pixels Per Block Horizontal (NPPBH), (that is, the number of columns of each block) and the Number of Pixels Per Block Vertical (NPPBV), (that is, the number of rows in each block) be positive integers that satisfy NPPBH$\leq$C and NPPBV$\leq$R. If R is an integer multiple of NPPBV and C is an integer multiple of NPPBH, then I may be viewed as an array B of sub arrays each having NPPBV rows and NPPBH columns. These sub arrays $B_{r,c}$ are called blocks. The block $B_{r,c}$ is in the rth row of blocks and the cth column of blocks. The number of columns of blocks (Number of Blocks Per Row (NBPR)) is the integer C/NPPBH, and the number of rows of blocks (Number of Blocks Per Column, (NBPC)) is the integer R/NPPBV.

b. For recording purposes, the image blocks are ordered and indexed sequentially by rows, i.e., B(1,1) B(1,NBPR); B(2,1) ... B(2,NBPR); B(NBPC,1) ... B(NBPC,NBPR). The relation of image blocks to image rows and columns is depicted on FIGURE 11 using the NITF display convention described in paragraph 5.4.2.1. Although the pixel values are placed in the file as a series of arrays (blocks), the coordinate used to reference any specific pixel remains the same as if the image were not blocked. For example, if R=C=2048 and NPPBV=NPPBH=1024, there will be four blocks in the image I. The second pixel value in B (1,2) has the coordinate I (0,1025) vice the internal index (0,1) of the subarray.

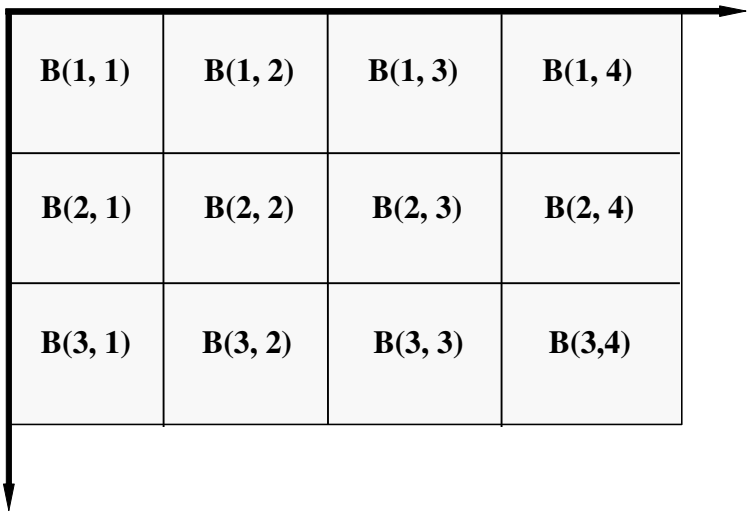


FIGURE 11. A Blocked Image.

c. If the number of rows in an image is not initially an integer multiple of NPPBV, or if the number of columns is not an integer multiple of NPPBH, an application that creates the blocked image construct in NITF shall "fill" the image to an appropriate number of rows and columns so the divisibility condition is met by adding rows to the bottom and/or columns to the right side of the image, as viewed. The result is that an image may have a block(s) (subarray(s)) composed of pixel values from the original image, and “fill” pixels inserted to meet block boundary conditions (FIGURE 12). If R (the number of rows in an image) is not initially an integer multiple of NPPBV, then NBPC is the integer [R/NPPBV] + 1; if C (the number of columns in an image) is not initially an integer multiple of NPPBH, then NBPR is the integer [C/NPPBH] + 1 ([r]: = largest integer $\leq$r).

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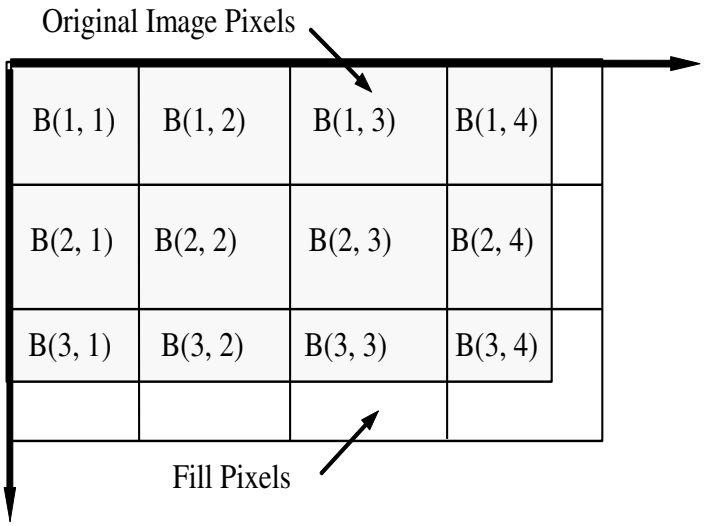


FIGURE 12. A Blocked, Filled Image.

d. For some varieties of large image arrays, the nature of the image data is such that it should be organized as a single block (un-blocked) or in large block sizes that exceed the 8192 pixel maximum range size of the NPPBH and NPPBV data fields. In these cases, the large block option provides increased flexibility for large arrays. In the large block option, when either or both of the image subheader fields, NBPR and/or NBPC are set to the value 0001, the respective value for the NPPBV and/or NPPBH field may be set to 0000. In this case, the block size defaults to the respective horizontal and/or vertical size identified in the NCOL and/or NROW fields. This option allows for an image array to be a single large block, or a single row of blocks that are large in the row dimension, or a single column of blocks that are large in the column dimension. See TABLE 3 description for NPPBH and NPPBV.

e. The Motion Imagery Extension for NITF 2.1 (MIE4NITF), NGA.STND.0044, provides an extension to MIL-STD-2500C that allows image segments and blocks within image segments to contain multiple, temporally related, images called frames. Within an MI dataset, the image data within an image segment block possess spatial (row, column) and temporal (time or frame number) dimensions. Blocks containing both spatial and temporal dimensions are called spatiotemporal blocks to distinguish them from blocks containing a single image (frame) and to emphasize they contain a temporal sequence of frames. The blocking concepts shown in FIGURES 11 and 12 may be extended to temporal sequences of frames.. NGA.STND.0044 provides more details on the defined motion imagery/data model, further definition of what is a frame, and the extension of blocks to spatiotemporal blocks.

5.4.2.3 Blocked Image Masking. In some instances, a blocked image may have a considerable number of empty blocks (blocks without meaningful pixel values). This might occur when a rectangular image is not north aligned when scanned or otherwise sampled, but has been rotated to a north up orientation (FIGURE 13) resulting in the need to insert “pad” pixels to maintain the rectangular raster pattern of the pixel array. In this case, it is sometimes useful to not record or transmit empty blocks within an NITF file. However, if empty blocks are not recorded/transmitted, the image loses its logical structure as an image with n x m blocks. In order to retain logical structure, and to allow the exclusion of empty blocks, an image data mask table (TABLE 3(A), Block n, Band m Offset (BMRnBNDm) field) identifies the location of non-empty blocks so that the using application can reconstruct the image correctly. On FIGURE 13, the recording order would be B(1,1); B(1,2); B(1,3); B(2,1); B(2,2); B(2,3); B(2,4); B(3,1); B(3,2); B(3,3); B(3,4); B(4,2); B(4,3); B(4,4). Blocks B(1,4) and B(4,1) would not be recorded in the file. The blocked image mask would identify the locations of the recorded image blocks. If the image is band sequential (IMODE=S), there will be multiple blocked image masks (one for each image band), with each mask containing Number of Blocks Per Row (NBPR) x Number of Blocks Per Column (NBPC) records. Blocked image masks can be used in conjunction with a pad pixel mask (TABLE 3(A), Pad Pixel n, Band m (TMRnBNDm) field), as described below. A blocked image mask may also be used to provide an index for random access within the blocked image data for large images even if all blocks are recorded in the file.

5.4.2.3.1 Blocked image mask records may be used within NGA.STND.0044 to provide random access to the start of the MI data within each spatiotemporal block. Offsets to individual frames within an image segment spatiotemporal block is not supported. If MI data is band sequential temporal interleave (IMODE=D), there will be multiple blocked

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image masks containing NBPR x NBPC records in the same manner as band sequential interleave for still images. See NGA.STND.0044 for further details.

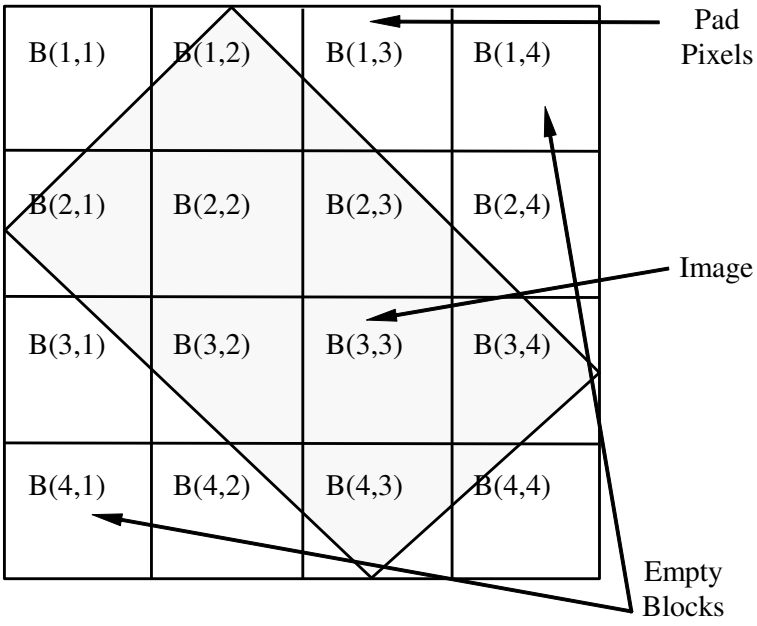


FIGURE 13. A Blocked, Padded Image with Empty Blocks.

5.4.2.4 Pad Pixel Masking. In addition to empty image blocks, FIGURE 13 also demonstrates that a significant number of pad pixels may be needed to "fill" an image to the nearest block boundary.

a. In the example on FIGURE 13, the locations of B(1,1); B(1,2); B(1,3); B(2,1); B(2,3); B(2,4); B(3,1); B(3,2); B(3,4); B(4,2); B(4,3); and B(4,4) would be recorded indicating that those blocks have pad pixels. B(1,4); B(2,2); B(3,3), and B(4,1) do not have pad pixels because B(1,4) and B(4,1) are empty and B(2,2) and B(3,3) are full image blocks.

b. If the image is band sequential (IMODE=S), there will be pixel masks that will be arranged in the same order as the image bands, with each mask containing NBPR x NBPC records.

c. The output pixel code that represents pad pixels is identified within the image data mask by the Pad Output Pixel Code (TPXCD) field. The length in bits of this code is identified in the Pad Output Pixel Code Length (TPXCDLNTH) field. Although this length is given in bits, the actual TPXCD value is stored in an integral number of bytes. When the number of bits used by the code is less than the number available in the TPXCD field (for example, a 12 bit code stored in two bytes), then the code will be justified in accordance with the Pixel Justification (PJUST) field in the Image Subheader.

d. When an application identifies pad pixel values, it may replace them with a user defined value (for example, a light blue background) at the time of presentation except when the value of TPXCD is zero (0x00). When the TPXCD value is zero (0x00), the pad pixel will be treated as “Transparent” for presentation. The application may choose to ignore pad pixels in histogram generation. In any case, pad pixels are not valid data, and should not be used for interpretation or exploitation.

e. Pad pixel masks are of limited use with MI. The compression algorithms commonly used with MI do not guarantee pixels values are preserved when a spatiotemporal block is decoded. Pad pixels are not recommended for compressed MI.

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5.4.3 NITF Image Information. In the NITF, the information describing an image is represented in a series of adjacent fields grouped into the image subheader followed by the image data. The field containing the image data shall follow immediately the last field of the corresponding image subheader with no intervening special characters to designate the beginning of the image. Similarly, the image subheader of the first image shall follow immediately the last byte of data of the last field in the NITF file header, and the image subheader of successive images shall follow immediately the last byte of the image of the preceding image.

5.4.3.1 Image Subheader. The data elements in the image subheader fields provide information about the image source, its identification, and characteristics needed to display and interpret it properly. The image subheader field definitions are detailed in TABLE 3.

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TABLE 3. NITF Image Subheader.

(TYPE "R" = Required, "C" = Conditional, "< >"= BCS spaces allowed for entire field)
("†" annotations are explained at the end of the table)

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
IM	File Part Type. This field shall contain the characters "IM" to identify the subheader as an image subheader.	2	BCS-A IM	R
IID1	Image Identifier 1. This field shall contain a valid alphanumeric identification code associated with the image. The valid codes are determined by the application.	10	BCS-A User-defined	R
IDATIM	Image Date and Time. This field shall contain the time (UTC) of the image acquisition in the format CCYYMMDDhhmmss, where CC is the century (00 to 99), YY is the last two digits of the year (00 to 99), MM is the month (01 to 12), DD is the day (01 to 31), hh is the hour (00 to 23), mm is the minute (00 to 59), ss is the second (00 to 59). UTC (Zulu) is assumed to be the time zone designator to express the time of day. Refer to Paragraph 5.1.7d when a portion of the date and/or time is unknown. When the acquisition is a span of time this value is the start of collection. A time span may be listed, by use of SDEs.	14	BCS-N CCYYMMDDhhmmss	R
TGTID	Target Identifier. This field shall contain the identification of the primary target in the format, BBBBBBBBBBOOOOOCC, consisting of ten characters of Basic Encyclopedia (BE) identifier, followed by five characters of facility OSUFFIX, followed by the bare two character country code as specified in Paragraph 5.1.7e.	17	BCS-A BBBBBBBBBB OOOOOCC (Default is BCS spaces (0x20) for all or any sub-part of this field)	<R>
IID2	Image Identifier 2. This field may contain the identification of additional information about the image.	80	ECS-A (Default is ECS spaces (0x20))	<R>
ISCLAS	Image Security Classification. This field shall contain a valid value representing the classification level of the image. Valid values are T (=Top Secret), S (=Secret), C (=Confidential), R (=Restricted), U (=Unclassified).	1	ECS-A T, S, C, R, or U	R
NOTE If ISCLAS is T, S, C, or R, then ISCLSY must be populated with a valid code for the security classification system used.				
ISCLSY	Image Security Classification System. This field shall contain valid values indicating the national or multinational security system using a geopolitical (country) code digraph to classify the image. For US producers a geopolitical code (Country Code) per Paragraph 5.1.7e shall be used to indicate their national security systems. The designator "XN" is for classified data generated by a component using NATO security system classification policy marking guidance. The designator "XF" indicates the de facto classification system represented by the security marking indicator NATO/ISAF." These codes were selected to not duplicate existing codes within the GEC codespace. If this field is populated with all ECS spaces (0x20), it shall imply that no security classification system applies to the image.	2	ECS-A (Default is ECS spaces (0x20)) The most up to date list of registered digraph geopolitical code values can be found on the NITF Field Value Registry, specified in section C.2.1 Online Resources.	<R>

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TABLE 3. NITF Image Subheader – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
NOTE If any of the following fields are populated with anything other than spaces, then ISCLSY must be populated with a valid code for the security classification system used: ISCODE, ISCTLH, ISREL, ISDCTP, ISDCDT, ISDCXM, ISDG, ISDGDT, ISCLTX, ISCATP, ISCAUT, ISCRSN, ISSRDT, and ISCTLN.				
ISCODE	Image Codewords. This field shall contain a word(s) that has been assigned a classification and a classified meaning to safeguard intentions and information regarding a specific classified plan or operation. This codeword may be a cryptonym used to identify sensitive intelligence data. Multiple entries shall be separated by a single ECS space (0x20). If this field is populated with all ECS spaces (0x20), it shall imply that no codewords apply to the image. Past codeword examples are located in the NITF registry, specified in section C.2.1 Online Resources.	11	ECS-A (Default is ECS spaces (0x20))	<R>
ISCTLH	Image Control and Handling. This field shall contain valid additional security control and/or handling instructions (caveats) associated with the image. Control and Handling is an indicator governing the distribution and use of information or material. Such indicators are the subject of intelligence community agreement and are specifically defined in appropriate regulations. Example values include digraphs found in the NITF Field Value Registry, specified in section C.2.1 Online Resources. The digraph may indicate single or multiple caveats. The selection of a relevant caveat(s) is application specific. If this field is populated with all ECS spaces (0x20), it shall imply that no control and handling instructions apply to the image.	2	ECS-A (Default is ECS spaces (0x20))	<R>
ISREL	Image Releasing Instructions. This field shall contain a valid list of bare country and/or multilateral entity codes to which countries and/or multilateral entities the image is authorized for release. Valid entities in the list are one or more bare country codes as specified by Paragraph 5.1.7e, separated by a single ECS space (0x20). If this field is populated with all ECS spaces (0x20), it shall imply that no image release instructions apply.	20	ECS-A (Default is ECS spaces (0x20))	<R>
ISDCTP	Image Declassification Type. This field shall contain a valid indicator of the type of security declassification or downgrading instructions which apply to the image. Valid values are DD (=declassify on a specific date), DE (=declassify upon occurrence of an event), GD (=downgrade to a specified level on a specific date), GE (=downgrade to a specified level upon occurrence of an event), O (=OADR), and X (= exempt from automatic declassification). If this field is populated with all ECS spaces (0x20), it shall imply that no image security declassification or downgrading instructions apply.	2	ECS-A DD, DE, GD, GE, O, X (Default is ECS spaces (0x20))	<R>
ISDCDT	Image Declassification Date. This field shall indicate the date on which an image is to be declassified if the value in the Image Declassification Type is DD. If	8	ECS-A CCYYMMDD (Default is ECS spaces	<R>

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TABLE 3. NITF Image Subheader – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
	this field is populated with all ECS spaces (0x20), it shall imply that no image declassification date applies.		(0x20))	
ISDCXM	Image Declassification Exemption. This field shall indicate the reason the image is exempt from automatic declassification if the value in Image Declassification Type is X. Valid values are X1 to X8 and 25X1 to 25X9. Values X1 to X8 correspond to the declassification exemptions found in DOD 5200.01-V1, paragraphs 4- 202b(1) to (8) for material exempt from the 10-year rule. Values 25X1 to 25X9 correspond to the declassification exemptions found in DOD 5200.01-V1, paragraphs 4-301a(1) to (9) for permanently valuable material exempt from the 25-year declassification system. If this field is populated with all ECS spaces (0x20), it shall imply that an image declassification exemption does not apply. Value DNIO is Exempt from automatic declassification and can be declassified only by the Director of National Intelligence (DNI).	4	ECS-A X1 to X8, 25X1 to 25X9, DNIO, (Default is ECS spaces (0x20))	<R>
ISDG	Image Downgrade. This field shall indicate the classification level to which an image is to be downgraded if the values in Image Declassification Type are GD or GE. Valid values are S (=Secret), C (=Confidential), R (= Restricted). If this field contains an ECS space (0x20), it shall imply that image security downgrading does not apply.	1	ECS-A S, C, R (Default is ECS space (0x20))	<R>
ISDGDT	Image Downgrade Date. This field shall indicate the date on which an image is to be downgraded if the value in Image Declassification Type is GD. If this field is populated with all ECS spaces (0x20), it shall imply that an image security downgrading date does not apply.	8	ECS-A CCYYMMDD (Default is ECS spaces (0x20))	<R>
ISCLTX	Image Classification Text. This field shall be used to provide additional information about image classification to include identification of a declassification or downgrading event if the values in Image Declassification Type are DE or GE. It may also be used to identify multiple classification sources and/or any other special handling rules. Values are user defined free text. If this field is populated with all ECS spaces (0x20), it shall imply that additional information about image classification does not apply. This field may also contain overflow values from the ISCTLH or ISREL fields. Consult with the NITF Field Value Registry, at for overflow population guidance, specified in section C.2.1 Online Resources.	43	ECS-A User-defined free text (Default is ECS spaces (0x20))	<R>
ISCATP	Image Classification Authority Type. This field shall indicate the type of authority used to classify the image. Valid values are O (= original classification authority), D (= derivative from a single source), and M (= derivative from multiple sources). If this field	1	ECS-A O, D, M (Default is ECS space (0x20))	<R>

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TABLE 3. NITF Image Subheader – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
	contains an ECS space (0x20), it shall imply that image classification authority type does not apply.			
ISCAUT	Image Classification Authority. This field shall identify the classification authority for the image dependent upon the value in Image Classification Authority Type. Values are user-defined free text which should contain the following information: original classification authority name and position or personal identifier if the value in Image Classification Authority Type is O; title of the document or security classification guide used to classify the image if the value in Image Classification Authority Type is D; and Derive-Multiple if the image classification was derived from multiple sources. In the latter case, the image originator will maintain a record of the sources used in accordance with existing security directives. One of the multiple sources may also be identified in Image Classification Text if desired. If this field is populated with all ECS spaces (0x20), it shall imply that no image classification authority applies.	40	ECS-A User-defined free text (Default is ECS spaces (0x20))	<R>
ISCRSN	Image Classification Reason. This field shall contain values indicating the reason for classifying the image. Valid values are A to H. These correspond to the reasons for original classification per E.O. 13526, Section 1.5.(a) through (h). If this field contains an ECS space (0x20), it shall imply that no image classification reason applies.	1	ECS-A A to H (Default is BCS space (0x20))	<R>
ISSRDT	Image Security Source Date. This field shall indicate the date of the source used to derive the classification of the image. In the case of multiple sources, the date of the most recent source shall be used. If this field is populated with all BCS spaces (0x20), it shall imply that an image security source date does not apply.	8	ECS-A CCYYMMDD (Default is ECS spaces (0x20))	<R>
ISCTLN	Image Security Control Number. This field shall contain a valid security control number associated with the image. The format of the security control number shall be in accordance with the regulations governing the appropriate security channel(s). If this field is populated with all ECS spaces (0x20), it shall imply that no image security control number applies.	15	ECS-A (Default is ECS spaces (0x20))	<R>
ENCRYP	Encryption. This field shall contain the value BCS zero (0x30) until such time as this specification is updated to define the use of other values.	1	BCS-N positive integer (Default is BCS zero (0x30)) 0 = not encrypted	R
ISORCE	Image Source. This field shall contain a description of the source of the image. If the source of the data is classified, then the description shall be preceded by the classification, including codeword(s) contained in the NITF Field Value Registry, specified in section C.2.1 Online Resources. If this field is populated with all ECS spaces (0x20), it shall imply that no image source data applies.	42	ECS-A (Default is ECS spaces (0x20))	<R>
NROWS	Number of Significant Rows in Image. This field	8	BCS-N positive	R

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TABLE 3. NITF Image Subheader – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
	shall contain the total number of rows of significant pixels in the image. When the product of the values of the NPPBV field and the NBPC field is greater than the value of the NROWS field (NPPBV * NBPC > NROWS), the rows indexed with the value of the NROWS field to (NPPBV * NBPC) minus 1 shall contain fill data. NOTE: Only the rows indexed 0 to the value of the NROWS field minus 1 of the image contain significant data. The pixel fill values are determined by the application.		integer 00000001 to 99999999	
NCOLS	Number of Significant Columns in Image. This field shall contain the total number of columns of significant pixels in the image. When the product of the values of the NPPBH field and the NBPR field is greater than the NCOLS field (NPPBH * NBPR > NCOLS), the columns indexed with the value of the NCOLS field to (NPPBH * NBPR) minus 1 shall contain fill data. NOTE: Only the columns indexed 0 to the value of the NCOLS field minus 1 of the image contain significant data. The pixel fill values are determined by the application.	8	BCS-N positive integer 00000001 to 99999999	R
PVTYPE	Pixel Value Type. This field shall contain an indicator of the type of computer representation used for the value for each pixel for each band in the image. Valid entries are INT for integer, B for bi-level, SI for 2's complement signed integer, R for real, and C for complex. The data bits of INT and SI values shall appear in the file in order of significance, beginning with the MSB and ending with the LSB. Except when the data is JPEG 2000 compressed, INT and SI data types shall be limited to 8, 12, 16, 32, or 64 bits (see field NBPP). R values shall be represented according to IEEE 32 or 64 bit binary floating-point representation (IEEE 754). C values shall be represented with the Real and Imaginary parts, each represented in IEEE 32 or 64-bit binary floating-point representation (IEEE 754) and appearing in adjacent four or eight-byte blocks, first Real, then Imaginary. B (bi-level) pixel values shall be represented as single bits with binary value 1 or 0.	3	BCS-A INT, B, SI, R, C	R
IREP	Image Representation. This field shall contain a valid indicator of the processing required in order to display an image. Valid representation indicators are MONO for monochrome; RGB for red, green, or blue true color, RGB/LUT for mapped color; MULTI for multiband imagery, NODISPLY for an image not intended for display, NVECTOR and POLAR for vectors with Cartesian and polar coordinates respectively, and VPH for SAR video phase history. In addition, compressed imagery can have this field set to YCbCr601 when compressed in the ITU-R Recommendation BT.601-5 color space using JPEG (IC field = C3). This field should be used in conjunction with the IREPBANDn field to interpret	8	BCS-A MONO, RGB, RGB/LUT, MULTI, NODISPLY, NVECTOR, POLAR, VPH, YCbCr601 (Please see TABLE 2 for Display Dependent Parameters)	R

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TABLE 3. NITF Image Subheader – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
	the processing required to display each band in the image.			
ICAT	Image Category. This field shall contain a valid indicator of the specific category of image, raster or grid data. The specific category of an IS reveals its intended use or the nature of its collector. For MI implementations, the string ".M" shall be added to a still imagery ICAT value. If adding ".M" causes the new ICAT value to exceed the allowed length of 8 characters, then the last letter(s) of the still imagery ICAT value shall be omitted so that the MI ICAT value is exactly 8 characters in length. (See NGA.STND.0044)	8	BCS-A For registered values, refer to the NITF Field Value Registry specified in section C.2.1 Online Resources. (Default is VIS)	R
ABPP	Actual Bits-Per-Pixel Per Band. This field shall contain the number of "significant bits" for the value in each band of each pixel without compression. Even when the image is compressed, ABPP contains the number of significant bits per pixel that were present in the image before compression. This field shall be less than or equal to Number of Bits Per Pixel (field NBPP). The number of adjacent bits within each NBPP is used to represent the value. These "representation bits" shall be left justified or right justified within the bits of the NBPP field, according to the value in the PJUST field. For example, if 11-bit pixels are stored in 16 bits, this field shall contain 11 and NBPP shall contain 16. The default number of significant bits to be used is the value contained in NBPP.	2	BCS-N positive integer 01 to 96	R
PJUST	Pixel Justification. When ABPP is not equal to NBPP, this field indicates whether the significant bits are left justified (L) or right justified (R). Nonsignificant bits in each pixel shall contain the binary value 0. Right justification is recommended.	1	BCS-A L or R (Default is R)	R
ICORDS	Image Coordinate Representation. This field shall contain a valid code indicating the type of coordinate representation used for providing an approximate location of the image in the Image Geographic Location field (IGEOLO). The valid values for this field are: U = UTM expressed in Military Grid Reference System (MGRS) form, N = UTM/UPS (Northern hemisphere), S = UTM/UPS (Southern hemisphere), P for UPS, G = Geographic (Degrees, Minutes, Seconds), and D = Decimal degrees. (Choice between N and S is based on hemisphere of northernmost point.) The default Geodetic reference system is WGS84 (appendix D, paragraph D.2.12 and FIGURE D-1). If no coordinate system is identified, the space (BCS 0x20) shall be used.	1	BCS-A U, G, N, S, P, D or BCS space (0x20)	<R>
IGEOLO	Image Geographic Location. This field, when present, shall contain an approximate geographic location which is not intended for analytical purposes (e.g., targeting, mensuration, distance calculation); it	60	BCS-A ±dd.ddd±ddd.ddd (four times) orddmmssXdddmmss	C

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TABLE 3. NITF Image Subheader – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
	is intended to support general user appreciation for the image location (e.g., cataloguing). The representation of the image corner locations is specified in the ICORDS field. The locations of the four corners of the (significant) image data shall be given in image coordinate order: (0,0), (0, MaxCol), (MaxRow, MaxCol), (MaxRow, 0). MaxCol and MaxRow shall be determined from the values contained, respectively, in the NCOLS field and the NROWS field. MaxCol = is equal to the value contained in the NCOLS field minus 1 (MaxCol = NCOLS -1). This field shall be omitted if the value of the ICORDS field is populated with BCS Space (0x20). Valid corner locations in geographic coordinates shall be expressed as latitude and longitude. The format ddmmssXdddmmssY represents latitude and longitude. The first half, ddmmssX, represents degrees, minutes, and seconds of latitude with X representing North or South (N for North, S for South). The second half, dddmmssY, represents degrees, minutes, and seconds of longitude with Y representing East or West (E for East, W for West), respectively. Coordinates shall only be populated in the IGEOLO field to the known precision of the corner coordinates. Non-significant digits of the field shall be replaced with BCS spaces (0x20). An example of the 60 character field with two spaces depicting the absence of arc seconds is ddmm Xdddmm Yddmm Xdddmm Yddmm Xdddmm Yddmm Xdddmm Y. Decimal degrees are expressed as ±dd.ddd±ddd.ddd (four times) where ±dd.ddd equals latitude (+ represents northern hemisphere, - represents southern hemisphere) and ±ddd.ddd equals longitude (+ represents eastern hemisphere, - represents western hemisphere). Non-significant digits of the field shall be replaced with BCS spaces (0x20). For the UTM coordinate representation, coordinates shall be expressed either in plain UTM coordinates or using MGRS. In either case, UTM coordinates should be in terms of the same zone, to ensure a unified image on the grid. Normally UTM/MGRS coordinates should be rounded to the nearest 10 meters to match the precision of the geographic coordinates. Plain UTM coordinates use the format zzeeeeeennnnnnn where zz represents the UTM zone number, and eeeeee, nnnnnnn represents Easting and Northing. Hemisphere (N or S) for plain UTM is expressed in the ICORDS field. Plain UPS coordinates use the format		Y (four times) or zzBJKeeeeennnnn (four times) or zzeeeeeennnnnnn (four times) or Peeeeeeennnnnnn (four times)	

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TABLE 3. NITF Image Subheader – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
	Peeeeeeennnnnnn where P identifies the hemisphere as being North or South (P = N or S) and the eeeeeee and nnnnnnn represent Easting and Northing. The presence of an initial N or S character identifies the corner point as being a UPS coordinate value. Note that the UPS coordinate system can only be used when the image footprint falls entirely within the Northern polar or entirely within the Southern polar regions. If any part of the image footprint is between 79°30' South and 83°30' North (the boundaries between the UPS overlap regions with UTM) then the UPS coordinate system cannot be used. UTM expressed in MGRS use the format zzBJKeeeeennnnn where zzBJK represents the zone, band and 100 km square within the zone and "eeeee," nnnnn represents residuals of Easting and Northing. UPS expressed in MGRS uses the format BJKeeeeennnnn, preceded by two blank spaces (zz = " ") since there is no zone number for UPS in MGRS. The omission of the zone number (blank-filled) and the band designations of A or B for the Southern hemisphere and Y or Z for the Northern hemisphere identifying the coordinates as UPS in MGRS. As with plain UTM/UPS coordinates, MGRS coordinates should be in terms of the same zone, to ensure a unified image on the grid. NOTE: Provide the value only to the decimal places (precision) warranted by the sources and methods used to determine the location. The remaining places will be (BCS spaces (0x20). There is no implied accuracy associated with the data in this field. Additional information associated with precise geo-referencing (e.g., accuracy, datums, etc.) are provided in geospatial related extensions if present in the file.			
NICOM	Number of Image Comments. This field shall contain the valid number of ICOMn field(s) that follow to be used as free text image comments.	1	BCS-N positive integer 0 to 9	R
. . . . . Start for each Image Comment ICOMn (if the value of the NICOM field is not equal to zero).				
ICOMn	Image Comment n. The field (ICOM1 to ICOMn), when present, shall contain free-form ECS text. They are intended for use as a single comment block and should be used that way. This field shall contain the nth free text image comment, where n is defined as follows: 1≤n≤ the value of the NICOM field. If the image comment is classified, it shall be preceded by the classification, including codeword(s). This field shall be omitted if the value in the NICOM field is populated with 0.	80	ECS-A User defined	C
. . . . . End for each ICOMn field; the number of loop repetitions is the value specified in the NICOMn field.				
IC	Image Compression. This field shall contain a valid code indicating the form of compression used in	2	BCS-A	R

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TABLE 3. NITF Image Subheader – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
	representing the image data. Valid values for this field are, C1 to represent bi-level, C3 to represent JPEG, C4 to represent Vector Quantization, C5 to represent lossless JPEG, I1 to represent down sampled JPEG, and NC to represent the image is not compressed. Also valid are M1, M3, M4, and M5 for compressed images, and NM for uncompressed images indicating an image that contains a block mask and/or a pad pixel mask. C6 and M6 are reserved values that will represent a future correlated multicomponent compression algorithm. C7 and M7 represent complex and magnitude-only SARZip compressed data. For C7 the image consists of a single NITF Image Block. For M7 the image consists of multiple NITF Image Blocks. Block C8 and M8 are the values for ISO standard compression JPEG 2000. The format of a mask image is identical to the format of its corresponding non-masked image except for the presence of an Image Data Mask at the beginning of the image data area. The format of the Image Data Mask is described in paragraph 5.4.3.2 and is shown in TABLE 3(A). (NOTE: C2 (ARIDPCM) is not valid in NITF 2.1.) See Appendix C for governing compression standards. For MI implementations: Use CB to represent JPEG 2000 with frames, MB to represent blocked JPEG 2000 with frames, C9 to represent H.264 with frames, M9 to represent blocked H.264 with frames, CA to represent H.265 with frames, and MA to represent blocked H.265 with frames.		Still Imagery values: NC, NM, C1, C3, C4, C5, C6, C7, C8, I1, M1, M3, M4, M5, M6, M7, M8 Motion Imagery values: CB, MB, C9, M9, CA, MA	
COMRAT	Compression Rate Code. If the IC field contains C1, C3, C4, C5, C7, C8, C9, CA, CB, I1, M1, M3, M4, M5, M7, M8, M9, MA, or MB, this field shall be present and contain a code indicating the compression rate for the image. If the value in IC is C1 or M1, the valid codes are 1D, 2DS, and 2DH, where: 1D represents One-dimensional Coding 2DS represents Two-dimensional Coding Standard Vertical Resolution (K=2) 2DH represents Two-dimensional Coding High Vertical Resolution (K=4) Explanation of these codes can be found in ITU-T T.4, AMD2. If the value in IC is C3, M3, C5, M5, or I1, the value of the field shall identify the embedded quantization table(s) used by the JPEG compression algorithm. In this case, the format of this field is XX.Y where XX is the image data type, and Y represents the quality	4	BCS-A Depending on the value of the IC field. (See description for constraints.)	C

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TABLE 3. NITF Image Subheader – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
	level 1 to 5. The image data types are represented by: 00 represents General Purpose 01 represents VIS 02 represents IR 03 represents SAR 04 represents Downsample (DS) JPEG Explanation of the optimized tables can be found in MIL-STD-188-198A and NGA N0106-97. The value of Y shall be 0 if customized tables are used. It is optional but highly recommended that the value of XX still be used for the image type with customized tables. If the value of IC is C5 or M5, then the value of Y shall be 0. It is optional but highly recommended that the value of XX still be used for the image type. If the value in IC is C4 or M4, this field shall contain a value given in the form n.nn representing the number of bits-per-pixel for the compressed image. Explanation of the compression rate for vector quantization can be found in MIL-STD-188-199. This field is omitted if the value in IC is NC or NM. If the value of IC is C7 or M7, this field shall contain a value representing the nominal compression rate number of bits-per-sample (BPS) of the compressed image. For BPS values less than 10.0, the BPS shall be written as N.NN, for BPS greater than or equal to 10.0, the BPS shall be written as NN.N. See USAF.RDUCE-001 for guidance on setting this field. If the value of IC is C8 or M8, this field shall contain a value representing the nominal compression rate (numbers of bits-per-pixel-per-band) of the compressed image. See the BIIF Profile for JPEG 2000 (BPJ2K01.10) for guidance in populating this field. For Motion Imagery/MIE4NITF implementations: If the value in the IC field is populated with CB or MB, this field shall contain a value representing the nominal compression rate (numbers of bits-per-pixel-per-band) of the compressed image. If the value in the IC field is populated with C9, M9, CA or MA, this field shall contain a value representing the nominal compression ratio (relative to the uncompressed data). A value of 12.3 would represent a compression ratio of 12.3:1. Any chroma subsampling is included in this number. If the value			

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TABLE 3. NITF Image Subheader – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
	is unknown or cannot be represented, the value is set to four hyphens "----".			
NBANDS	Number of Bands. This field shall contain the number of data bands within the specified image. This field and the IREP field are interrelated and independent of the IMODE field. The corresponding values for the IREP and NBANDS fields are NODISPLY, 0 to 9; MONO, 1; RGB, 3; RGB/LUT, 1; YCbCr601, 3; NVECTOR, 0 to 9; POLAR, 2; VPH, 2; MULTI, 1 to 9; and BCS zero (0x30) for multiple band images or matrices with greater than 9 bands.	1	BCS-N positive integer 0 to 9 BCS zero (0x30) (See description for details)	R
XBANDS	Number of Multispectral Bands. When NBANDS contains the value BCS zero (0x30), this field shall contain the number of bands or data points comprising the multiple band image. Otherwise this field shall be omitted if the value of the NBANDS field is 1 to 9.	5	BCS-N positive integer 00010 to 99999	C
. . . . . Start for each IREPBANDn to LUTDnm fields.				
NOTE: The fields IREPBANDn to LUTDnm fields repeat the number of times indicated in the NBANDS field or the XBANDS field.				
IREPBANDn	n^{th} Band Representation. This field shall contain a valid indicator of the processing required to display the nth band of the image with regard to the general image type as recorded in the IREP field. The significance of each band in the image can be derived from the combination of the ICAT, and ISUBCATn fields. Valid values of the IREPBANDn field depend on the value of the IREP field. The following standard values shall apply: • R, G, B respectively for a Red, Green, Blue representation of the band, • LU for a LUT representation of the band (e.g., a three table LUT for RGB and a single table LUT for shades of grey), • M for a monochrome representation of the band, BCS Spaces (0x20) for a band not designated for display, but may be displayed if desired, Y, Cb, Cr respectively for the Luminance, Chrominance blue (B minus Y), and Chrominance red (R minus Y) representation of a YCbCr601 (compressed case only) image. The only valid values when IREP contains MULTI are M, R, G, B, LU, and BCS Spaces (0x20): • It is strongly recommended that 3 of the multiple bands have the IREPBANDn fields populated with R, G, and B. When bands marked as LU, R, G, B, and M are present, the RGB designated bands are the default bands for display. If R, G, B are not present, the default displayable band is the LU	2	BCS-A (Default is BCS spaces (0x20)) Standard values are: LU, R, G, B, M, Y, Cb, Cr Additional values are allowed through the registration process. The most up to date list of registered values can be found on the NITF Field Value Registry, specified in section C.2.1 Online Resources.	<R>

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TABLE 3. NITF Image Subheader – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
	band. If R, G, B, or LU are not present, the default displayable band is the first M band. When no bands are marked with LU, R,G, B, or M the first three bands may be displayed as R, G, and B respectively. For consistency, multispectral images cannot have more than one band, each marked as R, G, and B. • IREPBANDn shall be filled with the M value, if the band is to be represented as monochrome. • IREPBANDn shall be filled with the LU value, if the band is to be represented using a LUT. • When IREPBANDn is filled with BCS Spaces (0x20), no specific representation is defined for the band, but it may be displayed if desired. Additional values are reserved for specific interpretations and shall be coordinated with the Custodian to regulate their use. The only valid values when IREP contains MONO are M, LU or BCS Spaces (0x20). The only valid values when IREP contains RGB images are R, G and B. The only valid value when IREP contains RGB/LUT images is LU. The only valid values when IREP contains YCbCr601 images are Y, Cb and Cr. • Note: There may be more than one band that contains M or LU where the default conditions are such that the first M or LU band is the band to be displayed. This is only the default display to be presented to the user. Any other band or combination of bands may be displayed by user intervention.			
ISUBCATn	n^{th} Band Subcategory. The purpose of this field is to provide the significance of the nth bands of the image with regard to the specific category (ICAT field) of the overall image. The use of this field is user-defined except for the following: (In all cases below where it says "ICAT is" followed by a value also applies to the MI version of the value.) For ICAT = IR or IR.M, multi- and hyperspectral ICATs (MS, MS.M, HS, HS.M), and EO ICATs used for multi- or hyperspectral imagery (CAVIS, EOVIS, LWIR, MWIR, NIR, SWIR, UV, VNIR or the MI equivalent), then ISUBCATn shall contain the representative band wavelength in nanometers.	6	BCS-A I, Q, M, P, SPEED, DIRECT, User-defined When (In all cases below where it says "ICAT is" and a value also applies to the MI version of the value.) When ICAT is MS, HS, or IR (or the MI equivalent) the value range is the wave length.	<R>

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TABLE 3. NITF Image Subheader – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
	When ICAT is CCD, ISAR, SAR or SARIQ, ISUBCATn contains: I for the in-phase band Q for the quadrature components band M for the magnitude band P for the phase components BCS spaces for all the other cases. When ICAT is WIND, CURRENT, ISUBCATn contains SPEED for wind or water speed, or DIRECT for wind or water direction. For location grids, the number of bands is strictly equal to 2, consequently, there are only 2 fields, the ISUBCAT1 field and the ISUBCAT2 field. Standard values of these fields of location grids are either CGX and CGY for the cartographic X (Easting) and Y (Northing) bands, or GGX and GGY with the geographic X representing the longitude band and Y representing the latitude band. Standard values for the matrix (ICAT is MATR) are FACC codes from DIGEST Part 4 – Annex B. Standard values for Digital Terrain Elevation Models (ICAT is DTEM) are units of length from DIGEST Part 3 - 7.		When ICAT is LOCG the value range is CGX, CGY (Cartographic), GGX, GGY (Geographic). (Default is BCS spaces (0x20)) The most up to date list of registered values can be found on the NITF Field Value Registry, specified in section C.2.2 ONLINE REGISTRIES.	
IFCn	n^{th} Band Image Filter Condition. This field shall contain the value N (to represent none). Other values are reserved for future use.	1	BCS-A N	R
IMFLTn	n^{th} Band Standard Image Filter Code. This field is reserved for future use. It shall be filled with BCS spaces (0x20).	3	BCS-A Fill with BCS spaces (0x20)	<R>
NLUTSn	Number of LUTS for the nth Image Band. This field shall contain the number of LUTs associated with the nth band of the image. LUTs are allowed only if the value of the PVTYPE field is INT or B. If the nth band of the image is monochromatic, this field can contain the value 1 or 2. If the value is 2, the first and second LUTs shall map respectively the most significant byte and the least significant byte of the 16-bit values. NOTE: If a system cannot support more than 256 different values, it may use only the values of the first LUT. In this case, the number of entries in the LUT (NELUTn) may exceed 256. If the nth band of the image is color-coded (the value of the IREPBANDn field is populated with LU), this field shall contain the value 3. The first, second, and third LUTs, in this case, shall map the image to the red, green, and blue display bands respectively.	1	BCS-N positive integer 0 to 4 (Default is BCS zero (0x30) if no LUTs are included.)	R

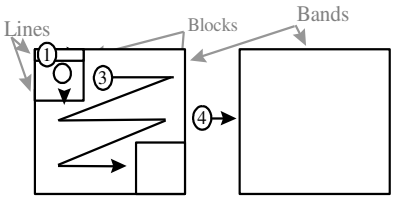
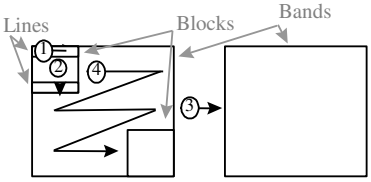
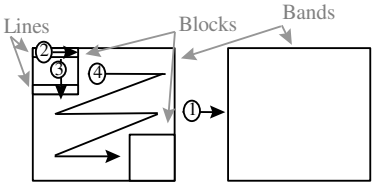
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TABLE 3. NITF Image Subheader – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
	The value 4 is reserved for future use.			
NELUTn	Number of LUT Entries for the nth Image Band. This field shall contain the number of entries in each of the LUTs for the nth image band. This field shall be omitted if the value in NLUTSn is BCS zero (0x30).	5	BCS-N positive integer 00001 to 65536	C
. . . . . Start for each LUT LUTDnm				
LUTDnm	n^{th} Image Band, m^{th} LUT. This field shall be omitted if the Number of LUTs (NLUTSn) is BCS zero (0x30). Otherwise, this field shall contain the data defining the mth LUT for the nth image band. Each entry in the LUT is composed of one byte, ordered from MSB to LSB, representing a binary value from zero (0x00) to 255 (0xFF). To use the LUT, for each integer k, $0 \leq k \leq$ (value of the NELUTn field) -1, the pixel value k in the nth image band shall be mapped to the value of the kth byte of this field (the LUT). NOTE: This is a repeating field based on the value of the NLUTSn field. When there are more than one LUT (value of the NELUTn field is greater than 1), the net effect is to have the LUT ordered in band sequential fashion, e.g., all the red values followed by the green values followed by the blue values.	†[3]	Unsigned binary integer LUT Values	C
. . . . . End for each LUTDnm field; the number of loop repetitions is the value specified in the NLUTSn field.				
. . . . . End for each IREPBANDn to LUTDnm fields; the number of loop repetitions is the value specified in the NBANDS field or the XBANDS field.				
ISYNC	Image Sync code. This field is reserved for future use. This field shall contain BCS zero (0x30).	1	BCS-N positive integer 0 = No Sync Code	R
IMODE	Image Mode. This field shall indicate how the Image Pixels are stored in the NITF file. (For additional information see Tables 3(B) thru 3(E).) Valid values are B, P, R, and S. The interpretation of IMODE is dependent on whether the image is JPEG compressed (IC = C3, C5, I1, M3, or M5), VQ compressed (IC = C4, or M4), SAR Compressed (IC = C7 or M7), or uncompressed (IC = NC or NM). a. Uncompressed Still Imagery. The value S indicates band sequential, where all blocks for the first band are followed by all blocks for the second band, and so on: [(block1, band1), (block2, band1), ... (blockM, band1)], [(block1, band2), (block2, band 2), ... (blockM, band2)] ... [(block1, bandN), (block2, bandN), ... (blockM, bandN)]. Note that, in each block, the pixels of the first line appear first, followed by the pixels of the second line, and so on.	1	BCS-A Still Imagery: B = Band Interleaved by Block. P = Band Interleaved by Pixel. R = Band Interleaved by Row. S = Band Sequential. Motion Imagery: T = Band Interleave by Block D = Band Sequential Temporal E = Temporal Band Interleave by Pixel F = Band Temporal Interleave by Block X = Band Interleave by Block Temporal Z = Band Sequential	R

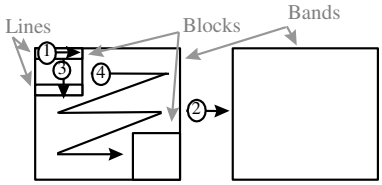
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TABLE 3. NITF Image Subheader – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
	 Band Sequential (IMODE = S) The value B indicates band interleaved by block. This implies that within each block, the bands follow one another: [(block1, band1), (block1, band2), ...(block1, bandN)], [(block2, band1), (block2, band2), ... (block2, bandN)], ... [(blockM, band1), (blockM, band2), ... (blockM, bandN)]. Note that, in each block, the pixels of the first line appear first and the pixels of the last line appear last.  Band Interleaved by block (IMODE = B) The value P indicates band interleaved by pixel within each block: such as, for each block, one after the other, the full pixel vector (all band values) appears for every pixel in the block, one pixel after another, the block column index varying faster than the block row index.  Band Interleaved by pixel (IMODE = P) The value R indicates band interleaved by row. The ordering mechanism for this case stores the pixel values of each band in row sequential order. Within each block, all pixel values of the first row of the first band are followed by pixel values of the first row of the second band continuing until all values of the first row are stored. The remaining rows are stored in a similar fashion until the last row of values has been stored. Each block shall be zero-filled to the next octet boundary when necessary.		Temporal	

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TABLE 3. NITF Image Subheader – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
	 Band Interleaved by row (IMODE = R) If the value of the NBANDS field is populated with 1, the cases B and S coincide. In this case, this field shall contain B. If the Number of Blocks is 1 (the NBPR field and the NBPC field contain 1), this field shall contain B for non-interleaved by pixel, and P for interleaved by pixel. The value S is only valid for images with multiple blocks and multiple bands. b. JPEG-compressed. The presence of B, P, or S implies specific ordering of data within the JPEG image data representation. For this case the interpretation of the various values of the IMODE field is specified in MIL-STD-188-198A. When IC contains C8, M8, or I1, IMODE contains B. c. Vector Quantization compressed. VQ compressed images are normally either RGB with a color look-up table or monochromatic. In either case, the image is single band, and the IMODE field defaults to B. d. Bi-Level Compressed. When the value of the IC field is populated with C1 or M1, the value of the IMODE field is populated with B e. Motion Imagery (MI). Six IMODEs are for MI files. If an NITF image segment (i.e., a temporal block) within an MI file contains only one temporal frame, then the S, P, B or R IMODEs are used. If there are multiple frames within a temporal block, the IMODEs T, D, E, F, X, and Z are to be used. (See section 5.4.3.3.1 and APPENDIX F below for details and examples regarding the IMODEs.) f. SARzip Compressed. When the value of the IC field is populated with C7 or M7, the value of the IMODE field is populated with P for Complex or B for Magnitude-only SAR images.			
NBPR	Number of Blocks Per Row. This field shall contain the number of image blocks in a row of blocks (paragraph 5.4.2.2) in the horizontal direction. If the image consists of only a single block, this field shall contain the value one.	4	BCS-N positive integer 0001 to 9999	R
NBPC	Number of Blocks Per Column. This field shall contain the number of image blocks in a column of blocks (paragraph 5.4.2.2) in the vertical direction. If the image consists of only a single block, this field shall contain the value one.	4	BCS-N positive integer 0001 to 9999	R
NPPBH	Number of Pixels Per Block Horizontal. This field shall contain the number of pixels horizontally in each block of the image. It shall be the case that the	4	BCS-N positive integer 0000 or 0001	R

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TABLE 3. NITF Image Subheader – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
	product of the values of the NBPR field and the NPPBH field is greater than or equal to the value of the NCOLS field (NBPR * NPPBH ≥ NCOLS). When NBPR is "0001", setting the NPPBH value to "0000" designates that the number of pixels horizontally is specified by the value in NCOLS.		to 8192	
NPPBV	Number of Pixels Per Block Vertical. This field shall contain the number of pixels vertically in each block of the image. It shall be the case that the product of the values of the NBPC field and the NPPBV field is greater than or equal to the value of the NROWS field (NBPC * NPPBV ≥ NROWS). When NBPC is "0001", setting the NPPBV value to "0000" designates that the number of pixels vertically is specified by the value in NROWS.	4	BCS-N positive integer 0000 or 0001 to 8192	R
NBPP	Number of Bits Per Pixel Per Band. If IC contains NC, NM, C4, or M4, this field shall contain the number of storage bits used for the value from each component of a pixel vector. The value in this field always shall be greater than or equal to Actual Bits Per Pixel (ABPP). For example, if 11-bit pixels are stored in 16 bits, this field shall contain 16 and Actual Bits Per Pixel shall contain 11. If IC = C3, M3, C5, M5, or I1 this field shall contain the value 8 or the value 12. If IC = C1, this field shall contain the value 1. If IC = C8 or M8, this field shall contain the number of bits of precision (01-38) used in the JPEG 2000 compression of the data and ABPP = NBPP.	2	BCS-N positive integer 01 to 96	R
IDLVL	Image Display Level. This field shall contain a valid value that indicates the display level of the image relative to other displayed file components in a composite display. The valid values are 001 to 999. The display level of each displayable segment (image or graphic) within a file shall be unique; that is, each number from 001 to 999 is the display level of, at most, one segment. Display level is discussed in paragraph 5.3.3. The image or graphic segment in the file having the minimum display level shall have attachment level 0 (ALVL000) (BCS Zeros (0x30)).	3	BCS-N positive integer 001 to 999	R
IALVL	Attachment Level. This field shall contain a valid value that indicates the attachment level of the image. Valid values for this field are BCS zeros (0x30), and the display level value of any other image or graphic in the file. The meaning of attachment level is discussed in paragraph 5.3.4. The image or graphic segment in the file having the minimum display level shall have attachment level 0 (ALVL000) (BCS zeros (0x30).	3	BCS-N positive integer 000 to 998 (Default is BCS zeros (0x30))	R
ILOC	Image Location. The image location is the location of the first pixel of the first line of the image. This field shall contain the image location offset from the ILOC or SLOC value of the segment to which the image is attached or from the origin of the CCS when	10	BCS-N RRRRRCCCCC For positive row and column values RRRRR and CCCCC	R

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TABLE 3. NITF Image Subheader – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
	the image is unattached (IALVL contains 000). A row or column value of 00000 indicates no offset. Positive row and column values indicate offsets down and to the right while negative row and column values indicate offsets up and to the left.		are both in the range 00000 to 99999. For negative row and column values RRRRR and CCCCC are both in the range -0001 to -9999	
IMAG	Image Magnification. This field shall contain the magnification (or reduction) factor of the image relative to the original source image. Decimal values are used to indicate magnification, and decimal fraction values indicate reduction. For example, "2.30" indicates the original image has been magnified by a factor of "2.30," while "0.5" indicates the original image has been reduced by a factor of 2. The default value is 1.0, indicating no magnification or reduction. In addition, the reductions can be represented as reciprocals of any non-negative integer: /2 (for 1/2), /3 (for 1/3), /4 (for 1/4), /5 (for 1/5), through /999 (for 1/999). The values are left justified and BCS spaces (0x20) filled to the right.	4	BCS-A decimal value, /x, where x = any non-negative integer ≤ 999 (Default is 1.0 followed by BCS space (0x20))	R
UDIDL	User Defined Image Data Length. This field shall be populated with all BCS zeros (0x30) if no TRE are included in the UDID field. If a TRE exists, the field shall contain the sum of the length of all the TRE (paragraph 5.8.1) appearing in the UDID field plus 3 bytes (length of UDOFL field). If a TRE is too long to fit in the UDID field or the IXSHD field, it shall be put in the TRE overflow DES with DESID set to the value TRE_OVERFLOW (paragraph 5.8.3.1)	5	BCS-N positive integer 00000 or 00003 to 99999	R
UDOFL	User Defined Overflow. If present, this field shall contain BCS zeros (0x30) if the TRE in UDID do not overflow into a DES, or shall contain the sequence number of the DES into which they do overflow. This field shall be omitted if the field UDIDL contains BCS zeros (0x30).	3	BCS-N positive integer 000 to 999	C
UDID	User Defined Image Data. If present, this field shall contain user-defined TRE (paragraph 5.8.1). The length of this field shall be the length specified by the field UDIDL minus 3. TRE in this field for an image shall contain information pertaining specifically to the image. TRE shall appear one after the other with no intervening bytes. The first byte of this field shall be the first byte of the first TRE appearing in the field. The last byte of this field shall be the last byte of the last TRE to appear in the field. This field shall be omitted if the field UDIDL contains BCS zeros (0x30).	††[3]	TRE(s)	C
IXSHDL	Image Extended Subheader Data Length. This field shall be populated with all BCS zeros (0x30) if no TRE are included in the IXSHD field. If a TRE exists, the field shall contain the sum of the length of all the TRE (paragraph 5.8.1) appearing in the IXSHD field	5	BCS-N positive integer 00000 or 00003 to 99999	R

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TABLE 3. NITF Image Subheader – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
	plus 3 (length of IXSOFL field) in bytes. If a TRE is too long to fit in the IXSHD field or the UDID field, it shall be put in the TRE overflow DES with DESID set to the value TRE_OVERFLOW (paragraph 5.8.3.1).			
IXSOFL	Image Extended Subheader Overflow. If present, this field shall contain BCS zeros (0x30) if the TRE in IXSHD do not overflow into a DES, or shall contain the sequence number of the DES into which they do overflow. This field shall be omitted if the field IXSHDL contains BCS zeros (0x30).	3	BCS-N positive integer 000 to 999	C
IXSHD	Image Extended Subheader Data. If present, this field shall contain TRE (paragraph 5.8.1) approved and under configuration management by the GWG NTB. The length of this field shall be the length specified by the field IXSHDL minus 3. TRE in this field for an image shall contain information pertaining specifically to the image. TRE shall appear one after the other in this field with no intervening bytes. The first byte of this field shall be the first byte of the first TRE appearing in the field. The last byte of this field shall be the last byte of the last TRE to appear in the field. This field shall be omitted if the field IXSHDL contains BCS zeros (0x30).	†††[3]	TRE(s)	C

†[3] A value as specified in the NELUTn field (in bytes).
††[3] A value as specified in the UDIDL field minus 3 (in bytes)
†††[3] A value as specified in the IXSHDL field minus 3(in bytes)

5.4.3.2 Image Data Mask. The image data mask table is a conditional data structure included in the image data stream for masked images when so indicated by the IC field value (IC values NM, M1, M3, M4, M5, M6, M7, M8, M9, MA and MB). The image data mask table is not recorded for non-masked images (IC values NC, C1, C3, C4, C5, C6, C7, C8, I1, C9, CA and CB). The image data field of a masked image is identical to that of non-masked images except for the following: the first byte of the image data is offset from the beginning of the image data field by the length of the image data mask table(s); and empty image blocks are not recorded/transmitted in the image data area. Please note that for SARZip compression C7 files will be single blocked and M7 files will be multi-block. If the image is band sequential (IMODE=S), there will be multiple blocked image and/or pad pixel masks (one for each band). All blocked image masks will be recorded first, followed by all pad pixel masks. Since the image data mask tables are in the image data area, the data recorded/transmitted there are binary. The structure of the image data mask table is defined in detail in TABLE 3(A).

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TABLE 3(A). NITF Image Data Mask Table.

(TYPE "R" = Required, "C" = Conditional, "< >"= BCS spaces allowed for entire field)
("†" annotations are explained at the end of the table)

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
IMDATOFF	Blocked Image Data Offset. This field is included if the IC value contains M. It identifies the offset from the beginning of the Image Data Mask to the first byte of the blocked image data. This offset, when used in combination with the offsets provided in the BMRnBND fields, can provide random access to any recorded image block in any image band.	4	Unsigned binary integer: range of values: 0 to 2^{32} - 1	C
BMRLNTH	Block Mask Record Length. This field is included if the IC value contains M. It identifies the length of each Block Mask Record (BMRnBNDm) in bytes. When present, the length of each Block Mask Record is 4 bytes. If all of the image blocks are recorded, this value may be set to 0x0000, and the conditional BMRnBNDm fields are not recorded/transmitted. Otherwise, the value may be set to 0x0004, and the conditional BMRnBNDm fields are recorded/transmitted and can be used as an offset index for each image block of the image. If this field is present, but coded as 0x0000, then only a pad pixel mask is included.	2	Unsigned binary integer; 0x0000 denotes No Block Mask Record; 0x0004 denotes Block Mask Records (4 bytes each) are present.	C
TMRLNTH	Pad Pixel Mask Record Length. This field is included if the IC value contains M. It identifies the length of each Pad Pixel Mask Record in bytes. When present, the length of each Pad Pixel Mask Record is 4 bytes. The total length of the Pad Pixel Mask Records is equal to TMRLN. If none of the image blocks contain pad pixels, this value is set to 0x0000, and the conditional TMRnBNDm fields are not recorded/transmitted. For IC value of M3, the value shall be set to 0x0000. If this field is present, but coded as 0x0000, then only a Block Mask is included.	2	Unsigned binary integer; 0x0000 denotes no Pad Pixel Mask Records; 0x0004 denotes Pad Pixel Mask Records (4 bytes each) are present.	C
TPXCDLNTH	Pad Output Pixel Code Length. This field is included if the IC value contains M. It identifies the length in bits of the Pad Output Pixel Code. If coded as 0x0000, then no pad pixels are present, and the TPXCD field is not recorded. For IC value of M3, the value shall be set to 0x000.	2	Unsigned binary integer; 0x0000 denotes no Pad Pixels; or Pad Pixel Code length in bits (Length must be as specified in NBPP)	C

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TABLE 3(A). NITF Image Data Mask Table – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
TPXCD	Pad Output Pixel Code. This field is included if the IC value contains M and TPXCDLNTH is not zeros (0x0000). It contains the output pixel code that represents a pad pixel in the image. This value is unique within the image, and allows the user to identify pad pixels. The pad output pixel code length is determined by TPXCDLNTH. If the number of bits used by TPXCD is less than the number of bits available for storage, the value shall be justified in accordance with the PJUST field in the image subheader (L for left, R for right justified.)	†[3A]	For PVTYPE INT Unsigned binary integer; range of values: 0 to 2^n -1 where n is the value contained by the TPXCDLNTH field For PVType SI: Signed binary integer; range of values: $-2^{(n-1)}$ to $2^{(n-1)}$ -1 where n is the value contained by the TPXCDLNTH field For PVType R and NBPP 32: IEEE 32-bit Floating Point For PVType R and NBPP 64: IEEE 64-bit Double-Precision Floating Point For PVType C: Two IEEE 32-bit Floating Point Numbers (Real followed by Imaginary)	C
. . . . .				
NOTE: The BMRnBNDm record repeats; one 4 byte record for each block, and for IMODE S, one record for each block of each band in the image. The number of BMRnBNDm records in the IMODE B, P, and R case will be NBPR * NBPC and the IMODE S case will be NBANDS (or XBANDS) * NBPR * NBPC.				
BMRnBNDm	Block n, Band m Offset. This field shall contain the nth Block Mask Record of band m. It is recorded/transmitted only if the BMRLNTH field does not contain zeros (0x0000). The field shall contain an offset in bytes from the beginning of the Blocked Image Data to the first byte of block n of band m. If block n of band m is not recorded/transmitted, the offset value is defaulted to 0xFFFFFFFF. Block mask records may be used with MI data under certain conditions. See NGA.STND.0044 for more details.	4	Unsigned binary integer For IMODEs B, P, R: Increment n only, m is always 1. 0≤ n≤ NBPR * NBPC-1 For IMODE S: Increment n prior to m 0≤ n≤ NBPR*NBPC -1 0≤ m≤ max(NBANDS,XBANDS) (Default is 0xFFFFFFFF if the block is not recorded)	C
. . . .				
NOTE: The TMRnBNDm record repeats; one 4 byte record for each block, and for IMODE S, one record for each block of each band in the image. The number of TMRnBNDm records in the IMODE B, P, and R case will be NBPR * NBPC and in the IMODE S case will be NBANDS (or XBANDS) * NBPR * NBPC.				
TMRnBNDm	Pad Pixel n, Band m. This field shall contain the nth Pad Pixel Mask Record for band m. It is recorded/transmitted only if the TMRLNTH field does not contain zeros (0x0000). The field shall contain an offset in bytes from the beginning of the Blocked Image Data to the first byte of block n of the image data of band m if block n contains pad pixels, or the default value 0xFFFFFFFF to	4	Unsigned binary integer For IMODEs B, P, R: Increment n only; m is always 1 0≤ n≤ NBPR*NBPC -1 For IMODE S: Increment n prior to m	C

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TABLE 3(A). NITF Image Data Mask Table – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
	indicate that this block does not contain pad pixels. Pad pixel mask records may not be used with compressed MI data. For uncompressed MI data, they may be used under certain conditions. See NGA.STND.0044 for more details.		0≤ n≤ NBPR*NBPC.-1 0≤ m≤ max(NBANDS,XBANDS) (Default is 0xFFFFFFFF if the block is not recorded)	

†[3A] The length of the TPXCD field is the next highest number of bytes that can contain the number of bits identified in the TPXCDLNTH field. For example, a TPXCDLNTH value of 12 would be stored in a TPXCD field of two bytes.

5.4.3.3 Image Data Format. Image data may be stored in an NITF file in either uncompressed or compressed form.

5.4.3.3.1 IMODE. The IMODE field value, along with other NITF image segment field values (i.e., NBPR, NBPC, NPPBH, NPPBV and IC), determine the ordering of pixel values (data samples for non-viewable data arrays) within an NITF image segment. Four IMODE field values (S, P, B and R) are defined for image segments containing a single still image and six IMODE field values (T, D, E, F, X, and Z) are defined for use with motion imagery contained within an image segment (see NGA.STND.0044).

5.4.3.3.1.1 Interleave. The pixel interleave orderings specified by IMODE may be thought of as a set of nested loops. Within the tables presented below, pixel row within a block (y) and pixel column within a block (x) represent spatial variables at the pixel value/data sample level within an image block or MI spatiotemporal block. The variables block row (Y) and block column (X) represent spatial variables at the image block/spatiotemporal block level within the NITF Image Segment. Finally, the variable (t) is used to represent the temporal dimension within a spatiotemporal block and (b) is used to represent image component or band within an image or MI frame. Note, for still imagery there is no temporal dimension to the pixel values.

5.4.3.3.1.1.1 Bolding is used in the following tables to represent variables that may be indexed (pointers) by the NITF image data mask table. Not all dimensions in MI data are indexed and therefore randomly addressable. Compression of still and motion imagery further reduces the ability to randomly address pixel values because the compression algorithm may jointly encode pixels, bands and frames together. In the compressed version of the tables, some of interleaving dimensions/variables will disappear as those dimensions of the imagery data have been subsumed into the compressed codestream by the compression algorithm. In the tables below, the outmost loop dimension/variable changes the slowest and the innermost loop changes fastest.

5.4.3.3.1.2 Still Imagery. The IMODE values in TABLE 3(B) apply when a single uncompressed still image is stored in an NITF image segment. Common names for these modes are: IMODE S – band sequential; IMODE P – band interleave by pixel; IMODE B – band interleave by block; IMODE R – band interleave by row.

TABLE 3(B). Uncompressed Still Imagery IMODE Values

IMODE S	IMODE P	IMODE B	IMODE R
b loop	**Y** loop	**Y** loop	**Y** loop
Y loop	**X** loop	**X** loop	**X** loop
X loop	y loop	b loop	y loop
y loop	x loop	y loop	b loop
x loop	b loop	x loop	x loop

5.4.3.3.1.2.1 As an example, consider IMODE B, band interleave by block, with an uncompressed multiple band image. Blocks of pixels in the image are ordered in the image segment data from the top left to the bottom right in the image. The number of image blocks and their dimensions are determined by the size of the image and by the image segment field values, NBPR, NBPC, NPPBH, and NPPBV. For a given image block, i.e., a specific value of block row (Y) and block column (X), uncompressed values are ordered by band (b), sample row (y), and sample column (x). Each image band appears in its entirety one after another constrained to the image block boundary. Within each image band, the sample values are arranged by rows, with all values in the first row appearing, then all values in the second row and so on until the band is complete. After the first band, the second band of sample values are given and so on until all bands are completed. Section 5.4.3.3.2 and its subsections further discuss the still imagery IMODEs.

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5.4.3.3.1.3 Motion Imagery. The IMODE values in TABLE 3(C) apply when uncompressed MI is stored in an NITF image segment. An additional loop over frame or time has been added to the still imagery IMODE constructs to accommodate MI. The time index corresponds to the different frames contained within the NITF image segment and spatiotemporal block. TABLE 3(C) shows the nested loop constructs corresponding to the six MI IMODE values. (Appendix F contains illustrative figures depicting the differences between the various still and MI IMODEs; see Section 8).

TABLE 3(C). Uncompressed Motion Imagery IMODE Values

IMODE T	IMODE D	IMODE E	IMODE F	IMODE X	IMODE Z
Y loop	**b** loop	**Y** loop	**Y** loop	**t** loop	**t** loop
X loop	**Y** loop	**X** loop	**X** loop	Y loop	b loop
t loop	**X** loop	t loop	**b** loop	X loop	Y loop
b loop	t loop	y loop	t loop	b loop	X loop
y loop	y loop	x loop	y loop	y loop	y loop
x loop	x loop	b loop	x loop	x loop	x loop

5.4.3.3.2 Uncompressed Still Image Data Format. The order in which pixel values of a single band still image are stored is fixed. When an image has more than one band, several options are available for the order in which pixel values are stored. The following subparagraphs describe the possibilities within this format. In describing the interleaving of image data, the NITF display convention is invoked freely for ease of expression. Let the image to be encoded be denoted by I, and assume I has R rows and C columns. Let I have n bands; that is, each pixel is an n-vector, the ith value of which is the value for that pixel location of the ith band of the image. Let N denote the number of bits-per-pixel-per-band. Thus, there are n*N bits-per-pixel. Let I be blocked with H blocks per row and V blocks per column. Note that special cases such as single band images and single blocked images are included in this general image by setting n=1, and H=V=1, respectively.

5.4.3.3.2.1 Single Band Still Image Uncompressed Data Format. For single band images, n=1, and there is only one order for storing pixels. The field IMODE in the image subheader shall be set to B for this case. The blocks (one or more) shall be stored, one after the other starting with the upper left block and proceeding first left to right across rows of blocks, one row of blocks after the other, top to bottom. Image data within each block shall be encoded as one continuous bit stream, one pixel value after another, beginning with the N bits of the upper left corner pixel, I(0,0), followed by the N bits of I(0,1) and so on until all pixels from the first row in the block are encoded. These shall be followed immediately by the N bits of data for pixel I(1,0) continuing from left to right along each row, one row after another from the top of the block to the bottom. The last byte of each block's data is zero-filled to the next byte boundary, but all other byte boundaries within the block are ignored. See the field PVTYPE description in TABLE 3 for the specification of the bit representation of pixel values.

5.4.3.3.2.2 Multiple Band Still Image Uncompressed Data Format. For multiple band images, there are four orders for storing pixels.

5.4.3.3.2.2.1 Band Sequential. The first case is "band sequential", in which each band is stored contiguously, starting with the first band, one after the other, until the last band is stored. Within each band the data shall be encoded as if it were a single band image with one or more blocks (paragraph 5.4.3.3.2.1). The field IMODE in the image subheader shall be set to S for this case. This case is only valid for images with multiple blocks and multiple bands. (For single blocked images, this case collapses to the “band interleaved by block” case where IMODE is set to B.)

5.4.3.3.2.2.2 Band Interleaved By Pixel. The ordering mechanism for this case stores the pixels in a block sequential order in which each block is stored contiguously, starting with the upper left block and proceeding first left to right across rows of blocks, one row of blocks after the other, top to bottom. For "band interleaved by pixel" the n*N bits of the entire pixel vector are stored pixel-by-pixel in the same left to right, top to bottom pixel order as described in paragraph 5.4.3.3.1.1. The n*N bits for a single pixel are stored successively in this order: The N bits of the first band followed by the N bits of the second band and, so forth, ending with the N bits of the last band. Each block shall be zero-filled to the byte boundary. The field IMODE in the image subheader shall be set to P for this storage option. See the field PVTYPE description in TABLE 3 for the specification of the bit representation of pixel values for each band.

5.4.3.3.2.2.3 Band Interleaved By Block. The ordering mechanism for this case stores the pixels in a block sequential order where each block is stored contiguously, starting with upper left block and proceeding first left to right across rows of blocks, one row of blocks after the other, top to bottom. For "band interleaved by block" the data from each block is stored starting with the first band, one after the other until the last band is stored. Each block shall be zero-filled to the

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next byte boundary. The field IMODE in the image subheader shall be set to B for this storage option. See the field PVTYPE description in TABLE 3 for the specification of the bit representation of pixel values for each band.

5.4.3.3.2.2.4 Band Interleaved By Row. The ordering mechanism for this case stores the pixel values of each band in row sequential order. Within each block, all pixel values of the first row of the first band are followed by pixel value of the first row of the second band continuing until all values of the first row are stored. The remaining rows are stored in a similar fashion until the last row of values has been stored. The field IMODE shall be set to R for this option.

5.4.3.3.3 Compressed Still Image Data Format. The format of the image data after compression is provided with the description of the NITFS image compression algorithms in ITU-T T.4 (1993.03), AMD2 08/95, ISO/IEC 10918-1, ISO/IEC 12087-5, ISO/IEC 15444-1/4, USAF.RDUCE-001, and NGA-N0106-97. Also found in these references are the conditions the data must meet before a given compression method can be applied meaningfully. See the entry for the Image Compression (IC) field in TABLE 3 for additional details.

5.4.3.3.4 Uncompressed Motion Imagery Data Format. The motion imagery IMODEs are derived from one of three still image IMODEs. There is no MI IMODE derived from IMODE R. Each MI IMODE is formed by adding a frame or time (t) index into the loop constructs shown in TABLE 3(B). For example, IMODE T (temporal band interleave by block) is derived from IMODE B buy inserting a temporal loop just outside of the loop over bands (b). IMODE F (band temporal interleave by block) is also derived from IMODE B, but the temporal loop is placed just inside of the loop over bands. This is a subtle difference, but it allows MI to be treated either as a temporal sequence of multi-band images or as multiple temporal sequences of monochrome images, one temporal sequence for each band. IMODE E (temporal band interleave by pixel) is derived from IMODE P and IMODE D (band sequential temporal interleave) is derived from IMODE S.

5.4.3.3.5 Compressed Motion Imagery Data Format. NGA.STND.0044 provides for the compression of motion imagery using encoders specifically designed for MI. Still image encoders may also be used to compress MI in a frame by frame mode. There are applications where it is advantageous to use a still image encoder vice a motion imagery encoder. For example, a still image encoder may support a greater number of color bands, a greater number of bits allowed per pixel and band, a multiresolution representation, a lossless encoding of the pixel values, or other features not commonly found in MI encoders. Application of compression to MI alters the IMODE interleaving. Motion imagery compression techniques attempt to exploit spatial and temporal similarities within and between MI frames to reduce the number of bytes required to represent the MI relative to its uncompressed state. Decoding compressed MI may require that an entire MI frame be decoded to access a desired region in the frame; or that multiple MI frames be decoded to access one specific MI frame. The following sections further discuss IMODE and compressed motion imagery.

5.4.3.3.5.1 Motion Imagery Encoders. Motion imagery compression algorithms typically fold pixel location, bands and time together into their compressed codestreams. This may be thought of in terms of IMODE as the elimination of certain loops or data dimensions. It is not possible to simply address a single sample or band for a given frame without decoding one or more MI frames. TABLE 3(D) shows the IMODEs when a MI encoder (e.g. H.264 or H.265) is used on the frames in a spatiotemporal block. Motion imagery encoders will compress temporally across frames thereby eliminating the time dimension and they may encode across bands for three band color motion imagery. Alternatively, a system may choose to treat MI bands independently and generate a compressed codestream for each individual band. This might be done if the number of bands is greater than three or if the bands do not correlate to visual color bands.

TABLE 3(D). IMODE Values for Motion Imagery Compressed Using MI Encoders

IMODE T	**IMODE D**	**IMODE F**
Y loop	**b** loop	**Y** loop
X loop	**Y** loop	**X** loop
	X loop	**b** loop

5.4.3.3.5.1.1 Consider IMODE T when a MI encoder is used on single band or three band color Motion Imagery. The encoder will operate across the bands and frames creating one compressed codestream for each spatiotemporal block. The only surviving loops in the IMODE T stack are therefore the loops over spatiotemporal block row and column (Y and X). Under these same conditions, IMODE E degenerates into IMODE T and there is no difference between the two. IMODEs D and F are used in situations where each band or color component in MI data is compressed into its own codestream. The MI data is treated as a set of concurrently captured bands each independently compressed. IMODEs D and F then represent different interleaves for the independently compressed bands.

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5.4.3.3.5.1.2 IMODEs X and Z are not to be used with MI encoders (see section 5.4.3.3.5.2.3 below for usage of IMODE X with still image encoders). These IMODEs are primarily used for the uncompressed MI collections. Motion imagery encoders have little or no support for spatial blocking of frames inside of the compression algorithm. Because of this lack of spatial tiling or blocking within the compression algorithm, any IMODE where the time index/loop (t) is outside of or above the spatial block loops (Y, X) cannot be used with MI encoders.

5.4.3.3.5.2 Still Imagery Encoders. TABLE 3(E) shows the IMODE interleaves for MI data compressed with a still imagery encoder such as JPEG 2000. Still imagery encoders treat each frame, or each sub frame within a spatiotemporal blocked frame, as an individual still image. They do not exploit the temporal redundancy between frames. When a spatiotemporal block contains multiple sub frames, the independently encoded codestreams are concatenated one after another within the NITF Image Segment according to their IMODE interleave. Each frame, or sub frame, is independently decodable. The concatenation of these individually encoded sub frames forms the spatiotemporal block codestream.

TABLE 3(E). IMODE Values for Motion Imagery Compressed Using Still Imagery Encoders

IMODE T	IMODE D	IMODE F
Y loop	**b** loop	**Y** loop
X loop	**Y** loop	**X** loop
t loop	**X** loop	**b** loop
	t loop	t loop

5.4.3.3.5.2.1 Using still imagery encoders to compress MI data on a frame by frame basis may seem an unwise choice at first. Failure to exploit the temporal redundancy between consecutive frames in an MI sequence will lead to increased file size. There are, however, reasons to employ a still image encoder. JPEG 2000 can compress more bands/components into its codestream (16,384) and it can handle higher bit depth (32 bits/sample/component) data than any MI encoder. There are no MI encoders with these capabilities. For scientific MI data applications, using JPEG 2000 is a viable alternative to MI encoders as it may better preserve the MI data at the expense of increased file size.

5.4.3.3.5.2.2 Consider IMODE T interleave in TABLE 3(E), each spatiotemporal block is encoded as a temporal sequence of still imagery block codestreams. The bands/components of each sub frame are included in a single codestream and the codestreams appear in the file as a contiguous sequence of sub frames, ordered by frame number, one after another, within the spatiotemporal block. IMODEs D and F are used in cases where each band/component is handled in its own separate codestream.

5.4.3.3.5.2.3 IMODE X may be used with the JPEG 2000 image encoder and MI data, see TABLE 3(E). When IMODE X is used with JPEG 2000 image compression, the MI frames are treated in the same manner as they are for still imagery. Each frame is independently encoded in its own JPEG 2000 codestream. The internal tiling mechanism for JPEG 2000 is used to perform spatial blocking (see ISO/IEC BIIF Profile BPJ2K01.10 for details). Each JPEG 2000 encoded MI frame is concatenated one after another within the image segment. Block mask records may be used with IMODE X to provide random access to each MI frame. Note this is the only case where block mask offsets may be used to provide random access to frames.

5.4.3.3.5.2.4 It is possible to use IMODE Z and JPEG 2000 to individually encode the bands in an MI frame. This would be done in a manner similar to using IMODE S when encoding a still image. The independently encoded bands from each frame would be ordered sequentially by band and each set of bands for the frames temporally sequenced within an image segment. At this time, however, IMODE Z is not allowed when compressing MI frames. It may only be used for uncompressed MI data.

5.4.3.4 Grey Scale Look-Up Tables (LUT). The grey scale to be used in displaying each pixel of a grey scale image is determined using the image's LUT, if present. A LUT for a grey scale image when present, shall comprise a one byte entry for each integer (the entry's index) in the range 0 to NELUTn-1 (Number of LUT Entries for the nth Image Band field). The bytes of the LUT shall appear in the file one after the other without separation. The entries shall occur in the index order, the first entry corresponding to index 0, the second to index 1 and so on, the last corresponding to index NELUTn-1. The display shade for a pixel in the image shall be determined by using the image pixel value as an index into the LUT. The LUT value shall correspond to the display grey shade in a way specific to the display device. NELUTn shall be equal to or greater than the maximum pixel value in the image to ensure that all image pixels are mapped to the display device.

5.4.3.5 Color Look-Up Tables (LUT). Color images are represented using the RGB color system notation. For color images, each LUT entry shall be composed of the output color components red, green, and blue, appearing in the file in

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that order. There shall be a LUT entry for each pixel value in a particular band of an NITF image (the entries index of the LUT will range from 0 to NELUTn-1). The LUT entries shall appear in the file in increasing index order beginning with index 0. The display color of an image pixel shall be determined by using the pixel value as an index into each LUT (red, green, blue). The corresponding values for red, green, and blue shall determine the displayed color in a manner specific to the display device. The color component values may be any of the 256 pixel values associated with the band. The presence of color LUTs is optional for 24 bit per pixel (true color) images. Pseudocolor (e.g. 8-bit per pixel color images) shall contain a LUT to correlate each pixel value with a designated true color value.

5.5 Graphic Data. Graphic data is used in the NITF to store two-dimensional information represented as a CGM. Each GS consists of a graphic subheader and data fields. A graphic may be black and white, grey scale, or color. Examples of graphics are circles, ellipses, rectangles, arrows, lines, triangles, logos, unit designators, object designators (ships, aircraft), text, and special characters. A graphic is stored as a distinct unit in the NITF file allowing it to be manipulated and displayed nondestructively relative to the images, and other graphics in the file. This standard does not preclude the use of n-dimensional graphics when future standards are developed.

5.5.1 Graphic Subheader. The graphic subheader is used to identify and supply the information necessary to display the graphic data as intended by the file builder. The format for a graphic subheader is detailed in TABLE 4.

TABLE 4. NITF Graphic Subheader.

(TYPE "R" = Required, "C" = Conditional, "< >" = BCS spaces allowed for entire field)
("†" annotations are explained at the end of the table)

FIELD	NAME/DESCRIPTION	SIZE	VALUERANGE	TYPE
SY	File Part Type. This field shall contain the characters SY to identify the subheader as a graphic subheader.	2	BCS-A SY	R
SID	Graphic Identifier. This field shall contain a valid alphanumeric identification code associated with the graphic. The valid codes are determined by the application.	10	BCS-A User defined	R
SNAME	Graphic name. This field shall contain an alphanumeric name for the graphic.	20	ECS-A (Default is ECS spaces (0x20))	<R>
SSCLAS	Graphic Security Classification. This field shall contain a valid value representing the classification level of the graphic. Valid values are T (=Top Secret), S (=Secret), C (=Confidential), R (=Restricted), U (=Unclassified).	1	ECS-A T, S, C, R, or U	R
NOTE: If SSCLAS is T, S, C, or R, then SSCLSY must be populated with a valid code for the security classification system used.				

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Table 4. NITF graphic subheader. – Continued

FIELD	NAME/DESCRIPTION	SIZE	VALUERANGE	TYPE
SSCLSY	Graphic Security Classification System. This field shall contain valid values indicating the national or multinational security system using a geopolitical (country) code digraph to classify the graphic. For US producers a geopolitical code (Country Code) as specified by Paragraph 5.1.7e shall be used to indicate their national security systems. The designator "XN" is for classified data generated by a component using NATO security system marking guidance. The designator "XF" indicates the de facto classification system policy represented by the security marking indicator NATO/ISAF". These codes were selected to not duplicate codes within the GEC codespace. If this field is populated with all ECS spaces (0x20), it shall imply that no security classification system applies to the graphic.	2	ECS-A (Default is ECS spaces (0x20)) The most up to date list of registered digraph geopolitical code values can be found on the NITF Field Value Registry, specified in section C.2.1 Online Resources.	<R>
NOTE: If any of the following fields are populated with anything other than spaces, then SSCLSY must be populated with a valid code for the security classification system used: SSCODE, SSCTLH, SSREL, SSDCTP, SSDCDT, SSDCXM, SSDG, SSDGDT, SSCLTX, SSCATP, SSCAUT, SSCRSN, SSSRDT, and SSCTLN.				
SSCODE	Graphic Codewords. This field shall contain a word(s) that has been assigned a classification and a classified meaning to safeguard intentions and information regarding a specific classified plan or operation. This codeword may be a cryptonym used to identify sensitive intelligence data. Multiple entries shall be separated by a single ECS spaces (0x20). If this field is populated with all ECS spaces (0x20), it shall imply that no codewords apply to the graphic. Past codeword examples are located in the NITF registry, specified in section C.2.1 Online Resources.	11	ECS-A (Default is ECS spaces (0x20))	<R>
SSCTLH	Graphic Control and Handling. This field shall contain valid additional security control and/or handling instructions (caveats) associated with the graphic. Control and Handling is an indicator governing the distribution and use of information or material. Such indicators are the subject of intelligence community agreement and are specifically defined in appropriate regulations. Examples values include digraphs found in the NITF Field Value Registry, specified in section C.2.1 Online Resources. The digraph may indicate single or multiple caveats. The selection of a relevant caveat(s) is application specific. If this field is populated with all ECS spaces (0x20), it shall imply that no control and handling instructions apply to the graphic.	2	ECS-A (Default is ECS spaces (0x20))	<R>

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Table 4. NITF graphic subheader. – Continued

FIELD	NAME/DESCRIPTION	SIZE	VALUERANGE	TYPE
SSREL	Graphic Releasing Instructions. This field shall contain a valid list of bare country and/or multilateral entity codes to which countries and/or multilateral entities the graphic is authorized for release. Valid entities in the list are one or more bare country codes as found in Paragraph 5.1.7e. If this field is populated with all ECS spaces (0x20), it shall imply that no graphic release instructions apply.	20	ECS-A (Default is ECS spaces (0x20))	<R>
SSDCTP	Graphic Declassification Type. This field shall contain a valid indicator of the type of security declassification or downgrading instructions which apply to the graphic. Valid values are DD (=declassify on a specific date), DE (=declassify upon occurrence of an event), GD (=downgrade to a specified level on a specific date), GE (=downgrade to a specified level upon occurrence of an event), O (=OADR), and X (= exempt from automatic declassification). If this field is populated with all ECS spaces (0x20), it shall imply that no graphic security declassification or downgrading instructions apply.	2	ECS-A DD, DE, GD, GE, O, X (Default is ECS spaces (0x20))	<R>
SSDCDT	Graphic Declassification Date. This field shall indicate the date on which a graphic is to be declassified if the value in the Graphic Declassification Type field is populated with DD. If this field is populated with all ECS spaces (0x20), it shall imply that no graphic declassification date applies.	8	ECS-A CCYYMMDD (Default is BCS spaces (0x20))	<R>
SSDCXM	Graphic Declassification Exemption. This field shall indicate the reason the graphic is exempt from automatic declassification if the value in Graphic Declassification Type is X. Valid values are X1 to X8 and 25X1 to 25X9. Values X1 to X8 correspond to the declassification exemptions found in DOD 5200.01-V1, paragraphs 4-202b(1) to (8) for material exempt from the 10-year rule. Values 25X1 to 25X9 correspond to the declassification exemptions found in DOD 5200.01-V1, paragraphs 4-301a(1) to (9) for permanently valuable material exempt from the 25-year declassification system. If this field is populated with all ECS spaces (0x20), it shall imply that a graphic declassification exemption does not apply. Value DNIO is Exempt from automatic declassification and can be declassified only by the Director of National Intelligence (DNI).	4	ECS-A X1 to X8, 25X1 to 25X9, DNIO, (Default is ECS spaces (0x20))	<R>
SSDG	Graphic Downgrade. This field shall indicate the classification level to which a graphic is to be downgraded if the values in Graphic Declassification Type are GD or GE. Valid values are S (=Secret), C (=Confidential), R (= Restricted). If this field contains an ECS space (0x20), it shall imply that graphic security downgrading does not apply.	1	ECS-A S, C, R Default is ECS space (0x20))	<R>

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Table 4. NITF graphic subheader. – Continued

FIELD	NAME/DESCRIPTION	SIZE	VALUERANGE	TYPE
SSDGDT	Graphic Downgrade Date. This field shall indicate the date on which a graphic is to be downgraded if the value in Graphic Declassification Type is GD. If this field is populated with all ECS spaces (0x20), it shall imply that a graphic security downgrading date does not apply.	8	ECS-A CCYYMMDD (Default is ECS spaces (0x20))	<R>
SSCLTX	Graphic Classification Text. This field shall be used to provide additional information about graphic classification to include identification of a declassification or downgrading event if the values in Graphic Declassification Type are DE or GE. It may also be used to identify multiple classification sources and/or any other special handling rules. Values are user defined free text. If this field is populated with all ECS spaces (0x20), it shall imply that additional information about graphic classification does not apply. This field may also contain overflow values from the SSCTLH or SSREL fields. Consult with the NITF Field Value Registry for overflow population guidance, specified in section C.2.1. Online Resources.	43	ECS-A User-defined free text (Default is ECS spaces (0x20))	<R>
SSCATP	Graphic Classification Authority Type. This field shall indicate the type of authority used to classify the graphic. Valid values are O (= original classification authority), D (= derivative from a single source), and M (= derivative from multiple sources). If this field contains an ECS space (0x20), it shall imply that graphic classification authority type does not apply.	1	ECS-A O, D, M (Default is ECS space (0x20))	<R>
SSCAUT	Graphic Classification Authority. This field shall identify the classification authority for the graphic dependent upon the value in Graphic Classification Authority Type. Values are user defined free text which should contain the following information: original classification authority name and position or personal identifier if the value in Graphic Classification Authority Type is O; title of the document or security classification guide used to classify the graphic if the value in Graphic Classification Authority Type is D; and Derive-Multiple if the graphic classification was derived from multiple sources. In the latter case, the graphic originator will maintain a record of the sources used in accordance with existing security directives. One of the multiple sources may also be identified in Graphic Classification Text if desired. If this field is populated with all ECS spaces (0x20), it shall imply that no graphic classification authority applies.	40	ECS-A User-defined free text (Default is ECS spaces (0x20))	<R>
SSCRSN	Graphic Classification Reason. This field shall contain values indicating the reason for classifying the graphic. Valid values are A to H. These correspond to the reasons for original classification per E.O. 13526, Section 1.5.(a) to (h). If this field contains an ECS space (0x20), it shall imply that no graphic classification reason applies.	1	ECS-A A to H (Default is ECS space (0x20))	<R>

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Table 4. NITF graphic subheader. – Continued

FIELD	NAME/DESCRIPTION	SIZE	VALUERANGE	TYPE
SSSRDT	Graphic Security Source Date. This field shall indicate the date of the source used to derive the classification of the graphic. In the case of multiple sources, the date of the most recent source shall be used. If this field is populated with all ECS spaces (0x20), it shall imply that a graphic security source date does not apply.	8	ECS-A CCYYMMDD (Default is ECS spaces (0x20))	<R>
SSCTLN	Graphic Security Control Number. This field shall contain a valid security control number associated with the graphic. The format of the security control number shall be in accordance with the regulations governing the appropriate security channel(s). If this field is populated with all ECS spaces (0x20), it shall imply that no graphic security control number applies.	15	ECS-A (Default is ECS spaces (0x20))	<R>
ENCRYP	Encryption. This field shall contain the value BCS zero (0x30) until such time as this specification is updated to define the use of other values.	1	BCS-N positive integer 0=Not Encrypted	R
SFMT	Graphic Type. This field shall contain a valid indicator of the representation type of the graphic. The valid value is C, which represents Computer Graphics Metafile. The graphic data contain a Computer Graphics Metafile in binary format that defines the graphic according to BPCGM01.00. Future versions of the NITF may include additional CGM profiles.	1	BCS-A C for CGM	R
SSTRUCT	Reserved for Future Use. Reserved.	13	BCS-N positive integer 0000000000000 to 9999999999999 (Default is BCS zeros (0x30))	R
SDLVL	Graphic Display Level. This field shall contain a valid value that indicates the graphic display level of the graphic relative to other displayed file components in a composite display. The valid values are 001 to 999. The display level of each displayable file component (image or graphic) within a file shall be unique; that is, each number from 001 to 999 is the display level of, at most, one item. The meaning of display level is discussed in paragraph 5.3.3. The graphic or image component in the file having the minimum display level shall have attachment level 0 (ALVL000) (BCS zeros (0x30)).	3	BCS-N positive integer 001 to 999	R
SALVL	Graphic Attachment Level. This field shall contain a valid value that indicates the attachment level of the graphic. Valid values for this field are 0 and the display level value of any other image or graphic in the file. The meaning of attachment level is discussed in paragraph 5.3.4. The graphic or image component in the file having the minimum display level shall have attachment level 0 (ALVL000) (BCS zeros (0x30)).	3	BCS-N positive integer 000 to 998 (Default is BCS zeros (0x30))	R

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Table 4. NITF graphic subheader. – Continued

FIELD	NAME/DESCRIPTION	SIZE	VALUERANGE	TYPE
SLOC	Graphic Location. The graphics location is specified by providing the location of the graphic's origin point relative to the position (location of the CCS, image, or graphic to which it is attached. This field shall contain the graphic location offset from the ILOC or SLOC value of the CCS, image, or graphic to which the graphic is attached or from the origin of the CCS when the graphic is unattached (SALVL000). A row and column value of 000 indicates no offset. Positive row and column values indicate offsets down and to the right, while negative row and column values indicate offsets up and to the left.	10	BCS-N RRRRRCCCCC For positive row and column values RRRRR and CCCCC are both in the range 00000 to 99999. For negative row and column values RRRRR and CCCCC are both in the range -0001 to -9999.	R
SBND1	First Graphic Bound Location. This field shall contain an ordered pair of integers defining a location in Cartesian coordinates for use with CGM graphics. It is the upper left corner of the bounding box for the CGM graphic. See paragraph 5.5.2.1 for a description. The format is rrrrrccccc, where rrrrr is the row and ccccc is the column offset from ILOC or SLOC value of the item to which the graphic is attached. If the graphic is unattached (value of the SALVL field is populated with BCS zeros (0x30)), rrrrr and ccccc represent offsets from the origin of the coordinate system that is common to all images and graphics in the file having the value of BCS zeros (0x30) in the SALVL field. The range for rrrrr and ccccc shall be -9999 to 99999.	10	BCS-N rrrrrccccc with -9999≤rrrrr≤99999 -9999≤ccccc≤99999 (Default is BCS zeros (0x30))	R
SCOLOR	Graphic Color. If SFMT = C, this field shall contain a C if the CGM contains any color pieces or an M if it is monochrome (i.e., black, white, or levels of grey).	1	BCS-A C, M	R
SBND2	Second Graphic Bound Location. This field shall contain an ordered pair of integers defining a location in Cartesian coordinates for use with CGM graphics. It is the lower right corner of the bounding box for the CGM graphic. See paragraph 5.5.2.1 for a description. The format is rrrrrccccc, where rrrrr is the row and ccccc is the column offset from ILOC or SLOC value of the item to which the graphic is attached. If the graphic is unattached (SALVL field value is BCS zeros(0x30)), rrrrr and ccccc represent offsets from the origin of the coordinate system that is common to all images and graphics in the file having the value of BCS zeros (0x30) in the SALVL field. The range for rrrrr and ccccc shall be -9999 to 99999.	10	BCS-N rrrrrccccc with -9999≤rrrrr≤99999 -9999≤ccccc≤99999 (Default is BCS zeros (0x30))	R
SRES2	Reserved for Future Use. This field is reserved for future use. The default value shall be BCS zeros (0x30).	2	BCS-N positive integer 00 to 99 (Default is BCS zeros (0x30))	R

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Table 4. NITF graphic subheader. – Continued

FIELD	NAME/DESCRIPTION	SIZE	VALUERANGE	TYPE
SXSHDL	Graphic Extended Subheader Data Length. A value of BCS zero (0x30) shall represent that no TRE are included in the graphic subheader. If a tagged record extension exists, the field shall contain the sum of the length of all the TRE (paragraph 5.8.1) appearing in the SXSHD field plus 3 bytes (length of SXSOFL field). If a tagged record extension is too long to fit in the SXSHD field, it shall be put in the TRE overflow DES with DESID set to the value TRE_OVERFLOW (paragraph 5.8.3.1).	5	BCS-N positive integer 00000 or 00003 to 09741 (Default is BCS zeros (0x30))	R
SXSOFL	Graphic Extended Subheader Overflow. If present, this field shall contain BCS zeros (0x30) if the TRE in SXSHD do not overflow into a DES or shall contain the sequence number of the DES into which they do overflow. This field shall be omitted if the field SXSHDL contains BCS zeros (0x30).	3	BCS-N positive integer 000 to 999	C
SXSHD	Graphic Extended Subheader Data. If present, this field shall contain TRE (paragraph 5.8.1) approved and under configuration management by the GWG. The length of this field shall be the length specified by the field SXSHDL minus 3 bytes. TREs in this field for a graphic shall contain information pertaining specifically to the graphic. TRE shall appear one after the other in this field with no intervening bytes. The first byte of this field shall be the first byte of the first TRE appearing in the field. The last byte of this field shall be the last byte of the last TRE to appear in the field. This field shall be omitted if the field SXSHDL contains BCS zeros (0x30).	†[5]	TRE(s)	C

†[5] A value as specified by the SHSHDL field minus 3 (in bytes)

5.5.2 Graphic Data Format. The graphic format is CGM as described in ISO/IEC 8632-1. The precise tailoring of the CGM standard to NITF is found in BPCGM01.00.

5.5.2.1 CGM Graphic Bounding Box. CGM graphic placement is defined by the SLOC field and the CGM graphic extent is given by the SBND1 (graphic bound 1) and SBND2 (graphic bound 2) fields. SLOC defines the origin for the CGM coordinate system. The area covered by the CGM graphic is defined by a bounding box. The bounding box is the smallest rectangle that could be placed around the entire CGM graphic. The first bounding box coordinate (SBND1) is the upper left corner of the rectangle. The second bounding box coordinate (SBND2) is the lower right corner of the rectangle. SBND1 and SBND2 are values in the coordinate system defined by the attachment level. For attachment level 0, this would be the CCS. The SBND1 and SBND2 values are calculated by adding SLOC to the coordinate values for the bounding box (upper left and lower right) corners as given in the CGM graphic coordinate system.

5.6 Reserved Segment (RS). The RS are place holders to support the expansion of the NUMX field within the NITF file header for a future standard data type that has yet to be defined.

5.7 Text Data. Text data shall be used to store textual data or unformatted text. Text is intended to convey information about an associated segment in the NITF file.

5.7.1 Representation of Textual Information. The Text Format (TXTFMT) field contains a three character code that indicates the type or format of text data contained in the TS. The allowable field values are STA, MTF, UT1, or U8S.

5.7.1.1 Standard (STA). STA designates BCS character codes in a simple format. Any BCS code may be used in the text data segment when STA is indicated in the TXTFMT field. All lines within a text data segment shall be separated

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by carriage return/line feed pairs. A carriage return followed by a line feed shall be used to delimit lines in the text where the first character from the next line immediately follows the line feed character.

5.7.1.2 Message Text Formatting (MTF). MTF indicates that the text data segment contains BCS-A characters formatted according to MIL-STD-6040.

5.7.1.3 ECS Text Formatting (UT1). This is a legacy formatting that is replaced by the U8S text formatting (U8S). UT1 text formatting uses ECS character codes. Any ECS code may be used in the Text Data Segment when UT1 is indicated in the TXTFMT field. All lines within the TS shall be separated by Carriage Return/Line Feed pairs. A Carriage Return followed by a Line Feed shall be used to delimit lines in the text where the first character from the next line immediately follows the Line Feed character.

5.7.1.4 U8S Text Formatting (U8S). The U8S text formatting replaces the legacy ECS text formatting (UT1). U8S text formatting uses U8S character codes. Any U8S character (either 1-byte or 2-byte encoded) may be used in the Text Data Segment when U8S is indicated in the TXTFMT field. All lines within Text Data Segment shall be separated by Carriage Return/Line Feed pairs. A Carriage Return followed by a Line Feed shall be used to delimit lines in the text where the first character from the next line immediately follows the Line Feed character.

5.7.2 Text Subheader. The text subheader is used to identify and supply the information necessary to read and display the text within the data field. The text subheader is detailed in TABLE 5.

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TABLE 5. NITF Text Subheader.

(TYPE "R" = Required, "C" = Conditional, "< >" = BCS spaces allowed for entire field)
("†" annotations are explained at the end of the table)

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
TE	File Part Type. This field shall contain the characters "TE" to identify the subheader as a text subheader.	2	BCS-A TE	R
TEXTID	Text Identifier. This field shall contain a valid alphanumeric identification code associated with the text item. The valid codes are determined by the application.	7	BCS-A User defined	R
TXTALVL	Text Attachment Level. This field shall contain a valid value that indicates the attachment level of the text. Valid values for this field are 000 (BCS zeros (0x30)) or the display level value of any image or graphic in the file.	3	BCS-N positive integer 000 to 998 (Default is BCS zeros (0x30))	R
TXTDT	Text Date and Time. This field shall contain the time (UTC) (Zulu) of origination of the text in the format CCYYMMDDhhmmss, where CC is the century (00 to 99), YY is the last two digits of the year (00 to 99), MM is the month (01 to 12), DD is the day (01 to 31), hh is the hour (00 to 23), mm is the minute (00 to 59), and ss is the second (00 to 59). UTC (Zulu) is assumed to be the time zone designator to express the time of day. Refer to Paragraph 53.1.7d when a portion of the date and/or time is unknown.	14	BCS-N CCYYMMDDhhmmss	R
TXTITL	Text Title. This field shall contain the title of the text item.	80	ECS-A (Default is ECS spaces (0x20))	<R>
TSCLAS	Text Security Classification. This field shall contain a valid value representing the classification level of the text. Valid values are T (=Top Secret), S (=Secret), C (=Confidential), R (=Restricted), U (=Unclassified).	1	ECS-A T, S, C, R, or U	R
NOTE: If TSCLAS is T, S, C, or R, then TSCLSY must be populated with a valid code for the security classification system used.				
TSCLSY	Text Security Classification System. This field shall contain valid values indicating the national or multinational security system using a geopolitical (country) code digraph to classify the text. For US producers a geopolitical code (Country Code), as specified by Paragraph 5.1.7e, shall be used to indicate their national security systems. The designator "XN" is for classified data generated by a component using NATO security system classification policy marking guidance. The designator "XF" indicates the de facto classification system policy represented by the security marking indicator NATO/ISAF". These codes were selected to not duplicate codes within the GEC codespace. If this field is populated with all ECS spaces (0x20), it shall imply that no security classification system applies to the file.	2	ECS-A (Default is ECS spaces (0x20)) The most up to date list of registered digraph geopolitical code values can be found on the NITF Field Value Registry, specified in section C.2.1 Online Resources.	<R>

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TABLE 5. NITF Text Subheader – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
NOTE: If any of the following fields are populated with anything other than spaces, then TSCLSY must be populated with a valid code for the security classification system used: TSCODE, TSCTLH, TSREL, TSDCTP, TSDCDT, TSDCXM, TSDG, TSDGDT, TSCLTX, TSCATP, TSCAUT, TSCRSN, TSSRDT, and TSCTLN.				
TSCODE	Text Codewords. This field shall contain a word(s) that has been assigned a classification and a classified meaning to safeguard intentions and information regarding a specific classified plan or operation. This codeword may be a cryptonym used to identify sensitive intelligence data. Multiple entries shall be separated by a single ECS spaces (0x20). If this field is populated with all ECS spaces (0x20), it shall imply that no codewords apply to the text. Past codeword examples are located in the NITF registry, specified in section C.2.1 Online Resources.	11	ECS-A (Default is ECS spaces 0x20))	<R>
TSCTLH	Text Control and Handling. This field shall contain valid additional security control and/or handling instructions (caveats) associated with the text. Control and Handling is an indicator governing the distribution and use of information or material. Such indicators are the subject of intelligence community agreement and are specifically defined in appropriate regulations. Example values include digraphs found in the NITF Field Value Registry, specified in section C.2.1 Online Resources. The digraph may indicate single or multiple caveats. The selection of a relevant caveat(s) is application specific. If this field is populated with all ECS spaces (0x20), it shall imply that no control and handling instructions apply to the text.	2	ECS-A (Default is ECS spaces (0x20))	<R>
TSREL	Text Releasing Instructions. This field shall contain a valid list of bare country and/or multilateral entity codes to which countries and/or multilateral entities the text is authorized for release. Valid entities in the list are one or more bare country codes as specified by Paragraph 5.1.7e, separated by a single ECS space (0x20). If this field is populated with all ECS spaces (0x20), it shall imply that no text release instructions apply.	20	ECS-A (Default is ECS spaces (0x20))	<R>
TSDCTP	Text Declassification Type. This field shall contain a valid indicator of the type of security declassification or downgrading instructions which apply to the text. Valid values are DD (=declassify on a specific date), DE (=declassify upon occurrence of an event), GD (=downgrade to a specified level on a specific date), GE (=downgrade to a specified level upon occurrence of an event), O (=OADR), and X (= exempt from automatic declassification). If this field is populated with all ECS spaces (0x20), it shall imply that no text security declassification or downgrading instructions apply.	2	ECS-A DD, DE, GD, GE, O, X (Default is ECS spaces (0x20))	<R>
TSDCDT	Text Declassification Date. This field shall indicate the date on which a text is to be declassified if the value in the Text Declassification Type field is populated with DD. If this field is populated with all ECS spaces (0x20), it shall imply that no text declassification date	8	ECS-A CCYYMMDD (Default is ECS spaces (0x20))	<R>

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TABLE 5. NITF Text Subheader – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
	applies.			
TSDCXM	Text Declassification Exemption. This field shall indicate the reason the text is exempt from automatic declassification if the value in Text Declassification Type is X. Valid values are X1 to X8 and 25X1 to 25X9. Values X1 to X8 correspond to the declassification exemptions found in DOD 5200.01-V1, paragraphs 4-202b(1) to (8) for material exempt from the 10-year rule. Values 25X1 to 25X9 correspond to the declassification exemptions found in DOD 5200.01-V1, paragraphs 4-301a(1) to (9) for permanently valuable material exempt from the 25-year declassification system. If this field is populated with all ECS spaces (0x20), it shall imply that a text declassification exemption does not apply. Value DNIO is Exempt from automatic declassification and can be declassified only by the Director of National Intelligence (DNI).	4	ECS-A X1 to X8, 25X1 to 25X9, DNIO, (Default is ECS spaces (0x20))	<R>
TSDG	Text Downgrade. This field shall indicate the classification level to which a text is to be downgraded if the values in Text Declassification Type are GD or GE. Valid values are S (=Secret), C (=Confidential), R (= Restricted). If this field contains an ECS space (0x20), it shall imply that text security downgrading does not apply.	1	ECS-A S, C, R (Default is ECS space (0x20))	<R>
TSDGDT	Text Downgrade Date. This field shall indicate the date on which a text is to be downgraded if the value in Text Declassification Type is GD. If this field is populated with all ECS spaces (0x20), it shall imply that a text security downgrading date does not apply.	8	ECS-A CCYYMMDD (Default is ECS spaces (0x20))	<R>
TSCLTX	Text Classification Text. This field shall be used to provide additional information about text classification to include identification of a declassification or downgrading event if the values in Text Declassification Type are DE or GE. It may also be used to identify multiple classification sources and/or any other special handling rules. Values are user defined free text. If this field is populated with all ECS spaces (0x20), it shall imply that additional information about text classification does not apply. This field may also contain overflow values from the TSCTLH or TSREL fields. Consult with the NITF Field Value Registry for overflow population guidance, specified in section C.2.1. Online Resources.	43	ECS-A User-defined free text (Default is ECS spaces (0x20))	<R>
TSCATP	Text Classification Authority Type. This field shall indicate the type of authority used to classify the text. Valid values are O (= original classification authority), D (= derivative from a single source), and M (= derivative from multiple sources). If this field contains an ECS space (0x20), it shall imply that text classification authority type does not apply.	1	ECS-A O, D, M (Default is ECS space (0x20))	<R>

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TABLE 5. NITF Text Subheader – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
TSCAUT	Text Classification Authority. This field shall identify the classification authority for the text dependent upon the value in Text Classification Authority Type. Values are user defined free text which should contain the following information: original classification authority name and position or personal identifier if the value in Text Classification Authority Type is O; title of the document or security classification guide used to classify the text if the value in Text Classification Authority Type is D; and Derive-Multiple if the text classification was derived from multiple sources. In the latter case, the text originator will maintain a record of the sources used in accordance with existing security directives. One of the multiple sources may also be identified in Text Classification Text if desired. If this field is populated with all ECS spaces (0x20), it shall imply that no text classification authority applies.	40	ECS-A User-defined free text (Default is ECS spaces (0x20))	<R>
TSCRSN	Text Classification Reason. This field shall contain values indicating the reason for classifying the text. Valid values are A to H. These correspond to the reasons for original classification per E.O. 13526, Section 1.5.(a) to (h). If this field contains an ECS space (0x20), it shall imply that no text classification reason applies.	1	ECS-A A to H (Default is ECS space (0x20))	<R>
TSSRDT	Text Security Source Date. This field shall indicate the date of the source used to derive the classification of the text. In the case of multiple sources, the date of the most recent source shall be used. If this field is populated with all ECS spaces (0x20), it shall imply that a text security source date does not apply.	8	ECS-A CCYYMMDD (Default is ECS spaces (0x20))	<R>
TSCTLN	Text Security Control Number. This field shall contain a valid security control number associated with the text. The format of the security control number shall be in accordance with the regulations governing the appropriate security channel(s). If this field is populated with all ECS spaces (0x20), it shall imply that no text security control number applies.	15	ECS-A (Default is ECS spaces (0x20))	<R>
ENCRYP	Encryption. This field shall contain the value BCS zero (0x30) until such time as this specification is updated to define the use of other values.	1	BCS-N positive integer (Default is BCS zero (0x30)) 0=Not Encrypted	R
TXTFMT	Text Format. This field shall contain a valid three-character code indicating the format or type of text data. Valid codes are MTF to indicate USMTF (Refer to MIL-STD-6040 for examples of the USMTF format), STA to indicate BCS, UT1 to indicate ECS text formatting, and U8S to indicate U8S text formatting. Refer to paragraph 5.7.1 for additional discussion of standards and the BCS.	3	BCS-A MTF, STA, UT1, U8S	R
TXSHDL	Text Extended Subheader Data Length. This field shall	5	BCS-N positive	R

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TABLE 5. NITF Text Subheader – Continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
	be populated with all BCS zeros (0x30) if no TRE are included in the text subheader. If a TRE exists, the field shall contain the sum of the length of all the TRE (paragraph 5.8.1) appearing in the TSXHD field plus 3 bytes (length of TSXOFL field). If a TRE is too long to fit in the TXSHD field, it shall be put in the TRE overflow DES with DESID set to the value TRE_OVERFLOW (paragraph 5.8.3.1).		integer 00000 or 00003 to 09717 (Default is BCS zero (0x30))	
TXSOFL	Text Extended Subheader Overflow. If present, this field shall contain BCS zeros (0x30) if TRE in TXSHD do not overflow into a DES, or shall contain the sequence number in the file of the DES into which they do overflow. This field shall be omitted if the field TXSHDL contains BCS zeros (0x30).	3	BCS-N positive integer 000 to 999	C
TXSHD	Text Extended Subheader Data. If present, this field shall contain TRE (paragraph 5.8.1) approved and under configuration management by the GWG NTB. The length of this field shall be the length specified by the field TXSHDL minus 3. TRE in this field shall contain information pertaining specifically to the text. TRE shall appear one after the other in this field with no intervening bytes. The first byte of this field shall be the first byte of the first TRE appearing in the field. The last byte of this field shall be the last byte of the last TRE to appear in the field. This field shall be omitted if the field TXSHDL contains BCS zeros (0x30).	†[6]	BCS-A	C

†[6] A value as specified by the value in the TXSHDL field minus 3 (in bytes).

5.8 Data Extensions. Data extensions are provided to extend NITF functionality with minimal impact on the underlying standard document. There are three types of Support data Extensions (SDE): TRE, DES, and RES. All these extensions may be incorporated into the NITF file while maintaining backward compatibility. The data extension identifier and byte count mechanisms allow applications developed prior to the addition of newly defined data, to skip over extension fields that they are not designed to interpret. Currently, most of the unclassified TREs and DESs can be found in STDI-0002 Volumes 1 and 2 maintained on the NSG Standards Registry: https://nsgreg.nga.mil/. All data extensions shall be controlled or registered (see 5.8.1.1 and 5.8.1.2 below) with the NGA NITF Technical Board (NTB). Any data extension not controlled or registered with the NTB is not, and shall not be, valid. The RES structure is designated for future use and the NTB currently does not maintain a registry for this extension type. For additional information on classified or extensions under development or maintained by other agencies contact the NTB or the JITC NITFS Compliance Test Facility (email the NTB Chair at NTBChair@nga.mil or see: http://jitc.fhu.disa.mil/projects/nitf/index.aspx).

5.8.1 Support Data Extension (SDE).An SDE is either a TRE, a DES, or a RES. An allowed and approved SDE shall be either Controlled or Registered by the NITF Technical Board (NTB)

5.8.1.1 Controlled Extension (CE). A Controlled SDE allows additional data constructs within an NITF file with NTB consensus. For a CE, both the Unique Extension Type Identifier (CETAG CETAG for TREs, DESID or DESs, and RESID for RESs) and the specification contained in the User-Defined Data (CEDATA CEDATA for TREs, DESDATA for DESs, and RESDATA for RESs) field are subject to full NTB registration and configuration control. Upon receipt of an NITF file that contains a CE, an NITF compliant implementation that is not designed to interpret that CE shall ignore it and properly interpret the other NITF file components.

5.8.1.2 Registered Extension (RE). A Registered SDE allows NITF users to establish user defined data constructs within an NITF file without NTB consensus. RE use is considered private in the sense that a specific RE is

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meaningful only to NITF users who have agreed to its use. The structure and content of the User-Defined Data (REDATA) field does not need to be configuration managed. However to prevent duplication, each newly defined Unique Extension Type Identifier (RETAG) shall be registered, along with its name and purpose, with the NTB. Upon receipt of an NITF file that contains an RE, an NITF compliant implementation that is not designed to interpret that RE shall ignore it and properly interpret the other NITF file components.

5.8.2 Tagged Record Extension (TRE). A TRE is a collection of data fields that provides space within the NITF file structure for adding, as yet unspecified, future capabilities to the standard. The TRE is used to extend NITF by adding additional attributes to designated fields in the NITF file header (TABLE 1) and in the image, text, and graphic subheaders (tables 3, 4, and 5). Each TRE consists of three required fields that are defined in TABLE 6. There are two similar, but different, TRE types: Controlled Extensions (CE) and Registered Extensions (RE). The principles are described below and illustrated on FIGURE 14.

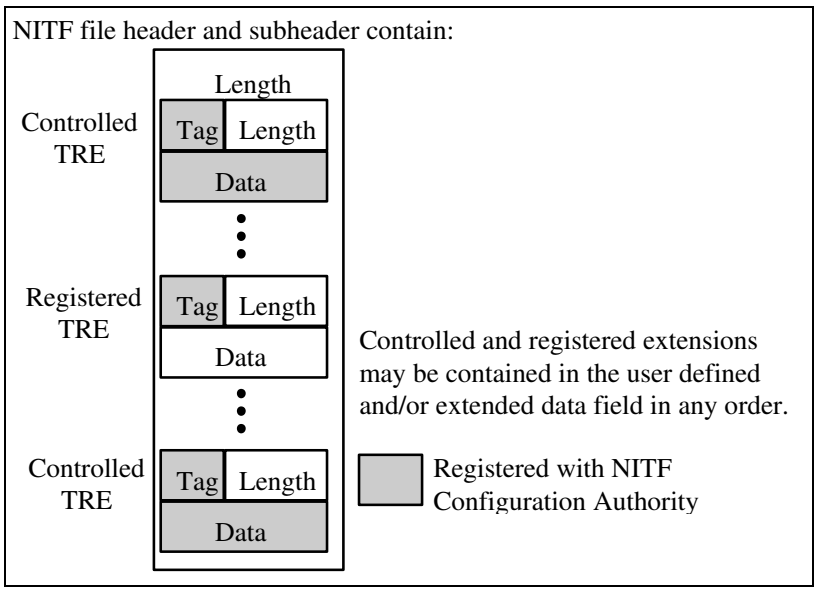


FIGURE 14. Tagged Record Extension (TRE).

5.8.2.1 TRE Placement. A sequence of TREs can be used in the NITF file header User Defined Data (UDHD) field, in any image subheader's User Defined Image Data (UDID) field, in Extended Header and Extended Subheader (XHD, IXSHD, SXSHD, TXSHD) fields, and in a DES that is designated to contain TRE Overflow (TRE_OVERFLOW). When the TRE carries data associated with the NITF file and sufficient room is available, it should appear in the NITF file header. If the TRE carries data associated with a segment and sufficient room is available in the segment's subheader, the data should appear in the segment's subheader. When sufficient room is not available in the NITF file header or the segment's subheader, the TRE may be placed in the TRE_OVERFLOW DES (paragraph 5.8.3.1). The entire TRE shall be included within the NITF file header, subheader, or DES that has been selected to contain it.

5.8.2.2 TRE Registry. A current listing of the TREs that are registered with the NTB is provided in the Data Extension Registry maintained by the JITC.

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TABLE 6. Registered and Controlled Tagged Record Extension Format.

(TYPE "R" = Required, "C" = Conditional, "< >" = BCS spaces allowed for entire field)
("†" annotations are explained at the end of the table)

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
RETAG or CETAG	Unique Extension Type Identifier. This field shall contain a valid alphanumeric identifier properly registered with the GWG NTB.	6	BCS-A	R
REL or CEL	Length of REDATA Field. This field shall contain the length in bytes of the data contained in REDATA or CEDATA. The full TRE length is the size of RETAG/CETAG field (6 bytes) plus REL/CEL field (5 bytes) plus the value (in bytes) of the REL/CEL field.	5	BCS-N positive integer 00001 to 99985	R
REDATA or CEDATA where appropriate	User-Defined Data. This field shall contain data of either binary or character data types defined by and formatted according to user specification. The length of this field shall not cause any other NITF field length limits to be exceeded, but is otherwise fully user-defined.	†7	User-defined	R

†7 A value as indicated in the REL field or the CEL field (in bytes).

5.8.3 Data Extension Segment (DES). The DES structure allows the format to include different data types within an NITF file. Each data type is encapsulated in its own DES. Each DES can carry only one data type, but an NITF file can contain multiple DES. Multiple DES contained in one NITF file can be of the same or different data types. Each encapsulated extension shall appear in its own DES and shall conform to the DES structure contained in TABLE 7. There is one DES type identifiers defined in the MIL-STD, the TRE Overflow (TRE_OVERFLOW). The extension segments are intended primarily to provide a vehicle for adding support for new kinds of data. Other DES data types are identified in the current listing of DES types provided in the DES Registry maintained by the JITC. All NITF implementations should handle the receipt of unknown DES extensions by at least recognizing that they are unknown extension types and ignoring them. This can be accomplished using the byte count, extension identifier, and data length field. Using these length offsets, the unknown extension can be ignored and the user can be informed that extension data has been skipped. Like a TRE, the DES can be a controlled extension subject to full NTB registration and configuration control, or can be a registered extension considered private in the sense that a specific registered DES is meaningful only to NITF users who have agreed to its use. However, to prevent duplication, each newly defined Unique DES Type Identifier must be registered, along with its name and purpose, with the NTB.

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TABLE 7. NITF Data Extension Segment (DES) Subheader.

(TYPE "R" = Required, "C" = Conditional, "< >" = BCS spaces allowed for entire field)
("†" annotations are explained at the end of the table)

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
DE	File Part Type. This field shall contain the characters "DE" to identify the subheader as a data extension.	2	BCS-A DE	R
DESID	Unique DES Type Identifier. This field shall contain a valid alphanumeric identifier properly registered with the GWG NTB.	25	BCS-A Registered value only	R
DESVER	Version of the Data Definition. This field shall contain the alphanumeric version number of the use of the tag. The version number is assigned as part of the registration process.	2	BCS-N positive integer 01 to 99	R
DECLAS	Data Extension File Security Classification. This field shall contain a valid value representing the classification level of the DES. Valid values are T for Top Secret, S for Secret, C for Confidential, R for Restricted, or U for Unclassified.	1	ECS-A T, S, C, R, or U	R
NOTE: If the value of the DECLAS field is populated with T, S, C, or R, then the DESCLSY field must be populated with a valid code for the security classification system used.				
DESCLSY	DES Security Classification System. This field shall contain valid values indicating the national or multinational security system using a geopolitical (country) code digraph to classify the DES. For US producers a geopolitical code (Country Code), as specified by Paragraph 5.1.7e, shall be used to indicate their national security systems. The designator "XN" is for classified data generated by a component using NATO security system classification policy marking guidance. The designator "XF" indicates the de facto classification system policy represented by the security marking indicator NATO/ISAF". These codes were selected to not duplicate codes within the GEC codespace. If this field is populated with all ECS spaces (0x20), it shall imply that no security classification system applies to the DES.	2	ECS-A (Default is ECS spaces (0x20)) The most up to date list of registered digraph geopolitical code values can be found on the NITF Field Value Registry, specified in section C.2.1 Online Resources.	<R>
NOTE: If any of the following fields are populated with anything other than spaces, then the DESCLSY field must be populated with a valid code for the security classification system used: DESCODE, DESCTLH, DESREL, DESDCTP, DESDCDT, DESDCXM, DESDG, DESDGDT, DESCLDES, DESCATP, DESCAUT, DESCRSN, DESSRDT, and DESCTLN.				
DESCODE	DES Codewords. This field shall contain a word(s) that has been assigned a classification and a classified meaning to safeguard intentions and information regarding a specific classified plan or operation. This codeword maybe a cryptonym used to identify sensitive intelligence data. Multiple entries shall be separated by a single ECS space (0x20). If this field is populated with all ECS spaces (0x20), it shall imply that no codewords apply to the DES. Past codeword examples are located in the NITF registry, specified in section C.2.1 Online Resources.	11	ECS-A (Default is ECS spaces (0x20))	<R>

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TABLE 7. NITF Data Extension Segment (DES) Subheader – continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
DESCTLH	DES Control and Handling. This field shall contain valid additional security control and/or handling instructions (caveats) associated with the DES. Control and Handling is an indicator governing the distribution and use of information or material. Such indicators are the subject of intelligence community agreement and are specifically defined in appropriate regulations. Example values include digraphs found in the NITF Field Value Registry, specified in section C.2.1 Online Resources. The digraph may indicate single or multiple caveats. The selection of a relevant caveat(s) is application specific. If this field is populated with all ECS spaces (0x20), it shall imply that no control and handling instructions apply to the DES.	2	ECS-A (Default is ECS spaces (0x20))	<R>
DESREL	DES Releasing Instructions. This field shall contain a valid list of bare country and/or multilateral entity codes to which countries and/or multilateral entities the DES is authorized for release. Valid entities in the list are one or more bare country codes as specified by Paragraph 5.1.7e, separated by a single ECS space (0x20). If this field is populated with all ECS spaces (0x20), it shall imply that no DES release instructions apply.	20	ECS-A (Default is ECS spaces (0x20))	<R>
DESDCTP	DES Declassification Type. This field shall contain a valid indicator of the type of security declassification or downgrading instructions which apply to the DES. Valid values are DD for declassify on a specific date, DE for declassify upon occurrence of an event, GD for downgrade to a specified level on a specific date, GE for downgrade to a specified level upon occurrence of an event, O for OADR, and X for exempt from automatic declassification. If this field is populated with all ECS spaces (0x20), it shall imply that no DES security declassification or downgrading instructions apply.	2	ECS-A DD, DE, GD, GE, O, X (Default is ECS spaces (0x20))	<R>
DESDCDT	DES Declassification Date. This field shall indicate the date on which a DES is to be declassified if the value in the DESDCTP field is populated with DD. If this field is populated with all ECS spaces (0x20), it shall imply that no DES declassification date applies.	8	ECS-A CCYYMMDD (Default is ECS spaces (0x20))	<R>
DESDCXM	DES Declassification Exemption. This field shall indicate the reason the DES is exempt from automatic declassification if the value in DESDCTP is X. Valid values are X1 to X8 and 25X1 to 25X9. Values X1 to X8 correspond to the declassification exemptions found in DOD 5200.01-V1, paragraphs 4-202b(1) to (8) for material exempt from the 10-year rule. Values 25X1 to 25X9 correspond to the declassification exemptions found in DOD 5200.01-V1, paragraphs 4-301a(1) to (9) for permanently valuable material exempt from the 25-year declassification system. If this field is populated with all ECS spaces (0x20), it	4	ECS-A X1 to X8, 25X1 to 25X9, DNIO, (Default is ECS spaces (0x20))	<R>

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TABLE 7. NITF Data Extension Segment (DES) Subheader – continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
	shall imply that a DES declassification exemption does not apply. Value DNIO is Exempt from automatic declassification and can be declassified only by the Director of National Intelligence (DNI).			
DESDG	DES Downgrade. This field shall indicate the classification level to which a DES is to be downgraded if the values in DESDCTP are GD or GE. Valid values are S for Secret, C for Confidential, R for Restricted. If this field contains an ECS space (0x20), it shall imply that DES security downgrading does not apply.	1	ECS-A S, C, R (Default is ECS space (0x20))	<R>
DESDGDT	DES Downgrade Date. This field shall indicate the date on which a DES is to be downgraded if the value in DESDCTP is GD. If this field is populated with all ECS spaces (0x20), it shall imply that a DES security downgrading date does not apply.	8	ECS-A CCYYMMDD (Default is ECS spaces (0x20))	<R>
DESCLTX	DES Classification Text. This field shall be used to provide additional information about DES classification to include identification of a declassification or downgrading event if the values in DESDCTP are DE or GE. It may also be used to identify multiple classification sources and/or any other special handling rules. Values are user-defined free text. If this field is populated with all ECS spaces (0x20), it shall imply that additional information about DES classification does not apply.	43	ECS-A User-defined free text (Default is ECS spaces (0x20))	<R>
DESCATP	DES Classification Authority Type. This field shall indicate the type of authority used to classify the DES. Valid values are O for original classification authority, D for derivative from a single source, and M for derivative from multiple sources. If this field contains an ECS space (0x20), it shall imply that DES DESCATP does not apply. This field may also contain overflow values from the DSCTLH or DSREL fields. Consult with the NITF Field Value Registry for overflow population guidance, specified in section C.2.1. Online Resources.	1	ECS-A O, D, M (Default is ECS space (0x20))	<R>
DESCAUT	DES Classification Authority. This field shall identify the classification authority for the DES dependent upon the value in DESCATP. Values are user-defined free text which should contain the following information: original classification authority name and position or personal ID if the value in DESCATP is O; title of the document or security classification guide used to classify the DES if the value in DESCATP is D; and Deriv- Multiple if the DES classification was derived from multiple sources. In the latter case, the DES originator will maintain a record of the sources used in accordance with existing security directives. One of the multiple sources may also be identified in DESCLTX if desired. If this field is populated with all ECS spaces (0x20), it shall imply that no DES file classification authority applies.	40	ECS-A User-defined free text (Default is ECS spaces (0x20))	<R>

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TABLE 7. NITF Data Extension Segment (DES) Subheader – continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
DESCRSN	DES Classification Reason. This field shall contain a value indicating the reason for classifying the DES. Valid values are A to H. These correspond to the reasons for original classification per E.O. 13526, Section 1.5.(a) to(h). If this field contains all ECS spaces (0x20), it shall imply that no DES classification reason applies.	1	ECS-A A to H (Default is ECS space (0x20))	<R>
DESSRDT	DES Security Source Date. This field shall indicate the date of the source used to derive the classification of the DES. In the case of multiple sources, the date of the most recent source shall be used. If this field is populated with all ECS spaces (0x20), it shall imply that a DES security source date does not apply.	8	ECS-A CCYYMMDD (Default is ECS spaces (0x20))	<R>
DESCTLN	DES Security Control Number. This field shall contain a valid security control number associated with the DES. The format of the security control number shall be in accordance with the regulations governing the appropriate security channel(s). If this field is populated with all ECS spaces (0x20), it shall imply that no DES security control number applies.	15	ECS-A (Default is ECS spaces (0x20))	<R>
DESOFLW	DES Overflowed Header Type. This field shall be present if DESID = TRE_OVERFLOW. Its presence indicates that the DES contains a TRE that would not fit in the file header or segment subheader where it would ordinarily be located. Its value indicates the segment type to which the enclosed TRE is relevant.	6	BCS-A XHD, IXSHD, SXSHD, TXSHD, UDHD, UDID	C
DESITEM	DES Data Item Overflowed. This field shall be present if DESOFLW is present. It shall contain the number of the data item in the file, of the type indicated in DESOFLW to which the TRE in the segment apply. For example, if DESOFLW = UDID and DESITEM = 003, then the TRE in the segment apply to the third image in the file. If the value of DESOFLW = UDHD, the value of DESITEM shall be BCS zeros (0x30).	3	BCS-N positive integer 000 to 999	C
DESSHL	DES User-defined Subheader Length. This field shall contain the number of bytes in the field DESSHF. If this field contains BCS zeros (0x30), DESSHF shall not appear in the DES subheader. This field shall contain BCS zeros (0x30) if DESID = “Registered Extensions” or “Controlled Extensions.”	4	BCS-N positive integer 0000 to 9999	R
DESSHF	DES User-defined Subheader Fields. This field shall contain user-defined fields. Data in this field shall be alphanumeric, formatted according to user specification.	†[8]	BCS-A User defined	C
DESDATA	DES User-Defined Data. This field shall contain data of either binary or character types defined by and formatted according to the user’s specification. However, if DESID=TRE_OVERFLOW, the TRE shall appear according to their definition with no intervening bytes. The length of this field shall not cause another NITF field length limits to be exceeded, but is otherwise fully user-defined.	††[8]	User defined	R

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†[8] Value of the DESSHL (in bytes)
††[8] Determined by user. If the DESID is set to the value TRE_OVERFLOW, this signifies the sum of the lengths of the included TRE.

5.8.3.1 DES Use. The following rules apply to DES usage.

a. Only those DES accepted and registered by the NTB shall be used.

b. Upon receipt of an NITF file that contains one or more DES, an NITF compliant implementation that is not designed to interpret that DES shall ignore it and properly interpret the other NITF file components.

c. NITF implementations that support a specific DES shall comply with the minimum conformance requirements specified in the DES description.

5.8.3.2 DES Structure. The NITF file header accommodates up to 999 DES. Each DES shall consist of a DES subheader and a DES User-Defined Data (DESDATA) field (similar to the way a standard data segment has a subheader and an adjacent associated data field). The DES group in the NITF file header contains the number of DES in the NITF file, the length (size) of each DES subheader, and the length (size) of the DESDATA field. The field size specifications in the NITF file header allow each DES to be just less than one gigabyte in length. The DES subheader shall contain the fields defined in TABLE 7. The DES structure includes a mechanism for defining additional DES subheader fields (DES User-Defined Subheader Length (DESSHL) field and DES User-Defined Subheader Fields (DESSHF) field), and for defining encapsulated data (DESDATA). This structure encourages the formation of a specific DES in a manner similar to the way standard data segments group fields (subheader fields) that describe the data and follow it with the data.

5.8.4. Defined DES. Additional DES, registered by the NTB, will be maintained in the JITC Data Extension Registry.

5.8.4.1 Tagged Record Extension Overflow (TRE-OVERFLOW) DES. The TRE_OVERFLOW DES is used for encapsulating a series of TRE in a DES as overflow from the NITF file header or any segment's subheader. A separate DES is used for each NITF file header or subheader field that overflows. Which NITF file header or subheader field overflowed is indicated in the DES Overflowed Header Type (DESOFLOW) field and DES Data Segment Overflowed (DESITEM) field contents. The TRE_OVERFLOW DES for encapsulating TRE is defined in TABLE 7(A).

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TABLE 7(A). TRE_OVERFLOW DES Subheader.

TYPE R = Required, C = Conditional, < > = BCS Spaces (0x20) are allowed for the entire field
(† annotations are explained at the end of the table)

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
DE	Data Extension Subheader. This field shall contain the characters DE to identify the subheader as a data extension.	2	BCS-A DE	R
DESID	Unique DES Type Identifier. This field shall contain TRE_OVERFLOW	25	BCS-A TRE_OVERFLOW	R
DESVER	Version of the Data Definition. This field shall contain the alphanumeric version number of the use of the Tag. The version number is assigned as part of the registration process.	2	BCS-N positive integer 01	R
DECLAS	Data Extension File Security Classification. This field shall contain a valid value representing the classification level of the DES. Valid values are T for Top Secret, S for Secret, C for Confidential, R for Restricted or U for Unclassified.	1	ECS-A T, S, C, R, or U	R
NOTE: If the value of the DECLAS field is populated with T, S, C, or R, then the DESCLSY field must be populated with a valid code for the security classification system used.				
DESCLSY	DES Security Classification System. This field shall contain valid values indicating the national or multinational security system using a geopolitical (country) code digraph to classify the DES. For US producers a geopolitical code (Country Code), as specified by Paragraph 5.1.7e, are used to indicate their national security systems. The designator "XN" is for classified data generated by a component using NATO security system classification policy marking guidance. The designator “XF” indicates the de facto classification system policy represented by the security marking indicator NATO/ISAF”. These codes were selected to not duplicate codes within the GEC codespace. If this field is populated with all ECS Spaces (0x20), it shall imply that no Security Classification System applies to the DES.	2	ECS-A (Default is ECS spaces (0x20)) The most up to date list of registered digraph geopolitical code values can be found on the NITF Field Value Registry, specified in section C.2.1 Online Resources.	<R>
NOTE: If any of the following fields are populated with anything other than spaces, then the DESCLSY field must be populated with a valid code for the security classification system used: DESCODE, DESREL, DESDCTP, DESDCDT, DESDCXM, DESDG, DESDGDT, DESCLDES, DESCATP, DESCAUT, DESCRSN, DESSRDT, and DESCTLN.				
DESCODE	DES Codewords. This field shall contain a word(s) that has been assigned a classification and a classified meaning to safeguard intentions and information regarding a specific classified plan or operation. This codeword may be a cryptonym used to identify sensitive intelligence data. Multiple entries shall be separated by a single ECS space (0x20). If this field is populated with all ECS spaces (0x20), it shall imply that no codewords apply to the DES. Past codeword examples are located in the NITF registry, specified in section C.2.1 Online Resources.	11	ECS-A (Default is ECS spaces (0x20))	<R>

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TABLE 7(A). TRE_OVERFLOW DES Subheader – continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
DESCTLH	DES Control and Handling. This field shall contain valid additional security control and/or handling instructions (caveats) associated with the DES. Values include digraphs found in the NITF Field Control and Handling is an indicator governing the distribution and use of information or material. Such indicators are the subject of intelligence community agreement and are specifically defined in appropriate regulations. Example value Registry, specified in section C.2.1 Online Resources. The digraph may indicate single or multiple caveats. The selection of a relevant caveat(s) is application specific. If this field is populated with all ECS spaces (0x20), it shall imply that no control and handling instructions apply to the DES.	2	ECS-A (Default is ECS spaces (0x20))	<R>
DESREL	DES Releasing Instructions. This field shall contain a valid list of bare country and/or multilateral entity codes to which countries and/or multilateral entities the DES is authorized for release. Valid entities in the list are one or more bare country codes as specified by Paragraph 5.1.7, separated by a single BCS space (0x20). If this field is populated with all ECS spaces (0x20), it shall imply that no DES release instructions apply.	20	ECS-A (Default is ECS spaces (0x20))	<R>
DESDCTP	DES Declassification Type. This field shall contain a valid indicator of the type of security declassification or downgrading instructions which apply to the DES. Valid values are DD for declassify on a specific date, DE for declassify upon occurrence of an event, GD for downgrade to a specified level on a specific date, GE for downgrade to a specified level upon occurrence of an event, O for OADR, and X for exempt from automatic declassification. If this field ispopulated with all ECS Spaces (0x20), it shall imply that no DES security declassification or downgrading instructions apply.	2	ECS-A DD, DE, GD, GE, O, X (Default is ECS spaces (0x20))	<R>
DESDCDT	DES Declassification Date. This field shall indicate the date on which a DES is to be declassified if the value DES DESDCTP field is populated with DD. If this field is populated with all ECS Spaces (0x20), it shall imply that no DES declassification date applies.	8	ECS-A CCYYMMDD (Default is ECS spaces (0x20))	<R>
DESDCXM	DES Declassification Exemption. This field shall indicate the reason the DES is exempt from automatic declassification if the value of the DESDCTP field is populated with X. Valid values are X1 to X8 and 25X1 to 25X9. Values X1 to X8 correspond to the declassification exemptions found in DOD 5200.01-V1, paragraphs 4-202b(1) to (8) for material exempt from the 10-year rule. Values 25X1 to 25X9 correspond to the declassification exemptions found in DOD 5200.01-V1, paragraphs 4-301a(1) to (9) for permanently valuable material exempt from the 25-year declassification system. If this field is populated with all ECS Spaces (0x20), it shall imply that a DES	4	ECS-A X1 to X8 25X1 to 25X9, DNIO, (Default is ECS spaces (0x20))	<R>

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TABLE 7(A). TRE_OVERFLOW DES Subheader – continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
	declassification exemption does not apply. Value DNIO is Exempt from automatic declassification and can be declassified only by the Director of National Intelligence (DNI).			
DESDG	DES Downgrade. This field shall indicate the classification level to which a DES is to be downgraded if the value of the DESDCTP field is populated with GD or GE. Valid values are S for Secret, C for Confidential, R for Restricted. If this field contains an ECS space (0x20), it shall imply that DES security downgrading does not apply.	1	ECS-A S, C, R (Default is ECS space (0x20))	<R>
DESDGDT	DES Downgrade Date. This field shall indicate the date on which a DES is to be downgraded if the value of the DESDCTP field is populated with GD. If this field is populated with all ECS Spaces (0x20), it shall imply that a DES security downgrading date does not apply.	8	ECS-A CCYYMMDD (Default is ECS spaces (0x20))	<R>
DESCLTX	DES Classification Text. This field shall be used to provide additional information about DES classification to include identification of a declassification or downgrading event if the value of the DESDCTP field is populated with DE or GE. It may also be used to identify multiple classification sources and/or any other special handling rules. Values are user-defined free text. If this field is populated with all ECS Spaces (0x20), it shall imply that additional information about DES classification does not apply. This field may also contain overflow values from the DSCTLH or DSREL fields. Consult with the NITF Field Value Registry for overflow population guidance, specified in section C.2.1 Online Resources.	43	ECS-A User-defined free text (Default is ECS spaces (0x20))	<R>
DESCATP	DES Classification Authority Type. This field shall indicate the type of authority used to classify the DES. Valid values are O for original classification authority, D for derivative from a single source, and M for derivative from multiple sources. If this field contains an ECS space (0x20), it shall imply that DES classification authority type does not apply.	1	ECS-A O, D, M (Default is ECS space (0x20))	<R>
DESCAUT	DES Classification Authority. This field shall identify the classification authority for the DES dependent upon the value of the DESCATP field. Values are user-defined free text which should contain the following information: original classification authority name and position or personal ID if the of the DESCATP field is populated with O; title of the document or security classification guide used to classify the DES if the of the DESCATP field is populated with D; and Deriv-Multiple if the DES classification was derived from multiple sources and the value of the DESCATP field is populated with M. In the latter case, the DES originator will maintain a record of the sources used in accordance with existing	40	ECS-A User-defined free text (Default is ECS spaces (0x20))	<R>

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TABLE 7(A). TRE_OVERFLOW DES Subheader – continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
	security directives. One of the multiple sources may also be identified by the DESCLTX field if desired. If this field is populated with all ECS Spaces (0x20), it shall imply that no DES classification authority applies.			
DESCRSN	DES Classification Reason. This field shall contain values indicating the reason for classifying the DES. Valid values are A to H. These correspond to the reasons for original classification per E.O. 13526, Section 1.5.(a) to (h). If this field contains all ECS Spaces (0x20), it shall imply that no DES classification reason applies.	1	ECS-A A to G (Default is ECS space (0x20))	<R>
DESSRDT	DES Security Source Date. This field shall indicate the date of the source used to derive the classification of the DES. In the case of multiple sources, the date of the most recent source shall be used. If this field is populated with all ECS spaces (code Spaces (0x20), it shall imply that a DES security source date does not apply.	8	ECS-A CCYYMMDD (Default is ECS spaces (0x20))	<R>
DESCTLN	DES Security Control Number. This field shall contain a valid security control number associated with the DES. The format of the security control number shall be in accordance with the regulations governing the appropriate security channel(s). If this field is populated with all ECS Spaces (0x20), it shall imply that no DES security control number applies.	15	ECS-A (Default is ECS spaces (0x20))	<R>
DESOFLW	Overflowed Header Type. This field shall be present if DESID contains TRE_OVERFLOW. Its presence indicates that the DES contains a TRE that would not fit in the NITF file header or segment subheader where it would ordinarily be located. Its value indicates the data type to which the enclosed TRE is relevant.	6	BCS-A UDHD, UDID, XHD, IXSHD, SXSHD, TXSHD; otherwise, field is omitted.	C
DESITEM	Data Item Overflowed. This field shall be present if the DESOFLW field is present. It shall contain the number of the data item in the NITF file, of the type indicated in the DESOFLW field to which the TRE in the segment apply. If the value of the DESOFLW field is populated with UDHD or XHD the value of the DESITEM field shall be populated with 000.	3	BCS-N positive integer 000 to 999	C
DESSHL	Length of DES-Defined Subheader Fields.	4	BCS-N positive integer 0000	R
DESDATA	DES-Defined Data Field. This field shall contain data of either binary or character types defined by and formatted according to the user's specification. The length of this field shall not cause any other NITF field length limits to be exceeded, but is otherwise fully user-defined.	†	User-defined TRE with no intervening octets.	R

† Profile defined.

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5.8.5 Reserved Extension Segment (RES). The RES structure is designated for future use and provides a mechanism for, yet further, expansion of the standard. A RES subheader shall contain the fields defined in TABLE 8. RES that are registered with the NTB will be maintained in the JITC's Data Extension Registry.

TABLE 8. NITF Reserved Extension Segment (RES) Subheader.

(TYPE "R" = Required, "C" = Conditional, "< >" = BCS spaces allowed for entire field)
("†" annotations are explained at the end of the table)

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
RE	File Part Type. This field shall contain the characters "RE" to identify the subheader as a reserved extension.	2	BCS-A RE	R
RESID	Unique RES Type Identifier. This field shall contain a valid alphanumeric identifier properly registered with the GWG NTB.	25	BCS-A Registered value only	R
RESVER	Version of the Data Definition. This field shall contain the alphanumeric version number of the use of the tag. The version number is assigned as part of the registration process.	2	BCS-N positive integer 01 to 99	R
RECLAS	Reserved Extension File Security Classification. This field shall contain a valid value representing the classification level of the RES. Valid values are T for Top Secret, S for Secret, C for Confidential, R for Restricted, or U for Unclassified.	1	ECS-A T, S, C, R, or U	R
NOTE: If the value of the RECLAS field is populated with T, S, C, or R, then the RECLSY field must be populated with a valid code for the security classification system used.				
RECLSY	RES Security Classification System. This field shall contain valid values indicating the national or multinational security system using a geopolitical (country) code digraph used to classify the RES. For US producers a geopolitical code (Country Code), as specified in Paragraph 5.1.7e, are used to indicate their national security systems. The designator "XN" is for classified data generated by a component using NATO security system classification policy marking guidance. The designator "XF" indicates the de facto classification system policy represented by the security marking indicator NATO/ISAF". These codes were selected to not duplicate codes within the GEC codespace. If this field is populated with all ECS spaces (0x20), it shall imply that no security classification system applies to the file.	2	ECS-A (Default is ECS spaces (0x20)) The most up to date list of registered digraph geopolitical code values can be found on the NITF Field Value Registry, specified in section C.2.1 Online Resources.	<R>
NOTE: If any of the following fields are populated with anything other than spaces, then the RECLSY field must be populated with a valid code for the security classification system used: RECODE, RESCTLH, REREL, REDCTP, REDCDT, REDCXM, REDG, REDGDT, RECLTX, RECATP, RECAUT, RECRSN, RESRDT, and RECTLN.				

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TABLE 8. NITF Reserved Extension Segment (RES) Subheader – continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
RECODE	RES Codewords. This field shall contain a word(s) that has been assigned a classification and a classified meaning to safeguard intentions and information regarding a specific classified plan or operation. This codeword may be a cryptonym used to identify sensitive intelligence data. Multiple entries shall be separated by a single ECS space (0x20).. If this field is populated with all ECS spaces 0x20), it shall imply that no codewords apply to the RES. Past codeword examples are located in the NITF registry, specified in section C.2.1 Online Resources.	11	ECS-A (Default is ECS spaces (0x20))	<R>
RECTLH	RES Control and Handling. This field shall contain valid additional security control and/or handling instructions (caveats) associated with the RES. Control and Handling is an indicator governing the distribution and use of information or material. Such indicators are the subject of intelligence community agreement and are specifically defined in appropriate regulations. Example values include digraphs found in the NITF Field Value Registry, specified in section C.2.1 Online Resources. The digraph may indicate single or multiple caveats. The selection of a relevant caveat(s) is application specific. If this field is populated with all ECS spaces (0x20), it shall imply that no control and handling instructions apply to the RES.	2	ECS-A (Default is ECS spaces (0x20))	<R>
REREL	RES Releasing Instructions. This field shall contain a valid list of bare country and/or multilateral entity codes to which countries and/or multilateral entities the RES is authorized for release. Valid entities include one or more bare country codes as specified in Paragraph 5.1.7.e, separated by a single BCS space (0x20). If this field is populated with all ECS spaces (0x20), it shall imply that no RES release instructions apply.	20	ECS-A (Default is ECS spaces (0x20))	<R>
REDCTP	RES Declassification Type. This field shall contain a valid indicator of the type of security declassification or downgrading instructions which apply to the RES. Valid values are DD for declassify on a specific date, DE for declassify upon occurrence of an event, GD for downgrade to a specified level on a specific date, GE for downgrade to a specified level upon occurrence of an event, O for OADR, and X for exempt from automatic declassification. If this field is populated with all ECS spaces (0x20), it shall imply that no RES security declassification or downgrading instructions apply.	2	ECS-A DD, DE, GD, GE, O, X (Default is ECS spaces (0x20))	<R>
REDCDT	RES Declassification Date. This field shall indicate the date on which a RES is to be declassified if the value in the REDCTP field is populated with DD. If this field is populated with all ECS spaces (0x20), it shall imply that no RES declassification date applies.	8	ECS-A CCYYMMDD (Default is ECS spaces (0x20))	<R>
REDCXM	RES Declassification Exemption. This field shall indicate the reason the RES is exempt from automatic declassification if the value in REDCTP is X. Valid	4	ECS-A X1 to X8, 25X1 to 25X9,	<R>

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TABLE 8. NITF Reserved Extension Segment (RES) Subheader – continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
	values are X1 to X8 and 25X1 to 25X9. Values X1 to X8 correspond to the declassification exemptions found in DOD 5200.01-V1, paragraphs 4-202b(1) to (8) for material exempt from the 10-year rule. Values 25X1 to 25X9 correspond to the declassification exemptions found in DOD 5200.01-V1, paragraphs 4-301a(1) to (9) for permanently valuable material exempt from the 25-year declassification system. If this field is populated with all ECS spaces (0x20), it shall imply that a file declassification exemption does not apply.		(Default is ECS spaces (0x20))	
REDG	RES Downgrade. This field shall indicate the classification level to which a RES is to be downgraded if the values in REDCTP are GD or GE. Valid values are S for Secret, C for Confidential, R for Restricted. If this field contains an ECS space (0x20), it shall imply that RES security downgrading does not apply.	1	ECS-A S, C, R (Default is ECS space (0x20))	<R>
REDGDT	RES Downgrade Date. This field shall indicate the date on which a RES is to be downgraded if the value in REDCTP is GD. If this field is populated with all BCS spaces (0x20), it shall imply that a RES security downgrading date does not apply.	8	ECS-A CCYYMMDD (Default is ECS spaces (0x20))	<R>
RECLTX	RES Classification Text. This field shall be used to provide additional information about the RES classification to include identification of a declassification or downgrading event if the values in REDCTP are DE or GE. It may also be used to identify multiple classification sources and/or any other special handling rules. Values are user-defined free text. If this field is populated with all ECS spaces (0x20), it shall imply that additional information about RES classification does not apply. This field may also contain overflow values from the RSCTLH or RSREL fields. Consult with the NITF Field Value Registry for overflow population guidance, specified in section C.2.1. Online Resources.	43	ECS-A User-defined free text (Default is ECS spaces (0x20))	<R>
RECATP	RES Classification Authority Type. This field shall indicate the type of authority used to classify the RES. Valid values are O for original classification authority, D for derivative from a single source, and M for derivative from multiple sources. If this field contains an ECS space (0x20), it shall imply RECATP does not apply.	1	ECS-A O, D, M (Default is ECS space (0x20))	<R>
RECAUT	RES Classification Authority. This field shall identify the classification authority for the RES dependent upon the value in RECATP. Values are user-defined free text which should contain the following information: original classification authority name and position or personal ID if the value in RECATP is O; title of the document or security classification guide used to classify the RES if the value in RECATP is D; and Deriv-Multiple if the RES classification was derived from multiple sources. In the latter case, the RES originator	40	ECS-A User-defined free text (Default is ECS spaces (0x20))	<R>

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TABLE 8. NITF Reserved Extension Segment (RES) Subheader – continued.

FIELD	NAME/DESCRIPTION	SIZE	VALUE RANGE	TYPE
	will maintain a record of the sources used in accordance with existing security directives. One of the multiple sources may also be identified in RECLTX if desired. If this field is populated with all ECS spaces (0x20), it shall imply that no RES classification authority applies.			
RECRSN	RES Classification Reason. This field shall contain values indicating the reason for classifying the RES. Valid values are A to H. These correspond to the reasons for original classification per E.O. 13526, Section 1.5.(a) to (h). If this field contains an ECS space (0x20), it shall imply that no RES classification reason applies.	1	ECS-A A to H (Default is ECS space (0x20))	<R>
RESRDT	RES Security Source Date. This field shall indicate the date of the source used to derive the classification of the RES. In the case of multiple sources, the date of the most recent source shall be used. If this field is populated with all ECS spaces (0x20), it shall imply that a RESRDT does not apply.	8	ECS-A CCYYMMDD (Default is ECS spaces (0x20))	<R>
RECTLN	RES Security Control Number. This field shall contain a valid security control number associated with the RES. The format of the security control number shall be in accordance with the regulations governing the appropriate security channel(s). If this field is populated with all ECS spaces (0x20), it shall imply that no RECTLN applies.	15	ECS-A (Default is ECS spaces (0x20))	<R>
RESSHL	RES User-defined Subheader Length. This field shall contain the number of bytes in the field RESSHF. If this field contains BCS zeros (0x30), RESSHF shall not appear in the RES subheader.	4	BCS-N positive integer 0000 to 9999	R
RESSHF	RES User-Defined Subheader Fields. This field shall contain user-defined fields. Data in this field shall be alphanumeric, formatted according to user specification.	†[9]	BCS-A User-defined	C
RESDATA	RES User-Defined Data. This field shall contain data of either binary or character types defined by and formatted according to the user's specification. The length of this field shall not cause any other NITF field length limits to be exceeded, but is otherwise fully user defined.	††[9]	User defined	R

†[9] Value of the RESSHL field (in bytes)
††[9] Determined by the definition of the specific RES as registered and controlled with the NTB.

5.8.5.1 RES Use. The following rules apply to RES usage.

a. Only those RES accepted and registered by the NTB shall be used.

b. Upon receipt of an NITF file that contains a RES, an NITF compliant implementation that is not designed to interpret that RES shall ignore it and properly interpret the other NITF file components.

c. NITF implementations that support a specific RES shall comply with the minimum conformance requirements specified in the RES description.

5.8.5.2 RES Structure. The NITF file header accommodates up to 999 RES. Each RES shall consist of a RES subheader and a RES User-Defined Data (RESDATA) field (similar to the way a standard data segment has a subheader

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and an adjacent associated data field). The RES group in the NITF file header contains the number of RES in the NITF file, the length (size) of each RES subheader, and length (size) of the RESDATA field. The field size specifications in the NITF file header allow each RES to be just less than ten megabytes in length. The RES subheader shall contain the fields defined in TABLE 8. The RES structure includes a mechanism for defining additional RES subheader fields (RES User-Defined Subheader Length (RESSHL) field and RES User-Defined Subheader Fields (RESSHF) field), and for defining encapsulated data (RESDATA). This structure encourages the formation of a specific RES in a manner similar to the way standard data segments group fields (Subheader fields) that describe the data and follow it with the data.

5.9 Complexity Level (CLEVEL). The CLEVEL concept allows NITFS to be implemented on a wide range of hardware platforms with various levels of internal resources while maintaining a baseline level of interoperability between all compliance tested systems. TABLE 9 defines the conditions of NITF file features used to determine the CLEVEL assignment for a given NITF file. The six key NITF features which differentiate CLEVEL are: CCS extent, file size (bytes), image size (rows/columns), number of multispectral bands, number of ISs per NITF file, and aggregate size of GSs. The other listed features provide the parameter, value, range conditions, and constraints for all the defined CLEVEL (03, 05, 06, and 07). Although an NITF file shall be marked at the lowest CLEVEL for which it qualifies, it shall be marked no lower than the highest CLEVEL feature condition included in the NITF file. For example, a 51 Mbyte file shall be marked at CLEVEL 05, even when all other features in the NITF file do not exceed the specified CLEVEL 03 conditions.

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TABLE 9. NITF 02.10 Still Imagery Complexity Level (CLEVEL)

NITF File Features	Still Imagery Complexity Level (CLEVEL)			
	3	**5**	**6**	**7**
Common Coordinate System Extent (Pixels)	(00000000, 00000000) to (00002047, 00002047)	(00000000, 00000000) to (00008191, 00008191)	(00000000, 00000000) to (00065535, 00065535)	(00000000, 00000000) to (99999999, 99999999)
Maximum File Size	50 Mbyte –1byte (52,428,799 bytes)	1 Gbyte –1 byte (1,073,741,823 bytes)	2 Gbyte –1 byte (2,147,483,647 bytes)	10 Gbyte –1 byte (10,737,418,239 bytes)
Image Size (Image(s) placed within CCS extent)	00000001 to 00002048 Rows X 00000001 to 00002048 Columns (R and C ≤ 2048)	00000001 to 00008192 Rows X 00000001 to 00008192 Columns (R or C > 2048)	00000001 to 00065536 Rows X 00000001 to 00065536 Columns (R or C > 8192)	00000001 to 99999999 Rows X 00000001 to 99999999 Columns (R or C > 65536)
Image Blocking (Rectangular Blocks allowed)	Single and Multiple Blocks 0001to 2048 Rows X 0001 to 2048 Columns	Single and Multiple Blocks 0001 to 8192 Rows X 0001 to 8192 Columns	Multiple blocking is mandatory for images that exceed 8192 Pixels per Row or Column. 0001 to 8192 Rows X 0001 to 8192 Columns	
Monochrome (MONO) No Compression	Single Band 1, 8, 12, 16, 32, and 64-Bits per Pixel (NBPP) With and without LUT IC = NC, NM IMODE = B			
Color 1 and 8-Bit (RGB / LUT) No Compression	Single Band 1 and 8-Bits per Pixel (NBPP) With LUT IC = NC, NM IMODE = B			
Color (RGB) No Compression	Three Band 8-Bits per Pixel (NBPP) No LUT IC = NC, NM IMODE = B, P, R, S		Three Band 8, 16, 32-Bits per Pixel (NBPP) No LUT IC = NC, NM IMODE = B, P, R, S	
Multiband (MULTI) No Compression	1 to 9 Bands, 1, 8, 16, 32, and 64-Bits per Pixel per Band With and without LUT in each Band IC = NC, NM IMODE = B, P, R, S	1 to 255 Bands, 1, 8, 16, 32, and 64-Bits per Pixel per Band With and without LUT in each Band IC = NC, NM IMODE = B, P, R, S		1 to 999 Bands, 1, 8, 16, 32, and 64-Bits per Pixel per Band With and without LUT in each Band IC = NC, NM IMODE = B, P, R, S
JPEG 2000 Compression Monochrome (MONO)	1 Band 1-32 bits per Pixel per Band With and without LUT IC = C8, M8 IMODE = B Note: LUTs are typically only useful when the data is compressed numerically lossless.			
JPEG 2000 Compression Mapped Color (RGB/LUT)	1 Band 1-32 bits per Pixel per Band With LUT IC = C8, M8 IMODE = B Note: LUTs are typically only useful when the data is compressed numerically lossless.			

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TABLE 9. NITF 02.10 Still Imagery Complexity Level (CLEVEL) - continued.

NITF File Features	Still Imagery Complexity Level (CLEVEL)			
	3	5	6	7
JPEG 2000 Compression Color (RGB)	3 Bands 1-32 bits per Pixel per Band No LUT IC = C8, M8 IMODE = B Note: The JPEG 2000 color transform may be used as part of the compression and decompression process when IREP=RGB.			
JPEG 2000 Compression Color (YCbCr601)	3 Bands 1-32 bits per Pixel per Band No LUT IC = C8, M8 IMODE = B Note: When IREP=YCbCr601, it signifies that the data representation was YCbCr prior to the JPEG 2000 compression process. The internal JPEG 2000 color transform shall not be used.			
JPEG2000 Compression Multiband (MULTI)	1 to 9 Bands 1-32 bits per Pixel per Band With and without LUT IC = C8, M8 IMODE = B	1 to 255 Bands 1-32 bits per Pixel per Band With and without LUT IC = C8, M8 IMODE = B		1 to 999 Bands 1-32 bits per Pixel per Band With and without LUT IC = C8, M8 IMODE = B
JPEG DCT Compression Monochrome (MONO)	Single Band 8 and 12-Bit Sample (NBPP) No LUT IC = C3, M3 IMODE = B			
JPEG DCT Compression 24-Bit Color (RGB)	Three Bands 8-Bit Sample per Band (NBPP) No LUT IC = C3, M3 IMODE = P			
JPEG DCT Compression 24-Bit Color (YCbCr601)	Three Bands 8-Bit Sample per Band (NBPP) No LUT IC = C3, M3 IMODE = P			
Downsampled JPEG DCT Monochrome (MONO)	Single Band Single Block Only 8-Bit Sample (NBPP) No LUT IC = I1 IMODE = B (Image size may not exceed 2048 Pixels per Row or Column.)			
JPEG Lossless Compression Monochrome (MONO)	Single Band 8, 12, and 16-Bit Sample per Band With and Without LUT IC = C5, M5 IMODE = B (This feature is optional for implementation.)			
JPEG Lossless Compression 24-Bit Color (RGB)	Three Bands 8-Bit Sample per Band (NBPP) No LUT IC = C5, M5 IMODE = P (This feature is optional for implementation.)			
Bi-Level Compression (MONO)	This feature is optional for implementations If this feature is used, the following elements are to be used: Single Band Single Block 1-Bit per Pixel (NBPP) With and without LUT IC = C1, M1 IMODE = B COMRAT = 1D, 2DS, 2DH (Image size may not exceed 2560 Pixels per Row by 8192 Pixels per Column.)			

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TABLE 9. NITF 02.10 Still Imagery Complexity Level (CLEVEL) - continued.

NITF File Features	Still Imagery Complexity Level (CLEVEL)			
	3	5	6	7
Bi-Level Compression (RGB/LUT)	This feature is optional for implementations If this feature is used, the following elements are to be used: Single Band Single Block Only 1-Bit per Pixel (NBPP) With LUT IC = C1, M1 IMODE = B COMRAT = 1D, 2DS, 2DH (Image size may not exceed 2560 Pixels per Row by 8192 Pixels per Column.)			
VQ Monochrome (MONO)	Single Band 8-Bits per Pixel (NBPP) 4 x 4 Kernel organized in 4 Tables With and without LUT IC = C4, M4 IMODE = B			
VQ 8-Bit Color (RGB/LUT)	Single Band 8-Bits per Pixel (NBPP) 4 x 4 Kernel organized in 4 Tables With LUT IC = C4, M4 IMODE = B			
Multiband (MULTI) Individual Band JPEG Compression	1 to 9 Bands 8 and 12-Bits per Pixel per Band No LUT IC = C3, M3 IMODE = B, S	1 to 255 Bands 8 and 12-Bits per Pixel per Band No LUT IC = C3, M3 IMODE = B, S		1 to 999 Bands 8 and 12-Bits per Pixel per Band No LUT IC = C3, M3 IMODE = B, S
Elevation Data (NODISPLY)	Single Band 8, 12, 16, 32, and 64-Bits per Pixel (NBPP) No LUT IC = NC, NM IMODE = B ICAT = DTEM, ISUBCATn code from DIGEST, Part 3, Annex B (or BCS Spaces (0x20) Applicable TRE: Geospatial Support Data Extensions (GEOSDE), DIGEST, Part 2, Annex D (This feature is optional for implementation.)			
Location Grid (NODISPLY)	Two Bands 8, 12, 16, 32, and 64-Bits per Pixel (NBPP) No LUT IC = NC, NM IMODE = B, P ICAT = LOCG, ISUBCATn = CGX, CGY, or GGX, GGY Applicable TRE: Geospatial Support Data Extensions (GEOSDE), DIGEST, Part 2, Annex D (This feature is optional for implementation.)			
Matrix Data (NODISPLY) No Compression	1 to 9 Bands 8, 16, 32, and 64-Bits per Pixel per Band No LUT in any Band IC = NC, NM IMODE = B, P, R, S (This feature is optional for implementation.)	1 to 255 Bands 8, 16, 32, and 64-Bits per Pixel per Band No LUT in any Band IC = NC, NM IMODE = B, P, R, S (This feature is optional for implementation.)		1 to 999 Bands 8, 16, 32, and 64-Bits per Pixel per Band No LUT in any Band IC = NC, NM IMODE = B, P, R, S (This feature is optional for implementation.)
Matrix Data (NODISPLY) JPEG 2000	1 to 9 Bands 1-32 Bits per Pixel per Band No LUT in any Band IC = C8, M8 Numerically Lossless only IMODE = B (This feature is optional for implementation.)	1 to 255 Bands 1-32 Bits per Pixel per Band No LUT in any Band IC = C8, M8 Numerically Lossless only IMODE = B (This feature is optional for implementation.)		1 to 999 Bands 1-32 Bits per Pixel per Band No LUT in any Band IC = C8, M8 Numerically Lossless only IMODE = B (This feature is optional for implementation.)
Vectors in Polar Coordinates (POLAR)	2 Bands 8, 16, 32, 64 – bits per pixel NBPP No LUT IC = NC IMODE = B, P, S			

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TABLE 9. NITF 02.10 Still Imagery Complexity Level (CLEVEL) - continued.

NITF File Features	Still Imagery Complexity Level (CLEVEL)			
	3	5	6	7
Number of Image Segments per File	0 to 20	0 to 100		
Number of CGM Graphic Segments per File	0 to 100			
Aggregate Size of Graphic Segments	1 Mbyte maximum	2 Mbyte maximum		
CGM Graphic Profile	BPCGM01.00			
Number of Text Segments per File	0 to 32 Segments			
Text Format Codes Supported	STA, MTF, UT1, U8S			
Text Data per Segment	00001 to 99999 Bytes			
Tagged Record Extensions (TRE)	TRE may appear in the UDHD, XHD, UDID, IXSHD, SXSHD, and TXSHD fields and TRE_OVERFLOW DES(s) regardless of CLEVEL.			
Number of Data Extension Segments (DES) per File	0 to 10		0 to 50	0 to 100
Currently Registered DES	TRE_OVERFLOW CSATTA CSSHPA XML_DATA_CONTENT LIDARA See Registry for others			
Number of Reserved Extension Segments (RES) per File	None			
Currently Approved RES	None			
SARZip C7, complex Compression	This feature is optional for implementations If this feature is used, the following elements are to be used: Two Band (ISUBCAT = 'I' & 'Q') Single Block PVTYPE SI, R Without LUT NBPP 16 and 32-Bits IMODE = P NBPR = NBPC =0001 (Image size may not exceed 8192 Pixels per Row by 8192 Pixels per Column, if exceeded NBPR and/or NBPC will be set to 0000 and CLEVEL set to "09".)			
SARZip C7, magnitude Compression	This feature is optional for implementations If this feature is used, the following elements are to be used: Single Band (ISUBCAT = 'M') Single Block PVTYPE INT Without LUT NBPP 16 and 32-Bits IMODE = B NBPR = NBPC = 0001 (Image size may not exceed 8192 Pixels per Row by 8192 Pixels per Column, if exceeded NBPR and/or NBPC will be set to 0000 and CLEVEL set to "09".)			

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TABLE 9. NITF 02.10 Still Imagery Complexity Level (CLEVEL) - continued.

NITF File Features	Still Imagery Complexity Level (CLEVEL)			
	3	5	6	7
SARZip M7 Multi Block complex Compression	This feature is optional for implementations If this feature is used, the following elements are to be used: Two Band (ISUBCAT = 'I' & 'Q') Multi Block PVTYPE SI, R Without LUT NBPP 16 and 32-Bits IMODE = P			
SARZip M7, magnitude Compression	This feature is optional for implementations If this feature is used, the following elements are to be used: Single Band (ISUBCAT = 'M') Multi Block PVTYPE INT Without LUT NBPP 16 and 32-Bits IMODE = B			

Note 1: CLEVEL 09 is used to designate NITF 2.1 files that exceed the CLEVEL 07 constraints in TABLE 9 but remain within the bounds of the standard. CLEVEL 09 designates that the file exceeds at least one of the CLEVEL 07 constraints: 1. Maximum File Size (10 Gbyte or greater), 2. Image segments with the block size exceeding 8192 pixels per row or column, (see paragraph 5.4.2.2 d for additional implementation guidance). 3. More than 999 Bands, 4. More than 100 images, 5. More than 100 graphic segments, 6. Graphic aggregate size exceeds 2 Mbytes, 7. Number of Text Segments exceeds 32, 8. The number of DES exceeds 100.
NOTE 2: CLEVEL 50 - 59 are reserved for future use by the Motion Imagery Extension for NTIF 2.1 (MIE4NITF) community of producers. These CLEVELs will address additional Image Modes and Image Compressions, as well as a modification of the current concept of display and attachment levels.

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TABLE 9(A). NITF 2.1 Motion Imagery Complexity Level (CLEVEL)

NITF File Features	51	54	57
Common Coordinate System Extent (pixels)	(00000000, 00000000) to (00008191, 00008191)	(00000000, 00000000) to (00065535, 00065535)	(00000000, 00000000) to (99999999, 99999999)
Maximum File Size (per NITF File)	214,748,364,800 (200 GiB)	429,496,729,600 (400 GiB)	999,999,999,998 (931.32 GiB)
Image Size (image(s) placed within CCS extent)	Rows: 1 to 8,192 Columns: 1 to 8,192 (R and C ≤ 8,192)	Rows: 1 to 65,536 Columns: 1 to 65,536 (R and C ≤ 32,768)	Rows: 1 to 99,999,999 Columns: 1 to 99,999,999 (R and C ≤ 65,536)
Image Blocking (spatiotemporal blocking)	Rows: 1 to 8,192 Columns: 1 to 8,192	Rows: 1 to 32,768 Columns: 1 to 32,768	Rows: 1 to 32,768 Columns: 1 to 32,768
Monochrome (MONO) (no compression)	Single Band 1, 8, 12, 16, 32, and 64-Bits per Pixel (NBPP) With and without LUT IC = NC, NM IMODE = F		
Color (RGB) (no compression)	Three bands 8, 16, 32-Bits per Pixel (NBPP) No LUT IC = NC, NM IMODE = T, D, E, F		
Multiband (MULTI) (no compression)	2 to 9 Bands 8, 16, 32, and 64-Bits per Pixel per Band With and without LUT in each Band IC = NC, NM IMODE = T, D, E, F	2 to 255 Bands 8, 16, 32, and 64-Bits per Pixel per Band With and without LUT in each Band IC = NC, NM IMODE = T, D, E, F	2 to 999 Bands 8, 16, 32, and 64-Bits per Pixel per Band With and without LUT in each Band IC = NC, NM IMODE = T, D, E, F
JPEG 2000 Compression Monochrome (MONO)	1 Band 1-32 bits per Pixel per Band With and without LUT IC = C8, M8 IMODE = B Note: There is only one image per image segment for this CLEVEL. Standard NITF still image rules for JPEG 2000 apply.		1 Band 1-32 bits per Pixel per Band With and without LUT IC = CB, MB IMODE = T, D, F Note: There are multiple compressed images stacked within each spatiotemporal block within an image segment in these CLEVELs.

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TABLE 9(A). NITF 2.1 Motion Imagery Complexity Level (CLEVEL) – continued.

NITF File Features	51	54	57
JPEG 2000 Compression Monochrome (RGB/LUT)	1 Band 1-32 bits per Pixel per Band With LUT IC = C8, M8 IMODE = B Note: There is only one image per image segment for this CLEVEL. Standard NITF still image rules for JPEG 2000 apply.		1 Band 1-32 bits per Pixel per Band With LUT IC = CB, MB IMODE = T, D, F Note: There are multiple compressed images stacked within each spatiotemporal block within an image segment in these CLEVELs.
JPEG 2000 Compression Color (RGB)	3 bands 1-32 bits per Pixel per Band No LUT IC = C8, M8 IMODE = B Note: The JPEG 2000 color transform may be used when IREP = RGB. There is only one image per image segment for this CLEVEL.		3 bands 1-32 bits per Pixel per Band No LUT IC = CB, MB IMODE = T, D, F Note: The JPEG 2000 color transform may be used when IREP = RGB. There are multiple compressed images stacked within each spatiotemporal block within an image segment in these CLEVELs.
JPEG 2000 Compression Color (YCbCr601)	3 bands 1-32 bits per Pixel per Band No LUT IC = C8, M8 IMODE = B Note: When IREP = YCbCr601 the JPEG 2000 color transform shall not be used. There is only one image per image segment for this CLEVEL. Standard NITF still image rules for JPEG 2000 apply.		3 bands 1-32 bits per Pixel per Band No LUT IC = CB, MB IMODE = T, D, F Note: When IREP = YCbCr601 the JPEG 2000 color transform shall not be used. There are multiple compressed images stacked within each spatiotemporal block within an image segment in these CLEVELs.

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TABLE 9(A). NITF 2.1 Motion Imagery Complexity Level (CLEVEL) – continued.

NITF File Features	51	54	57
JPEG 2000 Compression Multiband (MULTI)	2 to 9 bands 1-32 bits per Pixel per Band With and without LUT IC = C8, M8 IMODE = B Note: There is only one image per image segment for this CLEVEL. Standard NITF still image rules for JPEG 2000 apply.	2 to 255 bands 1-32 bits per Pixel per Band With and without LUT IC = C8, M8 IMODE = B Note: There is only one image per image segment for this CLEVEL. Standard NITF still image rules for JPEG 2000 apply.	2 to 999 bands 1-32 bits per Pixel per Band With and without LUT IC = CB, MB IMODE = T, D, F Note: There are multiple compressed images stacked within each spatiotemporal block within an image segment in this CLEVEL.
AVC/H.264 Compression Monochrome (MONO)	1 band 8 bits per Pixel per Band No LUT IC = C9, M9 IMODE = T Constrained Baseline Profile Level 4.0 Note: Represent monochrome imagery as three equal RGB values prior to 4:2:0 YCbCr conversion.	1 band 8 bits per Pixel per Band No LUT IC = C9, M9 IMODE = T Main Profile Level 5.1 Note: Represent monochrome imagery as three equal RGB values prior to 4:2:0 YCbCr conversion.	1 band 8 to 10 bits per Pixel per Band No LUT IC = C9, M9 IMODE = T High 4:2:2 Profile Level 5.1 Note: Represent monochrome imagery as three equal RGB values prior to 4:2:0 YCbCr conversion. There is no need to use 4:2:2 color for monochrome.
AVC/H.264 Compression Color (RGB)	3 bands 8 bits per Pixel per Band No LUT IC = C9, M9 IMODE = T Constrained Baseline Profile Level 4.0	3 bands 8 bits per Pixel per Band No LUT IC = C9, M9 IMODE = T Main Profile Level 5.1	3 bands 8 to 10 bits per Pixel per Band No LUT IC = C9, M9 IMODE = T High 4:2:2 Profile Level 5.1 Note: 4:2:0 or 4:2:2 color space may be used as desired.

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TABLE 9(A). NITF 2.1 Motion Imagery Complexity Level (CLEVEL) – continued.

NITF File Features	51	54	57
AVC/H.264 Compression Multiband (MULTI)	2 to 9 bands 8 bits per Pixel per Band No LUT IC = C9, M9 IMODE = D, F Constrained Baseline Profile Level 4.0 Note: Each band is encoded independently as a monochrome codestream (see above).	2 to 255 bands 8 bits per Pixel per Band No LUT IC = C9, M9 IMODE = D, F Main Profile Level 5.1 Note: Each band is encoded independently as a monochrome codestream (see above).	2 to 999 bands 8 to 10 bits per Pixel per Band No LUT IC = C9, M9 IMODE = D, F High 4:2:2 Profile Level 5.1 Note: Each band is encoded independently as a monochrome codestream (see above).
HEVC/H.265 Compression Monochrome (MONO)	1 band 8 to 10 bits per Pixel per Band No LUT IC = CA, MA IMODE = T Main 10 Profile Level 4.0 Note: Represent monochrome imagery as three equal RGB values prior to 4:2:0 YCbCr conversion.	1 band 8 to 10 bits per Pixel per Band No LUT IC = CA, MA IMODE = T Main 4:2:2 10 Profile Level 5.1 Note: Represent monochrome imagery as three equal RGB values prior to 4:2:0 YCbCr conversion. There is no need to use 4:2:2 color for monochrome.	1 band 8 to 12 bits per Pixel per Band No LUT IC = CA, MA IMODE = T Main 4:4:4 12 Profile Level 6 Note: Represent monochrome imagery as three equal RGB values prior to 4:2:0 YCbCr conversion. There is no need to use 4:2:2 or 4:4:4 color for monochrome.
HEVC/H.265 Compression Color (RGB)	3 bands 8 to 10 bits per Pixel per Band No LUT IC = CA, MA IMODE = T Main 10 Profile Level 4.0	3 bands 8 to 10 bits per Pixel per Band No LUT IC = CA, MA IMODE = T Main 4:2:2 10 Profile Level 5.1 Note: 4:2:0, 4:2:2 or color space may be used as desired.	3 bands 8 to 12 bits per Pixel per Band No LUT IC = CA, MA IMODE = T Main 4:4:4 12 Profile Level 6 Note: 4:2:0, 4:2:2 or 4:4:4 color space may be used as desired.

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TABLE 9(A). NITF 2.1 Motion Imagery Complexity Level (CLEVEL) – continued.

NITF File Features	51	54	57
HEVC/H.265 Compression Multiband (MULTI)	2 to 9 bands 8 to 10 bits per Pixel per Band No LUT IC = CA, MA IMODE = D, F Main 10 Profile Level 4.0 Note: Each band is encoded independently as a monochrome codestream (see above).	2 to 255 bands 8 to 10 bits per Pixel per Band No LUT IC = CA, MA IMODE = D, F Main 4:2:2 10 Profile Level 5.1 Note: Each band is encoded independently as a monochrome codestream (see above).	2 to 999 bands 8 to 12 bits per Pixel per Band No LUT IC = CA, MA IMODE = D, F Main 4:4:4 12 Profile Level 6 Note: Each band is encoded independently as a monochrome codestream (see above).
Number of Image Segments per File	0 to 999		
Number of CGM Graphic Segments per File	0 to 999		
Aggregate Size of Graphic Segments	45 MiB maximum		
CGM Graphic Profile	MIL-STD-2301A		
Number of Text Segments per File	0 to 999		
Text Data per Segment	0 to 99,998 bytes (97.65 KiB)		
Tagged Record Extensions (TRE)	TRE may appear in the UDHD, XHD, UDID, IXSHD, SXSHD, and TXSHD fields and TRE_OVERFLOW DES(s) regardless of CLEVEL.		
Number of Data Extension Segments (DES) per File	0 to 999		
Currently Registered DES	TRE_OVERFLOW STREAMING_FILE_HEADER		
Number of Reserved Extension Segments (RES) per File	None		

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Subject term (key word) listing.

Computer graphics metafile
Discrete cosine transform
Image compression
Image coordinate
Imagery
Joint Photographic Experts Group
Message Transfer Facility
Motion imagery extension
Still imagery
Tagged record extension

6.2 International Standardization Agreement Implementation. This standard implements STANAG 4545, NATO Secondary Imagery Format (NSIF). When changes to, revision, or cancellation of this standard are proposed, the preparing activity must coordinate the action with the U.S. National Point of Contact for the international standardization agreement, as identified in the ASSIST database at https://assist.dla.mil.

6.3 Changes From Previous Issue. The margins of this standard are marked with vertical lines to indicate modifications generated by this change. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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APPENDIX A. Acronyms & Abbreviations (Normative)

A.1. ACRONYMS & ABBREVIATIONS

A.1.1 Scope. The appendix details the acronyms defined in this document. This appendix is a mandatory part of this standard. The information contained herein is intended for guidance.

ALVL	-	Attachment LeVeL
AMD	-	AMenDment
AMSC	-	Acquisition Management Systems Control
As	-	GENC concept-type for administrative subdivisions
ASD (C^3I)	-	Assistant Secretary of Defense for Command, Control, Communications, and Intelligence
B	-	band interleaved by Block (IMODE value)
BARO	-	Barometric pressure (ICAT value)
BCKGDA	-	BaCKGround Data
BCS	-	Basic Character Set
BCS-A	-	Basic Character Set Alphanumeric
BCS-N	-	Basic Character Set Numeric
BE	-	Basic Encyclopedia
BGHIGHT	-	BackGround HeIGHT
BGWIDTH	-	BackGround WIDTH
BIIF	-	Basic Image Interchange Format (ISO/IEC IS 12087-5)
BMRnBNDm	-	n^{th} Block Mask Record of BaND m
BP	-	Black/white frame Photography (ICAT value)
C	-	(1) Column (2) Conditional
C1	-	Bi-level (IC value)
C3	-	JPEG (IC value)
C4	-	Vector Quantization (IC value)
C5	-	Lossless JPEG (IC value)
C7	-	SARZIP compression (IC value)
C8	-	JPEG 2000 compression (IC value)
C9	-	H264 compression with time (IC value)
CA	-	H265 compression with time (IC value)
CAT	-	Computerized Axial Tomography scan (ICAT value)
CB	-	JPEG 2000 compression with time (IC value)
CCD	-	Coherent Change Detection
CCS	-	Common Coordinate System
CE	-	Controlled Extension
CEDATA	-	Controlled Extension user-defined DATA
CETAG	-	Controlled unique Extension Type identifier
CGM	-	Computer Graphics Metafile
CLEVEL	-	Complexity LEVEL
CLOUD	-	Cloud
CP	-	Color frame Photography (ICAT value)
CRT	-	Cathode Ray Tube
CURRENT	-	water Current (ICAT value)
D	-	IMODE field value, temporal band sequential
DC	-	District of Columbia
DD	-	Day Day Declassify by Date
DED	-	Digital Elevation Data

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DEPTH	-	water Depth (ICAT value)
DES	-	Data Extension Segment
DESDATA	-	DES user-defined DATA
DESITEM	-	DES data segment overflowed
DESOFLW	-	DES OverFLoW header type
DESSHF	-	DES user-defined SubHeader Field
DESSHL	-	DES user-defined SubHeader Length
DIGEST	-	Digital Geographic Information Exchange Standard (https://www.dgiwg.org/digest/)
DLVL	-	Display LeVeL
DOD	-	Department Of Defense
DODISS	-	Department Of Defense Index of Specifications and Standards
DTEM	-	Digital Terrain Elevation Model (ICAT value)
DTV	-	Daylight Television
E	-	IMODE field value, temporal interleave by sample
ECS	-	Extended Character Set
ECS-A	-	Extended Character Set-Alphanumeric
EEI	-	Essential Elements of Information
ENCRYP	-	File ENCRYPtion
E.O.	-	Executive Order
EO	-	Electro-Optical (ICAT value)
F	-	IMODE field value, temporal band sequential by block
FACC	-	Features and Attribute Catalog Code
FBKGC	-	File BacKGround Color
FDT	-	File Date and Time
FHDR	-	File profile name
FL	-	(1) Forward Looking infrared (ICAT value) (2) File Length
FOUO	-	For Official Use Only
FP	-	Fingerprints (ICAT value)
FSCATP	-	File Classification Authority TyPe
FSCAUT	-	File Classification AUThority
FSCLAS	-	File Security CLASsification
FSCLSY	-	File Security Classification System
FSCLTX	-	File CLassification TeXt
FSCODE	-	File Security CODEwords
FSCRSN	-	File Classification ReaSoN
FSCTLH	-	File ConTroL and Handling
FSCTLN	-	File Security ConTroL Number
FSDCDT	-	File DeClassification DaTe
FSDCTP	-	File DeClassification TyPe
FSDCXM	-	File DeClassification eXeMption
FSDG	-	File DownGrade
FSDGDT	-	File DownGrade DaTe
FSREL	-	File REL instructions
FSSRDT	-	File Security SouRce DaTe
FTITLE	-	File TITLE
FVER	-	File VERsion
Ge	-	GENC concept-type for geopolitical entities
GEC	-	Geopolitical Entities and Codes
GENC	-	Geopolitical Entities, Names, and Codes
GINT	-	Geospatial Intelligence Technology
GS	-	Graphic Segment
GWG	-	Geospatial-Intelligence Standards Working Group

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HL	-	file Header Length
HR	-	High Resolution radar (ICAT value)
HS	-	HyperSpectral (ICAT value)
HTML	-	Hypertext Mark-up Language
I	-	Inphase
I1	-	Downsampled JPEG (IC value)
IC	-	(1) Intelligence Community (2) Image Compression
ICAT	-	Image Category
ICORDS	-	Image Coordinate Representation
IDLVL	-	Image Display LeVeL
IEC	-	International Electrotechnical Commission
IEEE	-	Institute of Electronic and Electrical Engineers
IGEOLO	-	Image Coordinate Location
ILOC	-	Image Location
IMODE	-	Image Mode
IPI	-	Image Processing and Interchange
IR	-	InfRared (ICAT value)
IREP	-	Image REPresentation
IREPBANDn	-	IREP nth BAND representation
IS	-	Image Segment
ISAR	-	Inverse Synthetic Aperture Radar
ISO	-	International Organization for Standardization
ISUBCATn	-	image nth band SUBCATegory
ITU	-	International Telecommunication Union
ITU-R	-	ITU Recommendation
ITU-T	-	ITU Telecommunications
IXSHD	-	Image eXtended SubHeader Data
JITC	-	Joint Interoperability Test Command
JPEG	-	Joint Photographic Experts Group
LDn	-	Length of nth Data extension segment
LEG	-	Legend (ICAT value)
LIn	-	Length of the nth Image segment
LISHn	-	Length of the nth Image SubHeader
LOC	-	Location
LOCG	-	LOCation Grid (ICAT value)
LSB	-	Least Significant Bit
LSn	-	Length of the nth graphic segment
LSSHn	-	Length of the nth graphic SubHeader
LUT	-	Look-Up Table
LVMI	-	Large Volume Motion Imagery
LVSD	-	Large Volume Streaming Data
LWIR	-	Longwave Infrared
M	-	Magnitude
M1	-	Compressed Bi-level (IC value)
M3	-	Compressed JPEG (IC value)
M4	-	Compressed Vector Quantization (IC value)
M5	-	Compressed Lossless JPEG (IC value)
M7	-	SARZIP compression (IC value)
M8	-	JPEG 2000 compression (IC value)
M9	-	Blocked H264 compression with time (IC value)
MA	-	Blocked H265 compression with time (IC value)
MB	-	Blocked JPEG 2000 compression with time (IC value)

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MAP	-	raster MAP (ICAT value)
MATR	-	MATRix data (ICAT value)
MGRS	-	Military Grid Reference System
MI	-	Motion Imagery
MIE4NITF	-	Motion Imagery Extension for NITF 2.1
MIL-STD	-	MILitary STandarD
MISB	-	Motion Imagery Standards Board
MISP	-	Motion Imagery Standards Profile
MMR	-	A 3-character code referring to the country that goes by either the name “Union of Burma” or “the Republic of the Union of Myanmar” depending on whether the application is following US or international guidance, respectively.
MONO	-	MONOchrome (IREP value)
MRI	-	Magnetic Resonance Imagery (ICAT value)
MS	-	MultiSpectral (ICAT value)
MSB	-	Most Significant Bit
MTF	-	Message Text Formatting (TXTFMT value)
MULTI	-	MULTIband Imagery (IREP value)
MWIR	-	Midwave Infrared
N	-	UTM/UPS Northern Hemisphere (ICORDS field value)
NATO	-	North Atlantic Treaty Organization
NBPC	-	Number of Blocks Per Column
NBPR	-	Number of Blocks Per Row
NC	-	No Compression
NELUTn	-	Number of LUT Entries for the nth image band
NGA	-	National Geospatial-Intelligence Agency (previously the National Imagery and Mapping Agency, NIMA)
NICOM	-	Number of Image COMments
NIR	-	Near-Infrared
NITF	-	National Imagery Transmission Format
NITFS	-	National Imagery Transmission Format Standard
NM	-	Uncompressed image indicating an image that contains a Block Mask or a Pad Pixel Mask (IC value)
NODISPLY	-	No DISPLaY (IREP value)
NPPBH	-	Number of Pixels Per Block Horizontal
NPPBV	-	Number of Pixels Per Block Vertical
NSIF	-	NATO Secondary Imagery Format
NSIL	-	NATO Standard Image Library
NTB	-	NITFS Technical Board
NUMDES	-	NUMber of Data Extension Segments
NUMI	-	NUMber of Image segments
NUMRES	-	NUMber of Reserved Extension Segments
NUMS	-	NUMber of graphic segments
NUMT	-	NUMber of text segments
NUMX	-	Reserved for Future Use
NVECTOR	-	VECTOR with Cartesian coordinates (IREP value)
NY	-	New York
OADR	-	Originating Agency Determination Required
ONAME	-	Originator’s NAME
OP	-	Optical (ICAT value)
OPHONE	-	Originator’s PHONE number
P	-	(1) Phase (2) band interleaved by Pixel (IMODE value)
PA	-	PennsylvaniA

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PAN	-	Panchromatic
PAT	-	color PATch (ICAT value)
PIXQUAL	-	PIXel QUALity image segment
PJUST	-	Pixel JUSTifcation
P.O.	-	Post Office
POLAR	-	Vector with POLAR coordinates (IREP value)
PROPIN	-	Proprietary Information
Q	-	Quadrature
R	-	(1) Row (2) band interleaved by Row (IMODE value) (3) Required
RD	-	RaDar (ICAT value)
RDUCE		Reduction of Data Using Compression Enhancements
RE	-	Registered Extension
REDATA	-	RES user-defined DATA
RES	-	Reserved Extension Segment
RESSHF	-	RES user-defined Subheader Fields
RESSHL	-	RES user-defined SubHeader Length
RETAG	-	Registered unique Extension Type identifier
RGB	-	Red, Green, Blue (IREP value)
RGB/LUT	-	Red, Green, Blue/Look-Up Table, mapped color (IREP value)
RS	-	Reserved Segment
Rset	-	Reduced resolution data set
RTF	-	Rich Text Format
S	-	(1) band Sequential (IMODE field value) (2) UTM/UPS Southern Hemisphere (ICORDS field value) (3) Secret (security fields value)
SAR	-	Synthetic Aperture Radar (ICAT value)
SARIQ	-	SAR radio hologram (ICAT value)
SARZip	-	Synthetic Aperture Radar (SAR) Compression
SBND1	-	Graphic BouND 1
SBND2	-	Graphic BouND 2
SDE	-	Support Data Extension (a TRE or a DES)
SDIF	-	SGML Document Interchange Format
SDLVL	-	Graphic Display LeVeL
SGML	-	Standard Generalized Mark-up Language
SI	-	International System of units (the modern metric system)
SID	-	(1) Secondary Imagery Dissemination System (2) Graphic Identifier
SIDS	-	Secondary Imagery Dissemination System
SL	-	Side-Looking radar (ICAT value)
SLOC	-	Graphic LOCation
STA	-	Standard (TXTFMT value)
STANAG	-	STANdardization AGreement
STYPE	-	Standard TYPE
SWIR	-	Shortwave Infrared
SXSHD	-	Graphic extended SubHeader Data
T	-	IMODE field value, temporal band interleave by block
TI	-	Thermal Infrared (ICAT value)
TMRnBNDm	-	n^{th} Pad Pixel Mask Record for BAND m
TPXCD	-	Pad Output Pixel Code
TPXCDLNTH	-	Pad Output Pixel Code Length
TRE	-	Tagged Record Extension
TRE_OVERFLOW	-	TRE overflow (DESID value)

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TS	-	(1) Text Segment
TXSHD	-	Text eXtended SubHeader Data
TXTFMT	-	TeXT ForMaT
U8S	-	UTF-8 Subset
UCS	-	Universal multiple octet coded Character Set
UDHD	-	User-Defined Header Data
UDHDL	-	User-Defined Header Data Length
URI	-	Uniform Resource Identifier
URL	-	UniversalUniform Resource Locator
URN	-	Uniform Resource Name
US	-	United States
USMTF	-	US Message Text Formatting
UTC	-	Coordinated Universal Time
UTF	-	UCS Transformation Format
UTM/UPS	-	Universal Transverse Mercator/Universal Polar Stereographic
UV	-	Ultraviolet
VA	-	VirginiA
VD	-	ViDeo (ICAT value)
VDC	-	Virtual Display Coordinates
VIS	-	VISible Imagery (ICAT value)
VNIR	-	Visible to Near-Infrared
VPH	-	Video Phase History (IREP value)
VQ	-	Vector Quantization
WGS	-	World Geodetic System

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APPENDIX A

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APPENDIX B. Definitions (Normative)

B.1. DEFINITIONS

B.1.1 ScopeThe appendix details the definitions in this document. The following terms and definitions are used for the purpose of this standard. All used concepts (file, field, segment, etc.) exclusively refer to the NITF standard. For concepts for which this is not correct a corresponding firm intention has been chosen (for example: system field, BIIF file, etc.). This appendix is a mandatory part of this standard. The information contained herein is intended for guidance. B.2 DEFINITIONS

- Associated Data. That related data required for completeness of the standard.
- Attachment Level (ALVL). A way to associate images and graphics to the same level during movement, rotation, or display.
- Band. A well-defined range of wavelengths, frequencies or energies of optical, electric, or acoustic radiation. At the pixel level, a band is represented as one of the vector values of the pixel.
- Bandwidth. 1. The difference between the limiting frequencies or wavelengths within which performance of a device, in respect to some characteristic, falls within specified limits. 2. The difference between the limiting frequencies or wavelengths of a continuous frequency or wavelength band.
- Base Image. A base image is the principle image of interest or focus for which other data may be inset or overlaid. The NITF file can have none, one, or multiple base images. For multiple base images in a single NITF file, the relative location of each base image is defined in the image location (ILOC) field in each image subheader. This location will be the offset within the Common Coordinate System (CCS).
- Basic Character Set (BCS). A subset of the Extended Character Set (ECS). The most significant bit of the BCS characters is set to 0. The range of valid BCS characters code is limited to 0x20 to 0x7E plus line feed (0x0A), form feed (0x0C), and carriage return, (0x0D).
- Basic Character Set-Alphanumeric (BCS-A). A subset of the Basic Character Set. The range of allowable characters consists of space to tilde, codes 0x20 to 0x7E.
- Basic Character Set-Numeric Integer (BCS-N integer). A subset of the Basic Character Set-Numeric (BCS-N) comprising the digits 0 to 9 (codes 0x30 to 0x39), plus sign (0x2B) and minus sign (0x2D).
- Basic Character Set – Numeric Positive Integer (BCS-N positive integer). A subset of the Basic Character Set Numeric (BCS-N) comprising the digits 0 to 9 (codes 0x30 to 0x39).
- BCS Space. BCS (and consequently ECS) code 0x20.
- Block. A block is a rectangular array of pixels. (Synonymous with tile.)
- Blocked Image. A blocked image is composed of the union of one or more non-overlapping blocks. (Synonymous with tiled image.)
- Blocked Image Mask. A structure which identifies the blocks in a blocked (tiled) image which contain no valid data, and which are not included in the file. The structure allows the receiver to recognize the offset for each recorded/transmitted block. For example, a 2x2 blocked image file which contains no valid data in the second block (block 1) would be recorded in the order: block 0, block 2, block 3. The blocked image mask would identify block 1 as a non-existing block, and would allow the receiving application to construct the image in the correct order.
- Brightness. 1. An attribute of visual perception, in accordance with which a source appears to emit more or less light. 2. Incoming radiant flux per solid angle per unit area projected in the direction of the radiation. Measurement units are often Watts per meter-squared per steradian (W m-2 sr-1)
- Byte. A sequence of 8 adjacent binary digits.

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- Character. 1. A letter, digit, or other graphic that is used as part of the organization, control, or representation of data. 2. One of the units of an alphabet.
- Common Coordinate System (CCS). The virtual two dimensional Cartesian-like coordinate space which will be common for determining the placement and orientation of displayable data.
- Complexity Level (CLEVEL). A code used in the file header which signals the degree of complexity an interpret implementation needs to support to adequately interpret the files. Items that differentiate complexity include: size of the common coordinate system, file size, image size, image blocking, color of imagery with greater than 8 bits per pixel, number of bands in an image segment, number of image segments, aggregate size of graphic segments, etc.
- Conditional (C). A state applied to an NITF header or subheader data field whose existence and content is dependent on the existence and/or content of another field.
- Controlled Extension (CE). A tagged record extension (TRE) or data extension segment (DES) submitted for approval by the NTB and maintained under formal configuration management control. For a TRE, the extension type identifier (six-character CETAG field), and user-defined data (CEDATA field) structure are under configuration management control. For a DES, the DES name, DES identifier (25-character DESID field), DES version (2-character DESVER field), and the user-defined data (DESDATA field) structure are under configuration management control.
- Coordinated Universal Time (UTC). The time scale maintained by the International Earth Rotation Service (having previously been maintained by the Bureau International de l'Heure that forms the basis of a coordinated dissemination of standard frequencies and time signals.
- Data. 1. A representation of facts, concepts or instruction in a formalized manner suitable for communication, interpretation, or processing by human or by automatic means. 2. The basic building block of information. 3. A collection of facts, figures, or statistics
- Data Communication. The transfer of information between functional units by means of data transmission according to a protocol.
- Data Extension Segment (DES). A type of extension segment with sub-header and data fields structured similarly to the standard data types in the NITF (e.g., image, label, symbol, text). The extension type identifier (25 character DESTAG field), the version (two character DESVER field), and the full underlying structure is under configuration management control as registered with the NTB.
- Date Time Group (DTG). A composite representation of date and time.
- Digraph. A two letter reference code.
- Extended Character Set (ECS). A set of 1-byte encoded characters. Valid ECS character codes range from 0x20 to 0x7E, and 0xA0 to 0xFF, as well as Line Feed (0x0A), Form Feed (0x0C) and Carriage Return (0x0D). The ECS characters are described in TABLE B-1. As an interim measure, because of inconsistencies between standards, it is strongly advised that character codes ranging from 0xA0 to 0xFF should never be used. Therefore, the use of ECS characters should be restricted to its BCS Subset.
- Extended Character Set (ECS) Space. See BCS Space definition.
- Extended Character Set - Alphanumeric (ECS-A). A subset of the Extended Character Set (ECS). Valid ECS-A character codes range from 0x20 to 0x7E, and 0xA0 to 0xFF. Line Feed (0x0A), Form Feed (0x0C) and Carriage Return (0x0D) are not valid ECS-A characters. As an interim measure, because of inconsistencies between standards, it is strongly advised that character codes ranging from 0xA0 to 0xFF should never be used. Therefore, the use of ECS-A characters should be restricted to its BCS-A Subset.
- Field. Elementary set of meaningful data.
- Fill Pixel. A pixel that has relevance to the storage and transmission of a blocked image, but no relevance to the display of the blocked image. Fill pixels are used when NROWS (the number of pixel rows in an image)

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is not an integer multiple of the number of pixel rows per image block, or when NCOLS (the number of pixel columns in an image) is not an integer multiple of the number of pixel columns per block. Fill pixels are always outside the boundary specified by NROWS and NCOLS, and therefore are never displayed.

- Frame: For Motion Imagery, A rectangular array of samples/pixel values and an associated temporal index or coordinate. The temporal index/coordinate of a frame is the temporal index/coordinate associated with the playback or viewing of a frame within a temporal sequence of frames.

- Graphic. Graphic data is used in the NITF to store two-dimensional information represented as a Computer Graphics Metafile (CGM). Each graphic segment consists of a symbol subheader and data. A graphic may be black and white, grey scale, or color. Examples of graphics are circles, ellipses, rectangles, arrows, lines, triangles, logos, unit designators, object designators (ships, aircraft), text, special characters**,** or combination thereof. A graphic is stored as a distinct unit in the NITF file allowing it to be manipulated and displayed nondestructively relative to the images, and other graphics in the file. This standard does not preclude the use of n-dimensional graphics when future standards are developed.

- Grey scale. An optical pattern consisting of discrete steps or shades of grey between black and white.

- Hyperspectral Imagery (HSI): Imagery acquired using tens to hundreds of spectral bands with narrow EO spectral bandwidth. Bands are acquired simultaneously or nearly so, and the spatial coverage for each band may differ slightly from band to band. Spectral bands are often but not necessarily contiguous, and may be acquired from one or more portions of the EO spectrum, enabling construction of an effectively continuous reflectance (emittance in the case of IR energy) spectrum for every pixel in the scene.

- Image. A two-dimensional rectangular array of pixels indexed by row and column. Also known as a Still Image.

- Image Codes. For a vector quantized image file, values in the image data section that are used to retrieve the v x h kernels from the image code book.

- Imagery. Collectively, the representations of objects reproduced electronically or by optical means on film, electronic display devices, or other media.

- Imagery Associated Data. Data which is needed to properly interpret and render pixels; data which is used to annotate imagery such as text, graphics, etc.; data which describes the imagery such as textual reports; and data which support the exploitation of imagery.

- Interface. 1. A concept involving the definition of the interconnection between two equipment items or systems. The definition includes the type, quantity, and function of the interconnecting circuits and the type, form, and content of signals to be interchanged via those circuits. Mechanical details of plugs, sockets, and pin numbers, etc., can be included within the context of the definition. 2. A shared boundary, e.g., the boundary between two subsystems or two devices. 3. A boundary or point common to two or more similar or dissimilar command and control systems, subsystems, or other entities against which or at which necessary information flow takes place. 4. A boundary or point common to two or more systems or other entities across which useful information flow takes place. (It is implied that useful information flow requires the definition of the interconnection of the systems which enables them to interoperate.) 5. The process of interrelating two or more dissimilar circuits or systems 6. The point of interconnection between user terminal equipment and commercial communication-service facilities.

- Interoperability. 1) The ability of systems, units, or forces to provide services to and accept services from other systems, units, or forces and to use the services so exchanged to enable them to operate effectively together. 2) The ability to operate in synergy in the execution of assigned tasks. (JP 3-0). 3) The condition achieved among communications-electronics systems or items of communications-electronics equipment when information or services can be exchanged directly and satisfactorily between them and/or their users. (JP 6-0).

- Kernel. For a vector quantized image file, a rectangular group of pixels used in the organization of quantizing image data.

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- Look-Up Table (LUT). A collection of values used for translating image samples from one value to another. The current sample value is used as an index into the look-up table(s); therefore, the number of entries in each look-up table for a binary image would contain two entries, and each look-up table for an 8-bit image would contain 256 entries. Multiple look-up tables allow for the translation of a 1-vector pixel value to an n-vector pixel value.
- Magnification. The multiplication factor which causes an apparent change in linear distance between two points in an image. Thus a magnification of 2 is a change which doubles the apparent distance between two points (multiplying area by 4), while a magnification of 0.5 is a change which halves the apparent distance.
- Military Grid Referencing System (MGRS). A means of expressing Universal Transverse Mercator (UTM) or Universal Polar Stereographic (UPS) coordinates as a character string, with the 100-kilometer components replaced by special letters (which depend on the UTM or UPS zone and ellipsoid).
- Motion Imagery. Motion imagery is a temporal sequence of frames that when viewed or processed provide information or intelligence value. Motion imagery frames are spatiotemporally related to one another. The spatial relationship is satisfied when each successive image maintains some recognizable visual overlap with its previous image. The temporal relationship dictates that the capture time (i.e., time stamp) of each successive frame be sequentially ordered. Both the space and time relationships exist in Motion Imagery.
- Multispectral Imagery (MSI): Imagery collected in at least 2 or more relatively wide bands of the EO spectrum. The spectral bands may or may not be contiguous in wavelength. Multispectral sensors often collect bands from discrete portions of the EO spectrum, which are often selected to take advantage of high atmospheric transmission. Bands are collected simultaneously or nearly so, and the spatial coverage for each band may differ slightly from band to band. Spectral bands may be acquired from one or more portions of the EO spectrum. Multispectral systems tend to collect tens of bands. Most multispectral images have fewer than 70 bands, and many satellite sensors collect fewer than 10 bands.
- Network. 1. An interconnection of three or more communicating entities and (usually) one or more nodes. 2. A combination of passive or active electronic components that serves a given purpose.
- Non-blank. Non-blank indicates that the field cannot be filled by the character space (0x20) but may contain the character space when included with other characters. (embedded blanks)
- Null. The field is populated entirely with spaces (0x20).
- 0x_. Hexadecimal notation.
- Pack Capable System. A system which is capable of generating an NITF file.
- Pad Pixel. A non-intelligent pixel that is within the significant image pixels defined by NROWS and NCOLS. The numerical value of the pad pixel is specified by the NITF Image Data Mask Table, Pad Output Pixel Code (TPXCD) field, which can only be present when the compression code (IC) includes the letter “M.” Pad pixels can be located anywhere inside the significant image pixels defined by NROWS and NCOLS, and therefore impact the display of the image.
- Pad Pixel Mask. A data structure which identifies recorded/transmitted image blocks which contain pad pixels. The pad pixel mask allows applications to identify image blocks which require special interpretation due to pad pixel content.
- Panchromatic (Pan) Imagery: A single band image covering a broad range of electro-optical wavelengths, usually used in context of collecting information from much of the visible spectrum and sometimes part of the near infrared spectrum.
- Parity. In binary-coded systems, the oddness or evenness of the number of ones in a finite binary stream. It is often used as a simple error-detection check and will detect (but not correct) the occurrences of any single bit error in the field.

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- Pixel. A pixel is represented by an n-vector of sample values, where n corresponds to the number of bands comprising the image.
- Primary Imagery. Unexploited, original imagery data that has been derived directly from a sensor. Elementary processing may have been applied at the sensor, and the data stream may include auxiliary data.
- Processed Imagery. Imagery that has been formatted into image pixel format, enhanced to remove detected anomalies, and converted to a format appropriate for subsequent disposition.
- Protocol. 1. [In general], A set of semantic and syntactic rules that determines the behavior of functional units in achieving communication. For example, a data link protocol is the specification of methods whereby data communication over a data link is performed in terms of the particular transmission mode, control procedures, and recovery procedures. 2. In layered communication system architecture, a formal set of procedures that are adopted to facilitate functional interoperation within the layered hierarchy. Note: Protocols may govern portions of a network, types of service, or administrative procedures.
- Pseudocolor. A user-defined mapping of N bits into arbitrary colors.
- Reconstruction. For a vector quantized image file, the process of transforming an image from a quantized form into a displayable and exploitable form.
- Registered Extension (RE). Those tagged record extensions for which the extension type identifier (six character RETAG field) and the user-defined data (REDATA field) structure is registered with the NTB. The user-defined data (REDATA field) structure is not controlled by the NTB.
- Reserved Extension Segment (RES). A type of extension segment with sub-header and data fields structured similarly to the standard data types in the NITF (e.g., image, label, symbol, text). The extension type identifier (25 character RESTAG field), the version (two character RESVER field), and the full underlying structure is under configuration management control as registered with the NTB. The RES construct provides the same mechanism as the DES construct for adding a variety of new data types for inclusion in NITF files. However, the RES is reserved for data types that need to be placed at or near the end of the file. For example, a digital signature that covered the whole file could be defined for placement in a RES to verify the bit level integrity of the NITF file.
- Required. An NITF header or subheader field that must be present and filled with valid data.
- Resolution. 1. The minimum difference between two discrete values that can be distinguished by a measuring device. 2. The degree of precision to which a quantity can be measured or determined. 3. A measurement of the smallest detail that can be distinguished by a sensor system under specific conditions. Note: High resolution does not necessarily imply high accuracy.
- Sample. The atomic element of an image pixel having a discrete value. One sample from the same location in each band comprising an image will combine to form a pixel.
- Secondary Imagery. Secondary Imagery is digital imagery and/or digital imagery products derived from primary imagery or from the further processing of secondary imagery.
- Secondary Imagery Dissemination (SID). The process of dispersing or distributing digital secondary imagery.
- Secondary Imagery Dissemination System (SIDS). The equipment and procedures used in secondary imagery dissemination.
- Segment. A header and data fields.
- Still Image. A two-dimensional rectangular array of pixels indexed by row and column.
- Tagged Record Extension (TRE). A set of fields to support user defined and extended data.
- Text. Information conveyed as characters.

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- Tile. Synonymous with Block
- Universal Multiple Octet Coded Character Set (UCS). The Universal Multiple Octet Coded Character Set is used for expressing text that must be human readable, potentially in any language of the world. It is defined in ISO/IEC 10646.
- Universal Multiple Octet Coded Character Set (UCS) Transformation Format 8 (UTF-8). UTF-8 is a coded representation form for all of the characters of the UCS. In the UTF-8 coded representation form each character from this UCS has a coded representation that comprises a sequence of octets of length 1, 2, 3, 4, 5, or 6 octets.
- Universal Multiple Octet Coded Character Set (UCS) Transformation Format 8 (UTF-8) Subset (U8S). A Subset of the UCS composed of 1-byte and 2-byte UTF-8 encoded characters (Basic Latin and Latin Supplement 1). The 1-byte encoded characters of the UTF-8 Subset (U8S) are the BCS characters. The 2-byte encoded characters of U8S are described in TABLE B-2.
- Universal Polar Stereographic (UPS). A pair of grids, one used north of 84° north and one used south of 80° south. Each grid is based on the polar stereographic projection. The actual grid depends on the choice of the geodetic datum.
- Universal Transverse Mercator (UTM). A system of grids for global use between latitudes 84 degrees North and 80 degrees South. The range of longitudes 180 degrees West to 180 degrees East is divided into 60 zones, each of which is a grid based on the Transverse Mercator projection. (Within each zone, there is a difference in coordinate systems either side of the Equator. On the northern side, northings start from zero at the Equator; on the southern side, northings are positive rising to 10 million at the Equator.) The actual grid depends on the choice of geodetic datum as well as the zone.
- Unpack Capable System. A system which is capable of receiving/processing an NITF file.
- Vector Quantization (VQ). A compression technique in which many groups of pixels in an image are replaced by a smaller number of image codes. A clustering technique is used to develop a code book of "best fit" pixel groups to be represented by the codes. Compression is achieved because the image codes can be recorded using fewer bits than the original groups of pixels they represent.
- vsize. For a vector quantized image file, the size of the kernel in pixels.
- v x h kernel. For a vector quantized image file, a rectangular group of pixels (kernels) with v-rows and h-columns.

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APPENDIX C. References (Normative)

C.1. REFERENCES

C.1.1 Scope. The documents listed in this appendix are specified in sections 3, 4, or 5 of this standard. This appendix does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

C.2. GOVERNMENT DOCUMENTS.

C.2.1 Specifications, Standards, and Handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NATO STANDARDIZATION AGREEMENTS

STANAG 4545	-	NATO Secondary Image Format (NSIF), Edition 2, 6 May 2013

(Copies of these standards are available online at http://www.gwg.nga.mil/ntb/baseline/documents.html)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-188-198A	-	Joint Photographic Experts Group (JPEG) Image Compression for the National Imagery Transmission Format Standard, through NOTICE 4
MIL-STD-188-199	-	Vector Quantization Decompression for the National Imagery Transmission Format Standard, through NOTICE 2
MIL-STD-6040D	-	United States Message Text Format (USMTF) Description

(Copies of these documents are available online at http://quicksearch.dla.mil.)

DEFENSE GEOSPATIAL INFORMATION WORKING GROUP

DIGEST	-	Digital Geographic Information Exchange Standard (DIGEST)

(Copies of these documents are available online at https://www.dgiwg.org/dgiwg/)

C.2.2. Other Government Documents, Drawings, and Publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

U.S. PRESIDENT EXECUTIVE ORDERS

E.O 13526	-	Classified National Security Information

DEPARTMENT OF DEFENSE JOINT PUBLICATIONS

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JP 1-02	-	Department of Defense Dictionary of Military and Associated Terms 8 November 2010 (As Amended Through 15 February 2016)

DEPARTMENT OF DEFENSE REGULATION

DOD 5200.01 Vol 1 -4	-	Department of Defense Information Security Program Regulation, February 24, 2012
DODI 8330.01		Interoperability of Information Technology (IT), Including National Security Systems (NSS), 21 May 2014

NATIONAL GEOSPATIAL-INTELLIGENCE AGENCY PUBLICATIONS

NGA.STND.0036	-	Department of Defense World Geodetic System 1984: Its Definition and Relationships with Local Geodetic Systems, Version 1.0.0, 8 July 2014
N-0105/98	-	National Imagery Transmission Format Standard (NITFS) Standards Compliance and Interoperability Test and Evaluation Program Plan, 25 August 1998
NGA.STND.0033_3.0.1_GENC	-	Geopolitical Entities, Name and Codes (GENC) Standard, 2016-07-08, Edition 3.0.1
GEC	-	Geopolitical Entities and Codes (formerly FIPS Publication 10-4), 15 April 2010; http://geonames.nga.mil/gns/html/PDFDocs/GEOPOLITICAL_CODES.pdf
GEC Update 18	-	All change notices to Geopolitical Entities and Codes:2010, through: Geopolitical Entities and Codes (formerly FIPS Publication 10-4) Update 18, 31 December 2014: http://geonames.nga.mil/gns/html/PDFDocs/UPDATE18.pdf
NGA.STND.0044	-	Motion Imagery Extension for NITF 2.1.
NGA.RP.0001	-	Recommended Practices for Universally Unique Identifiers
STDI-0002-1	-	The Compendium of Controlled Extensions (CE) for the National Imagery Transmission Format (NITFS) Volume 1 Tagged Record Extensions, Version 4.0, 01 August 2011
STDI-0002-2	-	The Compendium of Controlled Extensions (CE) for the National Imagery Transmission Format (NITFS) Volume 2 Data Extension Segments, Version 1.0, 03 December 2012

(The NGA standards listed above are normative in this document. The current version of these documents may be found at https://nsgreg.nga.mil/ntb.jsp, https://nsgreg.nga.mil/misb.jsp, and https://nsgreg.nga.mil/mfg.jsp or through the NSG Standards Registry search at https://nsgreg.nga.mil/doc/search/index.jsp)

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ISMCAT	-	Information Security Marking Country Codes and Tetragraphs http://w3id.org/ic/standards/public/ismcat

UNITED STATE AIR FORCE PUBLICATIONS

USAF.RDUCE-001	-	Synthetic Aperture RADAR Image Compression Standard (SARzip) https://intelshare.intelink.gov/sites/rduce/

ONLINE REGISTRIES

NSG Standards Registry	-	https://nsgreg.nga.mil
NITF DES Registry	-	http://jitc.fhu.disa.mil/projects/nitf/registers/desreg.aspx
NITF Field Value Registry	-	http://jitc.fhu.disa.mil/projects/nitf/registers/nitfcte/trereg.aspx
NITF TRE Registry	-	http://jitc.fhu.disa.mil/projects/nitf/registers/trereg.aspx
NATO NSIF Registry	-	https://nsgreg.nga.mil/doc/registers.jsp?register=NATO
NATO Repository	-	http://nso.nato.int/nso/

C.2.3 Non-Government Publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

INTERNATIONAL TELECOMMUNICATION UNION

ITU-R BT.6/270 (Rev1)	-	Studio encoding parameters of digital television for standard 4:3 and wide-screen 16:9 aspect ratios, 03/2011
ITU-T T.4	-	Standardization of group 3 facsimile apparatus for document transmission, AMD2 07/2003

(Copies of these documents can be obtained online through http://www.itu.int.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION

ISO 80000-1: 2009/Cor 1:2011	-	Quantities and Units – Part 1: General
ISO/IEC 7498-1	-	Information technology - Open systems interconnection; Basic reference model - Part 1: The basic model, 1994
ISO/IEC 10918-1/Cor 1: 2005	-	Information technology - Digital compression and coding of continuous-tone still images - Part 1: Requirements and guidelines; 1994
ISO/IEC 12087-5:1998/Cor 1: 2001/Cor 2: 2002	-	Information technology - Computer graphics and image processing -Image processing and interchange (IPI) - Functional specification - Part 5: Basic image interchange format (BIIF), 1 December 1998

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ISO/IEC 15444-1:2004/Amd 7: 2015	-	Information technology - JPEG 2000 image coding system - Part 1: Core Coding System
ISO/IEC 15444-4:2004/Cor 1: 2009	-	Information technology - JPEG 2000 image coding system - Part 4: Image Coding System: Conformance testing
BPCGM01.00	-	ISO/IEC BIIF Profile for Computer Graphics Metafile
BPJ2K01.10	-	Information technology - Computer graphics and image processing - registered graphical item - Class: BIIF Profile - BIIF Profile for JPEG 2000 Version 01.00 (BPJ2K01.10), 15 April 2009
ISO/IEC BIIF PROFILE NSIF01.01	-	Information Technology - Computer Graphics and Image Processing -Registered Graphical Item, Class: BIIF Profile- NATO Secondary Imagery Format Version 01.01 (NSIF01.01)

(Copies of these documents can be obtained online through http://www.ansi.org, http://www.iso.org, http://www.iec.ch, and http://www.itu.int. or from American National Standards Institute, 4th Floor, 25 West 43nd Street, New York, NY 10036.)

INSTITUTE OF ELECTRONIC AND ELECTRICAL ENGINEERS STANDARD

IEEE 754-2008	-	IEEE Standard for binary floating-point arithmetic

(Copies of IEEE documents can be ordered online through http://ieee.org or from Customer Service, 445 and 501 Hoes Lane, Piscataway, NJ 08854-4141.)

C.2.4 Order of Precedence Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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APPENDIX D. Implementation Considerations (Normative)

D.1. IMPLEMENTATION CONSIDERATIONS

D.1.1 Scope. The appendix details the implementation considerations in this document. This appendix is a mandatory part of this standard. The information contained herein is intended for guidance.

D.2 GENERAL REQUIREMENTS

D.2.1 Scope of NITF Implementation. NITF describes the format of images and graphics and text within the NITF file only. It does not define the image or text requirements of the host system. The host system is responsible for the handling of unpacked image and text files, as well as image and text display capabilities.

D.2.2 Creating Headers and Subheaders. This standard specifies legal values for the header and subheader fields. The NITF preprocessor for any particular host system will be responsible for enforcing the field values as stated in this standard.

D.2.3 Character Counts. The NITF uses explicit byte counts to delimit fields. No end-of-field characters are used. These byte counts are critical for the proper interpretation of an NITF file. The NITF preprocessor should compute these byte counts based on file contents to insure accuracy. All fields in the NITF header and subheaders must be present exactly as specified in the NITF header and subheader descriptions, and no additional fields may be inserted. The NITF uses various conditional fields whose presence is determined by previous fields and counts. If an expected conditional field is missing, the remainder of the file will be misinterpreted. A similar result will occur if a conditional field is inserted when it is not required. For these reasons, the item count fields are critical, and every effort must be made to ensure their accuracy. The NITF preprocessor should compute these item counts based on file contents whenever possible.

D.2.4 Data Entry. To reduce any operator workload imposed by the preprocessor, each preprocessor should provide for the automatic entry of data. Global default values for the particular NITF version should be inserted automatically in the file. System default values, such as the standard size parameters for a base image, also should be entered automatically by the preprocessor. Values that are known to the system, such as the time or the computed size of an overlay, also should be entered automatically.

D.2.5 Out of Bounds Field Values The file creator is responsible for ensuring that all NITF field values are within the bounds specified by the NITF document.

D.2.6 Use of Images in NITF. The NITF specifies a format for images contained within an NITF file only. An NITF implementation must be capable of translating this format to and from the host system's local format. Some host systems have multiple formats for binary data. In these cases, the NITF implementation must use the appropriate host format to provide the necessary data exchange services with other system packages. When imagery data of less than M bits-per-pixel is displayed on an M-bit (2^M grey shades) display device, it must be transformed into the dynamic range of the device. One way to do this is to modify the LUTs of the display device. However, if M-bit and less than M-bit imagery is displayed simultaneously, the M-bit image will appear distorted. The recommended method is to convert the less than M-bit imagery into M-bit imagery, then use the standard LUTs. The following equation will transform a less than M-bit pixel into an M-bit pixel:

N = number of bits-per-pixel

P_N = N-bit pixel value

P_M = M-bit pixel value

$$P_M = \frac{2^M - 1}{2^N - 1} P_N$$

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D.2.7 Use of Text Files in the NITF. The TXTFMT field (TABLE 6) is provided to help the NITF file reader determine how to interpret the text data received. The NITF file reader is responsible for interpreting the various text data formats and associated character sets. Character set designations explicitly supported by the NITF are addressed in paragraphs 5.1.7 and 5.7.

D.2.7.1 Formatted Documents. The TS is intended to convey plain text, not marked up text typical of word processed documents. In the future, formatted documents (e.g., Standardized Graphic Mark-up Language (SGML), Hypertext Mark-up Language (HTML), Rich Text Format (RTF), etc.) may be accommodated using a specialized DES. However, at the time of publication, a DES to contain formatted documents has not been defined. Should such DES be developed, they must be submitted through the registration process.

D.2.8 Converting Color to Grey ScaleFull color may be specified as the file background and for various attributes of segments within an NITF file (e.g., color imagery and color annotations). Color items for receiving systems unable to support the presentation of full colors must be mapped to colors that are able to be supported and displayed.

D.2.8.1 Eight-Bit Grey Scale Presentation. For 8-bit grey scale systems an appropriate conversion is:

GREY (8-bit) = 0.299*RED + 0.587*GREEN + 0.114*BLUE

D.2.8.2 One-Bit Grey Scale Presentation. For 1-bit bi-tonal (e.g. black and white) systems, an appropriate conversion is to first calculate the grey scale conversion as shown above. Then,

BITONE(1-bit) = 1 (white), when GREY (8-bit) > 127

BITONE(1-bit) = 0 (black), when GREY (8-bit) <= 127

D.2.8.3 Greater Than Eight-Bit Grey Scale Presentation. For 8+ bit grey scale systems, color components can first be converted to 8-bit grey scale followed by a dynamic range adjustment to the bit range supported by the presentation device.

D.2.8.4 Washout. The potential exists for overlays to be inadvertently hidden or washed out when compared to the background over which they are placed, particularly when converting from color to grey scale. The application developer should take a design approach that obviates the potential for a recipient to inadvertently overlook presentation material caused by inadequate lack of contrast in the presentation.

D.2.9 File System Constraints. An NITF file is presented as a stream of contiguous bytes. This format may not be suitable for some file systems (e.g., those that store files on block boundaries vice byte boundaries). The translation of files to and from the local file format for a system should be examined for potential incompatibilities before an implementation is attempted.

D.2.10 Security Considerations. An NITF file contains sufficient security information in the file header, and subheaders to allow implementers to meet virtually any security requirement for displaying classification data. Exact security information handling requirements generally are specified by appropriate accreditation authorities or specific user requirements. It is recommended that implementors extract the classification data from the header and ensure that the information is always displayed whenever the NITF file or any of its segments is displayed. Implementations should not rely on graphic overlays alone to present security and handling instructions. Panning, roaming, zooming, and other imagery manipulation operations may cause security label graphics to move off the screen or not be printed.

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APPENDIX D

TABLE D-1. NITF 1-Byte Coded Characters

		Code			Character Set							
Char	Name	Dec	Hex	Binary	U8S	ECS	ECS-A	BCS	BCS-A	BCS-N	BCS-N Integer	BCS-N Positive Integers
	NOT USED	000	00	0000 0000								
	NOT USED	001	01	0000 0001								
	NOT USED	002	02	0000 0010								
	NOT USED	003	03	0000 0011								
	NOT USED	004	04	0000 0100								
	NOT USED	005	05	0000 0101								
	NOT USED	006	06	0000 0110								
	NOT USED	007	07	0000 0111								
	NOT USED	008	08	0000 1000								
	NOT USED	009	09	0000 1001								
	LINE FEED	010	0A	0000 1010	X	X	X	X				
	NOT USED	011	0B	0000 1011								
	FORM FEED	012	0C	0000 1100	X	X	X	X				
	CARRIAGE RETURN	013	0D	0000 1101	X	X	X	X				
	NOT USED	014	0E	0000 1110								
	NOT USED	015	0F	0000 1111								
	NOT USED	016	10	0001 0000								
	NOT USED	017	11	0001 0001								
	NOT USED	018	12	0001 0010								
	NOT USED	019	13	0001 0011								
	NOT USED	020	14	0001 0100								
	NOT USED	021	15	0001 0101								
	NOT USED	022	16	0001 0110								
	NOT USED	023	17	0001 0111								
	NOT USED	024	18	0001 1000								
	NOT USED	025	19	0001 1001								

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APPENDIX D

TABLE D-1. NITF 1-Byte Coded Characters – continued.

		Code			Character Set							
Char	Name	Dec	Hex	Binary	U8S	ECS	ECS-A	BCS	BCS-A	BCS-N	BCS-N Integer	BCS-N Positive Integers
	NOT USED	026	1A	0001 1010								
	NOT USED	027	1B	0001 1011								
	NOT USED	028	1C	0001 1100								
	NOT USED	029	1D	0001 1101								
	NOT USED	030	1E	0001 1110								
	NOT USED	031	1F	0001 1111								
	SPACE	032	20	0010 0000	X	X	X	X	X			
!	EXCLAMATION MARK	033	21	0010 0001	X	X	X	X	X			
"	QUOTATION MARK	034	22	0010 0010	X	X	X	X	X			
#	NUMBER SIGN	035	23	0010 0011	X	X	X	X	X			
$	DOLLAR SIGN	036	24	0010 0100	X	X	X	X	X			
%	PERCENT SIGN	037	25	0010 0101	X	X	X	X	X			
&	AMPERSAND	038	26	0010 0110	X	X	X	X	X			
'	APOSTROPHE	039	27	0010 0111	X	X	X	X	X			
(	LEFT PARENTHESIS	040	28	0010 1000	X	X	X	X	X			
)	RIGHT PARENTHESIS	041	29	0010 1001	X	X	X	X	X			
*	ASTERISK	042	2A	0010 1010	X	X	X	X	X			
+	PLUS SIGN	043	2B	0010 1011	X	X	X	X	X	X	X	
,	COMMA	044	2C	0010 1100	X	X	X	X	X			
-	HYPHEN-MINUS	045	2D	0010 1101	X	X	X	X	X	X	X	
.	FULL STOP	046	2E	0010 1110	X	X	X	X	X	X		
/	SOLIDUS	047	2F	0010 1111	X	X	X	X	X	X		

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APPENDIX D

TABLE D-1. NITF 1-Byte Coded Characters – continued.

		Code			Character Set							
Char	Name	Dec	Hex	Binary	U8S	ECS	ECS-A	BCS	BCS-A	BCS-N	BCS-N Integer	BCS-N Positive Integers
0	DIGIT ZERO	048	30	0011 0000	X	X	X	X	X	X	X	X
1	DIGIT ONE	049	31	0011 0001	X	X	X	X	X	X	X	X
2	DIGIT TWO	050	32	0011 0010	X	X	X	X	X	X	X	X
3	DIGIT THREE	051	33	0011 0011	X	X	X	X	X	X	X	X
4	DIGIT FOUR	052	34	0011 0100	X	X	X	X	X	X	X	X
5	DIGIT FIVE	053	35	0011 0101	X	X	X	X	X	X	X	X
6	DIGIT SIX	054	36	0011 0110	X	X	X	X	X	X	X	X
7	DIGIT SEVEN	055	37	0011 0111	X	X	X	X	X	X	X	X
8	DIGIT EIGHT	056	38	0011 1000	X	X	X	X	X	X	X	X
9	DIGIT NINE	057	39	0011 1001	X	X	X	X	X	X	X	X
:	COLON	058	3A	0011 1010	X	X	X	X	X			
;	SEMICOLON	059	3B	0011 1011	X	X	X	X	X			
<	LESS-THAN SIGN	060	3C	0011 1100	X	X	X	X	X			
=	EQUALS SIGN	061	3D	0011 1101	X	X	X	X	X			
>	GREATER-THAN SIGN	062	3E	0011 1110	X	X	X	X	X			
?	QUESTION MARK	063	3F	0011 1111	X	X	X	X	X			
@	COMMERCIAL AT	064	40	0100 0000	X	X	X	X	X			
A	LATIN CAPITAL LETTER A	065	41	0100 0001	X	X	X	X	X			
B	LATIN CAPITAL LETTER B	066	42	0100 0010	X	X	X	X	X			
C	LATIN CAPITAL LETTER C	067	43	0100 0011	X	X	X	X	X			
D	LATIN CAPITAL LETTER D	068	44	0100 0100	X	X	X	X	X			
E	LATIN CAPITAL LETTER E	069	45	0100 0101	X	X	X	X	X			
F	LATIN CAPITAL LETTER F	070	46	0100 0110	X	X	X	X	X			

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APPENDIX D

TABLE D-1. NITF 1-Byte Coded Characters – continued.

		Code			Character Set							
Char	Name	Dec	Hex	Binary	U8S	ECS	ECS-A	BCS	BCS-A	BCS-N	BCS-N Integer	BCS-N Positive Integers
G	LATIN CAPITAL LETTER G	071	47	0100 0111	X	X	X	X	X			
H	LATIN CAPITAL LETTER H	072	48	0100 1000	X	X	X	X	X			
I	LATIN CAPITAL LETTER I	073	49	0100 1001	X	X	X	X	X			
J	LATIN CAPITAL LETTER J	074	4A	0100 1010	X	X	X	X	X			
K	LATIN CAPITAL LETTER K	075	4B	0100 1011	X	X	X	X	X			
L	LATIN CAPITAL LETTER L	076	4C	0100 1100	X	X	X	X	X			
M	LATIN CAPITAL LETTER M	077	4D	0100 1101	X	X	X	X	X			
N	LATIN CAPITAL LETTER N	078	4E	0100 1110	X	X	X	X	X			
O	LATIN CAPITAL LETTER O	079	4F	0100 1111	X	X	X	X	X			
P	LATIN CAPITAL LETTER P	080	50	0101 0000	X	X	X	X	X			
Q	LATIN CAPITAL LETTER Q	081	51	0101 0001	X	X	X	X	X			
R	LATIN CAPITAL LETTER R	082	52	0101 0010	X	X	X	X	X			
S	LATIN CAPITAL LETTER S	083	53	0101 0011	X	X	X	X	X			
T	LATIN CAPITAL LETTER T	084	54	0101 0100	X	X	X	X	X			
U	LATIN CAPITAL LETTER U	085	55	0101 0101	X	X	X	X	X			
V	LATIN CAPITAL LETTER V	086	56	0101 0110	X	X	X	X	X			
W	LATIN CAPITAL LETTER W	087	57	0101 0111	X	X	X	X	X			
X	LATIN CAPITAL LETTER X	088	58	0101 1000	X	X	X	X	X			
Y	LATIN CAPITAL LETTER Y	089	59	0101 1001	X	X	X	X	X			
Z	LATIN CAPITAL LETTER Z	090	5A	0101 1010	X	X	X	X	X			
[	LEFT SQUARE BRACKET	091	5B	0101 1011	X	X	X	X	X			
\	REVERSE SOLIDUS	092	5C	0101 1100	X	X	X	X	X			
]	RIGHT SQUARE BRACKET	093	5D	0101 1101	X	X	X	X	X			

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APPENDIX D

TABLE D-1. NITF 1-Byte Coded Characters – continued.

		Code			Character Set							
Char	Name	Dec	Hex	Binary	U8S	ECS	ECS-A	BCS	BCS-A	BCS-N	BCS-N Integer	BCS-N Positive Integers
^	CIRCUMFLEX ACCENT	094	5E	0101 1110	X	X	X	X	X			
_	LOW LINE	095	5F	0101 1111	X	X	X	X	X			
`	GRAVE ACCENT	096	60	0110 0000	X	X	X	X	X			
a	LATIN SMALL LETTER A	097	61	0110 0001	X	X	X	X	X			
b	LATIN SMALL LETTER B	098	62	0110 0010	X	X	X	X	X			
c	LATIN SMALL LETTER C	099	63	0110 0011	X	X	X	X	X			
d	LATIN SMALL LETTER D	100	64	0110 0100	X	X	X	X	X			
e	LATIN SMALL LETTER E	101	65	0110 0101	X	X	X	X	X			
f	LATIN SMALL LETTER F	102	66	0110 0110	X	X	X	X	X			
g	LATIN SMALL LETTER G	103	67	0110 0111	X	X	X	X	X			
h	LATIN SMALL LETTER H	104	68	0110 1000	X	X	X	X	X			
i	LATIN SMALL LETTER I	105	69	0110 1001	X	X	X	X	X			
j	LATIN SMALL LETTER J	106	6A	0110 1010	X	X	X	X	X			
k	LATIN SMALL LETTER K	107	6B	0110 1011	X	X	X	X	X			
l	LATIN SMALL LETTER L	108	6C	0110 1100	X	X	X	X	X			
m	LATIN SMALL LETTER M	109	6D	0110 1101	X	X	X	X	X			
n	LATIN SMALL LETTER N	110	6E	0110 1110	X	X	X	X	X			
o	LATIN SMALL LETTER O	111	6F	0110 1111	X	X	X	X	X			
p	LATIN SMALL LETTER P	112	70	0111 0000	X	X	X	X	X			
q	LATIN SMALL LETTER Q	113	71	0111 0001	X	X	X	X	X			
r	LATIN SMALL LETTER R	114	72	0111 0010	X	X	X	X	X			
s	LATIN SMALL LETTER S	115	73	0111 0011	X	X	X	X	X			
t	LATIN SMALL LETTER T	116	74	0111 0100	X	X	X	X	X			

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APPENDIX D

TABLE D-1. NITF 1-Byte Coded Characters – continued.

		Code			Character Set							
Char	Name	Dec	Hex	Binary	U8S	ECS	ECS-A	BCS	BCS-A	BCS-N	BCS-N Integer	BCS-N Positive Integers
u	LATIN SMALL LETTER U	117	75	0111 0101	X	X	X	X	X			
v	LATIN SMALL LETTER V	118	76	0111 0110	X	X	X	X	X			
w	LATIN SMALL LETTER W	119	77	0111 0111	X	X	X	X	X			
x	LATIN SMALL LETTER X	120	78	0111 1000	X	X	X	X	X			
y	LATIN SMALL LETTER Y	121	79	0111 1001	X	X	X	X	X			
z	LATIN SMALL LETTER Z	122	7A	0111 1010	X	X	X	X	X			
{	LEFT CURLY BRACKET	123	7B	0111 1011	X	X	X	X	X			
|	VERTICAL LINE	124	7C	0111 1100	X	X	X	X	X			
}	RIGHT CURLY BRACKET	125	7D	0111 1101	X	X	X	X	X			
~	TILDE	126	7E	0111 1110	X	X	X	X	X			
	NOT USED	127	7F	0111 1111								
	NOT USED	128	80	1000 0000								
	NOT USED	129	81	1000 0001								
	NOT USED	130	82	1000 0010								
	NOT USED	131	83	1000 0011								
	NOT USED	132	84	1000 0100								
	NOT USED	133	85	1000 0101								
	NOT USED	134	86	1000 0110								
	NOT USED	135	87	1000 0111								
	NOT USED	136	88	1000 1000								
	NOT USED	137	89	1000 1001								
	NOT USED	138	8A	1000 1010								
	NOT USED	139	8B	1000 1011								

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APPENDIX D

TABLE D-1. NITF 1-Byte Coded Characters – continued.

		Code			Character Set							
Char	Name	Dec	Hex	Binary	U8S	ECS	ECS-A	BCS	BCS-A	BCS-N	BCS-N Integer	BCS-N Positive Integers
	NOT USED	140	8C	1000 1100								
	NOT USED	141	8D	1000 1101								
	NOT USED	142	8E	1000 1110								
	NOT USED	143	8F	1000 1111								
	NOT USED	144	90	1001 0000								
	NOT USED	145	91	1001 0001								
	NOT USED	146	92	1001 0010								
	NOT USED	147	93	1001 0011								
	NOT USED	148	94	1001 0100								
	NOT USED	149	95	1001 0101								
	NOT USED	150	96	1001 0110								
	NOT USED	151	97	1001 0111								
	NOT USED	152	98	1001 1000								
	NOT USED	153	99	1001 1001								
	NOT USED	154	9A	1001 1010								
	NOT USED	155	9B	1001 1011								
	NOT USED	156	9C	1001 1100								
	NOT USED	157	9D	1001 1101								
	NOT USED	158	9E	1001 1110								
	NOT USED	159	9F	1001 1111								
	NO BREAK SPACE	160	A0	1010 0000		X	X					
¡	INVERTED EXCLAMATION MARK	161	A1	1010 0001		X	X					
¢	CENT SIGN	162	A2	1010 0010		X	X					

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APPENDIX D

TABLE D-1. NITF 1-Byte Coded Characters – continued.

		Code			Character Set							
Char	Name	Dec	Hex	Binary	U8S	ECS	ECS-A	BCS	BCS-A	BCS-N	BCS-N Integer	BCS-N Positive Integers
£	POUND SIGN	163	A3	1010 0011		X	X					
¤	CURRENCY SIGN	164	A4	1010 0100		X	X					
¥	YEN SIGN	165	A5	1010 0101		X	X					
¦	BROKEN BAR	166	A6	1010 0110		X	X					
§	SECTION SIGN	167	A7	1010 0111		X	X					
¨	DIAERESIS	168	A8	1010 1000		X	X					
©	COPYRIGHT	169	A9	1010 1001		X	X					
ª	FEMININE ORDINAL INDICATOR	170	AA	1010 1010		X	X					
«	LEFT-POINTING DOUBLE ANGLE QUOTATION MARK	171	AB	1010 1011		X	X					
¬	NOT SIGN	172	AC	1010 1100		X	X					
-	SOFT HYPHEN	173	AD	1010 1101		X	X					
®	REGISTERED SIGN	174	AE	1010 1110		X	X					
¯	MACRON	175	AF	1010 1111		X	X					
°	DEGREE SIGN	176	B0	1011 0000		X	X					
±	PLUS-MINUS SIGN	177	B1	1011 0001		X	X					
²	SUPERSCRIPT TWO	178	B2	1011 0010		X	X					
³	SUPERSCRIPT THREE	179	B3	1011 0011		X	X					
´	ACUTE ACCENT	180	B4	1011 0100		X	X					
µ	MICRO SIGN	181	B5	1011 0101		X	X					
¶	PILCROW SIGN	182	B6	1011 0110		X	X					
·	MIDDLE DOT	183	B7	1011 0111		X	X					
¸	CEDILLA	184	B8	1011 1000		X	X					
¹	SUPERSCRIPT ONE	185	B9	1011 1001		X	X					

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APPENDIX D

TABLE D-1. NITF 1-Byte Coded Characters – continued.

		Code			Character Set							
Char	Name	Dec	Hex	Binary	U8S	ECS	ECS-A	BCS	BCS-A	BCS-N	BCS-N Integer	BCS-N Positive Integers
º	MASCULINE ORDINAL INDICATOR	186	BA	1011 1010		X	X					
»	RIGHT POINTING DOUBLE ANGLE QUOTATION MARK	187	BB	1011 1011		X	X					
¼	VULGAR FRACTION ONE QUARTER	188	BC	1011 1100		X	X					
½	VULGAR FRACTION ONE HALF	189	BD	1011 1101		X	X					
¾	VULGAR FRACTION THREE QUARTERS	190	BE	1011 1110		X	X					
¿	INVERTED QUESTION MARK	191	BF	1011 1111		X	X					
À	CAP A W/GRAVE	192	C0	1100 0000		X	X					
Á	CAP A W/ACUTE	193	C1	1100 0001		X	X					
Â	CAP A W/CIRCUMFLEX	194	C2	1100 0010		X	X					
Ã	CAP A W/TILDE	195	C3	1100 0011		X	X					
Ä	CAP A W/DIAERESIS	196	C4	1100 0100		X	X					
Å	CAP A WITH RING ABOVE	197	C5	1100 0101		X	X					
Æ	CAP LIGATURE AE	198	C6	1100 0110		X	X					
Ç	CAP C W/CEDILLA	199	C7	1100 0111		X	X					
È	CAP E W/GRAVE	200	C8	1100 1000		X	X					
É	CAP E W/ACUTE	201	C9	1100 1001		X	X					
Ê	CAP E W/CIRCUMFLEX	202	CA	1100 1010		X	X					
Ë	CAP E W/DIAERESIS	203	CB	1100 1011		X	X					
Ì	CAP I W/GRAVE	204	CC	1100 1100		X	X					
Í	CAP I W/ACUTE	205	CD	1100 1101		X	X					
Î	CAP I W/CIRCUMFLEX	206	CE	1100 1110		X	X					
Ï	CAP I W/DIAERESIS	207	CF	1100 1111		X	X					
Ð	CAP ETH (ICELANDIC)	208	D0	1101 0000		X	X					

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APPENDIX D

TABLE D-1. NITF 1-Byte Coded Characters – continued.

		Code			Character Set							
Char	Name	Dec	Hex	Binary	U8S	ECS	ECS-A	BCS	BCS-A	BCS-N	BCS-N Integer	BCS-N Positive Integers
Ñ	CAP N W/TILDE	209	D1	1101 0001		X	X					
Ò	CAP O W/GRAVE	210	D2	1101 0010		X	X					
Ó	CAP O W/ACUTE	211	D3	1101 0011		X	X					
Ô	CAP O W/CIRCUMFLEX	212	D4	1101 0100		X	X					
Õ	CAP O W/TILDE	213	D5	1101 0101		X	X					
Ö	CAP O W/DIAERESIS	214	D6	1101 0110		X	X					
×	MULTIPLICATION SIGN	215	D7	1101 0111		X	X					
Ø	CAP O W/STROKE	216	D8	1101 1000		X	X					
Ù	CAP U W/GRAVE	217	D9	1101 1001		X	X					
Ú	CAP U W/ACUTE	218	DA	1101 1010		X	X					
Û	CAP U W/CIRCUMFLEX	219	DB	1101 1011		X	X					
Ü	CAP U W/DIAERESIS	220	DC	1101 1100		X	X					
Ý	CAP Y W/ACUTE	221	DD	1101 1101		X	X					
Þ	CAP THORN (ICELANDIC)	222	DE	1101 1110		X	X					
ß	CAP SHARP S (GERMAN)	223	DF	1101 1111		X	X					
à	SMALL A W/GRAVE	224	E0	1110 0000		X	X					
á	SMALL A W/ACUTE	225	E1	1110 0001		X	X					
â	SMALL A W/CIRCUMFLEX	226	E2	1110 0010		X	X					
ã	SMALL A W/TILDE	227	E3	1110 0011		X	X					
ä	SMALL A W/DIAERESIS	228	E4	1110 0100		X	X					
å	SMALL A W/RING ABOVE	229	E5	1110 0101		X	X					
æ	SMALL LIGATURE AE	230	E6	1110 0110		X	X					
ç	SMALL C W/CEDILLA	231	E7	1110 0111		X	X					

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APPENDIX D

TABLE D-1. NITF 1-Byte Coded Characters – continued.

		Code			Character Set							
Char	Name	Dec	Hex	Binary	U8S	ECS	ECS-A	BCS	BCS-A	BCS-N	BCS-N Integer	BCS-N Positive Integers
è	SMALL E W/GRAVE	232	E8	1110 1000		X	X					
é	SMALL E W/ACUTE	233	E9	1110 1001		X	X					
ê	SMALL E W/CIRCUMFLEX	234	EA	1110 1010		X	X					
ë	SMALL E W/DIAERESIS	235	EB	1110 1011		X	X					
ì	SMALL I W/GRAVE	236	EC	1110 1100		X	X					
í	SMALL I W/ACUTE	237	ED	1110 1101		X	X					
î	SMALL I W/CIRCUMFLEX	238	EE	1110 1110		X	X					
ï	SMALL I W/DIAERESIS	239	EF	1110 1111		X	X					
ð	SMALL ETH (ICLANDIC)	240	F0	1111 0000		X	X					
ñ	SMALL N W/TILDE	241	F1	1111 0001		X	X					
ò	SMALL O W/GRAVE	242	F2	1111 0010		X	X					
ó	SMALL O W/ACUTE	243	F3	1111 0011		X	X					
ô	SMALL O W/CIRCUMFLEX	244	F4	1111 0100		X	X					
õ	SMALL O W/TILDE	245	F5	1111 0101		X	X					
ö	SMALL O W/DIAERESIS	246	F6	1111 0110		X	X					
÷	DIVISION SIGN	247	F7	1111 0111		X	X					
ø	SMALL O W/STROKE	248	F8	1111 1000		X	X					
ù	SMALL U W/GRAVE	249	F9	1111 1001		X	X					
ú	SMALL U W/ACUTE	250	FA	1111 1010		X	X					
û	SMALL U W/CIRCUMFLEX	251	FB	1111 1011		X	X					
ü	SMALL U W/DIAERESIS	252	FC	1111 1100		X	X					
ý	SMALL Y W/ACUTE	253	FD	1111 1101		X	X					
þ	SMALL THORN (ICELANDIC)	254	FE	1111 1110		X	X					

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APPENDIX D

TABLE D-1. NITF 1-Byte Coded Characters – continued.

		Code			Character Set							
Char	Name	Dec	Hex	Binary	U8S	ECS	ECS-A	BCS	BCS-A	BCS-N	BCS-N Integer	BCS-N Positive Integers
ÿ	SMALL Y W/DIAERESIS	255	FF	1111 1111		X	X					

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TABLE D-2. NITF 2-Byte Coded Characters

CHAR	NAME	Hex	Binary	U8S
	NOT USED	C2 80	11000010 10000000	
	NOT USED	C2 81	11000010 10000001	
	NOT USED	C2 82	11000010 10000010	
	NOT USED	C2 83	11000010 10000011	
	NOT USED	C2 84	11000010 10000100	
	NOT USED	C2 85	11000010 10000101	
	NOT USED	C2 86	11000010 10000110	
	NOT USED	C2 87	11000010 10000111	
	NOT USED	C2 88	11000010 10001000	
	NOT USED	C2 89	11000010 10001001	
	NOT USED	C2 8A	11000010 10001010	
	NOT USED	C2 8B	11000010 10001011	
	NOT USED	C2 8C	11000010 10001100	
	NOT USED	C2 8D	11000010 10001101	
	NOT USED	C2 8E	11000010 10001110	
	NOT USED	C2 8F	11000010 10001111	
	NOT USED	C2 90	11000010 10010000	
	NOT USED	C2 91	11000010 10010001	
	NOT USED	C2 92	11000010 10010010	
	NOT USED	C2 93	11000010 10010011	
	NOT USED	C2 94	11000010 10010100	
	NOT USED	C2 95	11000010 10010101	
	NOT USED	C2 96	11000010 10010110	
	NOT USED	C2 97	11000010 10010111	
	NOT USED	C2 98	11000010 10011000	
	NOT USED	C2 99	11000010 10011001	
	NOT USED	C2 9A	11000010 10011010	
	NOT USED	C2 9B	11000010 10011011	
	NOT USED	C2 9C	11000010 10011100	
	NOT USED	C2 9D	11000010 10011101	
	NOT USED	C2 9E	11000010 10011110	
	NOT USED	C2 9F	11000010 10011111	
	NO BREAK SPACE	C2 A0	11000010 10100000	X
¡	INVERTED EXCLAMATION MARK	C2 A1	11000010 10100001	X
¢	CENT SIGN	C2 A2	11000010 10100010	X
£	POUND SIGN	C2 A3	11000010 10100011	X

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TABLE D-2. NITF 2-Byte Coded Characters - Continued.

CHAR	NAME	Hex	Binary	U8S
¤	CURRENCY SIGN	C2 A4	11000010 10100100	X
¥	YEN SIGN	C2 A5	11000010 10100101	X
¦	BROKEN BAR	C2 A6	11000010 10100110	X
§	SECTION SIGN	C2 A7	11000010 10100111	X
¨	DIAERESIS	C2 A8	11000010 10101000	X
©	COPYRIGHT	C2 A9	11000010 10101001	X
ª	FEMININE ORDINAL INDICATOR	C2 AA	11000010 10101010	X
«	LEFT-POINTING DOUBLE ANGLE QUOTATION MARK	C2 AB	11000010 10101011	X
¬	NOT SIGN	C2 AC	11000010 10101100	X
-	SOFT HYPHEN	C2 AD	11000010 10101101	X
®	REGISTERED SIGN	C2 AE	11000010 10101110	X
¯	MACRON	C2 AF	11000010 10101111	X
°	DEGREE SIGN	C2 B0	11000010 10110000	X
±	PLUS-MINUS SIGN	C2 B1	11000010 10110001	X
²	SUPERSCRIPT TWO	C2 B2	11000010 10110010	X
³	SUPERSCRIPT THREE	C2 B3	11000010 10110011	X
´	ACUTE ACCENT	C2 B4	11000010 10110100	X
µ	MICRO SIGN	C2 B5	11000010 10110101	X
¶	PILCROW SIGN	C2 B6	11000010 10110110	X
·	MIDDLE DOT	C2 B7	11000010 10110111	X
¸	CEDILLA	C2 B8	11000010 10111000	X
¹	SUPERSCRIPT ONE	C2 B9	11000010 10111001	X
º	MASCULINE ORDINAL INDICATOR	C2 BA	11000010 10111010	X
»	RIGHT POINTING DOUBLE ANGLE QUOTATION MARK	C2 BB	11000010 10111011	X
¼	VULGAR FRACTION ONE QUARTER	C2 BC	11000010 10111100	X
½	VULGAR FRACTION ONE HALF	C2 BD	11000010 10111101	X
¾	VULGAR FRACTION THREE QUARTERS	C2 BE	11000010 10111110	X
¿	INVERTED QUESTION MARK	C2 BF	11000010 10111111	X
À	CAP A W/GRAVE	C3 80	11000011 10000000	X
Á	CAP A W/ACUTE	C3 81	11000011 10000001	X
Â	CAP A W/CIRCUMFLEX	C3 82	11000011 10000010	X
Ã	CAP A W/TILDE	C3 83	11000011 10000011	X
Ä	CAP A W/DIAERESIS	C3 84	11000011 10000100	X
Å	CAP A WITH RING ABOVE	C3 85	11000011 10000101	X
Æ	CAP LIGATURE AE	C3 86	11000011 10000110	X
Ç	CAP C W/CEDILLA	C3 87	11000011 10000111	X
È	CAP E W/GRAVE	C3 88	11000011 10001000	X

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TABLE D-2. NITF 2-Byte Coded Characters - Continued.

CHAR	NAME	Hex	Binary	U8S
É	CAP E W/ACUTE	C3 89	11000011 10001001	X
Ê	CAP E W/CIRCUMFLEX	C3 8A	11000011 10001010	X
Ë	CAP E W/DIAERESIS	C3 8B	11000011 10001011	X
Ì	CAP I W/GRAVE	C3 8C	11000011 10001100	X
Í	CAP I W/ACUTE	C3 8D	11000011 10001101	X
Î	CAP I W/CIRCUMFLEX	C3 8E	11000011 10001110	X
Ï	CAP I W/DIAERESIS	C3 8F	11000011 10001111	X
Ð	CAP ETH (ICELANDIC)	C3 90	11000011 10010000	X
Ñ	CAP N W/TILDE	C3 91	11000011 10010001	X
Ò	CAP O W/GRAVE	C3 92	11000011 10010010	X
Ó	CAP O W/ACUTE	C3 93	11000011 10010011	X
Ô	CAP O W/CIRCUMFLEX	C3 94	11000011 10010100	X
Õ	CAP O W/TILDE	C3 95	11000011 10010101	X
Ö	CAP O W/DIAERESIS	C3 96	11000011 10010110	X
×	MULTIPLICATION SIGN	C3 97	11000011 10010111	X
Ø	CAP O W/STROKE	C3 98	11000011 10011000	X
Ù	CAP U W/GRAVE	C3 99	11000011 10011001	X
Ú	CAP U W/ACUTE	C3 9A	11000011 10011010	X
Û	CAP U W/CIRCUMFLEX	C3 9B	11000011 10011011	X
Ü	CAP U W/DIAERESIS	C3 9C	11000011 10011100	X
Ý	CAP Y W/ACUTE	C3 9D	11000011 10011101	X
Þ	CAP THORN (ICELANDIC)	C3 9E	11000011 10011110	X
ß	CAP SHARP S (GERMAN)	C3 9F	11000011 10011111	X
à	SMALL A W/GRAVE	C3 A0	11000011 10100000	X
á	SMALL A W/ACUTE	C3 A1	11000011 10100001	X
â	SMALL A W/CIRCUMFLEX	C3 A2	11000011 10100010	X
ã	SMALL A W/TILDE	C3 A3	11000011 10100011	X
ä	SMALL A W/DIAERESIS	C3 A4	11000011 10100100	X
å	SMALL A W/RING ABOVE	C3 A5	11000011 10100101	X
æ	SMALL LIGATURE AE	C3 A6	11000011 10100110	X
ç	SMALL C W/CEDILLA	C3 A7	11000011 10100111	X
è	SMALL E W/GRAVE	C3 A8	11000011 10101000	X
é	SMALL E W/ACUTE	C3 A9	11000011 10101001	X
ê	SMALL E W/CIRCUMFLEX	C3 AA	11000011 10101010	X
ë	SMALL E W/DIAERESIS	C3 AB	11000011 10101011	X
ì	SMALL I W/GRAVE	C3 AC	11000011 10101100	X
í	SMALL I W/ACUTE	C3 AD	11000011 10101101	X

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TABLE D-2. NITF 2-Byte Coded Characters - Continued.

CHAR	NAME	Hex	Binary	U8S
î	SMALL I W/CIRCUMFLEX	C3 AE	11000011 10101110	X
ï	SMALL I W/DIAERESIS	C3 AF	11000011 10101111	X
ð	SMALL ETH (ICLANDIC)	C3 B0	11000011 10110000	X
ñ	SMALL N W/TILDE	C3 B1	11000011 10110001	X
ò	SMALL O W/GRAVE	C3 B2	11000011 10110010	X
ó	SMALL O W/ACUTE	C3 B3	11000011 10110011	X
ô	SMALL O W/CIRCUMFLEX	C3 B4	11000011 10110100	X
õ	SMALL O W/TILDE	C3 B5	11000011 10110101	X
ö	SMALL O W/DIAERESIS	C3 B6	11000011 10110110	X
÷	DIVISION SIGN	C3 B7	11000011 10110111	X
ø	SMALL O W/STROKE	C3 B8	11000011 10111000	X
ù	SMALL U W/GRAVE	C3 B9	11000011 10111001	X
ú	SMALL U W/ACUTE	C3 BA	11000011 10111010	X
û	SMALL U W/CIRCUMFLEX	C3 BB	11000011 10111011	X
ü	SMALL U W/DIAERESIS	C3 BC	11000011 10111100	X
ý	SMALL Y W/ACUTE	C3 BD	11000011 10111101	X
þ	SMALL THORN (ICELANDIC)	C3 BE	11000011 10111110	X
ÿ	SMALL Y W/DIAERESIS	C3 BF	11000011 10111111	X

D.2.11 NITF Printer Incompatibilities. Some printers do not allow for transparent pixels in imagery (e.g., Postscript level 1 and 2). If an NITF composition uses CGM elements under images with NITF image padding (transparent pixels) the CGM will not be visible in any areas under the pad pixels. This problem can be avoided in two different ways.

a. Instead of using a CGM element, the background color may be specified with the FBKGC field in the main NITF file header. The background shape and size may be specified with the BGWIDTH and BGHIGHT fields of the BCKGDA TAG. (The BCKGDA TAG also specifies the background color and pixel size. The color specified in FBKGC must be the same as the color specified in the BCKGDA. If they are not the same, the BCKGDA TAG colors take precedence.) If the BCKGDA TAG is not present, the color specified in FBKGC applies to the entire CCS up to the size of the C-level.

b. The CGM rectangle can be broken down into four (or more if the composition is complex) CGM rectangles that do not coincide with (obscure or lie under) the imagery. These CGM rectangles would then be specified with higher display levels than any of the images.

D.2.12 Universal Transverse Mercator (UTM) Coordinate Hemisphere Resolution. The Image Coordinate Representation (ICORDS) field allows one value for an image's UTM hemispheric designation, N for northern or S for southern. When the Image Coordinate Location (IGEOLO) field is populated with UTM coordinates, image presentations that cross the Equator have northing values that are ambiguous, and it is not immediately obvious which corners are on which side of the Equator. Given their UTM coordinates and zone, the following method resolves the hemispheric designation of four individual corners of an image.

D.2.12.1 North (N)/South (S) Method. When used for pure UTM coordinates, the ICORDS field contains one of two values: N if the northernmost corner is on or north of the Equator, S if the northernmost point is south of the

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Equator. When the ICORDS field contains S, all 4 corners are south of the Equator. When the ICORDS field contains N, the following process resolves the hemispheres of the individual corners:

a. Assume that the north-south spread of the image is within 5000km (about 45 degrees).

b. Compute the smallest northing (nmin) from the 4 corners. (The corner with that northing value is certain to be one of the corners on or north of the Equator.)

c. Then any corner with a northing in excess of [nmin + 5 million] is located on or north of the Equator.

FIGURE D-1 depicts a single example UTM zone from the default geodetic reference system, WGS84.

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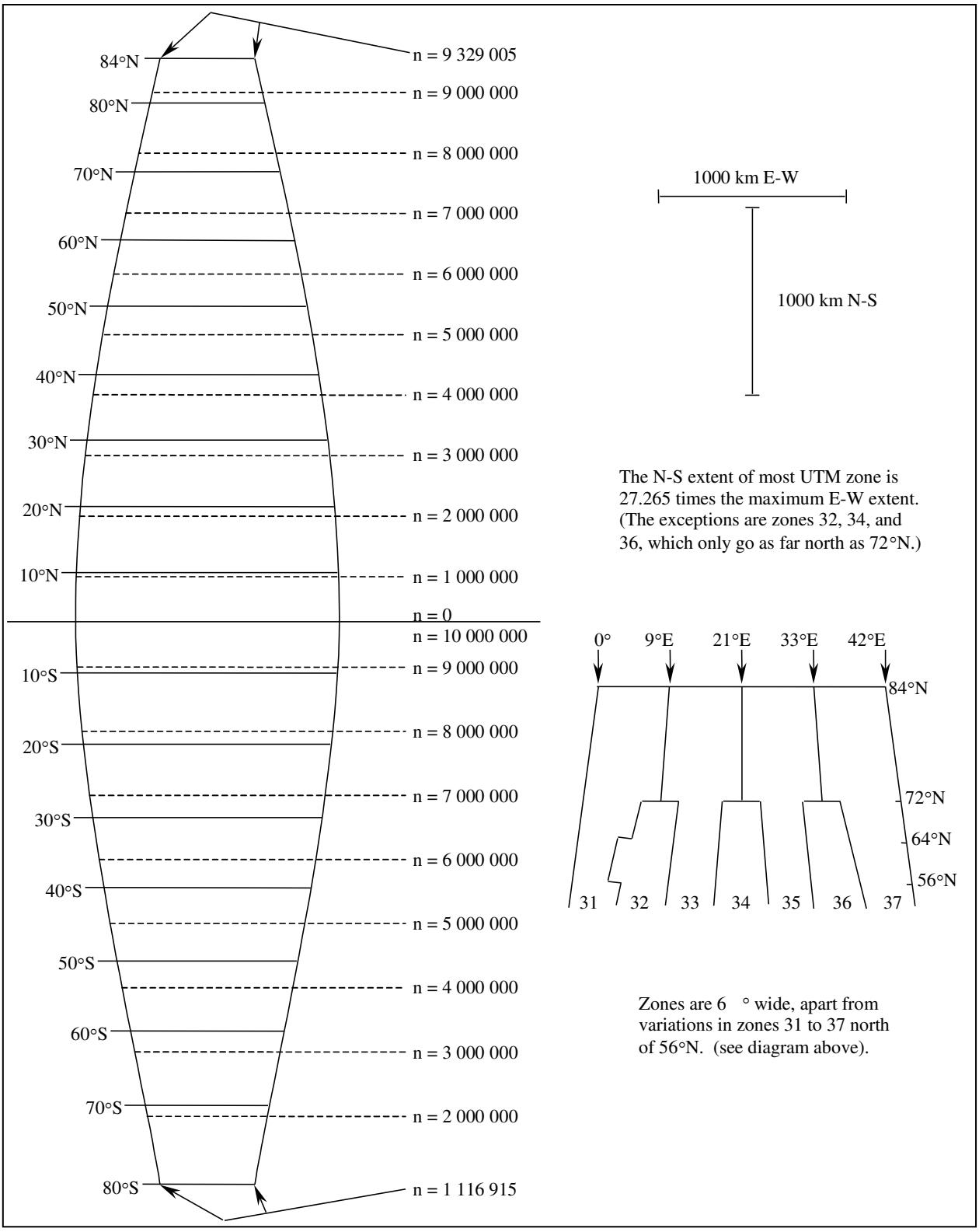


FIGURE D-1. A typical World Geodetic System 1984 (WGS 84) UTM zone (compressed).

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APPENDIX E. Example NITF File (Informative)

E.1 EXAMPLE NITF FILE

E.1.1 Scope. The appendix details an example NITF file. This appendix is not a mandatory part of this standard. The information contained herein is intended for guidance.

E.2 Use of NITF. Though the NITF was conceived initially to support the transmission of a file composed of a single base image, image insets (subimage overlays), graphic overlays, and text, its current form makes it suitable for a wide variety of file exchange needs. One of the flexible features of the NITF is that it allows several segments of each data type to be included in one file, yet any of the data types may be omitted. Thus, for example, the NITF may equally well be used for the storage of a single portion of text, a single image or a complex composition of several images, graphics, and text. The following section discusses an example NITF file of moderate complexity.

E.2.1 Example File. TABLE E-1 shows the contents of the fields in the header of an example NITF file composed of two ISs (an image with an inset image), five GS overlays, (two of which are multi-displayable element graphic segments), and five TSs. FIGURE E-1 shows part of the sample file as a composite image with its overlay graphics. In an NITF file, the data for each segment is stored in a data field preceded by the segment subheader. The subheader for a data type is omitted if no data of that type are included in the file. Segment subheader field contents in the sample file are shown in tables E-1 to E-10.

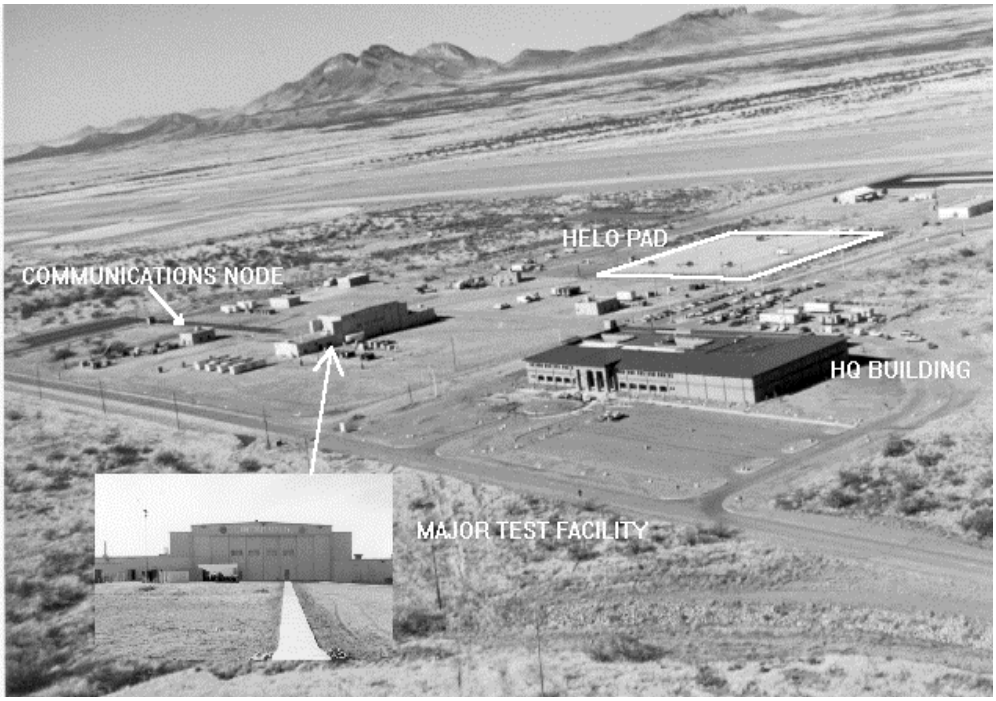


FIGURE E-1. Sample file composite image.

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TABLE E-1. Example NITF File Header.

NITF HEADER FIELD	FORMAT	COMMENT
File Profile Name (FHDR)	NITF	4 characters
File Version (FVER)	02.10	5 characters
Complexity Level (CLEVEL)	05	2 characters -- images less than or equal to 8k x 8k but greater than 2k x 2k in either or both rows and columns
Standard Type (STYPE)	BF01	4 characters
Originating Station ID (OSTAID)	U21SOO90	8 characters followed by 2 BCS Spaces (0x20) – 10 characters
File Date and Time (FDT)	19960930224632	14 digits
File Title (FTITLE)	MAJOR TEST FACILITY	19 characters followed by 61 ECS Spaces (0x20) - 80 characters
File Security Classification (FSCLAS)	U	1 character
File Classification Security System (FSCLSY)		2 spaces
File Codewords (FSCODE)		11 BCS Spaces (0x20)
File Control and Handling (FSCTLH)		2 BCS Spaces (0x20)
File Releasing Instructions (FSREL)		20 BCS Spaces (0x20)
File Declassification Type (FSDCTP)		2 BCS Spaces (0x20)
File Declassification Date (FSDCDT)		8 BCS Spaces (0x20)
File Declassification Exemption (FSDCXM)		4 BCS Spaces (0x20)
File Downgrade (FSDG)		1 BCS space (0x20)
File Downgrade Date (FSDGDT)		8 BCS Spaces (0x20)
File Classification Text (FSCLTX)		43 ECS Spaces (0x20)
File Classification Authority Type (FSCATP)		1 BCS space (0x20)
File Classification Authority (FSCAUT)		40 ECS Spaces (0x20)
File Classification Reason (FSCRSN)		1 BCS space (0x20)
File Security Source Date (FSSRDT)		8 BCS Spaces (0x20)
File Security Control Number (FSCTLN)		15 BCS Spaces (0x20)
File Copy Number (FSCOP)	00000	5 digits - all zeros indicate there is no tracking of NITF file copies
File Number of Copies (FSCPYS)	00000	5 digits – all zeros indicate there is no tracking of NITF file copies
Encryption (ENCRYP)	0	1 digit - required default
File Background Color (FBKGC)	0x000000	3 bytes (binary)
Originator's Name (ONAME)	W.TEMPEL	9 characters followed by 15 ECS Spaces (0x20) - 24 characters

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NITF HEADER FIELD	FORMAT	COMMENT
Originator's Phone Number (OPHONE)	44 1480 84 5611	15 characters followed by 3 BCS Spaces (0x20) - 18 characters
File Length (FL)	000002905629	12 digits
NITF File Header Length (HL)	000515	6 digits
Number of Image Segments (NUMI)	002	3 digits
Length of 1st Image Subheader (LISH001)	000679	6 digits
Length of 1st Image Segment (LI001)	0002730600	10 digits
Length of 2nd Image Subheader (LISH002)	000439	6 digits
Length of 2nd Image Segment (LI002)	0000089600	10 digits
Number of Graphic Segments (NUMS)	005	3 digits
Length of 1st Graphic Subheader (LSSH001)	0258	4 digits
Length of 1st Graphic Segment (LS001)	000122	6 digits
Length of 2nd Graphic Subheader (LSSH002)	0258	4 digits
Length of 2nd Graphic Segment (LS002)	000122	6 digits
Length of 3rd Graphic Subheader (LSSH003)	0258	4 digits
Length of 3rd Graphic Segment (LS003)	000150	6 digits
Length of 4th Graphic Subheader (LSSH004)	0258	4 digits
Length of 4th Graphic Segment (LS004)	000112	6 digits
Length of 5th Graphic Subheader (LSSH005)	0258	4 digits
Length of 5th Graphic Segment (LS005)	000116	6 digits
Reserved for future use (NUMX)	000	3 digits
Number of Text Files (NUMT)	005	3 digits
Length of 1st Text Subheader (LTSH001)	0282	4 digits
Length of 1st Text Segment (LT001)	20000	5 digits
Length of 2nd Text Subheader (LTSH002)	0282	4 digits
Length of 2nd Text Segment (LT002)	20000	5 digits
Length of 3rd Text Subheader (LTSH003)	0282	4 digits
Length of 3rd Text Segment (LT003)	20000	5 digits
Length of 4th Text Subheader (LTSH004)	0282	4 digits
Length of 4th Text Segment (LT004)	20000	5 digits
Length of 5th Text Subheader (LTSH005)	0282	4 digits
Length of 5th Text Segment (LT005)	20000	5 digits
Number of Data Extension Segments (NUMDES)	000	3 digits

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NITF HEADER FIELD	FORMAT	COMMENT
Number of Reserved Extension Segments (NUMRES)	000	3 digits
User Defined Header Data Length (UDHDL)	00000	5 digits
Extended Header Data Length (XHDL)	00000	5 digits

E.3 Explanation of the File Header The File Profile Name (FHDR), NITF, and File Version (FVER), 02.10, are listed first. The next field contains the file's CLEVEL, in this case 05. A four character reserved field for the Standard Type (STYPE), defaulted to blanks, appears next. An identification code (OSTAID) containing ten characters for the station originating the primary information in the file is given next. The file origination date and time (FDT) follow this and are given in Coordinated Universal Time (UTC) (Zulu) time format. This is followed by the File Title (FTITLE) field containing up to 80 characters of free form text. The title of the sample file contains less than 80 characters, and therefore, the remainder of the field is padded with blanks. After this a number of security-related fields occur. The File Security Classification (FSCLAS) field is mandatory and contains one character. The remaining fields - File Security Classification System (FSCLSY), File Codewords (FSCODE), File Control and Handling (FSCTLH), File Releasing Instructions (FSREL), File Declassification Type (FSDCTP), File Declassification Date (FSDCDT), File Declassification Exemption (FSDCXM), File Downgrade (FSDG), File Downgrade Date (FSDGDT), File Classification Text (FSCLTX), File Classification Authority Type (FSCATP), File Classification Authority (FSCAUT), File Classification Reason (FSCRSN), File Security Source Date (FSSRDT), and File Security Control Number (FSCTLN) - will be filled in if the file is classified in accordance with existing security directives. Which fields are actually populated will depend on the security system used and the security parameters which apply to the specific file. Some unclassified files may also require an entry in FSCTLH (e.g. Proprietary Information (PROPIN), For Official Use Only (FOUO)). In the example above, the file is unclassified and no handling caveats apply, so all following security-related fields are blank. File Encryption (ENCRYP) follows and is given a "0" indicating that the file is not encrypted. The next field, File Background Color (FBKGC), defines the background color behind displayable segments. It eliminates the potential to visually lose information if the originator selects a presentation color that is the same as the receiver’s selected background color. The originator's name (ONAME) and phone number (OPHONE) are given next. These fields may be left blank. Then the length in bytes of the entire file (FL) is given, including all headers, subheaders, and data. This is followed by the length in bytes of the NITF file header (HL). The Number of Image Segments (NUMI) field contains the characters 002 to indicate two images are included in the file. This is followed by six characters to specify the length of the first image subheader (LISHn), then ten characters for the length of the first image (LIn). The length of the second image subheader and the length of the second image follow. The next item in the file header is the Number of Graphics (NUMS), which contains 005 to indicate that five graphics are present in the file. The next ten characters contain the Length of Graphic Subheader (LSSHn) and Length of Graphic (LSn) (four and six characters respectively) for the first to fifth graphic, one after the other. The Number of Text Files (NUMT) field is given as 005 and is followed by four characters specifying the length of the text subheader and five characters specifying the number of characters in the TS for each of the five TSs. The Number of Data Extension Segments (NUMDES) and Number of Reserved Extension Segments (NUMRES) fields are given as "000." This completes the "road map" for separating the data subheaders from the actual data to follow. The next two fields in the header are the User Defined Header Data Length (UDHDL) and the User Defined Header Data (UDHD). User defined data could be used to include registered TRE that provide additional information about the file. In this example, however, the UDHDL is given as zero; therefore, the UDHD is omitted. The last field in the header is the Extended Header Data Length (XHDL). The XHDL is given as zero; therefore, the Extended Header Data (XHD) field is omitted, indicating that no CE are included in the file header.

E.4 Explanation of the Image Subheaders.

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TABLE E-2. Example of the first image subheader.

NITF IMAGE SUBHEADER FIELD	FORMAT	COMMENT
File Part Type (IM)	IM	2 characters
Image Identifier 1 (IID1)	0000000001	10 characters
Image Date and Time (IDATIM)	19960825203147	14 digits
Target Identifier (TGTID)		17 BCS Spaces (0x20)
Image Identifier 2 (IID2)	1996238CY02123456 78ABCD25AUG1995 2031F	40 characters followed by 40 ECS spaces - 80 characters
Image Security Classification (ISCLAS)	U	1 character
Image Classification Security System (ISCLSY)		2 spaces
Image Codewords (ISCODE)		11 BCS Spaces (0x20)
Image Control and Handling (ISCTLH)		2 BCS Spaces (0x20)
Image Releasing Instructions (ISREL)		20 BCS Spaces (0x20)
Image Declassification Type (ISDCTP)		2 BCS Spaces (0x20)
Image Declassification Date (ISDCDT)		8 BCS Spaces (0x20)
Image Declassification Exemption (ISDCXM)		4 BCS Spaces (0x20)
Image Downgrade (ISDG)		1 BCS space (0x20)
Image Downgrade Date (ISDGDT)		8 BCS Spaces (0x20)
Image Classification Text (ISCLTX)		43 ECS Spaces (0x20)
Image Classification Authority Type (ISCATP)		1 BCS space (0x20)
Image Classification Authority (ISCAUT)		40 ECS Spaces (0x20)
Image Classification Reason (ISCRSN)		1 BCS space (0x20)
Image Security Source Date (ISSRDT)		8 BCS Spaces (0x20)
Image Security Control Number (ISCTLN)		15 BCS Spaces (0x20)
Encryption (ENCRYP)	0	1 digit - required default
Image Source (ISORCE)	Hand-held digital camera model XYZ.	35 characters followed by 7 ECS Spaces (0x20) - 42 characters
Number of Significant Rows in image (NROWS)	00001332	8 digits
Number of Significant Columns in image (NCOLS)	00002050	8 digits
Pixel Value Type (PVTYPE)	INT	3 characters - indicating pixel values as integers
Image Representation (IREP)	MONO	4 characters followed by 4 BCS Spaces (0x20)– grey scale imagery

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NITF IMAGE SUBHEADER FIELD	FORMAT	COMMENT
Image Category (ICAT)	VIS	3 characters followed by 5 BCS Spaces (0x20)- visible imagery
Actual Bits-Per-Pixel Per Band (ABPP)	08	2 digits
Pixel Justification (PJUST)	R	1 character
Image Coordinate Representation (ICORDS)		BCS space (0x20)- indicates no geo-location coordinates
Number of Image Comments (NICOM)	3	1 digit
† Image Comment 1 (ICOM1)	This is a comment on Major Test Facility base and associated inset. This file w	80 characters
† Image Comment 2 (ICOM2)	as developed at Fort Huachuca, Arizona. It shows the Joint Interoperability Test	80 characters
† Image Comment 3 (ICOM3)	t Command Building and associated range areas.	44 characters followed by 36 ECS Spaces (0x20)- 80 characters
Image Compression (IC)	NC	2 characters - indicates no compression
Number of Bands (NBANDS)	1	1 digit
1st Band Representation (IREPBAND1)		2 BCS Spaces (0x20)
1st Band Subcategory (ISUBCAT1)		6 BCS Spaces (0x20)
1st Band Image Filter Condition (IFC1)	N	1 character - required default value
1st Band Standard Image Filter Code (IMFLT1)		3 BCS Spaces (0x20) - reserved
Number of LUTs for the 1st Image Band (NLUTS1)	0	1 digit
Image Sync Code (ISYNC)	0	1 digit
Image Mode (IMODE)	B	1 character - B required for 1 band
Number of Blocks per Row (NBPR)	0001	4 digits
Number of Blocks per Column (NBPC)	0001	4 digits
Number of Pixels Per Block Horizontal (NPPBH)	2050	4 digits
Number of Pixels Per Block Vertical (NPPBV)	1332	4 digits
Number of Bits Per Pixel (NBPP)	08	2 digits
Image Display Level (IDLVL)	001	3 digits - minimum DLVL requires this value
Image Attachment Level (IALVL)	000	Required 3 digit value since minimum DLVL.
Image Location (ILOC)	0000000000	10 digits upper left pixel located at origin of CCS

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NITF IMAGE SUBHEADER FIELD	FORMAT	COMMENT
Image Magnification (IMAG)	1.0	3 characters followed by a BCS space (0x20) - 4 characters
User Defined Image Data Length (UDIDL)	00000	5 digits
Image Extended Subheader Data Length (IXSHDL)	00000	5 digits

† According to the standard - this should look like a single continuous comment of up to 3 x 80 characters.

E.4.1 Explanation of the First Image Subheader. There are two images in this sample file. The first image has IDVLV001. Its subheader is shown in TABLE 2. It is an unclassified, single band, single block, grey scale image with 8 bits per pixel and does not have an associated LUT. There are three associated comments. It is visible imagery, does not have geo-location data, and is stored as an uncompressed image. It is located at the origin of the CCS within which all the displayable file components are located. It is 1332 rows by 2050 columns. FIGURE E-1 illustrates the image printed at approximately three hundred pixels per inch.

TABLE E-3. Example of the Second Image Subheader.

NITF IMAGE SUBHEADER FIELD	FORMAT	COMMENT
File Part Type (IM)	IM	2 characters
Image Identifier 1 (IID1)	Missing ID	10 characters
Image Date and Time (IDATIM)	19960927011729	14 digits
Target Identifier (TGTID)		17 BCS Spaces (0x20)
Image Identifier 2 (IID2)	1996271cy02123456 78 ABCD27SEP199620 56 ~~bbb~~F	40 characters followed by 40 ECS Spaces (0x20)- 80 characters total
Image Security Classification (ISCLAS)	U	1 character
Image Classification Security System (ISCLSY)		2 characters
Image Codewords (ISCODE)		11 BCS Spaces (0x20)
Image Control and Handling (ISCTLH)		2 BCS Spaces (0x20)
Image Releasing Instructions (ISREL)		20 BCS Spaces (0x20)
Image Declassification Type (ISDCTP)		2 BCS Spaces (0x20)
Image Declassification Date (ISDCDT)		8 BCS Spaces (0x20)
Image Declassification Exemption (ISDCXM)		4 BCS Spaces (0x20)
Image Downgrade (ISDG)		1 BCS space (0x20)
Image Downgrade Date (ISDGDT)		8 BCS Spaces (0x20)
Image Classification Text (ISCLTX)		43 ECS Spaces (0x20)
Image Classification Authority Type (ISCTP)		1 BCS space (0x20)
Image Classification Authority (ISCAUT)		40 ECS Spaces (0x20)
Image Classification Reason (ISCRSN)		1 BCS space (0x20)
Image Security Source Date (ISSRDT)		8 BCS Spaces (0x20)
Image Security Control Number (ISCTLN)		15 BCS Spaces (0x20)

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NITF IMAGE SUBHEADER FIELD	FORMAT	COMMENT
Encryption (ENCRYP)	0	1 digit - required default
Image Source (ISORCE)	Cut of original image.	22 characters followed by 20 ECS Spaces (0x20)- 42 characters
Number of Significant Rows in image (NROWS)	00000350	8 digits
Number of Significant Columns in image (NCOLS)	00000600	8 digits
Pixel Value Type (PVTYPE)	INT	3 characters - indicating pixel values as integers
Image Representation (IREP)	MONO	4 characters followed by 4 BCS Spaces (0x20) - grey scale imagery
Image Category (ICAT)	VIS	3 characters followed by 5 BCS Spaces (0x20) - visible imagery
Actual Bits-Per-Pixel Per Band (ABPP)	08	2 digits
Pixel Justification (PJUST)	R	1 character
Image Coordinate Representation (ICORDS)		BCS space (0x20) - indicates no geo-location coordinates
Number of Image Comments (NICOM)	0	1 digit
Image Compression (IC)	NC	2 characters - indicates uncompressed
Number of Bands (NBANDS)	1	1 digit
1st Band Representation (IREPBAND1)		2 BCS Spaces (0x20)
1st Band Subcategory (ISUBCAT1)		6 BCS Spaces (0x20)
1st Band Image Filter Condition (IFC1)	N	1 character - required default value
1st Band Standard Image Filter Code (IMFLT1)		3 BCS Spaces (0x20) - reserved
1st Band Number of LUTS (NLUTS1)	0	1 digit
Image Sync Code (ISYNC)	0	1 digit
Image Mode (IMODE)	B	1 character - B required for 1 band
Number of Blocks Per Row (NBPR)	0001	4 digits
Number of Blocks Per Column (NBPC)	0001	4 digits
Number of Pixels Per Block Horizontal (NPPBH)	0600	4 digits
Number of Pixels Per Block Vertical (NPPBV)	0350	4 digits
Number Bits Per Pixel (NBPP)	08	2 digits
Image Display Level (IDLVL)	002	3 digits
Image Attachment Level (IALVL	001	3 digits
Image Location (ILOC)	0088000205	10 digits, located at row 880 column 205 of base image
Image Magnification (IMAG)	1.0	3 characters followed by a BCS space (0x20) - 4 characters

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NITF IMAGE SUBHEADER FIELD	FORMAT	COMMENT
User Defined Image Data Length (UDIDL)	00000	5 digits
Image Extended Subheader Data Length (IXSHDL)	00000	5 digits

E.4.2 Explanation of the Second Image Subheader. This image is the second image in the file. As is the first image, this image is an 8 bit visible, grey scale image. It is much smaller (400 columns x 224 rows) and is not compressed. Also, unlike the first image, it has no associated comment fields, indicated by the fact that the Number of Image Comments (NICOM) = is equal to 0. Since it is attached to the base image (IALVL = 001), the ILOC field reveals that this image is located with its upper left corner positioned at Row 880, Column 205 with respect to the upper left corner of the base image. Since it has a display level greater than that of the base image, it will obscure part of the base image when they are both displayed.

E.5 Explanation of the Graphic Subheaders.

TABLE E-4. Graphic Subheader for the First Graphic.

NITF GRAPHIC SUBHEADER FIELD	FORMAT	COMMENT
File Part Type (SY)	SY	2
Graphic Identifier (SID)	0000000001	10
Graphic Name (SNAME)	HELO PAD RECTANGLE	18 characters followed by 2 ECS Spaces (0x20) - 20 characters
Graphic Security Classification (SSCLAS)	U	1 character
Graphic Classification Security System (SSCLSY)		2 characters
Graphic Codewords (SSCODE)		11 BCS Spaces (0x20)
Graphic Control and Handling (SSCTLH)		2 BCS Spaces (0x20)
Graphic Releasing Instructions (SSREL)		20 BCS Spaces (0x20)
Graphic Declassification Type (SSDCTP)		2 BCS Spaces (0x20)
Graphic Declassification Date (SSDCDT)		8 BCS Spaces (0x20)
Graphic Declassification Exemption (SSDCXM)		4 BCS Spaces (0x20)
Graphic Downgrade (SSDG)		1 BCS space (0x20)
Graphic Downgrade Date (SSDGDT)		8 BCS Spaces (0x20)
Graphic Classification Text (SSCLTX)		43 ECS Spaces (0x20)
Graphic Classification Authority Type (SSCATP)		1 BCS space (0x20)
Graphic Classification Authority (SSCAUT)		40 ECS Spaces (0x20)
Graphic Classification Reason (SSCRSN)		1 BCS space (0x20)
Graphic Security Source Date (SSSRDT)		8 BCS Spaces (0x20)
Graphic Security Control Number (SSCTLN)		15 BCS Spaces (0x20)

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NITF GRAPHIC SUBHEADER FIELD	FORMAT	COMMENT
Encryption (ENCRYP)	0	1 digit - required default
Graphic Type (SFMT)	C	1 character - indicates CGM
Reserved for Future Use (SSTRUCT)	0000000000000	13 BCS Zeros (0x30) - reserved
Graphic Display Level (SDLVL)	003	3 digits
Graphic Attachment Level (SALVL)	001	3 digits
Graphic Location (SLOC)	0041501160	10 digits
First Graphic Bound Location (SBND1)	0041501160	10 digits
Graphic Color (SCOLOR)	M	indicates CGM file contains no color components
Second Graphic Bound Location (SBND2)	0051001812	10 digits
Reserved for Future Use (SRES2)	00	2 BCS Zeros (0x30) - reserved
Graphic Extended Subheader Data Length (SXSHDL)	00000	5 digits

E.5.1 Explanation of the First Graphic Subheader This graphic is a computer graphics metafile graphic (HELO PAD RECTANGLE). The graphic is attached to the base image, and its location is recorded in SLOC (row 392, column 1110) and is measured as an offset from the origin at the upper left corner of that image.

TABLE E-5. Graphic Subheader for the Second Graphic.

NITF GRAPHIC SUBHEADER FIELD	FORMAT	COMMENT
File Part Type (SY)	SY	2
Graphic Identifier (SID)	0000000002	10
Graphic Name (SNAME)	ARROW	5 characters followed by 15 ECS Spaces (0x20) - 20 characters
Graphic Security Classification (SSCLAS)	U	1 character
Graphic Classification Security System (SSCLSY)		2 characters
Graphic Codewords (SSCODE)		11 BCS Spaces (0x20)
Graphic Control and Handling (SSCTLH)		2 BCS Spaces (0x20)
Graphic Releasing Instructions (SSREL)		20 BCS Spaces (0x20)
Graphic Declassification Type (SSDCTP)		2 BCS Spaces (0x20)
Graphic Declassification Date (SSDCDT)		8 BCS Spaces (0x20)
Graphic Declassification Exemption (SSDCXM)		4 BCS Spaces (0x20)
Graphic Downgrade (SSDG)		1 BCS space (0x20)

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NITF GRAPHIC SUBHEADER FIELD	FORMAT	COMMENT
Graphic Downgrade Date (SSDGDT)		8 BCS Spaces (0x20)
Graphic Classification Text (SSCLTX)		43 ECS Spaces (0x20)
Graphic Classification Authority Type (SSCATP)		1 BCS space (0x20)
Graphic Classification Authority (SSCAUT)		40 ECS Spaces (0x20)
Graphic Classification Reason (SSCRSN)		1 BCS space (0x20)
Graphic Security Source Date (SSSRDT)		8 BCS Spaces (0x20)
Graphic Security Control Number (SSCTLN)		15 BCS Spaces (0x20)
Encryption (ENCRYP)	0	1 digit - required default
Graphic Type (SFMT)	C	1 character - indicates CGM
Reserved for Future Use (SSTRUCT)	0000000000000	13 BCS Zeros (0x30) - reserved
Graphic Display Level (SDLVL)	004	3 digits
Graphic Attachment Level (SALVL)	002	3 digits
Graphic Location (SLOC)	0000000430	10 digits relative to origin of second image
First Graphic Bound Location (SBND1)	0022500430	10 digits relative to origin of second image
Graphic Color (SCOLOR)	M	indicates CGM file contains no color components
Second Graphic Bound Location (SBND2)	0000000300	10 digits relative to origin of second image
Reserved for Future Use (SRES2)	00	2 BCS Zeros (0x30) - reserved
Graphic Extended Subheader Data Length (SXSHDL)	00000	5 digits

E.5.2 Explanation of the Second Graphic Subheader. The second graphic is also a CGM graphic. It is the arrow pointing to the test facility. It is attached to the subimage. Therefore, its location as recorded in SLOC is measured as an offset from the upper left corner of the subimage.

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TABLE E-6. Graphic Subheader for the Third Graphic.

NITF GRAPHIC SUBHEADER FIELD	FORMAT	COMMENT
File Part Type (SY)	SY	2
Graphic Identifier (SID)	0000000003	10
Graphic Name (SNAME)	HQ BUILDING	11 characters followed by 9 ECS Spaces (0x20) - 20 characters
Graphic Security Classification (SSCLAS)	U	1 character
Graphic Classification Security System (SSCLSY)		2 characters
Graphic Codewords (SSCODE)		11 BCS Spaces (0x20)
Graphic Control and Handling (SSCTLH)		2 BCS Spaces (0x20)
Graphic Releasing Instructions (SSREL)		20 BCS Spaces (0x20)
Graphic Declassification Type (SSDCTP)		2 BCS Spaces (0x20)
Graphic Declassification Date (SSDCDT)		8 BCS Spaces (0x20)
Graphic Declassification Exemption (SSDCXM)		4 BCS Spaces (0x20)
Graphic Downgrade (SSDG)		1 BCS space (0x20)
Graphic Downgrade Date (SSDGDT)		8 BCS Spaces (0x20)
Graphic Classification Text (SSCLTX)		43 ECS Spaces (0x20)
Graphic Classification Authority Type (SSCATP)		1 BCS space (0x20)
Graphic Classification Authority (SSCAUT)		40 ECS Spaces (0x20)
Graphic Classification Reason (SSCRSN)		1 BCS space (0x20)
Graphic Security Source Date (SSSRDT)		8 BCS Spaces (0x20)
Graphic Security Control Number (SSCTLN)		15 BCS Spaces (0x20)
Encryption (ENCRYP)	0	1 digit - required default
Graphic Type (SFMT)	C	1 character - indicates CGM
Reserved for Future Use (SSTRUCT)	0000000000000	13 BCS Zeros (0x30) - reserved
Graphic Display Level (SDLVL)	005	3 digits
Graphic Attachment Level (SALVL)	001	3 digits
Graphic Location (SLOC)	0000000000	10 digits
First Graphic Bound Location (SBND1)	0066001705	10 digits
Graphic Color (SCOLOR)	M	indicates CGM file contains no color components
Second Graphic Bound Location (SBND2)	0070001990	10 digits
Reserved for Future Use (SRES2)	00	2 BCS Zeros (0x30) - reserved

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NITF GRAPHIC SUBHEADER FIELD	FORMAT	COMMENT
Graphic Extended Subheader Data Length (SXSHDL)	00000	5 digits

E.5.3 Explanation of the Third Graphic Subheader. The third graphic is a CGM annotation (HQ Building). It is attached to the base image. Its location as recorded in SLOC is measured as an offset from the upper left corner of the base image, in this case SLOC is 0,0 and the offsetting for this graphic is actually done within the CGM construct itself.

TABLE E-7. Graphic Subheader for the Fourth Graphic.

NITF GRAPHIC SUBHEADER FIELD	FORMAT	COMMENT
File Part Type (SY)	SY	2
Graphic Identifier (SID)	0000000004	10
Graphic Name (SNAME)	MAJOR TEST FACILITY	19 characters followed by 1 ECS Spaces (0x20) - l 20 characters
Graphic Security Classification (SSCLAS)	U	1 character
Graphic Classification Security System (SSCLSY)		2 characters
Graphic Codewords (SSCODE)		11 BCS Spaces (0x20)
Graphic Control and Handling (SSCTLH)		2 BCS Spaces (0x20)
Graphic Releasing Instructions (SSREL)		20 BCS Spaces (0x20)
Graphic Declassification Type (SSDCTP)		2 BCS Spaces (0x20)
Graphic Declassification Date (SSDCDT)		8 BCS Spaces (0x20)
Graphic Declassification Exemption (SSDCXM)		4 BCS Spaces (0x20)
Graphic Downgrade (SSDG)		1 BCS space (0x20)
Graphic Downgrade Date (SSDGDT)		8 BCS Spaces (0x20)
Graphic Classification Text (SSCLTX)		43 ECS Spaces (0x20)
Graphic Classification Authority Type (SSCATP)		1 BCS space (0x20)
Graphic Classification Authority (SSCAUT)		40 ECS Spaces (0x20)
Graphic Classification Reason (SSCRSN)		1 BCS space (0x20)
Graphic Security Source Date (SSSRDT)		8 BCS Spaces (0x20)
Graphic Security Control Number (SSCTLN)		15 BCS Spaces (0x20)
Encryption (ENCRYP)	0	1 digit - required default
Graphic Type (SFMT)	C	1 character - indicates CGM
Reserved for Future Use (SSTRUCT)	0000000000000	13 BCS Zeros (0x30) - reserved
Graphic Display Level (SDLVL)	006	3 digits

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NITF GRAPHIC SUBHEADER FIELD	FORMAT	COMMENT
Graphic Attachment Level (SALVL)	002	3 digits
Graphic Location (SLOC)	0008500655	10 digits relative to origin of second image
First Graphic Bound Location (SBND1)	0008500655	10 digits relative to origin of second image
Graphic Color (SCOLOR)	M	indicates CGM file contains no color components
Second Graphic Bound Location (SBND2)	0012001120	10 digits relative to origin of second image
Reserved for Future Use (SRES2)	00	2 BCS Zeros (0x30) - reserved
Graphic Extended Subheader Data Length (SXSHDL)	00000	5 digits

E.5.4 Explanation of the Fourth Graphic Subheader. The fourth graphic is a CGM graphic. It is the MAJOR TEST FACILITY text. It is attached to the subimage. Therefore, its location as recorded in SLOC is measured as an offset from the upper left corner of the subimage.

TABLE E-8. Graphic Subheader for the Fifth Graphic.

NITF GRAPHIC SUBHEADER FIELD	FORMAT	COMMENT
File Part Type (SY)	SY	2
Graphic Identifier (SID)	0000000005	10
Graphic Name (SNAME)	COMMUNICATION ARROW	19 characters followed by 1 ECS space (0x20) - 20 characters
Graphic Security Classification (SSCLAS)	U	1 character
Graphic Classification Security System (SSCLSY)		2 characters
Graphic Codewords (SSCODE)		11 BCS Spaces (0x20)
Graphic Control and Handling (SSCTLH)		2 BCS Spaces (0x20)
Graphic Releasing Instructions (SSREL)		20 BCS Spaces (0x20)
Graphic Declassification Type (SSDCTP)		2 BCS Spaces (0x20)
Graphic Declassification Date (SSDCDT)		8 BCS Spaces (0x20)
Graphic Declassification Exemption (SSDCXM)		4 BCS Spaces (0x20)
Graphic Downgrade (SSDG)		1 BCS space (0x20)
Graphic Downgrade Date (SSDGDT)		8 BCS Spaces (0x20)
Graphic Classification Text (SSCLTX)		43 ECS Spaces (0x20)
Graphic Classification Authority Type (SSCATP)		1 BCS space (0x20)

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NITF GRAPHIC SUBHEADER FIELD	FORMAT	COMMENT
Graphic Classification Authority (SSCAUT)		40 ECS Spaces (0x20)
Graphic Classification Reason (SSCRSN)		1 BCS space (0x20)
Graphic Security Source Date (SSSRDT)		8 BCS Spaces (0x20)
Graphic Security Control Number (SSCTLN)		15 BCS Spaces (0x20)
Encryption (ENCRYP)	0	1 digit - required default
Graphic Type (SFMT)	C	1 character - indicates CGM
Reserved for Future Use (SSTRUCT)	0000000000000	13 BCS Zeros (0x30) - reserved
Graphic Display Level (SDLVL)	007	3 digits
Graphic Attachment Level (SALVL)	001	3 digits
Graphic Location (SLOC)	0047000040	10 digits
First Graphic Bound Location (SBND1)	0047000040	10 digits
Graphic Color (SCOLOR)	M	indicates CGM file contains no color components
Second Graphic Bound Location (SBND2)	0059000600	10 digits
Reserved for Future Use (SRES2)	00	2 zeros (0x30) - reserved
Graphic Extended Subheader Data Length (SXSHDL)	00000	5 digits

E.5.5 Explanation of the Fifth Graphic Subheader. The fifth graphic is a CGM graphic. It is the COMMUNICATIONS NODE annotation with associated arrow. It is attached to the base image. Therefore, its location as recorded in SLOC is measured as an offset from the upper left corner of the base image.

E.6 Explanation of the Text Subheaders. There are 5 text segments included in the file. Other than the text data they contain, text files 1 to 4 differ only in matters such as title, date-time of creation, and ID. Therefore, only the first is discussed, since the subheaders of the other three are essentially the same, TABLE E-9. Text file 5 is a US Message Text Formatting (USMTF) file, TABLE E-10.

E.6.1 Explanation of the First Text Subheader. The first text segment is unclassified and was created on September 30, 1996 at 22:45 hours.

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TABLE E-9. Text Subheader for the Text Document.

NITF TEXT SUBHEADER FIELD	FORMAT	COMMENT
File Part Type (TE)	TE	2 characters
Text Identifier (TEXTID)	0000001	7 characters
Text Attachment Level (TXTALVL)	001	3 characters
Text Date and Time (TXTDT)	19960930224530	14 characters
Text Title (TXTITL)	First sample text file.	22 characters followed by 58 ECS Spaces (0x20) - 80 characters
Text Security Classification (TSCLAS)	U	1 character
Text Classification Security System (TSCLSY)		2 characters
Text Codewords (TSCODE)		11 BCS Spaces (0x20)
Text Control and Handling (TSCTLH)		2 BCS Spaces (0x20)
Text Releasing Instructions (TSREL)		20 BCS Spaces (0x20)
Text Declassification Type (TSDCTP)		2 BCS Spaces (0x20)
Text Declassification Date (TSDCDT)		8 BCS Spaces (0x20)
Text Declassification Exemption (TSDCXM)		4 BCS Spaces (0x20)
Text Downgrade (TSDG)		1 BCS space (0x20)
Text Downgrade Date (TSDGDT)		8 BCS Spaces (0x20)
Text Classification Text (TSCLTX)		43 ECS Spaces (0x20)
Text Classification Authority Type (TSCATP)		1 BCS space (0x20)
Text Classification Authority (TSCAUT)		40 ECS Spaces (0x20)
Text Classification Reason (TSCRSN)		1 BCS space (0x20)
Text Security Source Date (TSSRDT)		8 BCS Spaces (0x20)
Text Security Control Number (TSCTLN)		15 BCS Spaces (0x20)
Encryption (ENCRYP)	0	1 digit - required default
Text Format (TXTFMT)	STA	3 characters
Text Extended Subheader Data Length (TXSHDL)	00000	5 digits

E.6.1.1 Sample USMTF Message. The following is a sample USMTF message that is a data portion associated with a text subheader for MTF.

EXER/GRAU MESSER//
OVERLAY/A/420TH MI BDE/24153000ZFEB98/OP AREA 3//
GENTEXT/OVERLAY DESCRIPTION/THIS OVERLAY IDENTIFIES AN
APACHE HELICOPTER IN OPERATION AREA 3//

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IMG/DTE:970223/PRJ:MI/MSN:C031/FR/56-61,68/-/TOT:1322Z/50000
/BEN:0173-99999/SFX:A123/CAT:80000//
ICONID/A/421ST MI BDE/24190000ZFEB98/001/001/AFAPMHA00000000/E//
EQUIP/APACHE/AIRCRAFT/MAIN//
ICONLOC/313448.0N1102032W/-/-/-/-/ELE:00370M/273T/0.OKPH//

TABLE E-10. Text Subheader for USMTF.

NITF TEXT SUBHEADER FIELD	FORMAT	COMMENT
File Part Type (TE)	TE	2 characters
Text Identifier (TEXTID)	0000005	7 characters
Text Attachment Level (TXTALVL)	001	3 characters
Text Date and Time (TXTDT)	19980224153000	14 characters
Text Title (TXTITL)	Fifth sample text file.	23 characters followed by 57 ECS Spaces (0x20) - 80 characters
Text Security Classification (TSCLAS)	U	1 character
Text Classification Security System (TSCLSY)		2 characters
Text Codewords (TSCODE)		11 BCS Spaces (0x20)
Text Control and Handling (TSCTLH)		2 BCS Spaces (0x20)
Text Releasing Instructions (TSREL)		20 BCS Spaces (0x20)
Text Declassification Type (TSDCTP)		2 BCS Spaces (0x20)
Text Declassification Date (TSDCDT)		8 BCS Spaces (0x20)
Text Declassification Exemption (TSDCXM)		4 BCS Spaces (0x20)
Text Downgrade (TSDG)		1 BCS space (0x20)
Text Downgrade Date (TSDGDT)		8 BCS Spaces (0x20)
Text Classification Text (TSCLTX)		43 ECS Spaces (0x20)
Text Classification Authority Type (TSCATP)		1 BCS space (0x20)
Text Classification Authority (TSCAUT)		40 ECS Spaces (0x20)
Text Classification Reason (TSCRSN)		1 BCS space (0x20)
Text Security Source Date (TSSRDT)		8 BCS Spaces (0x20)
Text Security Control Number (TSCTLN)		15 BCS Spaces (0x20)
Encryption (ENCRYP)	0	1 digit - required default
Text Format (TXTFMT)	MTF	3 characters
Text Extended Subheader Data Length (TXSHDL)	00000	5 digits

E.7 Product Considerations. The NITF provides a very flexible means to package imagery products. One of the main objectives of NITF is to provide increased interoperability among potentially disparate imagery systems. For the

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purposes of NITF, interoperability means the ability to exchange NITF formatted imagery products among NITF capable systems in a manner that is meaningful and useful to the end users. This places a significant burden on NITF read capable implementations to be able to interpret and use potentially any combination of NITF file format options that may be created by NITF file producers. Consequently, significant care should be taken when defining product specifications for NITF formatted imagery products. The objective of the following discussion is to describe several generalized product configurations that can be used as the basis for defining specific imagery products. These product configurations are typical of those successfully used within the imagery and mapping community to date.

E.7.1 NITF Product Configurations.

E.7.1.1 General. An imagery product may potentially be produced under one of the following concepts.

E.7.1.2 Single File, Single Base Image. This is the most common use of the NITF format. In this product concept, the NITF file is produced with a focus on a single image, commonly called the 'base image'. All other segments and extended data within the file are focused on amplifying the information portrayed in the base image.

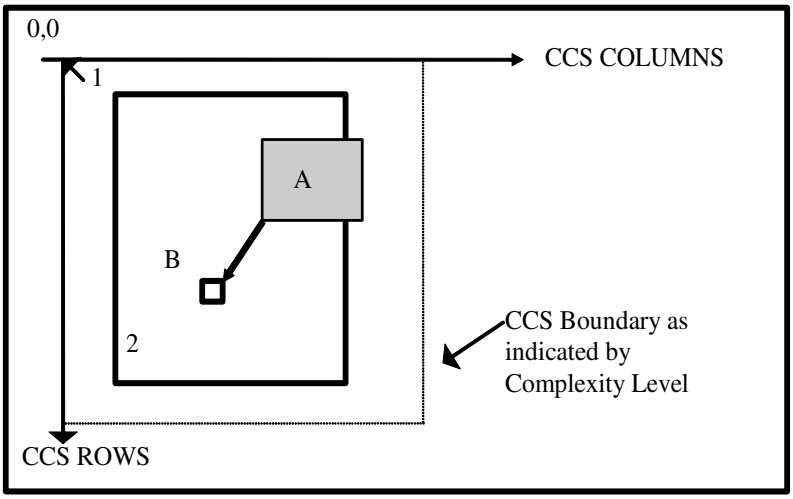
E.7.1.3 Single File, Multiple Images. In this product concept, the NITF file is produced containing multiple images, all of which have equal or similar significance to the value of the product. Other segments and extended data within the file are focused on amplifying the information portrayed in the image(s) to which they are associated.

E.7.1.4 Single File, No Image. This type of product may only have GSs, or only TSs, or only extension segments, or any combination of these segments. The significance of the data within the file may pertain only to that file, or it may pertain to one or more files with which it is associated.

E.7.1.5 Multiple correlated files. For this product concept, the product consists of multiple NITF files that are interrelated as defined in the governing product specification.

E.7.1.6 Single File, Single Base Image. For this type of product file, there is one image of central focus, the base image, placed on the CCS plane. Its first pixel may be located at the origin (0,0) of the CCS, or off-set from the CCS origin according to the row/column coordinate values placed in the location (ILOC) field of the image subheader. FIGURE E-2 provides a representative portrayal for the following discussion.

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1. Common Coordinate System Origin (0,0)
2. Image DLVL =100 ALVL =000 ILOC =45,50
A. Inset Image DLVL =110 ALVL =100 ILOC =50,300
B. Inset Graphic DLVL =120 ALVL =110 SLOC =250,-100
(Box & Arrow)

FIGURE E-2. Single File, Single Base Image.

E.7.1.7 Image Segment (IS) Overlays. Additional images, often called subimages or inset images, may be included as separate ISs in the file. The purpose of these images is to add information or clarity about the base image. Their placement in the CCS plane is controlled by the value of each segment's ALVL and Location (LOC) row/column value. When overlay images are attached to the base image, the LOC value represents a row/column off-set in the CCS from the location specified by the base image row/column LOC value. If the overlay image is unattached to any other segment (ALVL=000), the overlay's LOC value is a row/column off-set from the CCS origin (0,0).

E.7.1.8 Graphic Segment (GS) Overlays. GSs are used to provide graphical (lines, polygons, ellipses, etc.) and textual annotation to the base image. The graphic representation is done using CGM. In a manner similar to IS overlays, the placement of graphics in the CCS plane is controlled by the value of each segment's ALVL and LOC values. CGM has its own internal Cartesian coordinate space called "Virtual Display Coordinates (VDC)" that has its own defined origin (0,0) point. The row/column value in the GS LOC field identifies the placement of the graphic's VDC origin point relative to the CCS origin when ALVL=000, or relative to the segment LOC to which it is attached.

E.7.1.9 Non-Destructive Overlays. NITF IS and GS overlays are handled in a non-destructive manner. The overlays may be placed anywhere within the bounds of the CCS (defined for a specific NITF file by the CLEVEL. They may be placed totally on the base image, partially on the base image, or entirely off of the base image. Any IS or GS can be placed on or under any other segment, fully or partially. The visibility of pixel values of overlapping segments is determined by the DLVL assigned to that segment. Each displayable segment (images and graphics) is assigned a DLVL (ranging from 001 to 999) that is unique within the file. At any CCS pixel location shared by more than one image or graphic, the visible pixel value is the one from the segment having the greatest DLVL value. If the user of an NITF file opts to move an overlay, or turn off the presentation of an overlay, the next greatest underlying pixel value(s) will then become visible. This approach allows for the non-destructible nature of NITF overlays as opposed to the 'burned in' approach where overlay pixel values are used to replace pixels values of the underlying image.

E.7.1.10 Text Segment (TS). TSs allow inclusion in the NITF file of textual information related to the base image, perhaps a textual description of the activities portrayed in the image. For the purpose of this standard, segment refers to header or subheader and associated data. Below is a sample GRAPHREP text message:

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EXER/GRAU MESSER//
MSGID/GRAPHREP-OVERLAY/420TH MI BDE//
OVERLAY/A/420TH MI BDE/24153000ZFEB98/OP AREA 3//
GENTEXT/OVERLAY DESCRIPTION/THIS OVERLAY IDENTIFIES AN
APACHE HELICOPTER IN OPERATION AREA 3//
IMG/DTE:970223/PRJ:MI/MSN:C031/FR/56-61,68/-/TOT:1322Z/50000
/BEN:0173-99999/SFX:A123/CAT:80000//
ICONID/A/421ST MI BDE/24190000ZFEB98/001/001/AFAPMHA00000000/E//
EQUIP/APACHE/AIRCRAFT/MAIN//
GRD/24190000ZFEB98/-/ACRCVR//
ICONLOC/313448.0N1102032W/-/-/-/-/ELE:00370M/273T/0.OKPH//

E.7.1.11 Extension Data. The NITF file header and each standard data type sub-header have designated expandable fields to allow for the optional inclusion of extension data. The inclusion of extension data provides the ability to add data/information about the standard data type (metadata) that is not contained in the basic fields of the headers and subheaders. The additional data is contained within one or more NITF TRE that are placed in the appropriate field (user defined data field or extended data field) of the standard data type subheader for which the metadata applies. When TRE have application across multiple data types in the file, or otherwise apply to the entire NITF file in general, they are placed in the appropriate file header fields. Whereas general purpose NITF readers should always be able to portray IS and GS and act on standard header and subheader data, they may not always be able to act on product specific extension data. Upon receipt of a file that contains extension data, an NITF compliant implementation should at least ignore the extensions and properly interpret the other legal components of the NITF file. Exemplary use of TRE:

a. Data about people, buildings, places, landmarks, equipment or other objects that may appear in the image.

b. Data to allow correlation of information among multiple images and annotations within an NITF file.

c. Data about the equipment settings used to obtain the digital image, x-ray, etc.

d. Data to allow geo-positioning of items in the imagery or measurement of distances of items in the imagery.

E.7.1.12 Single File, Multiple Images Without Overlap. For this type of product file, multiple images of equal or similar focus (multiple 'base' images) are placed within the CCS plane. Each image is located at an off-set from the CCS origin such that there is no overlap among the images. The CLEVEL of the file must be chosen such that the bounds of the CCS for the file are sufficient to contain the extent of all segments within the file. FIGURE E-3 provides a representative portrayal for this product type.

E.7.1.13 Overlays. Each image may be overlaid with additional ISs and GSs in the same fashion as described for the single file, single image case above. All overlays associated with a specific image should be attached to that specific image. Display levels assigned to each image and graphic in the file must be unique within the file.

E.7.1.14 Text Segment (TS). Each TS should be clearly marked as to whether it applies to the file as a whole, or if it is associated with specific images within the file. For the purpose of this standard, segment refers to header or subheader and associated data.

E.7.1.15 Extension Data. TRE are placed in the file header extension fields when applicable to the file as a whole. Extensions specific to a segment are placed in that segment's subheader.

E.7.1.16 Single File, No Image. An NITF single file product does not always contain an image. It could contain one or more GSs, one or more TSs, one or more extension segments, or any combination of these non-ISs. The file may be useful as a stand-alone product, or it may be intended for use in conjunction with other NITF files. For example, the file could contain graphic overlays to be merged with or applied to another NITF file that was prepositioned or transmitted at an earlier time. Any general purpose NITF reader should at least be able to interpret and render the standard segments of no image NITF files on a stand-alone basis.

E.7.1.17 Multiple Correlated Files. An imagery product may be composed of multiple NITF files that are interrelated in a specified manner. This approach vastly increases the potential combination and permutation of options

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a general purpose NITF reader would need to support to maintain full interpret capability. Therefore, each NITF file in a multiple correlated file set must be structured such that a general purpose NITF reader can properly interpret and render the file as if it were a stand-alone product. The correlation of multiple NITF files in a single product must be explicitly and unambiguously defined in a product specification. NITF readers can then be further categorized according to specific multiple file product specifications that are supported. Representative use of multiple correlated NITF files includes:

E.7.1.18 Stereo Imagery. Some stereo image products are composed of separate NITF files for the stereo components of each image scene.

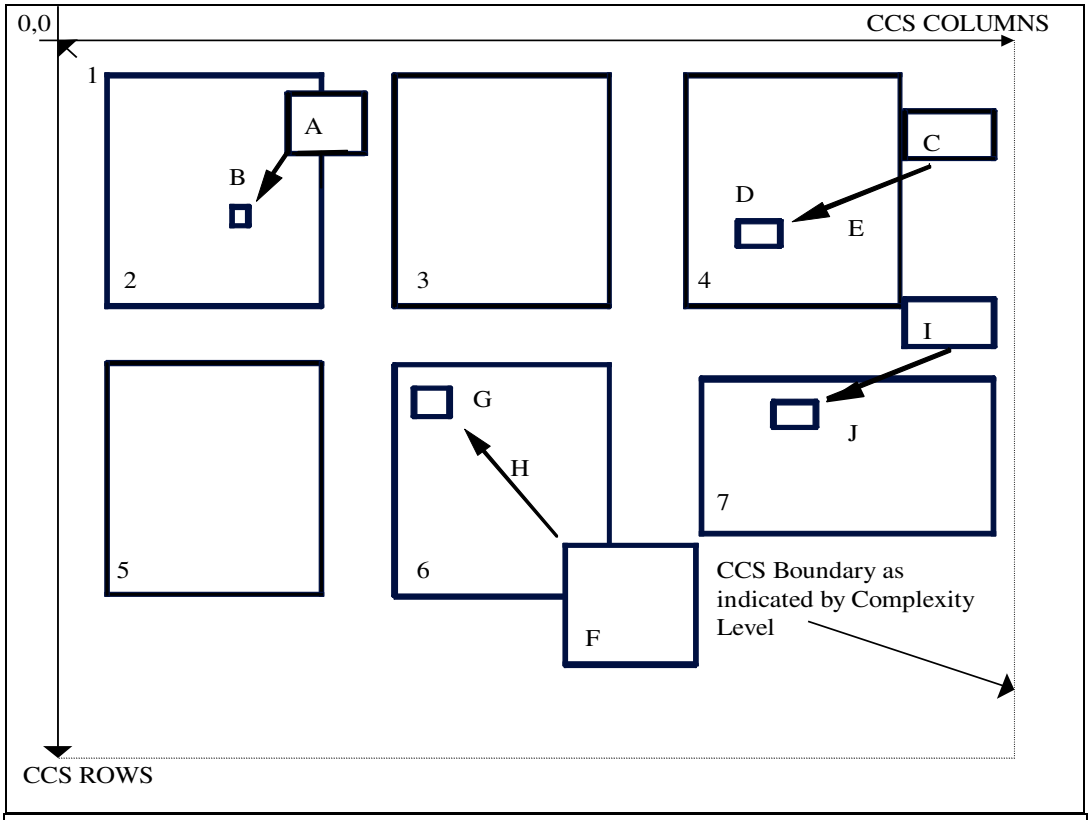
E.7.1.19 Imagery Mosaics. Some extremely large image and map products consist of multiple NITF files structured such that they can be pieced together in mosaic fashion by the interpret application as if the multiple files were a single larger image.

E.7.1.20 Reduced Resolution Data Sets (RSETS). Some Rset products are composed of multiple NITF files. One file contains a full resolution image and the other files contain the same image in a variety of lower resolutions.

E.7.1.21 Imagery and Maps. Some geo-positioning products exist which consist of multiple separate NITF files containing interrelated maps, images, graphics, legends, product indices, and geo-reference data.

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APPENDIX E



		Display Level	Attachment Level	Location
1.	Common Coordinate System Origin (0,0)			
2.	Image	DL = 100	AL = 000	50, 50
	(A) Image Segment Overlay	DL = 110	AL = 100	50, 300
	(B) Graphic Overlay (CGM Rectangle & Arrow)	DL = 120	AL = 110	100,-50
3.	Image	DL = 200	AL = 000	50, 550
4.	Image	DL = 300	AL = 000	50,1050
	(C) Image Segment Overlay	DL = 310	AL = 300	100, 450
	(D) Graphic Segment Overlay (CGM Rectangle)	DL = 320	AL = 300	300, 75
	(E) Graphic Segment Overlay (CGM Arrow)	DL = 330	AL = 300	300,200
5	Image	DL = 400	AL = 000	800, 50
6	Image	DL = 500	AL = 000	800, 550
	(F) Image Segment Overlay	DL = 510	AL = 500	300, 300
	(G) Graphic Segment Overlay (CGM Rectangle)	DL = 520	AL = 500	35, 35
	(H) Graphic Segment Overlay (CGM Arrow)	DL = 530	AL = 500	125, 125
7	Image	DL = 600	AL = 000	820,1050
	(I) Image Segment Overlay	DL = 610	AL = 600	-150, 400
	(J) CGM Graphic Overlay (Rectangle & Arrow)	DL = 620	AL = 600	50, 100

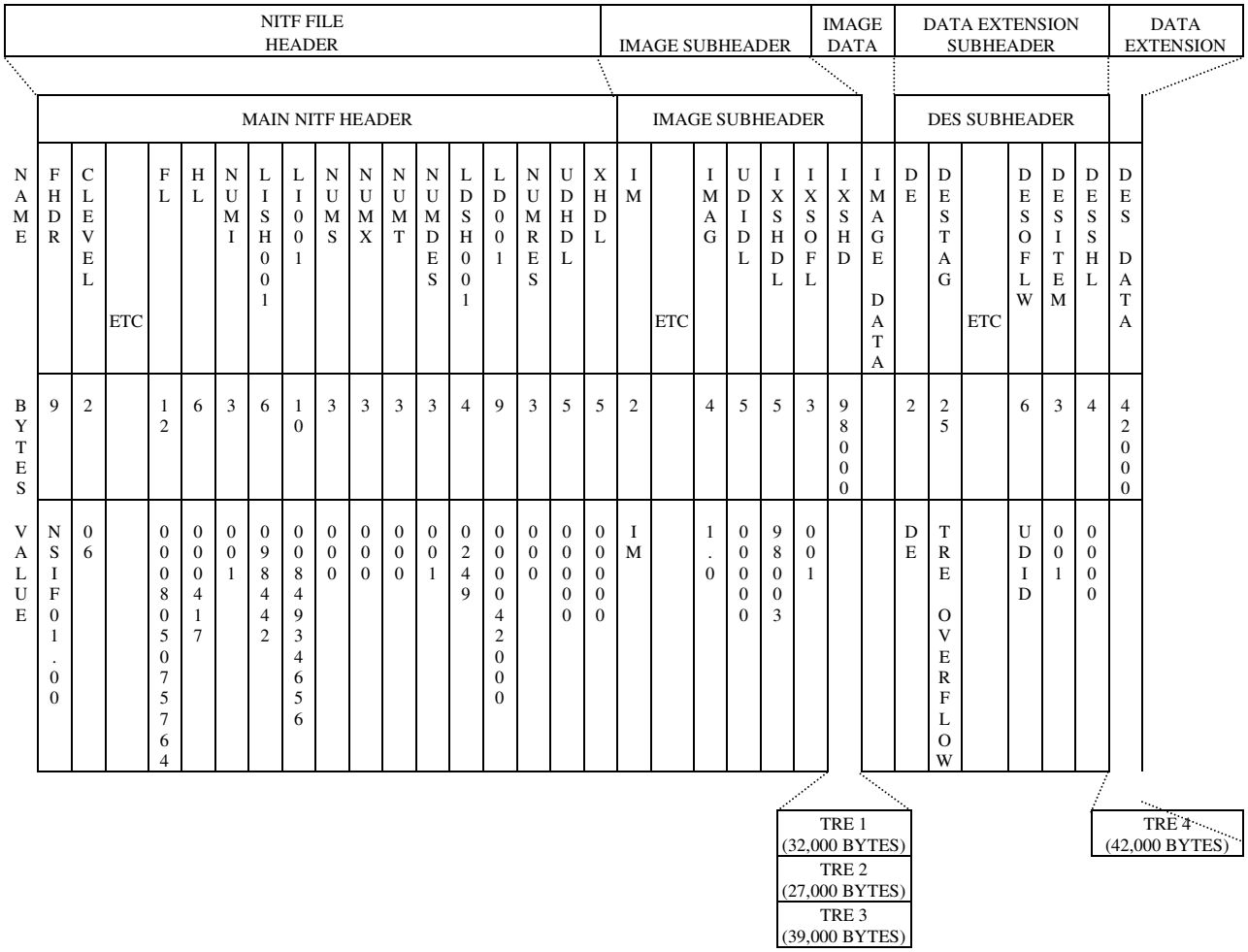
FIGURE E-3. Single File, Multiple Images.

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APPENDIX E

E.7.2 Sample NITF File Structure. The following is an example of handling a file that has control TRE with overflow. The file has a single image.

TABLE E-11. Sample NITF File Structure.



Note: Capacity of IXSHD is 99,999 bytes. You cannot split a TRE, therefore the first 3 TRE fit into the IXSHD and the 4th TRE is overflowed into the Data Extension Area.

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APPENDIX F. IMODE EXPLANATIONS (Informative)

F.1 IMODE Explanations

F.1.1 Scope. This appendix provides additional figures that illustrate the various still and motion imagery IMODEs. This appendix is not a mandatory part of this standard and the information contain herein is intended for guidance. In the event of any discrepancy between the information contained in this appendix and the IMODE descriptions found in section 5.4.3.3.1 and its subsections, the text in section 5.4.3.3.1 and its subsections take precedence.

F.2 Example Image Sequence. The following figures are provided to further understanding of IMODE concepts and interactions between IMODE, still image and MI compression, image blocks and MI spatiotemporal blocks. The "looping" tables presented in section 5.4.3.3.1. are a concise way to express data storage order for the various IMODEs but they require background knowledge of computer programming concepts. In this section the various IMODEs are explained using diagrams based on the following example "image sequence".

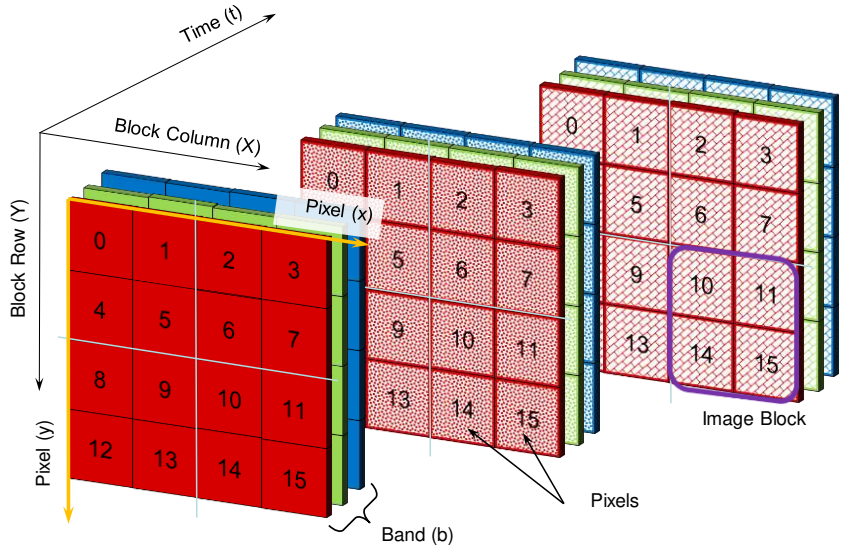


FIGURE F-1. IMODE Sample Image Sequence

F.2.1 IMODE Sample Image Sequence FIGURE F-1 illustrates a simple three frame MI sequence that will be used for the IMODE discussion and examples. Each frame is a three band color image comprised of 16 pixels. Spatiotemporal blocks have been used to block each frame into four blocks (two rows by two columns). Where still image IMODEs are discussed, only the first image/frame in the temporal sequence is considered (frame with solid color bands) and the remaining two frames are discarded. Where MI IMODEs are discussed, all three frames are used in the discussion. It is best to first consider uncompressed imagery data when discussing IMODE. Compression of still images or MI complicates the IMODE discussion due to the operation of compression algorithms on the imagery. Compression algorithms will cause some of the IMODE indices or dimensions to "disappear" as most compression algorithms will jointly encode pixels, bands and frames together.

F.3 Still Imagery IMODEs FIGURE F-2 illustrates the still image IMODE S. IMODE S is commonly called "band sequential" because pixel values for a given band are located contiguously together. On the left-hand side of the figure, a three band, 16 sample "image" is presented. The image has been blocked into 2x2 blocks which are indicated by the blue lines in the figure. On the bottom right-hand side of the figure, the looping table for IMODE S from section 5.4.3.3.1 has been reproduced. The top-right portion of the figure illustrates how the pixel/data samples would be sequentially ordered. The sample order is determined by reading from top left to bottom right. The pixel band values are ordered left to right, top to bottom within a block. Blocks within a band

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are ordered left to right, top to bottom as well. Once all values are written for a band, the pixel band values for the next band are given. This is repeated until all data has been processed.

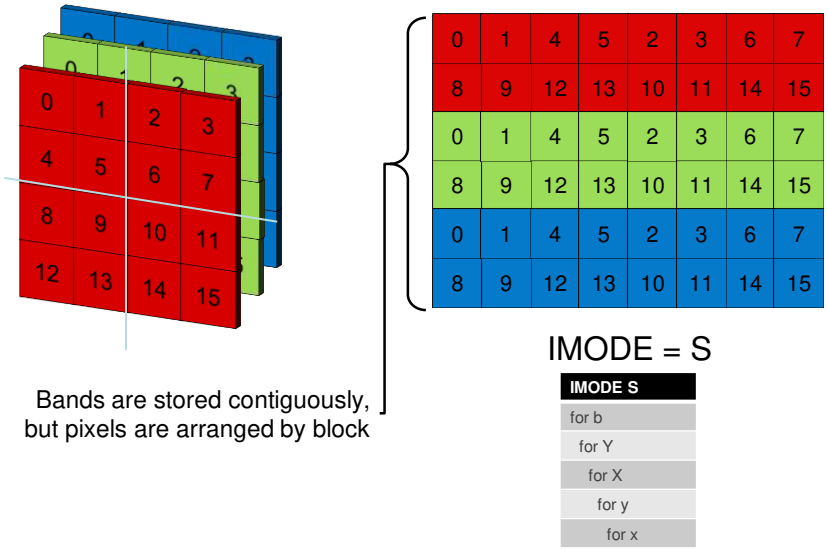


FIGURE F-2. IMODE S Example

F.3.1 IMODE S in TABLE F-1 illustrates how the indexing/looping construct presented in section 5.4.3.3.1 may be used to determine the sequential ordering of pixels band/sample values for IMODE S. The lower right-hand portion of FIGURE F-2 provides the nested loops that determine the ordering of the sample values. The table illustrates the usage of the nested loops to select and order the sample values from the three band image. The rows in TABLE F-1 have been color coded to match the band colors used in FIGURE F-2. The left most index (column b or band) in TABLE F-1 changes the slowest and the right-most index (column x or sample column) changes the fastest. This corresponds to the nesting of the loops in FIGURE F-2, where the innermost loops change the quickest. The pixel band data is ordered sequentially by row in the table.

TABLE F-1. IMODE S Pixel Ordering

b (band)	Y (block row)	X (block column)	y (sample row)	x (sample column)	Pixel number
1	1	1	1	1	0
1	1	1	1	2	1
1	1	1	2	1	4
1	1	1	2	2	5
1	1	2	1	1	2
1	1	2	1	2	3
1	1	2	2	1	6
1	1	2	2	2	7
1	2	1	1	1	8
1	2	1	1	2	9
1	2	1	2	1	12
1	2	1	2	2	13
1	2	2	1	1	10

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b (band)	Y (block row)	X (block column)	y (sample row)	x (sample column)	Pixel number
1	2	2	1	2	11
1	2	2	2	1	14
1	2	2	2	2	15
2	1	1	1	1	0
2	1	1	1	2	1
2	1	1	2	1	4
2	1	1	2	2	5
2	1	2	1	1	2
2	1	2	1	2	3
2	1	2	2	1	6
2	1	2	2	2	7
2	2	1	1	1	8
2	2	1	1	2	9
2	2	1	2	1	12
2	2	1	2	2	13
2	2	2	1	1	10
2	2	2	1	2	11
2	2	2	2	1	14
2	2	2	2	2	15
3	1	1	1	1	0
3	1	1	1	2	1
3	1	1	2	1	4
3	1	1	2	2	5
3	1	2	1	1	2
3	1	2	1	2	3
3	1	2	2	1	6
3	1	2	2	2	7
3	2	1	1	1	8
3	2	1	1	2	9
3	2	1	2	1	12
3	2	1	2	2	13
3	2	2	1	1	10
3	2	2	1	2	11
3	2	2	2	1	14
3	2	2	2	2	15

F.3.2 IMODE B in FIGURE F-3 illustrates the still image IMODE B. IMODE B is commonly called “band interleave by block”. Sample values within an image block are interleaved one band after another. Placing the loop over the image bands (b) inside of the loops over block row (Y) and block column (X), forces the pixel values to be ordered by band within each block before the next block of pixel values is presented.

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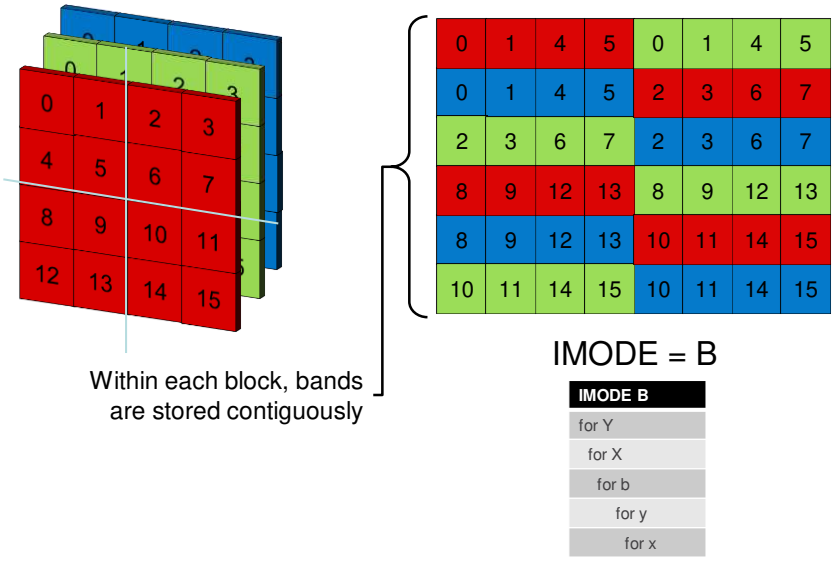


FIGURE F-3. IMODE B Example

F.3.3 IMODE R, see FIGURE F-4, is commonly known as “interleave by row” or “band interleave by line”. IMODE R orders pixel values with each NITF block by interleaving a row of values from each band. IMODE R is frequently used with uncompressed multi and hyperspectral images. This IMODE facilitates spectral processing across all bands, one row at a time.

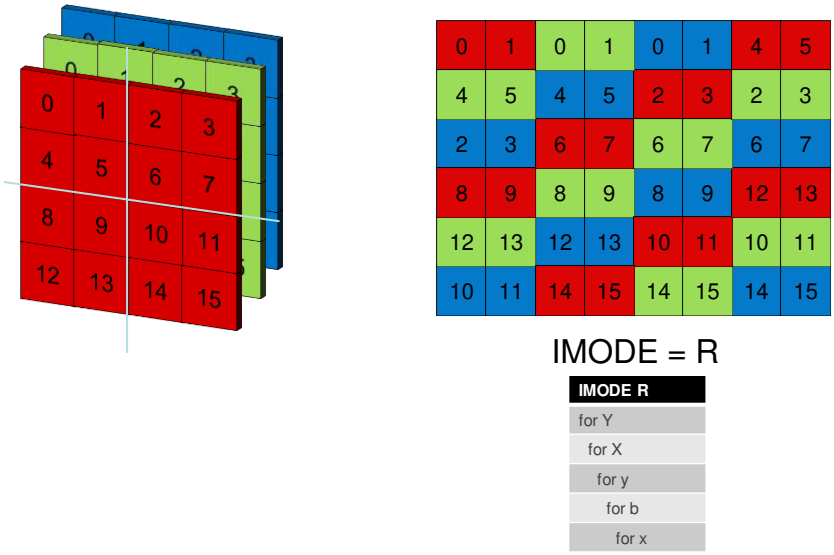


FIGURE F-4. IMODE R Example

F.3.4 IMODE P in FIGURE F-5 illustrates IMODE P. This IMODE is known as “interleave by pixel” or “band interleave by pixel”. In IMODE P the pixel values across the bands are contiguously stored together. IMODE P is another common way to store and process uncompressed multi and hyper spectral imagery.

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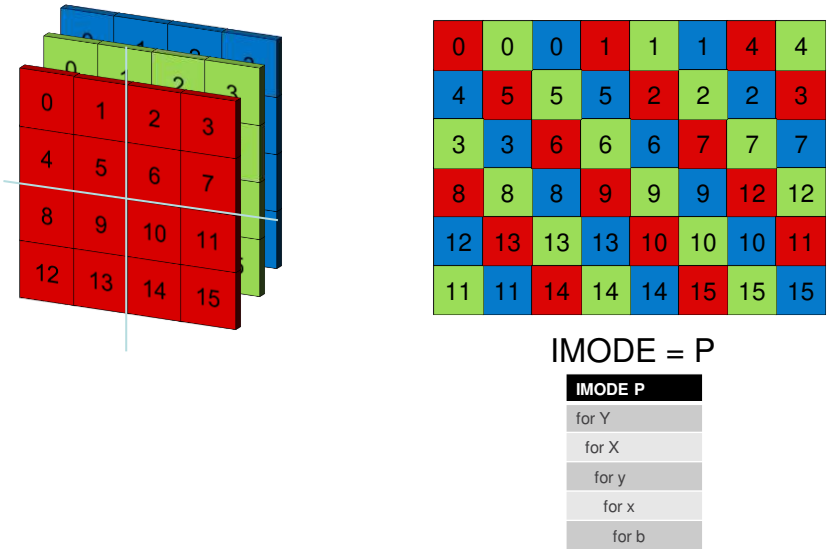


FIGURE F-5. IMODE P Example

F.3.5 The reader is encouraged to try to recreate TABLE F-1 for the other IMODES (B, R and P). This is an effective way to verify understanding of IMODE concepts.

F.4 Motion Imagery IMODEs. Adding a temporal dimension to the IMODEs adds an additional loop over time (t). In reality, the loop is over frame number even though it is thought of as time. Motion imagery is represented as a sequence of frames collected at different points in time over an area of common regard. Each frame in an MI collection has an associated time stamp indicating when it was collected. The temporal spacing between frames need not be uniform, but in many collections frames are generated at a fixed temporal rate. Between different MI collections, the frame rate may vary greatly, from 120 frames per second (or greater), to one frame per second (or slower). For example, if motion imagery were being used to capture the construction of building, frames might be collected at a very slow temporal rate, say once an hour or even once a day. The MI IMODEs are all derived from the still imagery IMODEs. There are six different MI IMODEs. In the following discussion, the IMODEs are demonstrated using the three frame sequence of "images" shown in FIGURE F-1.

F.4.1 IMODE D in FIGURE F-6 illustrates the MI IMODE D. IMODE D is derived from the still imagery IMODE S. A time loop has been placed into the IMODE S loop structure just inside the loop over spatiotemporal block column (X). For MI collections we use the term "spatiotemporal block" in place of "block" to make a distinction between MI and still imagery data and reinforce the temporal nature of MI data. In the top-left corner of the figure is a three frame sequence of three band imagery. Like bands in the three frames are given the same color (red, green, blue) and frames are differentiated from each other through the use of different textures (solid, speckle, brick).

F.4.2 IMODE D and S Similarity In the bottom-left corner of FIGURE F-6 the loop structures of IMODE D and IMODE S are presented to show their similarity. The right-half of the figure shows how the pixel values are sequentially ordered under IMODE D. If we were to read the pixels out of the MI data, one at a time, they would be read out as indicated starting in the top-left corner, proceeding left to right for each row, and top to bottom until the bottom-right corner of the figure is reached. IMODE D is called "temporal band sequential" in NGA.STND.0044. This IMODE is used in situations where the MI bands are to be treated individually. It is useful where each band is to be compressed into its own MI codestream, see NGA.STND.0044 for more information.

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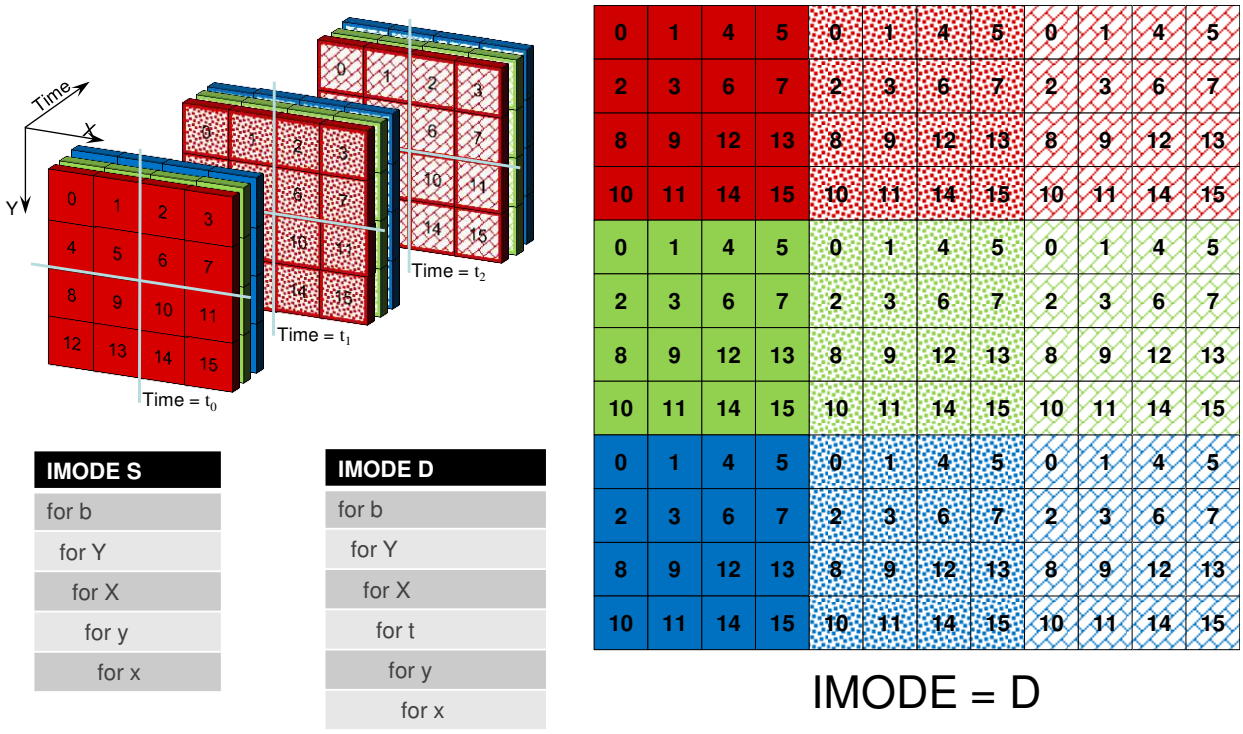


FIGURE F-6. IMODE D Example

F.4.3 IMODE Z in FIGURE F-7 illustrates IMODE Z. IMODE Z is based upon the still imagery IMODE S. It differs from IMODE D by placing the time loop on the very outermost level, outside of the bands loop. IMODE Z is simply IMODE S wrapped by a time loop. IMODE Z is called "band sequential temporal" within NGA.STND.0044. It is used for uncompressed data. Usage of IMODE Z with MI compression algorithms (IC equal to M9, MA, C9 or CA) or still image compression algorithms is not allowed.

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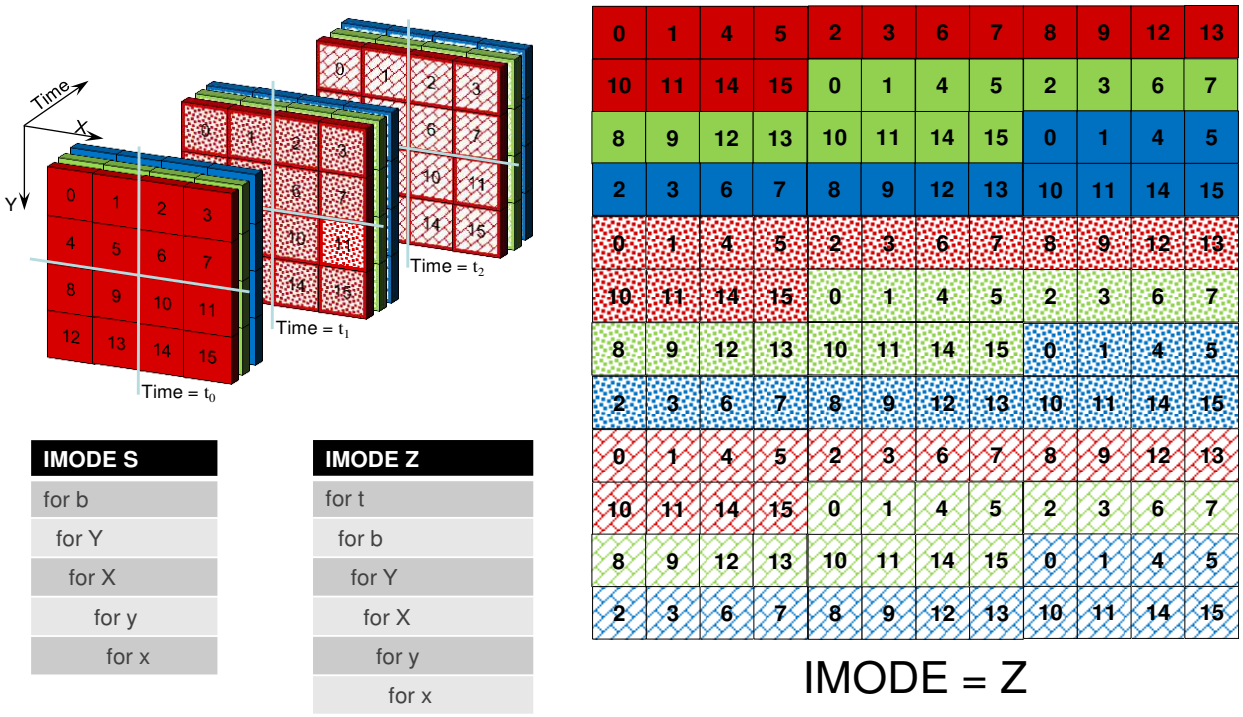


FIGURE F-7. IMODE Z Example

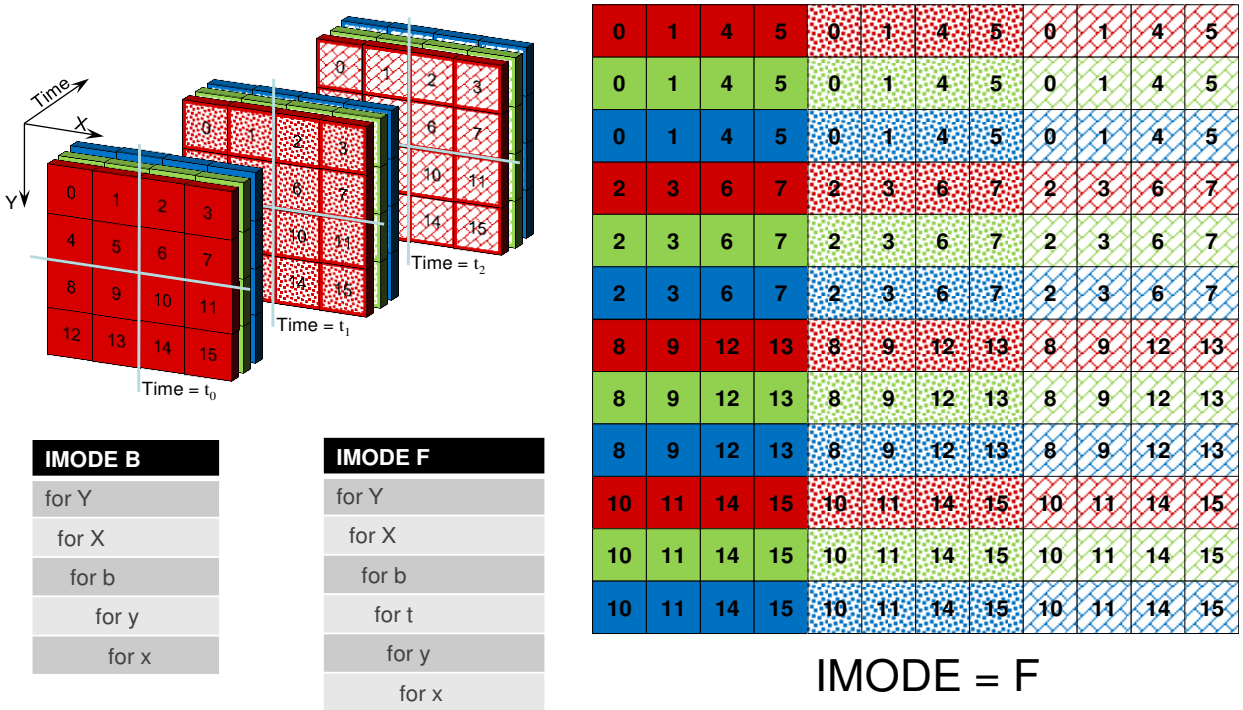


FIGURE F-8. IMODE F Example

F.4.4 IMODE F in FIGURE F-8 is based upon still image IMODE B. IMODE F is called “temporal band sequential by block” within NGA.STND.0044. A time loop has been inserted inside of the bands loop to create

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APPENDIX F

IMODE F. IMODE F is used in a similar fashion as IMODE D. It is used whenever the same band across different frames are collected together. In IMODE F, this behavior is constrained within each spatiotemporal block. IMODEs D and F are useful for independently compressing MI data on a band by band basis. The H.264/H.265 motion imagery compression algorithms are limited in the number of bands or color components they allow. Applications requiring more than three bands may compress each band separately as an individual elementary codestream. The codestreams are then interleaved together according to the IMODE used. This technique is similar to some of the JPEG/JPEG 2000 approaches used with multi and hyper spectral still imagery.

F.4.5 IMODE D and F Usage The usage of IMODEs D and F with compressed MI codestreams is illustrated in FIGURE F-9. The individual band/spatiotemporal block codestreams are indicated by "Codestream (Y,X)" where Y is the spatiotemporal block row and X represents the spatiotemporal block column. As can be readily seen, IMODE D collects together all codestreams for a band. IMODE F collects together all codestreams for a spatiotemporal block. See NGA.STND.0044 for more information.

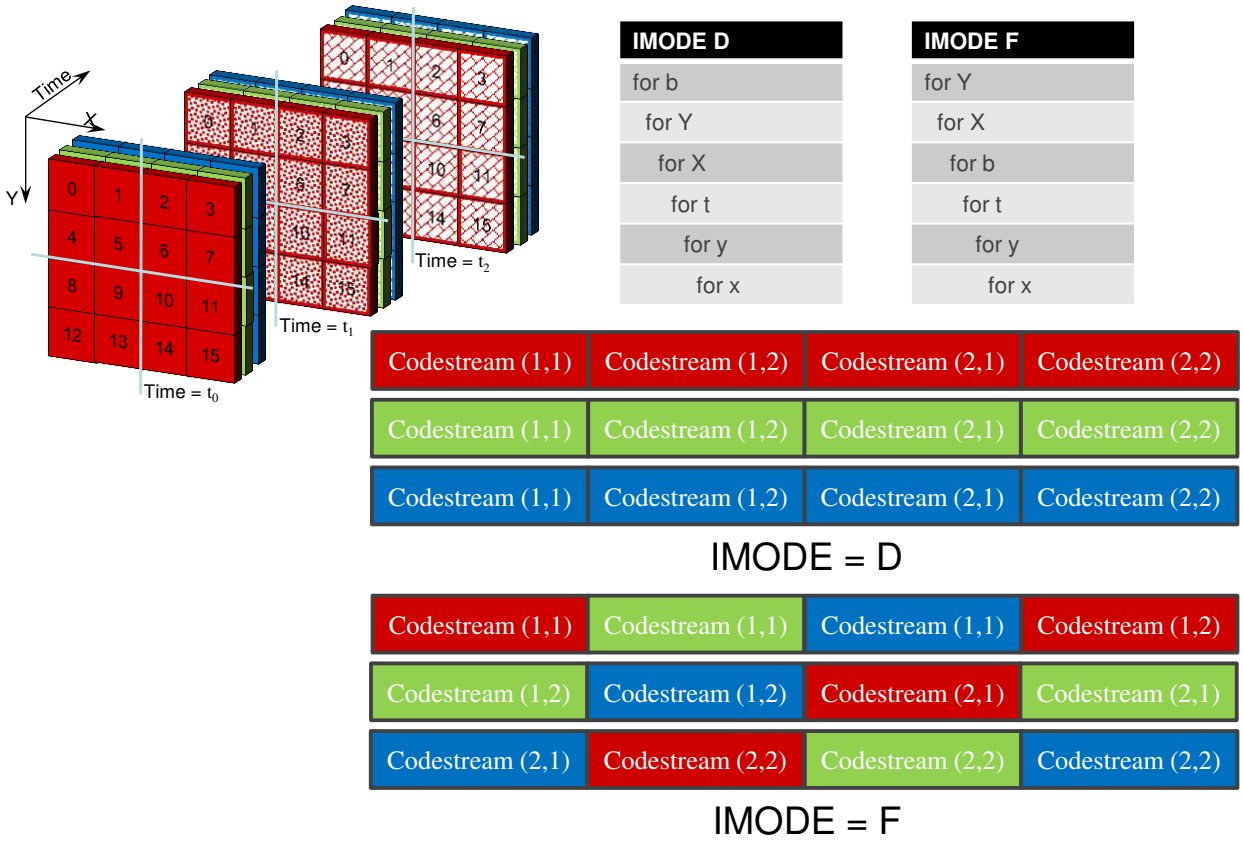


FIGURE F-9. IMODE D and F Compressed MI Codestream Usage

F.4.6 IMODE T in FIGURE F-10, is called "temporal band interleave by block" within NGA.STND.0044. This IMODE is derived from IMODE B by insertion of a temporal loop just outside of the bands loop. IMODE T would commonly be used for monochrome or color (three band) motion imagery, either compressed or uncompressed. It is common with MI compression techniques to treat monochrome motion imagery as a special case of color motion imagery where the three color bands (red, green and blue) contain the same pixel band values. It may seem that we are encoding three times the amount of data and should therefore expect a larger compressed file size. In practice, the file size does not increase by an appreciable amount. MI compression algorithms typically transform RGB (red, green, blue) into YCrCb (luma, chroma red, chroma blue) prior to compression. The RGB to YCrCb transform results in the luma band being equal to the monochrome motion

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imagery band and the chroma bands are zeroed out and contain no information. The chroma bands may there be encoded with a small number of bytes since all samples in the chroma bands are the same value.

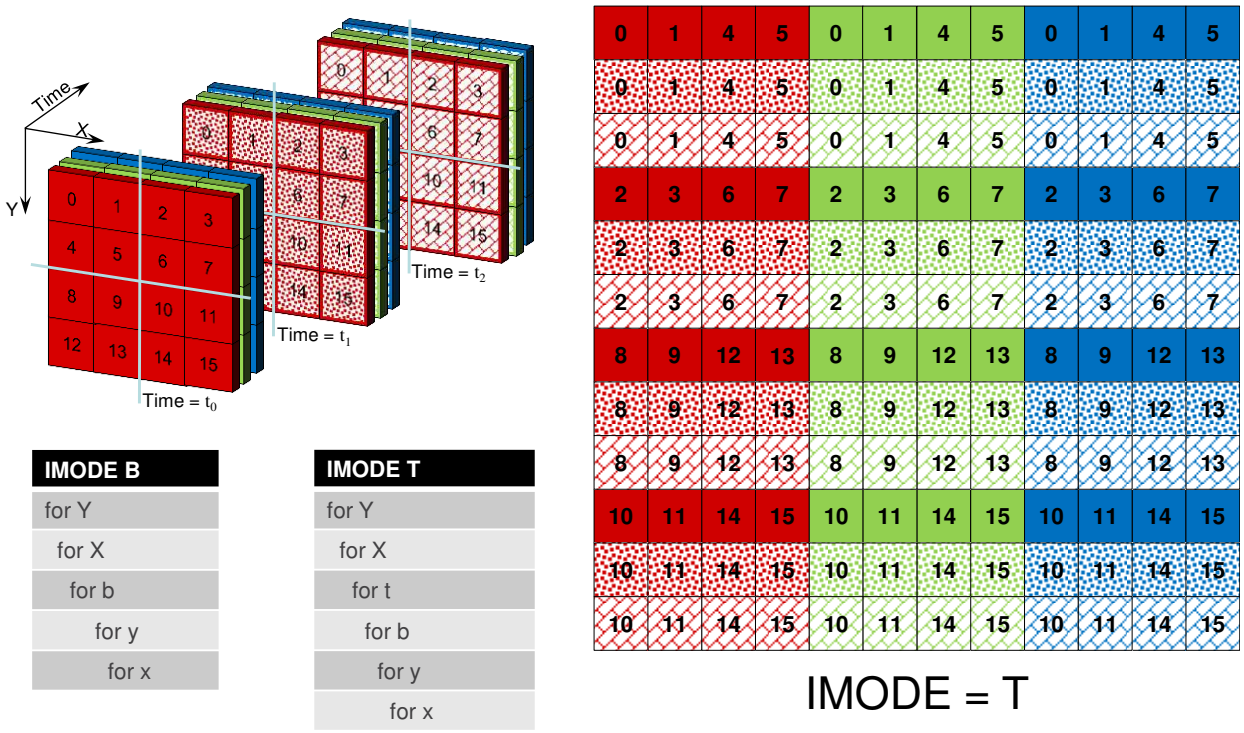


FIGURE F-10. IMODE T Example

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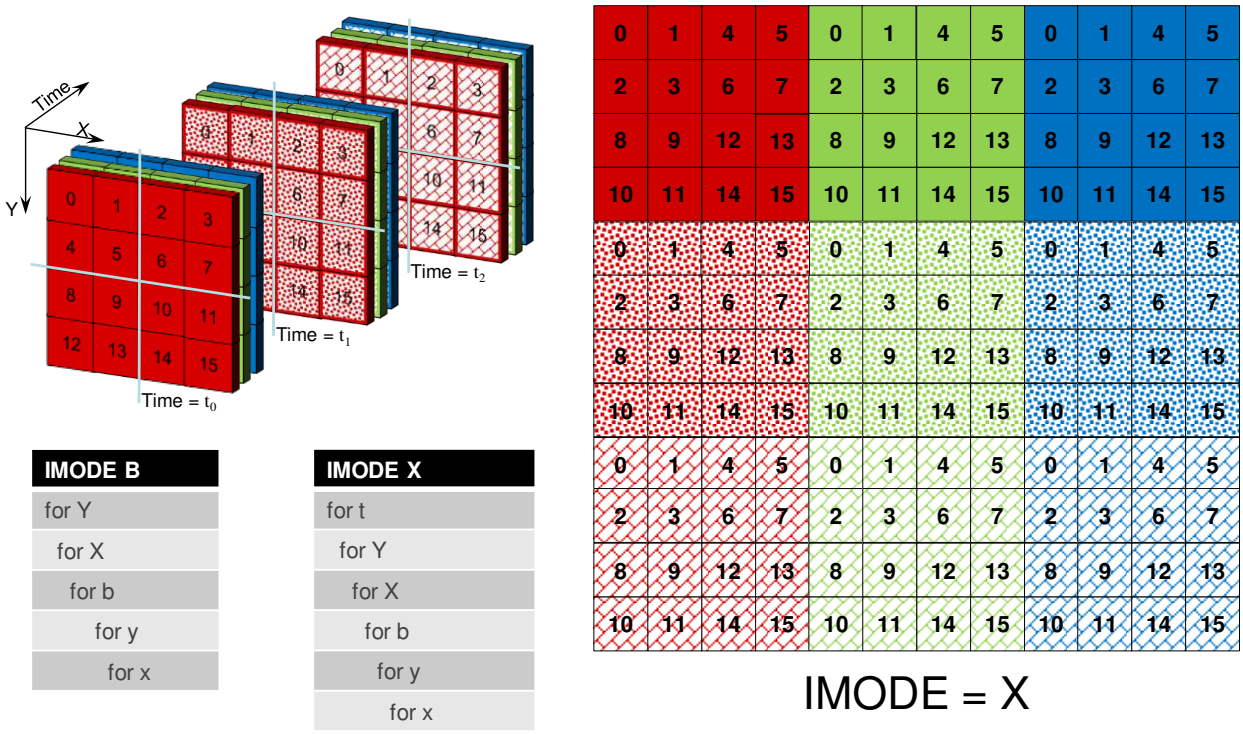


FIGURE F-11. IMODE X Example

F.4.7 IMODE X IMODE X is another MI IMODE based on IMODE B. IMODE X, see FIGURE F-11, wraps a temporal loop around IMODE B. IMODE X is useful for uncompressed MI and it enables usage of JPEG 2000 in the "traditional" IC = C8 or M8 fashion with temporal stacking of JPEG 2000 codestreams. IMODE X is not allowed for use with current MPEG-based MI compression algorithms (i.e. H.264/AVC and H.265/HEVC). The MI algorithms do not fully support tiling within their compressed codestreams.

F.4.8 JPEG 2000 with IMODE T in FIGURE F-12 illustrates the use of JPEG 2000 with the example MI sequence. Each frame or spatiotemporal block is a separately encoded JPEG 2000 codestream. In the case where IMODE T is used with JPEG 2000, each spatiotemporal block within a frame is treated as a separate JPEG 2000 codestream. To enable this capability and differentiate between the still imagery method of using JPEG 2000, the IC field values of "CB" and "MB" were introduced. These field values indicate that each spatiotemporal block within a frame is a distinct JPEG 2000 codestream. This change allows the stacking of multiple JPEG 2000 codestreams within a spatiotemporal block, one for each frame, in temporal order. This is illustrated in FIGURE F-12 under the IMODE T portion of the figure. In the figure, the label "J2K (t_0, 1,1)" indicates the JPEG 2000 codestream for spatiotemporal block (1,1) from the frame associated with time t_0. Note that it is possible to determine the start of each frame's JPEG 2000 codestream without fully decoding all frames in the spatiotemporal block, but it requires searching through the stacked codestreams for a specific two byte sequence (0xFFD9, the JPEG 2000 end of codestream marker, see ISO/IEC 15444-1).

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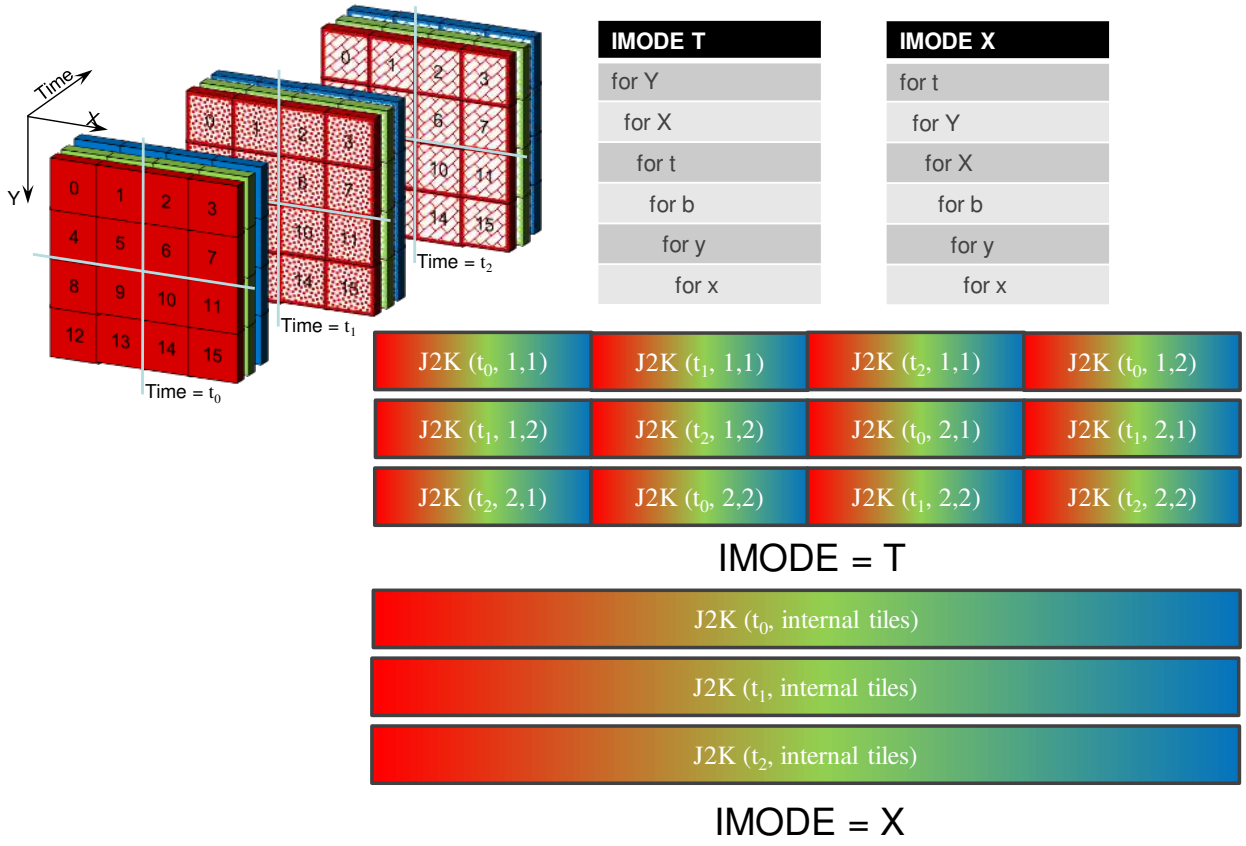


FIGURE F-12. IMODE T and X JPEG 2000 Compressed Codestream Usage

F.4.9 JPEG 2000 with IMODE X The bottom portion of FIGURE F-12 illustrates the use of JPEG 2000 with IMODE X. In this case, the internal tiling mechanism of JPEG 2000 is used to implement NITF blocks. This is analogous to the usage of IMODE B with JPEG 2000 still imagery. There is only one JPEG 2000 codestream per frame and these codestreams are stacked within an image segment as shown. Since the still imagery treatment of blocking is used, the appropriate IC field value would be either “C8” or “M8”.

F.4.10 IMODE E is illustrated in FIGURE F-13. IMODE E is derived from the still imagery IMODE P as shown. A time/frame loop (t) has been added just inside of the spatial block loops (Y, X). IMODE E is used only with uncompressed MI data.

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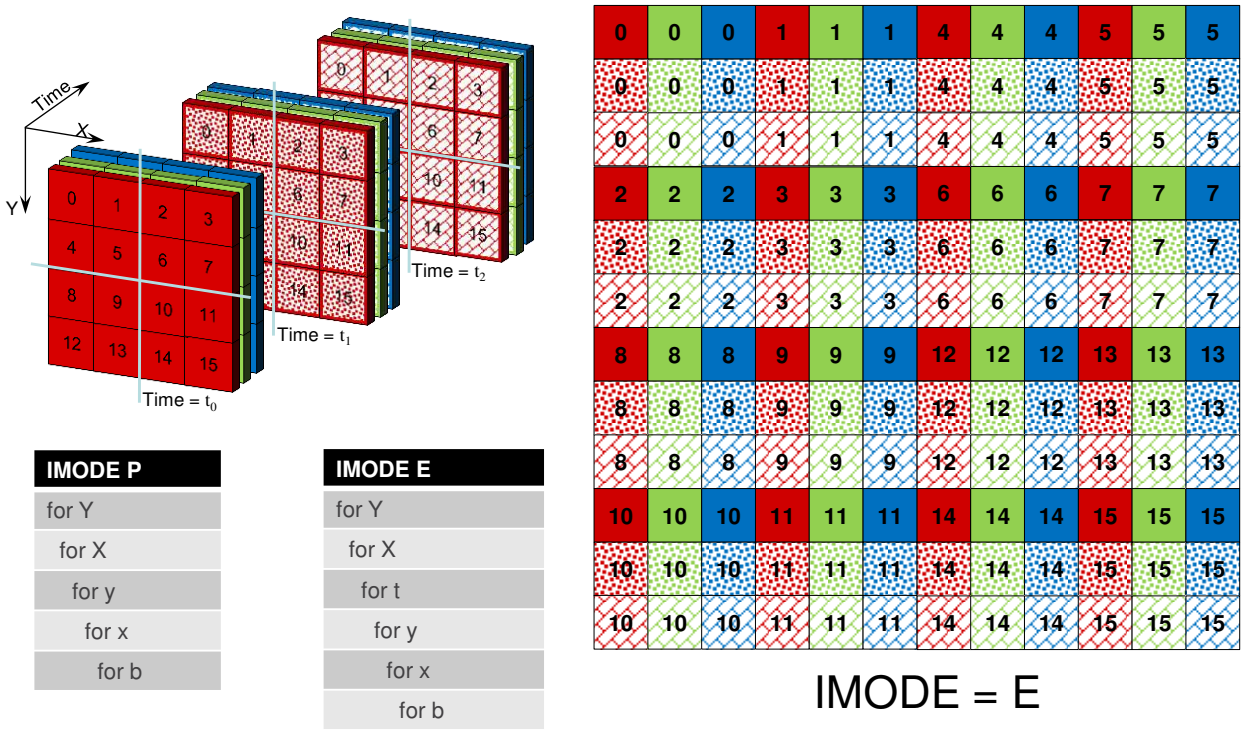


FIGURE F-13. IMODE E Example

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Custodians:
Army - CR
Navy - OM
Air Force – 02
NGA - MP

Preparing activity:
NGA - MP

Review activities:
Army - AC, PT, TM
Air Force - 13
DLA - DH
OASD - SO
OTHER - DC2, NS

(Project GINT-2018-002)

Civil agency coordinating activities:
COM - NIST
DOE
EPA
GPO
HHS - NIH
DOI - BLM, GES, MIN
DOT – CGCT

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